

P2P (PEER TO PEER): In a network, the peers are computer systems which are connected to each other via the Internet. Files can be shared directly between systems on the network without the need of a central server. In other words, each computer on a P2P network becomes a file server as well as a client.

P680 (PHOTOSYSTEM II PRIMARY DONOR): A special form of chlorophyll that is most reactive and best in absorbing light at a wavelength of 680 nm. It is one of the constituents of the photosystem II light-gathering centre of photosynthesis. The absorption of light energy by the P680 molecules causes, among other things, the formation of a strong oxidant that can then oxidise water. This results in the evolution of molecular oxygen and initiates electron flow, which continues, via photosystem I, through to the production of reduced NADP.

P700 (PHOTOSYSTEM I PRIMARY DONOR): A special form of chlorophyll of photosystem I that is most reactive and best in absorbing light at a wavelength of 700 nm. It is found within the photosystem I light-gathering centre of photosynthesis. Other ordinary chlorophyll *a* molecules are also present and these absorb light energy and pass it to P700, creating a charged form, P700+. Electron transfer occurs from photosystem II, electrons being passed on ultimately to produce NADPH, the reduced form of NADP.

PACEMAKER: 1. A group of specialised muscle fibres in the mammalian heart, found in the wall of the right atrium near the opening of the vena cava that send out impulses to regulate the heartbeat. These cells contract spontaneously and rhythmically setting the pace for the contractions of the rest of the heart. The cells themselves are controlled by the autonomic nervous system, which determines the heart rate. 2. A medical device that can be implanted surgically into the chest or worn outside to produce and maintain the heart beat. This device is used if the heart's own pacemaker is defective or diseased.

PACHYCLADOUS: Bearing thick branches.

PACHYTENE: The period in the first prophase of meiosis, when paired homologous chromosomes become short and thick and divide into four distinct chromatids.

PACIFIC OCEAN: The largest division of the hydrosphere and the largest ocean in the world. It covers an area of 166,242,500 sq. km. It extends from the Arctic to Antarctic regions between North and South America on the east and Asia and Australia on the west. It is connected with the Arctic Ocean by the Bering Strait; with the Atlantic Ocean by the Drake Passage, Straits of Magellan and the Panama Canal; and with the Indian Ocean by passages in the Malay Archipelago and between Australia and Antarctica. It occupies about one-third of the surface of the Earth. It is bigger than the total size of the landmass of all the continents put together. Its average depth is 4,188 m.

PACINIAN CORPUSCLES: One of the four major types of mechanoreceptor located deep in the epidermis that detect pressure and vibration. Each consists of a nerve ending surrounded by an oval capsule made of connective tissue, responsible for sensitivity to pain and pressure. The Pacinian corpuscle is oval shaped and approximately 1 mm in length. It has 20 to 60 concentric lamellae composed of fibrous connective tissue and fibroblasts, separated by gelatinous material. The lamellae are very thin, flat, modified Schwann cells. In the centre of the corpuscle is the inner bulb, a fluid-filled cavity with a single afferent unmyelinated nerve ending.

PACKED CELL VOLUME (PCV): The ratio of the volume occupied by packed red blood cells to the volume of the whole blood as measured by a hematocrit. Packed cell volume (PCV) has traditionally been determined by measuring the height of the red cell volume in a micro-haematocrit capillary filled with whole blood, after centrifugation. PCV is a directly measured value, whereas the haematocrit (Hct) is the corresponding calculated value, calculated by multiplying a red blood cell count (RBC count) with the mean erythrocyte volume (MCV). For most practical purposes, PCV and haematocrit are interchangeable

and are frequently used as such by many scientists, but typically PCV is slightly higher than the more accurate haematocrit due to plasma trapping between the packed cells in a centrifuged capillary.

PACKET: In data communication, a sequence of binary digits, including data and control signals that is transmitted and switched as a composite whole. The packets are numbered and transmitted independently and reassembled using their numbers. The data, control signals and, possibly, error control information are arranged in a specific format.

PACKET INTERNET GROPER (PING): In computer science, it is computer network administration software that is built into Windows and UNIX operating systems that can send short messages over a network by one computer to another. It is used by system administrators to test network response time and performance and generally to diagnose problems.

PACKET RADIO: In computing, the use of amateur (ham) radio, instead of telephones to communicate between computers. A terminal node controller (TNC) replaces the modem, a radio transceiver takes the place of the telephone and the phone system is replaced by radio waves. Packet radio, which works on several different frequencies, has a complete network of its own, complete with satellite links and terrestrial relays. It cannot be used on the Internet, but it can be connected through the Internet.

PACKET SWITCHING: A method of transmitting digitally encoded information in a communication network from a source to a destination. The information messages may be divided into smaller entities called **packets**. Each packet contains the address of its origin and destination and information that connects it to the related packets being sent.

PACKHOUSE: A building used for grading, cleaning and packing produce on farm, before it is sent to the customer.

PACKING: 1. In column chromatography, it is the active solid, stationary liquid plus solid support or swollen gel put in the column. It refers to the conditions existing before the chromatographic run is started (i.e. to the material introduced into the column), whereas the stationary phase refers to the conditions during the run. 2. A way to place objects of the same kind so that they touch in some specified way, often inside a container with specified properties. The objects to be packed may be polyhedra, polygons, spheres, ellipsoids, hyperspheres or any other type of shape and the number of dimensions involved may range from two upwards. The fraction of a space filled by a given collection of objects is called the **packing density**. The densest packing of circles in the plane is the hexagonal lattice of the bee's honeycomb, which has a packing density of 0.9069. 3. In computer science, it is a term used to describe the process of bunching information together into one file, causing all the data to occupy less space on the storage device. Once the data has been packed, it can be unpacked back to its original location.

PACKING DENSITY: 1. The quantity of information stored in a specific space of a storage system as characters per inch on magnetic tape. 2. A measure of the extent to which the grains of a sedimentary rock occupy the gross volume of the rock in contrast to the spaces between the grains. It is equal to the cumulative grain-intercept length along a traverse in a thin section.

PACKING FRACTION: The algebraic difference between the relative mass of an isotope and its mass number, divided by the mass number, often interpreted as a measure of stability.

PAD: In computing, to expand a vector, matrix or other quantity in order to bring it to a desired length, dimension or the like. It is usually arranged by adding zeros or empty elements. For example, padding with zeros permits the assumption that a matrix is square of size 2^n .

PADÉ APPROXIMATION: In mathematics, the rational function analogue of the Taylor series derived by expanding a function as a ratio of two power series and determining both the numerator and denominator coefficients.

PAD-PRINTED KEYCAP: A keyboard keycap that has its legend printed, using pads dipped into ink to transfer lettering onto the keycaps and has a protective layer that is baked onto the keyboard. It is one of the main printing technologies used for keyboard keycap printing.

PADDOCK: 1. A small enclosed field usually near a farm building. 2. A small field near a stable or house with grazing for horses.

PADDOCK GRAZING: A rotational grazing system which uses paddocks of equal area for grazing, followed by a rest period.

PAEDIATRICS: The branch of medicine concerned with the care of children. An important aspect is the recognition and treatment of congenital disease in which structural or functional defects occur due to inherited disease such as Down's syndrome or disease acquired during the development of the embryo and foetus such as spina bifida. A medical practitioner who specialises in this area is known as a **paediatrician**.

PAEDOGENESIS: Reproduction by an organism that is still in the larval stage of maturity. Paedogenesis is a form of neoteny (the existence of juvenile features in an adult animal). It is common in insects in which the larval stage reproduces without achieving maturity and occurs in the females of certain beetles, Strepsiptera, bagworms and gall midges. In certain species such as midges, the larvae are essentially born pregnant and reproduce quickly. Often, these insects will devour the mother in the process (matrphagy). It is also found in some amphibians (salamanders).

PAEDOMORPHOSIS (JUVENIFICATION): An alternative term for neoteny, which is the existence of juvenile traits in an adult animal, for example, the retention of gills in salamanders. It is an evolutionary process in which larval or juvenile features of an ancestral organism are retained to the adult forms of its descendants. Paedomorphosis is common in many animal species domesticated by humans, including dogs, chickens, pigs and cattle. It is believed to be a side effect of the selective pressure of human-directed breeding for juvenile behavioural characteristics such as docility. The various types of paedomorphosis include: (i) neoteny, in which somatic (or physical) development is slowed, resulting in a sexually mature juvenile or larval form; (ii) progenesis, in which development is halted before full maturity; (iii) postdisplacement, in which the start of development is delayed.

PAGE: 1. A leaf in a book, magazine, newspaper, manuscript, etc or the writing or printing on one side of a leaf or one or both sides of a sheet of paper in a book. 2. In computer science, it is a quantity of memory storage, which is equal to between 512 and 4,096 bytes. 3. An acronym for **polyacrylamide gel electrophoresis**, a technique used in DNA fingerprinting and other processes to determine the size and composition of proteins. Fragments of DNA are placed on a matrix of polyacrylamide gel and an electric field with negative and positive poles is applied. The protein molecules migrate from the negative pole towards the positive pole, the smaller molecules moving at a faster rate through the pore of the gel. After a time, the fragments have separated enough to form a series of separated lines like a bar code that characterises the DNA. Electrophoresis can be used to identify genes in humans that have already been identified in other organisms and also used to compare the DNA of sample of hair or blood found at crime scenes.

PAGE BREAK (HARD PAGE BREAK): A code inserted by a software program such as Word processor that tells the printing device where to end the current page and begin the next page. After a page break is inserted into a document, a symbol indicating the page break is displayed.

PAGE DESCRIPTION LANGUAGE, PDL: In computing, it is any instruction sent from a computer to a printer to describe the contents and layout of a printed page. PDLs are frequently used to control the operation of laser printers. The most popular PDLs are Adobe Postscript and Hewlett-Packard Printer Control Language.

PAGE FAULT: An interruption that occurs when a software program attempts to access an invalid page in memory in Microsoft Windows. It is caused by a software issue or confliction, driver issue or a hardware related issue.

PAGE FILE (SWAP FILE; PAGING FILE): A file stored on the computer hard drive that is used as a temporary location to store information that is not currently being used by the computer RAM. By using a swap file, a computer has the ability to use more memory than what is physically installed in the computer. Users with low hard drive space may have slower computer speed because of the inability of the swap file to grow in size.

PAGE LAYOUT: The arrangement of text, images and other objects on a page. Page layout techniques are used to customise the appearance of magazines, newspapers, books, websites and other types of publications.

PAGE PRINTER: A printer that prints a complete page of text and graphics at a time. Laser, LED and solid ink printers are examples.

PAGE SETUP: The parameters such as size, margins, page orientation or quality of print defined by the user that help determine how a printed page will appear. The page setup dialogue box is usually available from the File drop down menu.

PAGE-DESCRIPTION LANGUAGE (PDL): A high-level computer language used in computing to describe the contents and layout of a complete printed page, which can be translated into specific codes by any printer that supports the language.

PAGER (BEEPER): A small short-range wireless telecommunications device that receives alert signals or short messages.

PAGET'S DISEASE (OSTEITIS DEFORMANS): A chronic disease caused by excessive breakdown and formation of bone tissue that can cause bones to weaken. It commonly affects the pelvis, lumbar spine (lower back), skull and long bones. It normally occurs in people of over 40 years. Depending on which part of the body is affected, signs and symptoms include bone pain, headaches and hearing loss, damage to cartilage of joints and hip pain. Paget's disease is named after English surgeon James Paget (1814-1899).

PAGINA: The blade of a leaf.

PAGINATION: The process of taking a complete document and breaking that document into separate pages for printing or viewing. In a word processor, pagination is automatically created by creating new pages; to create a new page, a page break is inserted into the document. When viewing a document such as help menu in a console window, a user may experience improper pagination if that window is an abnormal size or if the columns or rows are set improperly.

PAGING: 1. The transfer of program segments (pages) between faster main and slower auxiliary memories to optimise performance. 2. A communications service system, began in the 1980s, which included a one-way beeper, to a one-way numeric text, to a one-way alphanumeric text, to a two-way text service. It is now made obsolete by the cell-phone.

PAIN: Severe physical or mental discomfort or distress associated with a bodily disorder such as a disease or injury. Pain receptors, found in the skin and other tissues, are nerve fibres that react to mechanical, thermal and chemical stimuli. Pain of a physical origin arises through detection of noxious stimuli or tissue injury by specific sensory neurons in the affected part of the body, which relay signals to the thalamus in the brain. However, perception of and susceptibility to pain are determined by higher centres in the cortex and can be influenced by such factors as emotional state. Pain can act as a warning mechanism that helps protect an organism by influencing it to withdraw from harmful stimuli, such as dipping a hand into hot water.

PAINT: Any of the various materials with viscosity, drying time and flowing properties dictated by formulation, normally consisting of a vehicle or binder, a pigment, a solvent or thinner and a drier, which

may be applied in relatively thin layers and which changes to a solid in time. It is used to give a protective and decorative finish to surfaces or for making pictures. Paint can be classified into three main categories, taking into account the performance requirements. They are: architectural paints, commercial finishes and industrial coatings.

PAINT PROGRAM: Graphics software program that allows a user to draw or paint bitmapped images.

PAIR: 1. In mathematics, a set with two members, denoted by $\{a, b\}$. Alternatively, an ordered set with two members, enclosed in angle brackets as $\langle a, b \rangle$. 2. In biology, a male and female animal of the same species that mate with each other or two things of the same kind, usually used together or objects consisting of two parts joined together.

PAIR ATTENUATION COEFFICIENT: The attenuation coefficient, when only the pair production process is taken into account.

PAIR CORRELATION LENGTH: A quantity which approximates to the characteristic length associated with certain number density profiles of diffuse layers. It is defined as the separation between two interacting entities (molecules, macromolecular segments, ions, spins), either in bulk or interfacial regions, beyond which an appropriate function, referred to as pair correlation function, either decays, for example, to an 'initial' value or vanishes.

PAIR DISTRIBUTION FUNCTION (PDF): The probability of finding a second particle as a function of distance from an initial particle. This probability is found from a statistical average over the system concerned. For practical applications, the radial pair distribution function, which is independent of orientation, is a major descriptor for the atomic structure of amorphous materials such as glasses or polymers and liquids.

PAIR PRODUCTION: The simultaneous creation of an electron and a positron from a photon in a strong electric field such as surrounding an atomic nucleus. The electron and the positron, each have a mass of about 9×10^{-31} kg.

PAIRED VECTOR SPACES: In mathematics, two vector spaces X and Y over a given field that are endowed with a bilinear mapping, $(,)$, from $X \times Y$ into the scalar field. Often the space Y is already a dual vector space for X and $\langle x, y \rangle = y(x)$.

PAIRED-SAMPLE PROBLEM: In statistics, any problem requiring a test to be applied to two related samples.

PAIRING (SYNAPSIS): The close association between homologous chromosomes that develops during the first prophase of meiosis. The two chromosomes move together and a synaptonemal complex of proteins forms between them, ensuring exact pairing of corresponding points along their lengths occurs as they lie side by side, resulting in a structure called bivalent.

PAIRING ENERGY: The energy required to pair two electrons in the same orbital. It results in nuclei with odd numbers of neutrons and protons having a lower binding energy and being less stable than nuclei with even numbers of neutrons and protons.

PAIRWISE: In mathematics, taken two at a time from source to a destination; for example, Cartesian coordinate axes are pairwise perpendicular. Also, taking all unordered 2-subsets of a given set. For example, the pairwise products of the set $\{a, b, c\}$ are ab , ac and bc .

PAL: 1. Abbreviation for **Phase Alternation by Line**, which is the coding system for colour television broadcasting developed in Germany and adopted by the United Kingdom, the Netherlands and Switzerland in 1967. It broadcasts 25 interlaced frames per second (50 half frames per second) at 625 lines of resolution. Brazil uses PAL M, which broadcasts 30 frames per second (fps). PAL is used throughout Europe and China as well as in various African, South American and Middle Eastern countries. PAL's colour signals are maintained automatically and the TV set does not have a user-adjustable hue control. 2. **Programmable Array Logic (PAL)** is a type of programmable logic chip (PLD) that contains arrays of programmable AND gates and

predefined OR gates. PALs are defined by their number of inputs and outputs. For example, a 22v10 PAL means 22 inputs and 10 outputs. The inputs are connected by fuses to logic circuits, which themselves are connected by fuses to the output lines. Often used for glue logic, the fuses are blown apart by a programmer device to create the custom configuration. 3. **Paradox Application Language (PAL)** is a programming language used within Borland's Paradox relational database. Paradox is part of Corel's WordPerfect suite.

PALAEOBOTANY: The study of plants through geological time as revealed by their fossil remains such as charred, waterlogged or desiccated seeds, plant fibre, leaves, wood or fruit. It overlaps with other aspects of plant study including anatomy, ecology, evolution and taxonomy.

PALAEOCENE (PALEOCENE): The first geological epoch of the Tertiary Period. It began about 65–56.5 million years ago and marked by the appearance of placental mammals and continental collisions leading to the formation of the Rocky Mountains and the Himalayas. Many types of mammals spread rapidly after the disappearance of the great reptiles of the Mesozoic era.

PALAEOCLIMATOLOGY (PALEOCLIMATOLOGY): The scientific study of the extended climatic conditions of earlier geological periods. This is based largely on the study of sediments and fossils that were laid during these periods and of fossils. Speculative attempts at explanation have come largely from astronomy, atmospheric physics, meteorology and geophysics. The changes in the positions of the continents as a result of continental drift and plate tectonics complicate the study.

PALAEOECOLOGY: The study of the relationships of fossil organisms to each other and to their environments. It involves the study of both the fossils and of the surrounding rocks in which they are found. The origin, development and extinction of species may also be studied in order to test ecological hypotheses.

PALAEOLITHIC: The Old Stone Age, lasting in Europe from about 2.5 million to 9,000 years ago, during which man used primitive stone tools, made by chipping stones and flints. It is sub-divided into the Lower, Middle and Upper phases.

PALAEOMAGNETIC STRATIGRAPHY (MAGNETIC STRATIGRAPHY): The use of the change in the direction of the remanent magnetisation of rocks, caused by reversals in the polarity of the Earth's magnetic field magnetism retained in rocks at the time of their formation to date them.

PALAEOMAGNETISM: The science of the reconstruction of the direction and intensity of the Earth's ancient magnetic field and the former positions of the continents from the increase of remnant magnetisation in ancient rocks (traces left by the earth's magnetic field in igneous rocks before cooling). It is an important tool in unravelling the past movements of the Earth's tectonic plates.

PALAEONTOLOGY (PALEONTOLOGY): The scientific study of extinct organisms including their structure, environment, evolution, taxonomic relationships with each other and with modern species, geographic distribution and interrelationships with the environment as revealed by their fossil remains. Palaeontology has been instrumental in reconstructing the Earth's history and has provided evidence to support the theory of evolution and has also guided petroleum geologists in locating deposits of oil and natural gas, which are frequently associated with the remains of certain ancient life forms.

PALAEOZOIC (PALEOZOIC ERA): The era of geological time between about 570 and 225 million years ago that succeeds the Precambrian and precedes the Mesozoic. It is subdivided into six periods, the Cambrian ordovician and Silurian (lower Palaeozoic) and the Devonian, Carboniferous and Permian (upper Palaeozoic).

PALAEOZOOLOGY (PALEOZOOLOGY): The branch of palaeontology concerned with the study, the recovery and identification of multicellular animal remains throughout geological

time. Palaeontological work also makes important contributions to geology in revealing stratigraphic relationships between rock strata and determining the physical appearance and climate of the Earth during past geological ages and in the reconstruction of prehistoric environments and ancient ecosystems.

PALATE: 1. The roof of the mouth of vertebrates separating the oral and nasal cavities. It consists of the hard palate in the front, which is a plate of bone covered by mucous membrane and the soft palate in the rear, which is made of muscle and connective tissue and ends in the uvula, a fleshy projection. The hard palate gives the tongue a surface against which to make speech, sounds and shape food during chewing and keeps pressures in the mouth from closing off the nasal passage. The soft palate blocks the nasal and upper pharynx off from the mouth and the lower pharynx for swallowing or to create a vacuum for drinking. 2. An appendage that is raised on the lower lip of a bilabiate corolla, which partially or completely closes the throat. This is usually seen in the snapdragon.

PALE: 1. A pointed piece of wood used for fencing. 2. A husk or grass or cereal seed. 3. Lacking colour or intensity or reflecting little light. 4. Unusually light in skin complexion due to illness, shock or worry.

PALE LEAF SPOT: A white spot which forms on leaves of clover plants due to potash deficiency.

PALE, SOFT, EXUDATIVE MEAT: A condition where an animal's meat becomes pale and loses its firmness. It is caused by a rapid postmortem pH decline while the muscle temperature is too high.

PALEA: 1. The upper of the pair of bracts beneath each floret (flower) in the inflorescence of a grass, the other being the lemma. The palea is usually thin, narrow and parallel sided. It usually has two ribs, which may project as prominent keels. 2. The chaffy scales on the receptacle of a flower head found in the family Asteraceae. **Paleaceous** means having chaffy scales.

PALEOGEOGRAPHY: The science that studies ancient geography. It seeks to determine the physical and biological conditions of the Earth throughout geological time. Paleogeography also seeks to determine how conditions in one period led to conditions in the next and succeeding periods. Much of the data needed for good analysis is lacking and most studies in this area are general.

PALEOLA: A minute palea; a lodicule.

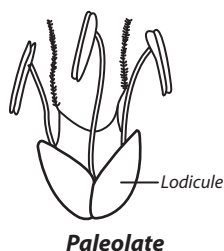
Paleolate means having a lodicule.

PALEOMAGNETISM: The study of past changes in the Earth's magnetic field by examination of rocks containing certain iron-bearing minerals such as haematite and magnetite. Reversals of the field and movements of the magnetic poles can be charted and information on continental drift may be obtained.

PALEOMORPHIC: Having no symmetry.

PALINDROME: 1. A word, phrase, verse, number or sentence that reads the same in either direction, for example, Anna or 24842 are palindromic. 2. A segment of double-stranded DNA in which the nucleotide sequence of one strand reads the same on both strands of DNA. For example, the following sequence: -ACTTGCAAGT- is palindromic because its nucleotide-by-nucleotide complement is TGGATCCA and reversing the order of the nucleotides in the complement gives the original sequence.

PALINDROMIC NUMBER: A number such as 1234321 that reads the same forwards and backwards. More generally, it is a symmetrical number written in some base a as $a_1 a_2 a_3 \dots | \dots a_3 a_2 a_1$. In the base 10 system, there are nine two-digit palindromic numbers: 11, 22, 33, 44, 55, 66, 77, 88, 99; there are 90 palindromics with three digits: 101, 111, 121, 131, 141, 151, 161, 171, 181, 191, ..., 909, 919, 929, 939, 949, 959, 969, 979, 989, 999; and there are 90 palindromics



with four digits: 1001, 1111, 1221, 1331, 1441, 1551, 1661, 1771, 1881, 1991, ..., 9009, 9119, 9229, 9339, 9449, 9559, 9669, 9779, 9889, 9999, giving a total of 199 palindromic numbers below 104. Below 105 there are 1099 palindromics and for other exponents of 10^n there are 1999, 10999, 19999, 109999, 199999, 1099999, etc. With the exception of 11, palindromic primes must have an odd number of digits.

PALISADE CELL: Elongated parenchyma cell of the leaf mesophyll arranged perpendicular to the surface of the leaf. It contains the largest number of chloroplasts per cell in plants, making them the primary site of photosynthesis in a plant's leaves. Palisade cells have a cylindrical shape, allowing a large amount of light to be absorbed by the chloroplasts, which convert the energy stored in photons to chemical energy through photosynthesis.

PALISADE LAYER: Leaf mesophyll parenchyma characterised by elongated form of cells arranged perpendicular to the surface of the leaf. Many plants have leaves with well-differentiated palisade parenchyma on adaxial side of the leaf. Certain plant species such as *Atriplex* have palisade parenchyma on both sides of the leaf, which is termed **isobilateral mesophyll**.

PALISADE MESOPHYLL: Cylindrical cells lying immediately beneath the epidermis of a leaf and at right angles to the leaf surface. They contain a large number of chloroplasts and their principal function is photosynthesis.

PALISOTA: A genus in the family Commelinaceae in the major group Angiosperms (Flowering plants) with about 20 accepted species. One of the species, *Palisota hirsuta* is a robust herb in forest re-growths, growing between 1.2 and 4 high with distinct main stem, large leaves in terminal rosettes and white to purplish flowers. Inflorescences open from 4 p.m. until dusk. It has a blackish glossy fruit. The whole plant and its various parts are used in West Africa for the treatment of several conditions such as for pain-relieving and antiseptic effects and for the treatment of gonorrhoea. Some of the species include *Palisota ambigua*, *Palisota congolana*, *Palisota brachythyrso* and *Palisota laxiflora*.

PALLADIUM: A lightweight, ductile and malleable, silvery white, metallic element with a face-centred cubic crystalline structure having the chemical symbol Pd, atomic number 46, relative atomic mass 106.42, melting point 1,554°C (1827 K), boiling point 2,970°C (3243 K) and specific gravity 12.02 at 20°C. It is strongly resistant to corrosion in air and to the action of acids (except nitric acid) at ordinary temperatures, but is attacked by hot acids. It dissolves in aqua regia. Palladium has the uncommon ability to absorb up to 900 times its own volume of hydrogen at room temperatures. When heated, it allows hydrogen to diffuse rapidly through it; it is thus used to purify hydrogen gas. It is used as a catalyst in alloys of gold to make white gold and silver and in dentistry. It may be alloyed with platinum or substituted for it. It is used in watch bearings, springs and balance wheels and also for mirrors in scientific instruments. In chemistry, it is a catalyst in the manufacture of sulfuric acid and in hydrogenation processes. Palladium salts are used in electroplating. Although palladium is not as abundant as platinum, it is less expensive.

PALLASITE: One of the two main classes of stony-iron meteorite, the other being mesosiderites. Pallasites are igneous in nature and characterised by crystals of olivine, often peridots (clear, green olivine crystals of gem quality), embedded in a matrix of iron-nickel. They are believed to have come from the core/mantle boundary of differentiated asteroids that were broken apart by impact. In most cases, they have chemical, elemental and isotopic features that link them to specific chemical groups of iron meteorites, suggesting that they come from the same parent bodies as these irons.

PALLIATIVE: A medicine that provides temporary relief without removing the cause or something that reduces the harmful effects of something without removing its cause.

PALLID: Pale; feeble or insipid.

PALLIUM (CEREBRAL CORTEX): The grey colour layer of neural tissue that forms the outer layer of the hemispheres of the cerebrum

of the mammalian brain, protecting the brain. It plays very important roles in memory, attention, perceptual awareness, thought, language and consciousness. The human cerebral cortex is 2 - 4 mm thick.

PALM: 1. Any of about 2,800 species of flowering, subtropical trees, shrubs and vines that make up the family Arecaceae or Palmae in about 210 genera. It has an unbranched, cylindrical trunk encircled by scars of fallen leaves, terminated by a crown of large leaves at the apex. When branching does occur, as in *Hyphaene* (dour palm), it is dichotomous. The leaves may be pinnate, as in the feather palms or the leaflets may all arise from the tip of the midrib, as in fan palms. Palm inflorescences are large and may contain thousands of flowers in a panicle or spike. Many palms are of economic importance, examples being the coconut palm (*Cocos nucifera*), the date palm (*Phoenix dactylifera*) and the oil palm (*Elaeis guineensis*). Palm fruits such as coconut, date and palm nuts are edible or give oil valuable as food and in the making of soap. The trunk is used for building and many yield a sweet juice used as beverage. When fermented, the beverage is called palm wine or toddy, which is alcoholic. Palm wine may be distilled into a very potent liquor, known by various local names. 2. The inner surface of the hand between the wrist and the fingers or the underside of a mammal's forefoot that is often in contact with the ground.

PALM COMPUTER (HANDTOP COMPUTER; PALMTOP): Hand-held computers or personal digital assistant (PDA) that combine character recognition screen and pen (called stylus) input instead of a keyboard, although some have a built-in mini-keyboard or a collapsible full-size keyboard. It is used mainly as a personal organiser, for example, as an address book, store personal information and to send and receive e-mail or faxes.

PALM OS: An operating system that provides a software platform for use in handheld personal digital assistants (PDAs).

PALMATE (DIGITATE): 1. A term used to describe a compound leaf having four or more leaflets arising from a single point. The leaves of the horse chestnut (*Aesculus hippocastanum*) and of lupins (*Lupinus*) are examples, as shown in the diagram. 2. Describing a form of venation in which several equally prominent veins branch out from the base of the leaf blade.

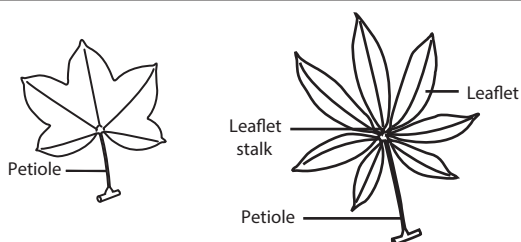


Fig. I: Palmately-lobed, simple leaf

Fig. II: Palmate compound leaf

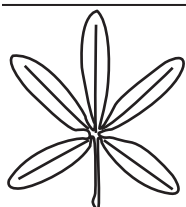
PALMATE-PINNATE: A plant that possesses primary and secondary leaflets. The primary leaflets have nerves that arise from a common or main point and the secondary leaflets are arranged in pairs on either side of a long central midrib.

PALMATIFID: Of leaves, palmately cleft or lobed about halfway to the base, as shown in the diagram.

PALMATISECT: Of leaves, having incisions reaching nearly to the petiole.

PALMELLOID: A term used to describe a type of colonial growth form in which an indefinite number of cells are held together after division in a mucilaginous matrix. Palmelloid forms are characteristic of certain blue-green algae such as *Merismopedia* and various eukaryotic algae.

PALMITATE (HEXADECANOATE ACID): A 16-carbon saturated fatty acid with the chemical formula $\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$. It is the first



Palmatisect

product of lipogenesis. Glycerides of palmitic acid occur widely in plant and animal oils and fats and the major component of palm oil. Retinyl palmitate is an antioxidant and a source of vitamin A and added to low fat milk to replace the vitamin content lost through the removal of milk fat.

PALMITIC ACID (HEXADECANOIC ACID): A common saturated fatty acid with the chemical formula $\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$, density 0.853 g/cm³ at 62 °C (335 K), melting point 63 °C (336 K) and boiling point 351 (624 K). It occurs widely in plant and animal oils and fats. It is a major component of the oil from palm trees such as palm oil, palm kernel oil and coconut oil. It is also found in butter, cheese, milk and meat. It is used in the manufacture of soaps, candles and food additives.

PALOUR SYSTEMS: The four basic designs of milking parlour: (i) the **herringbone parlour** with no stalls and the operator works from a central pit and the cows stand side by side at an angle on each side of the central pit. (ii) The **abreast parlour**, in which the cows stand side by side with their heads facing away from the milker. (iii) The **tandem parlour** in which the cows stand in line with their sides to the milker. (iv) The **rotary parlour** in which the cows stand on a rotating platform with the milker in the middle.

PALPUS (PALP): Any of a pair of elongated often segmented sensory organ, usually located near the mouth in many invertebrates. It is used for sensation, locomotion and feeding. Examples include the distal part of the mandibles of crustaceans, the tactile head appendages of polychaeta worms, the ciliated flap of tissue that produces feeding currents in bivalve molluscs and the olfactory parts of the first and second maxillae of some insects.

PALUDOSE (PALUSTRINE): Thriving or living in marshy places or wet meadows.

PALYNOLOGY: The scientific study of pollen grains and spores, especially pollen found in peat and coal deposits and other organic microfossils. Palynology can be classified into two broad fields: paleopalynology and neopalynology. Pollen and spores are very resistant to decay and, therefore, their fossils are found in sedimentary rocks and palynologists study microscopic bodies generally known as palynomorphs. They are extracted by various methods, such as boiling with potassium hydroxide solution, washing with strong oxidising mixtures and centrifuging repeatedly. Spores and pollen are classified according to shape, form of aperture and both internal and external details of the outer coat called exine. They indicate the nature of the dominant flora and therefore, the climate and conditions of the period in which they lived.

PALYNOMORPH: An organic microfossil consisting of a particle of a size between 5 and 500 micrometres, found in sedimentary rock deposits. Palynomorph is composed of organic materials such as chitin, pseudochitin and sporopollenin.

PAMPAS: A vast plain, covered with grass in temperate south-central South America. It covers an area of more than 750,000 km² from the lower Paraná River to south-central Argentina southwest of Buenos Aires. It is very fertile and supports pastoral and farming activities.

PAMPINUS: A tendril, foliage of a vine. **Pampiniform** means looking like a tendril.

PAN BALANCE: A type of scale used to weigh objects or compare their weights.

PAN BALANCE PROBLEMS: Algebra problems with equations represented by a pan balance. One weight (real or symbolic) represents the variable, and another weight represents a single unit. Exchanges that keep the pans balanced correspond to mathematical operations on both sides of an equation until, eventually, a single variable weight balances with a number of units representing the solution. The pan balance picture provides a visual cue for understanding the purpose of the equal sign.

PAN-BASED CARBON FIBRES: Carbon fibres obtained from polyacrylonitrile (PAN) precursor fibres by stabilisation treatment, carbonisation and final heat treatment.

PANCHROMATIC: Sensitive to all visible colours. For example, a panchromatic film is highly sensitive to all visible colours.

PANCREAS: A composite gland containing exocrine and endocrine cells of the digestive system located between the duodenum and the stomach. Under the influence of the hormone secretin, it secretes pancreatic juice, containing digestive enzymes such as trypsin, chymotrypsin, amylase and lipase into the duodenum through the pancreatic duct. The endocrine cells are arranged in separate islets (the islet of Langerhans), which produce the hormones insulin and glucagon which play roles in metabolism, especially in regulating blood sugar levels.

PANCREATIC ISLETS (ISLETS OF LANGERHANS): Clusters of endocrine cells in the pancreas that secrete insulin and glucagon. They make up about 1 to 2% of the mass of the pancreas. A healthy human adult has about one million islets. Paul Langerhans, a German pathological anatomist, discovered it in 1869 at the age of 22.

PANCREATIC POLYPEPTIDE: A polypeptide hormone secreted by the pancreatic polypeptide (PP) cells of the islets of Langerhans in the endocrine portion of the pancreas. Its release is triggered in humans by protein-rich meals, fasting, exercise and acute hypoglycaemia and is inhibited by somatostatin and intravenous glucose. It increases gastric emptying and gut motility and also relaxes the pyloric and ileocecal sphincters, the colon and gallbladder. PP levels increase after ingestion of food and remain elevated from 4-8 hours. A molecule of pancreatic polypeptide consists of a chain of 36 amino acids.

PANCREATIN: A mixture of digestive enzymes such as amylase, lipase and trypsin that has been extracted from the pancreas of domestic animals such as cattle, hogs and dogs and used as a digestive aid.

PANCREOZYMIN: The former name for **cholecystokinin** (CCK), a hormone of the duodenal mucosa of the small intestine, that stimulates the gall bladder to contract and eject bile into the intestine and stimulates the pancreas to secrete its digestive enzymes, especially amylase.

PANDEMIC: An epidemic infectious disease such as HIV, smallpox, Ebola and tuberculosis, which affects human populations across a large region such as a continent and even across the world.

PANDIAGONAL MAGIC SQUARE (PANMAGIC SQUARE; DIABOLIC SQUARE; DIABOLICAL SQUARE; DIABOLICAL MAGIC SQUARE): A peculiar square in which all the broken diagonals as well as the main diagonals add up to the magic constant. A panmagic square remains a panmagic square not only under rotation or reflection, but also if a row or column is moved from one side of the square to the opposite side. As such, an $n \times n$ panmagic square can be regarded as having $8n^2$ orientations.

PANDIGITAL NUMBER: An integer that contains each of the digits from 0 to 9 and whose leading digit is nonzero. The first few pandigital numbers are 1023456789, 1023456798, 1023456879, 1023456897 and 1023456978. A ten-digit pandigital number is always divisible by 9. If zeros are excluded, the first few nonzero pandigital numbers are 123456789, 123456798, 123456879, 123456897, 123456978 and 123456987 and the first few nonzero pandigital primes are 1123465789, 1123465879, 1123468597, 1123469587 and 1123478659. A **pandigital product** is one in which the digits of the multiplicand, multiplier and product, taken together, form a pandigital number.

PANDORA (SATURN XVII): One of the inner moons of Saturn and the outer shepherd moon for the F-ring. It was discovered in 1980 from Voyager 1 photos. Pandora is heavily cratered and, as revealed by the Cassini probe during its passage of the moon at a distance of 52,000 km in 2005, there are small grooves and ridges in the dust-fine

icy material that has collected over the craters. The two largest craters identified measure about 30 km across.

PANDURATE (PANDURIFORM): 1. Shaped like a fiddle. 2. Having rounded ends and a contracted centre, as shown in the diagram.



Pandurate leaf

PANEL STUDY: A study in which a group of people are interviewed or surveyed with reference to specific topic of interest on different occasions. This study is equivalent to longitudinal study although there may be several response variables observed at each point in time.

PANEMONE: A type of windmill in which a flat surface spins round a vertical axis. It has a rotating axis positioned at 90 degrees to the direction of the wind, while the wind-catching blades move parallel to the wind.

PANGAEA (PANGAEA): A hypothetical supercontinent made up of all the present constituents, believed to have existed between 250 and 200 million years ago. Pangaea broke apart during the Triassic and Jurassic Periods, separating into Laurasia and Gondwanaland.

PANGENESIS: A theory formulated by Charles Darwin that suggested a mechanism by which acquired characteristics may be inherited. He postulated that there were particles (pangenes) carried in the body fluids from all the organs of the body to the reproductive cells. These particles then influenced the gametes that in turn influenced the characteristics of the succeeding generation. The theory has been rejected.

PANGOLIN: Mammals of the genera *Smutsia*, *Uromanis* and *Phataginus* belonging to the order Pholidota, which is native to Africa and Asia. It has horny scales, a long sticky tongue for catching termites and ants, a long tapering snout and tail, toothless, short legs (60-180 cm long) and a weight of 5-27 kg. They are nocturnal animals and locate prey, mainly termites, by smell and rip open nests with their front claws.

PANICLE (COMPOUND RACEME): A type of flowering shoot common in the grass family, including species such as *Panicum maximum* and *Digitaria insularis*. It has an elongated axis and many one-flowered lateral pedicels. The primary axis bears groups of branched racemes, with the youngest groups of flowers at the top and each branch having a smaller raceme of flowers. **Paniculate** means having flowers in panicles. **Paniculiform** means an inflorescence with the appearance of a panicle but not necessarily meeting the strict definition of a panicle.



Panicle inflorescence

PANICUM: A large genus with over 400 species of grasses which are often, annual or perennial plants growing throughout the tropics with a few species extending into the northern temperate region. They can grow up to 3 metres tall with panicles that produce flowers. The flowers are made up of spikelets that develop into fruits. Only the upper floret of each spikelet is fertile whereas the lower one is sterile or staminate making the plant monoecious. Examples include *Panicum maximum* with the common name Guinea grass, a tufted perennial, usually with a shortly creeping rhizome; and *Panicum virgatum* with the common name switchgrass, native to North America prairies used primarily to protect soil from erosion, for forage, as an ornamental grass and may be used for the production of biofuels such as ethanol and butanol.

PANMIXIS (PANMICTIC): Random mating in a population. Mating is entirely random and any two, male and female) breeding individuals are equally likely to mate and assumes that there are no mating restrictions. They display neither genetic nor behavioural characteristics, therefore, all recombination is possible.

PANNAGE: 1. Pasturage in the wood or forest for swine. 2. The corn and beech mast on which pigs feed.

PANNOSE (PANNIFORM): 1. Covered with a short, dense tomentum. 2. Having an appearance similar to felt or woollen cloth.

PANSEXUAL (BISEXUAL; OMNISEXUAL): In humans, it is being romantically or sexually attracted to both males and females.

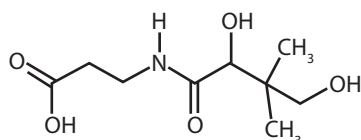
PANTHALASSA: The hypothetical ocean that surrounded the continent, Pangaea, between 250 and 200 million years ago.

PANTOCOLPATE: A term used to describe a pollen grain having many colpi.

PANTOGRAPH: 1. An instrument used in technical drawing and map-making for enlarging or reducing a geometric figure or motion. It consists of four hinged bars forming a parallelogram, one vertex being fixed. 2. A spring-loaded sliding type of current-collector for use on traction systems transferring current from an overhead wire to a vehicle such as an electric locomotive, employing an overhead contact wire. The contact-strip of the collector is mounted on a hinged diamond-shaped structure, so that it can move vertically to follow variations in the contact wire height, without interfering with the smooth passage of the current collector.

PANTONEMATIC FLAGELLUM (TINSEL FLAGELLUM; FLIMMER FLAGELLUM): A threadlike projection arising from motile algal and fungal cells that has longitudinal rows of fine hairs (mastigonemata) along its outer membrane. Internally, the fibrillar structure is typical of a flagellum.

PANTOTHENIC ACID (PANTOTHENATE; VITAMIN B₅): One of the water soluble vitamins of the B complex with the chemical formula C₉H₁₇NO₅. Pantothenic acid is the amide between pantoate and beta-alanine. It is synthesised by plants and bacteria but not by vertebrates, which consequently require it as a vitamin in the diet. It is the precursor of coenzyme A (CoA), which performs a crucial role in the oxidation of fats, carbohydrates and certain amino acids. It occurs in many foods, especially whole-grain cereals, legumes, peas, egg yolk, liver and yeast.



Chemical Structure
of Pantothenic Acid

PAP SMEAR TEST (CERVICAL SMEAR TEST; PAPANICOLAOU TEST): A cancer screening test in which cells scraped from the cervix of the uterus (womb) are examined for abnormality under the microscope.

PAPAIN: A protein-digesting enzyme occurring in the fruit of the papaya (*Carica papaya*). The enzyme is obtained as the dried latex on the skin of the fruit by scratching it, while still on the tree and collecting the flow. It is used as a digestant, in the manufacture of most tenderisers and in medicine for healing wounds. At room temperature, it reacts slowly but increases to maximum activity at 80°C (353 K) and rapidly inactivated at higher temperatures. Therefore, papain tenderises meat only during the early stages of cooking.

PAPAYA PLANT (PAWPAW PLANT; *Carica papaya*): A giant herbaceous plant with a non-woody, hollow green or deep-purple stem that is thicker at the base with fruit and leaf scars that is in the genus *Carica* of the plant family Caricaceae. It can reach 6-9 metres, when fully matured, with a stem diameter of 30-40 centimetres. It has spirally arranged leaves on nearly horizontal, hollow petioles emerging from the stem apex. The blade is deeply divided into 5 to 9 main segments, each irregularly subdivided and varies from 30-60 centimetres in width and has prominent yellowish ribs and veins. The stem and leaves contain copious white milky latex. The flowers are 5-petalled and slightly fragrant, occurring in leaf axils. Papaya

plants are generally dioecious, bearing only short-stalked pistillate or female flowers, waxy and ivory-white. Some bear hermaphrodite or perfect flowers, having both pistils and stamens, ivory-white with bright-yellow anthers, borne on short stalks. Other plants may bear only staminate or male flowers, clustered on panicles. Some other plants are monoecious, producing separate male and female flowers on the same plant. The fruits vary in shape and size depending on the variety, type of plant and flower type. Female flowers are relatively large and rounded at the base, have a stigma but lack stamens and must receive pollen carried by wind or insects to set fruit. Male flowers are thin and tubular, usually borne on a long flower stalk or peduncle; and contain both male and female organs, but the small, vestigial ovary is nonfunctional. Hermaphrodite flowers are intermediate between female and male flowers in size and shape; they are less bulbous than female flowers, but not as thin as male flowers. They have perfect structure with functional stigma and stamens and usually are self-pollinating. Fruits from female flowers are usually oval to round and smaller than those from perfect flowers (they are usually cylindrical or club-shaped). The fruits, called papaya or pawpaw can weigh from 0.5-9 kg each. Young fruits are green and hard, containing white latex. When the fruit matures and ripens to light or dark yellow, the white latex disappears. The flesh of the fruit varies from yellow to orange to red and is thick and juicy, with a central cavity filled with many small black seeds. The ripe or green fruits are eaten as food, rich in nutrients such as Vitamin C, folate, carotenoids and dietary fibres. It may be also processed into juice. The leaves are used to tenderise tough meat and the stem and bark could be used in rope making. Traditionally, the root, unripe fruit and seeds have been used to induce abortion. The seeds are also taken as emmenagogue to induce menstrual flow and as a vermifuge to expel worms from the stomach.

PAPAVERACEOUS: 1. Looking like a poppy. 2. Belonging to the Papaveraceae, a family of plants that have large, showy flowers and a cylindrical seed capsule with pores beneath the lid. Examples are poppy, celadine and bloodroot.

PAPER: Felted or matted sheets of pulp of wood or other fibrous substances, formed on a wire screen from a water suspension and used for printing or writing on.

PAPER CHROMATOGRAPHY: A technique for separating and identifying mixtures in which a liquid sample flows down a vertical strip of adsorbent paper. A spot of the mixture to be investigated is placed near one edge of the paper and the sheet is suspended vertically in a solvent, which rises through the paper by capillarity action carrying the components with it. The components move at different rates, partly because they are absorbed to different extents on the cellulose and partly because of the partition between the solvent and the moisture in the paper. The paper is removed and dried and the different components form a line of spots along the paper. The components can be identified by the distance they move in a given time. Chromatography is used in the separation of pure substances from complex mixtures and is widely used in the analysis of foods, drugs, blood, petroleum products and radioactive-fission products.

PAPILIONACEOUS: An irregular corolla of a distinct shape, which looks like the butterfly, as seen in the diagram. Papilionaceous corolla has five petals which have different names. The upper and larger petal, which is generally wrapped round the rest in the bud is called the standard or banner. It has two side petals called the wings. It also possesses two anterior blades which commonly stick together a little and encloses the stamens and pistils in the flower and they are called keels. This is seen in the flowers of the pea and bean.



Papilionaceous

PAPILLA: Any small nipple-like protuberance projecting from the surface of an organ or organism, for example, on the skin, at the root of a hair or feather, at the base of a developing tooth, on the tongue, on the surface of petals, leaf and stigma. **Papillary** means looking like a papillae. **Papilate** means possessing tiny or minute papillae.

PAPILLOMA: A small solid benign tumour that projects from skin, mucous membranes or glandular ducts. An example is wart.

PAPILLOSE-HISPID: Having stiff hairs that are borne on swollen, nipple-like bases.

PAPOVAVIRUS: Any member of the former virus family of Papovaviridae, now split into the Papillomaviridae and the Polyomaviridae, containing DNA viruses which are double-stranded DNA. They are icosahedral in shape and do not have a lipoprotein envelope and produce tumours in their hosts. Papilloma viruses produce nonmalignant tumours such as warts in all vertebrates and also certain types of cancer such as cervical cancer in humans.

PAPPUS: A modified calyx, often in the form of a ring of silky hairs or feathery hairs, teeth, scales and bristles that surrounds the fruit of plants such as thistles and dandelions and helps to disperse the fruits by wind by acting like a parachute. **Pappiferous (Pappose)** means bearing pappus.

PAPPUS' THEOREMS: Two theorems named after the Greek mathematician, Pappus of Alexandria (320 A.D) that concern the rotation of a curve or a plane shape about a line that lies in the same plane. The first theorem states that the surface area generated by a curve revolving about a line that does not cross it, is equal to the length of the curve times the circumference of the circle traced out by its centroid. The second theorem states that the volume of a solid of revolution generated by a plane area that rotates about a line not crossing it is equal to the area times the circumference of the circle traced out by the centroid of the area.

PAPYRACEOUS: Having a papery texture and colour.

PARA: 1. A prefix denoting the form of diatomic molecules in which the nuclei have opposite spins, for example, para-hydrogen. 2. An isomer, modification, polymer, derivative, etc, of a specified substance, for example, paraformaldehyde. 3. A prefix usually italicised and hyphenated in chemical names for designating a benzene compound characterised by substitutions in the 1,4 positions (opposite sides in the benzene ring). Examples are *p*-xylene is 1,4-dimethylbenzene, *para*-isomers of benzene ring means 1,4 - substitution and *para*-aminobenzoic acid.

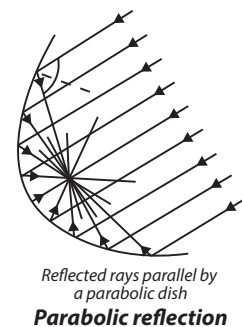
PARABOLA: A symmetrical open plane curve formed by the intersection of a cone with a plane parallel to its side. It is defined formally as a set of all points in a plane that is equidistant from a fixed point, called the **focus** and a fixed line called the **directrix**. The focus may not be on the directrix. The line through the focus and perpendicular to the directrix is called the **axis of symmetry** or **axis** of the parabola. The line segment through the focus and perpendicular to the axis of symmetry, which is cut by the parabola, is called the **focal chord** and its length is called the **focal width**. The **vertex** is the point of intersection of the axis of symmetry and the parabola on the X-axis. The graph of any quadratic equation is a parabola of the form: $y = ax^2 + bx + c$. Any line parallel to its axis of symmetry reflects through its focus and vice versa. Rotating a parabola about its axis produces a surface, called a paraboloid with the same reflection property and therefore, is an ideal shape for satellite dishes and reflectors in headlights. Parabolas occur naturally as the paths of projectiles such as a bullet or rocket. The shape is also seen in architectural designs such as bridges and arches.

PARABOLIC: 1. Of, relating to or shaped like a parabola or paraboloid. 2. In mathematics, of a second order partial differential equation, having discriminant $b^2 - 4ac$ zero, where

$au_{xx} + bu_{xy} + cu_{yy} + du_x + eu_y + fu = h$, is the general form of a second order partial differential equation.

PARABOLIC ORBIT: An open orbit around a central mass which is followed by an object that, at any point on its path, has the minimum velocity, which is needed to escape from the gravitating mass.

PARABOLIC REFLECTOR (PARABOLOIDAL REFLECTOR): A dish antenna having a concave reflective surface, which is generated either by translating a parabola perpendicular to the plane in which it lies or rotating it about its axis of symmetry. It is used to reflect electromagnetic radiation for such purposes as telescopes and satellite communication. The diagram shows a parabolic reflection.



PARABOLIC SPIRAL (FERMAT'S SPIRAL): In mathematics, it is the curve traced by a point moving such that the square of its distance from the pole is proportional to its polar angle. Its equation in polar coordinates is $r^2 = k\theta$.

PARABOLIC VELOCITY: The velocity at which an object at a given point would require in order to describe a parabola about the centre of attraction. This is also the escape velocity since it is the upper limit of velocity on a closed curve. Smaller values give an ellipse, larger values give a hyperbola. Parabolic velocity is obtained by multiplying the velocity of an object moving in a circular orbit by the square root of 2, which is approximately 1.414. For example, if the Earth's mean velocity around the Sun of 29.8 km/sec, in a near circular orbit, were increased to 42 km/sec, the Earth would move to a parabolic path and escape from the solar system.

PARABOLOID: The solid formed by rotating a parabola about its axis of symmetry. There are two kinds of paraboloids: **elliptic** and **hyperbolic**. The **elliptic paraboloid** is shaped like an oval cup and can have a maximum or minimum point. In a suitable coordinate system with three axes x , y and z , it can be represented by the equation

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}, \text{ where, } a \text{ and } b \text{ are constants that dictate the level of}$$

curvature in the $x - z$ and $y - z$ planes respectively. This is an elliptic paraboloid which opens upward. The **hyperbolic paraboloid** is a doubly ruled surface shaped like a saddle. In a suitable coordinate system, a hyperbolic paraboloid can be represented by the equation

$$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}. \text{ This is a hyperbolic paraboloid that opens up along the}$$

x -axis and down along the y -axis.

PARACELULAR PATHWAY: The route between cells that lack any means of active transport and substances move passively by simple diffusion. Substances can travel through the epithelia by a paracellular pathway if the tight junctions between constituent cells are not continuous, as in proximal tubule of a kidney nephron.

PARACETAMOL [ACETAMINOPHEN; 4-ACETAMIDOPHENOL; N-(4-HYDROXYPHENYL)ETHANAMIDE; N-(4-HYDROXYPHENYL)ACETAMIDE]: A synthetic analgesic drug similar to aspirin having the chemical formula $C_8H_9NO_2$, molecular mass 151.163 g/mol, density 1.263 g/cm³, melting point 169 °C (442.15 K) and boiling point of 420 °C (693.15 K). It is effective for relieving pain such as headache, muscle pain and lower back pain; and also used in reducing fever. It has weak anti-inflammatory properties. An overdose can cause severe, often irreversible or even fatal liver and kidney damage.

PARACHUTE: Any canopied fabric (such as canvas, silk and nylon) device, made by sewing separate panels together to form a canopy attached by suspension lines to a harness strapped to a person or

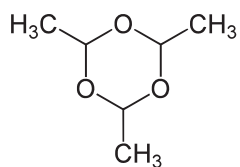
a package, which is used to slow down descent from an altitude or returning spent missiles or parts to a safe speed for landing or sometimes to aid the landing of a plane or missile.

PARADOX: A statement that seems to lead to a logical self-contradiction or to a situation that contradicts common intuition. The identification of a paradox based on seemingly simple and reasonable concepts has often led to significant advances in science, philosophy and mathematics.

PARAFFIN: 1. Also known as **kerosine**, it is a clear liquid with a density of 0.78–0.81 g/cm³ and boiling point of about 300 °C. It is immiscible in water, but miscible in petroleum solvents. It is made by the fractional distillation of petroleum. It is used as a fuel for jet engines and rockets, for heating and in lamps as fuel. 2. Also known as **paraffin hydrocarbon** or **alkane**, it is any member of the alkane series having the general formula C_nH_{2n+2}, where *n* is an integer between 22 and 27. Paraffins containing fewer than 5 carbon atoms per molecule are usually gaseous at room temperature, those having 5 to 15 carbon atoms are usually liquids and the straight-chain paraffins having more than 15 carbon atoms per molecule are solids. 3. Also called **paraffin wax**, it is a flammable, whitish, translucent, waxy solid consisting of a mixture of saturated hydrocarbons, obtained by distillation from petroleum or shale and used in candles, cosmetics, polishes and sealing and waterproofing compounds.

PARAKERATOSIS: A skin disease similar in appearance to mange caused by a deficiency of zinc in the diet.

PARALDEHYDE (ETHANAL TRIMER; 2,4,6-TRIMETHYL-1,3,5-TRIOXANE): A colourless flammable liquid with a pungent odour and unpleasant taste having the chemical formula C₆H₁₂O₃, molar mass 132.16 g mol⁻¹, melting point 12 °C (285 K) and boiling point 124 °C (397 K). It is produced by treating acetaldehyde with a small amount of sulfuric acid. It is slightly soluble in water and highly soluble in alcohol. It is used as a sedative and hypnotic drug (the safest hypnotic drug); in the manufacture of synthetic resins; as a preservative; and in preparing leather.



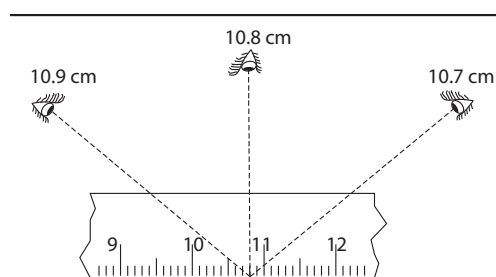
Chemical Structure of Paraldehyde

PARALLACTIC ANGLE (POSITION ANGLE): The direction of an imaginary arrow in the sky, measured in degrees from north through east; in other words, the angle between a body's hour circle and its vertical circle. Applied to a binary star system, it is the angle between the primary and secondary components of a binary system. Applied to a galaxy, it measures the object's tilt, that is, the angle between the galaxy's long axis and a line from its centre headed north.

PARALLACTIC DISPLACEMENT (PARALLACTIC MOTION): The apparent change in a star's position caused by the movement of Earth in its orbit. Over the course of a year, a star appears to follow a small path in the sky called a parallactic ellipse or a parallactic orbit, the shape of which ranges from a circle for a star at the ecliptic pole to a line for a star on the ecliptic.

PARALLAX: 1. An effect used by astronomers to measure distances to nearby stars. It is the apparent displacement of an object because of a change in the observer's point of view, when viewed from two different positions. Parallax can be used to measure the distances to nearby stars. 2. The angular displacement in the apparent position of a celestial body, when observed from two different points. With a baseline of known length between two observations, the distance to the object can be determined directly.

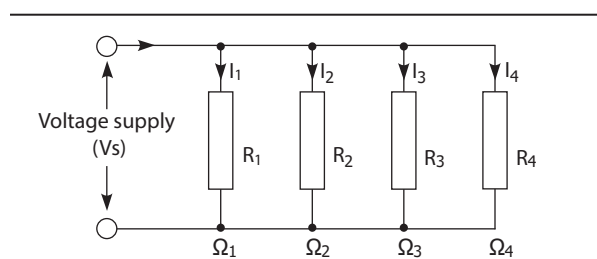
PARALLAX ERROR: The random error that results from identifying the position of a point, when observing it obliquely. Systematic error is the error due to the poor arrangement of an apparatus.



Parallax Error

PARALLEL: 1. Two or more lines having the same perpendicular distance between them at every point and never meeting or intersecting; or two lines with the same gradient. 2. Two vectors such as parallel forces are parallel if they extend in the same or opposite direction and if one is a scalar multiple of the other.

PARALLEL CIRCUIT: An electrical circuit in which the current is split between two or more parallel paths or conductors. In a parallel circuit, the total resistance, *R*, is given by $\frac{1}{R} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \dots$, where *r*₁, *r*₂ and *r*₃ are the resistances of the individual elements. For capacitors in parallel, the total capacitance, *C* is given by $C = c_1 + c_2 + c_3 \dots$. In a parallel circuit the potential difference (voltage) across each component is the same but the current through each branch of the parallel circuit may be different, depending on the power rating of the component.



Parallel Circuit

PARALLEL COMPUTING: The use of multi processors to perform a common task. This is achieved by dividing the task into units with each processor working on its section of the task.

PARALLEL EVOLUTION: The development of related organisms separated geographically along similar evolutionary paths due to strong selection pressures acting on all of them in the same way. An example is the similarity between the marsupial mammals of Australia and placental mammals elsewhere, which have arrived at remarkably similar forms through the separate courses of their evolution.

PARALLEL LINES AND PARALLEL PLANES: Straight lines or planes that do not intersect or always maintain a constant distance from one another no matter how far they are extended. Its symbol is ||.

PARALLEL OF LATITUDE (CIRCLE OF LATITUDE): A series of imaginary east-west running circles north and south of the equator, which are parallel to it and become smaller in circumference, the farther they are from the equator and closer to the poles.

PARALLEL PLANES: Two planes in space that do not intersect and are always the same distance from each other. Two planes that are not parallel always intersect in a line. One-dimensional (1-dimensional) shapes such as lines, segments, and rays in one plane are not necessarily parallel to 1-dimensional shapes in a parallel plane.

PARALLEL PLATE CAPACITOR: A device for storing charges that is constructed from two parallel conducting plates, placed a short distance apart with an insulating material called dielectric of permittivity *e* filling the space between them. The plates of a charged parallel capacitor each carry charges of the same size but of opposite

sign. The capacitance, C is given by: $C = \frac{k\epsilon_0 d}{A}$, where k_e is the dielectric material, d is the distance between the plates, and A is the area of the plates.

PARALLEL PORTS: 1. A hardware interface that communicates binary data to another device by sending the bits that represent each character simultaneously along a set of separate data lines, unlike a serial device. PC parallel ports are usually designated LPT1, LPT2, etc. 2. A socket on a computer used to connect a printer or other device via a parallel interface (eight data bits transferring simultaneously). It is now superseded by USB.

PARALLEL POSTULATE (EUCLID'S FIFTH POSTULATE): In mathematics, the axiom of Euclidean geometry stating that if two straight lines are cut by a third, the two will meet on the side of the third on which the sum of the interior angles is less than two right angles.

PARALLEL PROCESSING: A technique that allows more than one computation or process-stream of activity to be running at speeds far exceeding those of a conventional computer at any given moment in a computer system, hence processes can be executed in parallel. This means that two or more processors are active among a group of processes at any instant and access data through shared memory.

PARALLEL SECTION: In mathematics, a section of a surface of revolution, such as a paraboloid, that is perpendicular to the axis of revolution.

PARALLEL SPINS: Neighbouring spinning electrons in which the spins and hence the magnetic moments, of the electrons are aligned in the same direction.

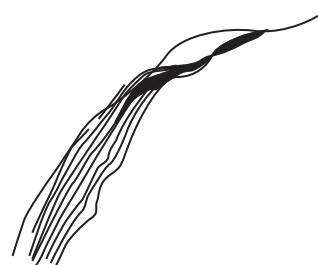
PARAMETRIC STATISTICS: The branch of statistics concerned with data measurable on interval or ratio scales, so that arithmetic operations are applicable to them, enabling parameters such as the mean of the distribution to be defined.

PARALLEL-CHAIN CRYSTAL: A type of crystal resulting from parallel packing of stems, irrespective of the stems' directional sense.

PARALLELEPIPED (PARALLELOPIPED): A polyhedron with six faces bounded by three pairs of parallel planes, so that all its faces are parallelograms. It is also a prism, whose base is a parallelogram. A rectangular parallelepiped has the shape of a cuboid.

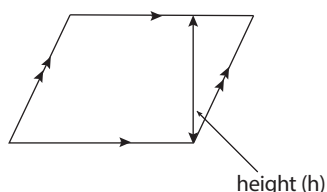
PARALLELODROMOUS (PARALLEL-VEINED):

Of a leaf venation, having two or more primary veins originating at the base of the lamina, which run to the apex in an essentially parallel manner. The leaves of many monocotyledons have this type of venation.



Parallelodromous leaf

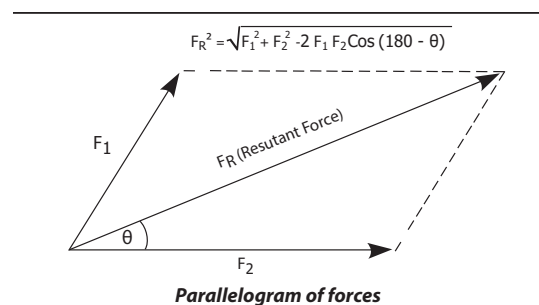
PARALLELOGRAM: A quadrilateral (four sided plane figure) with opposite pairs of sides equal in length and parallel. The opposite angles are equal, with the diagonals bisecting each other. The rhombus, rectangle and square are special types of parallelograms. The area (A) of a parallelogram is equal to the product of the lengths of its base and altitude: $A = b \times h$, where b is the base and h is the height.



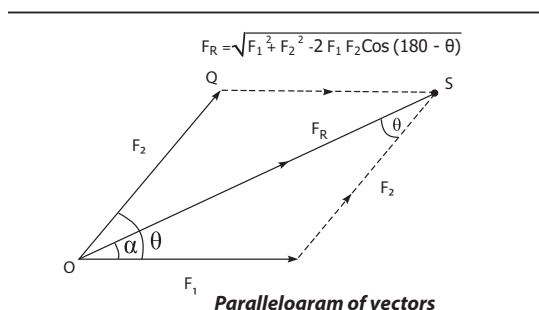
Parallelogram

PARALLELOGRAM LAW: 1. In mathematics, the equation that relates norms in an inner product space, expressed as $\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$, for all vectors x and y . 2. Also known as **parallelogram rule**, it permits vector addition of two vectors in two-dimensional space by constructing a parallelogram with the two adjacent sides representing the magnitudes and directions of the vectors and the resultant vector is the diagonal through the point of intersection of the vectors. This rule is applied in parallelogram of forces to determine the resultant force where two forces act on a particle at a point.

PARALLELOGRAM OF FORCES: A method for solving or visualising the results of applying several different forces to an object. In physics and applied mathematics, it is a method where two forces acting on a particle at a point can be represented both in magnitude and direction by the sides of a parallelogram taken from that point and are equivalent to a single force or resultant represented in magnitude and direction taken from that point as shown in the diagram below. Forces, being vectors are observed to obey the laws of vector addition.



PARALLELOGRAM OF VECTORS: A method of determining the resultant of two vector quantities. The two quantities are represented by two adjacent sides of a parallelogram and the resultant is the diagonal through their point of intersection. The method can be used either with careful scale drawing or with trigonometry. The trigonometrical relations is expressed as $F = \sqrt{F_1^2 + F_2^2 + 2F_1 F_2 \cos \theta}$ or $\sqrt{F_1^2 + F_2^2 - 2F_1 F_2 (180 - \theta)}$, where θ is the angle between F_1 and F_2 .



PARALLELOGRAM OF VELOCITIES: A method of calculating the resultant of two velocities acting on a body. If a body has two component velocities, represented in magnitude and direction by two adjacent sides of a parallelogram drawn from a point, the resultant velocity of the body is represented by the diagonal of the parallelogram drawn from that point.

PARALLELOPHYLY: The independent emergence of the same trait in two or more lines with a close common ancestor because of a genetic predisposition inherited from the common ancestor. Such a phenomenon could account for certain instances of apparent homoplasy. For example, the Lake Tanganyikan cichlid fish developed the same feeding method in six different lines.

PARALOGOUS GENE (PARALOGOUS LOCUS): Homologous genes (a matched pair of chromosomes) at different chromosomal locations

in the same organism that have structural similarities indicating that they are derived from a common ancestral gene and have since diverged from the parent copy by mutation and selection or drift. The copies evolve side by side in a genome of subsequent lineages, irrespective of speciation events and hence display similarity of base sequence and possibly, function. The class of homeotic genes known as Hox genes consists of 13 highly conserved paralogues, some of which have been identified in organisms as diverse as nematodes and mice.

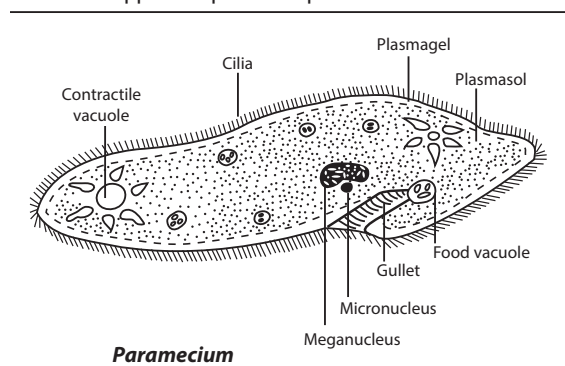
PARALYSIS: The loss or impairment of the voluntary use of one or more muscles either through the use of a drug or by damaging muscle or nerve function. Paralysis may be flaccid, in which muscle tone is lost or spastic (stiff). **Hemiplegia**, a one-sided paralysis is usually caused by stroke or brain tumour on the opposite side. **Diplegia**, a two-sided paralysis, results from generalised brain disease. An example of diplegia is cerebral palsy.

PARALYSIS AGITANS (PARKINSONISM; PARKINSON'S DISEASE): A chronic and progressive movement disorder disease of the nervous system characterised by a progressive loss of mobility, muscular rigidity, tremor and speech difficulties. It is caused by the gradual breakdown or death of brain cells that make dopamine, a chemical messenger in the brain that helps neurons or nerve cells control movement. The condition affects mainly people over the age of 50. It was first described by British neurologist, James Parkinson (1755-1824) in 1817.

PARAMAGNETIC MATERIAL: A material which becomes magnetised when an external magnetic field is applied in a similar way to ferromagnetic materials but which loses its magnetism when the external magnetic field is removed. Examples of such materials are aluminium and platinum. The permeability of a paramagnetic material is greater than 1. When the permeability is less than 1, the material is a diamagnetic material. Paramagnetic materials usually contain transition metals or rare earth elements that possess unpaired electrons. In non-metallic substances, induced magnetic moment varies inversely to the temperature.

PARAMAGNETISM: A property exhibited by substances in which the atoms or molecules have net orbital or spin magnetic moments that are capable of being aligned in the direction of the applied field. Paramagnetism occurs due to unpaired electrons in the atomic or molecular orbitals. Examples of paramagnetic substances are organic free radicals and gaseous nitric oxide, transition, rare-earth and actinide elements and many of their salts and molecular oxygen and organic biradicals.

PARAMECIUM: A single-celled microscopic organism or protozoan in the genus *Paramecium*, of the ciliate phylum Ciliophora. It is found in freshwater all over the world. It is covered with cilia that it uses to swim in water. The paramecium has a stiff outer covering that gives it a permanent slipper shape called pellicle.



PARAMETER: 1. In mathematics, a variable factor or characteristic, for example, length is one parameter of a rectangle and its base is another. 2. A constant in an equation that varies in other equations

of the same general form, especially such a constant in the equation of a curve or surface that can be varied to represent a family of curves or surfaces. 3. In statistics, a quantity, such as a mean, that is calculated from data and describes a population. 4. Any value passed to a programme by the user or by another programme in order to customise the programme for a particular purpose. Examples are a file name, a coordinate, a range of values and an amount of money. 5. A value passed to a subroutine or function for processing. Programming today's graphical applications in languages such as C, C++ and Java requires knowledge of hundreds, if not thousands, of parameters. 6. One of a set of measurable factors, such as temperature and pressure that define a system and determine its behaviour and are varied in an experiment.

PARAMETER RAM (PRAM): A small, battery-powered section in Apple Macintosh computers used to store basic settings such as the time and the data. PRAM is similar to IBM compatible computers' CMOS.

PARAMETER VARIABLE: A variable in terms of which other interrelated variables can be expressed.

PARAMETERISATION/PARAMETERIZATION: 1. The process of describing a phenomenon, problem, curve, surface, etc., by the use of parameters. 2. In computing, it is the representation of physical effects by simplified parameters in a computer model rather than by computing them dynamically. 3. In climate models, it is a method of representing processes, that cannot be explicitly resolved at the spatial or temporal resolution of the model (sub-grid scale processes), by relationships between the area- or time-averaged effect of such sub-grid-scale processes and the larger scale flow.

PARAMETRIC AMPLIFICATION: Amplification of a signal based on parametric processes.

PARAMETRIC EQUATION: 1. An equation of a curve expressed in the form of the parameters that locate points on the curve. For example, the parametric equation of a parabola is given by $x = at^2$, $y = 2at$. 2. An equation in which the coordinates are individually written as functions or parameters.

PARAMETRIC PROCESSES (OPTICAL PARAMETRIC PROCESSES): Interaction of photons in a non-linear medium satisfying the energy and momentum conservation laws. During the interaction, the photon frequencies are mixed and photon(s) with different frequencies are created. Optical parametric processes of three interacting beams, called frequency mixing processes, are: second-harmonic generation (SHG), sum-frequency generation (SFG), difference-frequency generation (DFG), optical parametric generation and amplification (OPG and OPA) and optical parametric oscillation (OPO).

PARAMORPH: 1. A variant form within a species for which a more specific description does not exist because of inadequate data and its taxonomic status cannot be determined. 2. A crystal that is chemically identical to the original crystal, but the atoms have been rearranged so that they no longer conform to the original outline. An example is the change of brookite to rutile or of aragonite to calcite.

PARAMYLUM: A polysaccharide that is the assimilation product of both green and colourless algae of the Euglenophyta.

PARANASAL SINUS: Any of the paired sinuses in the bones of the face adjacent to the nasal cavity that are lined with mucous membrane that is continuous with the lining of the nasal cavities. In humans, there are four groups, which include the frontal, ethmoid and sphenoid sinuses. Each group of sinuses takes its name from the bone of the skull in which it is found.

PARANOIA: A form of schizophrenia characterised by a slowly progressive deterioration of the personality, involving delusions and often hallucinations. It also refers to a mental disorder characterised

by any of several types of delusions, in which the personality otherwise remains relatively intact. The delusions may form a self-consistent system which replaces reality.

PARANORMAL: Denoting events or phenomena that cannot be explained by established science. Examples include extrasensory perception (ESP) which takes in clairvoyance, precognition and telepathy; telekinesis, the movement of objects from one position to another by human mental concentration; and mediumship, supposed contact with the spirits of the dead, usually via an intermediate 'guide' in the other world. **Parapsychology** is the study of such phenomena.

PARAPATRIC SPECIATION: Organisms, whose ranges do not significantly overlap but are immediately adjacent to each other but live in environmentally different habitats, in which there is free exchange of genes between the two. Although individuals from the two populations can interbreed, the offspring is not very successful in either habitat.

PARAPHYLETIC: Of or relating to a taxonomic group of organisms that do not contain all of the descendants of a particular single common ancestor. For example, the taxonomic group 'reptiles' is paraphyletic because it excludes birds and mammals, which share the same common ancestor as the reptiles. In evolutionary systematic, paraphyletic groups or evolutionary grades, are sometimes permitted in order to reflect biological similarities.

PARAPHYSIS: Any of the sterile filamentous hyphal end cells that are found around the sex organs of mosses, brown algae and certain fungi and not sufficiently differentiated into specialised, swollen, often protruding cells to be called cystidia.

PARAPLEGIA: Paralysis involving the lower part of the body, particularly the legs. Injury to the spinal cord is often the cause.

PARAPSYCHOLOGY (PSYCHIC RESEARCH): The study of the evidence for psychological phenomena, such as telepathy, clairvoyance and psychokinesis, which are inexplicable by science.

PARAQUAT: The trade name for *N,N'*-dimethyl-4, 4'-bipyridinium dichloride, a quick-acting and non-selective herbicide that destroys a wide range of plants by killing their foliage and becoming inert on contact with the soil. It is poisonous to humans and mammals. It is one of the most widely used herbicides in the world.

PARASEXUAL CYCLE: A reproduction process that occurs in fungi and single-celled organisms in which genes of haploid nuclei are transferred without meiosis.

PARASEXUAL RECOMBINATION: A type of reproduction occurring in vegetative or somatic cells or in the progeny of certain heterokaryotic fungi that results in genetic recombination, without the formation and fusion of gametes or which does not involve meiotic segregation. The process is carried out in three steps: **heterokaryosis**, which is the fusion of two haploid nuclei in the heterokaryon to yield heterozygous diploid nuclei; **recombination** and **segregation** at mitosis by two independent processes (mitotic crossing-over and loss of chromosomes). The asexual spores produced during the cycle will be genetically different from the original parent mycelium as a result of the crossing over that has occurred.

PARASITE: An organism that lives on or in another organism (called the host) and depends on it for nutrition, often at the expense of the host's welfare. Examples of parasites are helminth, arachnid, crustacean (copepod), arthropod parasites, multicelled and single-celled animals, fungi and bacteria. Viruses are sometimes considered to be parasites. The parasites of humans include fleas and lice, various bacteria, protozoans and fungi and tapeworms (*Taenia solium*) which live in the gut. **Ectoparasites** such as lice live on the body surface of the host. **Endoparasites** such as tapeworms live in their hosts' organs, tissues or cells and often rely on a third organism, the carrier or vector, to transmit

them to the host. Some parasites cause little harm to the host, but many cause diseases, which may not be immediately fatal to the host as the parasite may lose a feeding ground if the host dies.

Certain organisms such as cuckoo and cowbird practice brood parasitism, laying eggs in other birds' nests to be raised by the foster parents. Obligate parasites can only survive and reproduce as parasites. Facultative parasites can also live as saprotrophs. On the other hand, an Epiphyte (Air plant) is a plant which is one of the subdivisions of the Raunkiaer system, that lives and grows on another plant, such as a tree, for physical support, but is neither parasitic on it nor rooted in the ground or soil. It sometimes grows upon objects such as buildings or telegraph wires, derives its moisture and nutrients from the air and rain and sometimes from debris accumulating around it. It is found in the tropics (many ferns, cacti orchids and bromeliads) and in the temperate zones (many mosses, liverworts, lichens and algae). There are many aquatic species of algae, including seaweeds that are also epiphytes on other aquatic plants (seaweeds or aquatic angiosperms). **Parasitism** is an intimate relationship in which one organism called the parasite lives on or in the body of another (the host).

PARASITIC CAPTURE (PARASITIC ABSORPTION): The absorption of a neutron by a nuclide that does not result in either fission or a useful artificial element.

PARASITIC GASTRO-ENTERITIS: Acute infection of the stomach lining and intestines. It is caused by microorganisms such as roundworm secreting toxins or by invading the gut walls. Some of the symptoms are diarrhoea, vomiting and abdominal cramps.

PARASITOID: An organism that is parasitic only at a certain stage of its development. It lives in (endoparasites) or, on (ectoparasite) another organism, which it consumes and eventually kills in the process, for example, hymenopterans (the wasp, sawfly, ant, etc.) and some flies which are parasitoids for part of their life cycle, laying their eggs commonly in or on the larvae of other insects. The hatched parasitoids use the tissues of their host to obtain nutrition during development into free-living adults.

PARASITOLOGY: The scientific study of parasites and of parasitism.

PARASYMPATHETIC NERVOUS SYSTEM: Part of the autonomic nervous system originating in the brain stem and the lower part of the spinal cord. Its nerve endings release acetylcholine as a neurotransmitter and its actions tend to antagonise those of the sympathetic nervous system. For example, its actions may constrict the pupils, while the action of the sympathetic nervous system may dilate it, stimulate digestive secretions and slow the heart.

PARATHION [PARATHION-ETHYL; DIETHYL PARATHION; O,O-DIETHYL O-(4-NITROPHENYL) PHOSPHOROTHIOATE]:

A colourless and highly toxic organophosphorous insecticide, having the chemical formula, $C_{10}H_{14}NO_5PS$. Originally developed by IG Farben in the 1940s, it is not approved for use in most countries due to its toxicity to non-target organisms, including humans and insects.

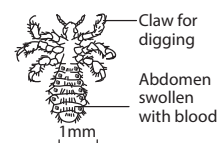


Fig. I: Dorsal view of a louse

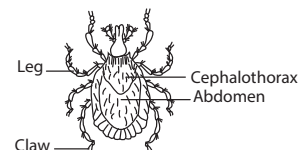


Fig. II: Dorsal view of a tick

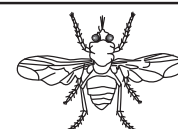
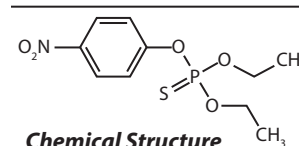


Fig. III: Tsetsefly

Ectoparasites



Chemical Structure of Parathion

PARATHYROID GLANDS: Two pairs of endocrine glands located behind the thyroid gland in all vertebrates except the fishes. They produce parathyroid hormone which regulates calcium by increasing the rate of resorption of calcium from bone and phosphorus concentrations in the blood. It also regulates the phosphorus concentrations in the blood by increasing the excretion of phosphorus by the kidneys. Humans generally have four glands, although the number can vary between three and six.

PARATHYROID HORMONE (PTH PARATHORMONE, PARATHYRIN): A peptide hormone, which is a single-chain polypeptide, composed of 84 amino acids secreted by the parathyroid glands that regulate calcium and phosphorus levels in the blood. It acts in opposition to calcitonin. It regulates normal calcium levels by increasing the number of osteoclasts, which break down the bone matrix and release calcium into the blood; increasing the reabsorption of calcium and magnesium ions in the kidney tubules and converting vitamin D to its active form, which increases calcium absorption in the intestine. The sequences of human, bovine and porcine parathyroid hormone have been identified and the gene for human parathyroid hormone has been cloned and sequenced.

PAROTID GLAND: The largest of the three main pairs of salivary glands located on either side of the face, just below and in front of the ears with ducts opening on the inside of the cheeks.

PARATYPHOID FEVERS (ENTERIC FEVERS): A disease of pigs caused by infection with three species of *Salmonellae*: *Salmonella paratyphi A* and *S. Salmonella paratyphi B* also called *S. schottmuelleri* and *S. paratyphi C* (*S. hirschfeldii*). They are transmitted by contaminated water or food. Young pigs have high fever and may die within 24 hours.

PARAXIAL RAY: A ray of light that falls on a reflecting or refracting surface close to and almost parallel to the axis so that sines of angles between the rays and the optical axis may be replaced by the angles themselves in calculations.

PARAZOA: Members of the subkingdom with only the single phylum Porifera. It consists of multicellular organisms including the sponges. They have less-specialised cells than in the Metazoa.

PARENCHYMA: Plant tissue mostly concerned with the manufacture and storage of food and composed of loosely-packed, cells with thin cellulose walls or loose connective tissues formed of large cells. It is frequently found as a homogeneous tissue in stems, roots, leaves and flower parts. The cortex and pith are composed of parenchyma cells. The mesophyll is also a specially differentiated parenchyma called chlorenchyma. Photosynthesis, assimilation, respiration, storage, secretion and excretion take place mainly in parenchymal cells.

PARENT: 1. A male or female that has given birth to an offspring. 2. An organism or cell that gives rise to new organisms or cells, for example, a **parent plant** is any plant that has produced one or more plants similar to itself by sexual or asexual reproduction.

PARENT HYDRIDE: An unbranched acyclic or cyclic structure or an acyclic/cyclic structure having a semi-systematic or trivial name to which only hydrogen atoms are attached.

PARENT ION: An electrically charged molecular moiety which may dissociate to form fragments, one or more of which may be electrically charged and one or more neutral species. A parent ion may be a molecular ion or an electrically charged fragment of a molecular ion.

PARENT MATERIAL: The horizon of unconsolidated material from which soil horizons develops. Soils inherit their structure and minerals from their parent material and often classified based upon their contents of consolidated or unconsolidated mineral material that has undergone some degree of physical or chemical weathering and the mode by which the materials were most recently transported.

PARENT ROCK: A rock formation that is the source of soil. Parent rocks can be sedimentary, igneous or metamorphic.

PARENT NODE: In a hierarchical network or database, it is a mode that is one level above the current node. It may have second child nodes, but a child node can only have one parent node.

PARENT NUCLIDE (MOTHER NUCLIDE): Nuclide that undergoes nuclear decay to form the daughter nuclide.

PARENTAL CARE: The process of promoting and supporting the physical, emotional, social and intellectual development of a child from infancy to maturity by its parents. Parental care is a form of altruism since this type of behaviour involves increasing the fitness of the offspring at the expense of the parents. The degree of parental care differs widely. For example, most fishes show no parental care but mammals and some birds show parental care up to a certain age. In most mammals, parenting is done mainly by biological parents, but may also be done by close relations, foster parents and even by the entire society to which they belong or by other species. An example of parenting by other species is a human parenting an orphaned animal.

PARENTAL GENERATION (P₁): The generation of individuals of different genotypes that is mated, usually for scientific purposes, to produce hybrids from which subsequent generations, known as the F₁, F₂, etc, will arise. Crosses between P₁ yield the F₁ Generation (First Filial Generation). Only pure-breeding (homozygous) individuals are selected for the P generation.

PARENTHESES: 1. A symbol () used to enclose a sum, product or other expression considered or treated as a collective entity in a mathematical operation. 2. Either or both of the upright curved lines, (), used to mark off explanatory, amplifying or qualifying remarks in writing or printing.

PARESIS: 1. Known as **partial paralysis**, it is a condition of muscular weakness caused by nerve damage or disease. 2. Inflammation of the brain in the later stages of syphilis, causing progressive dementia and paralysis.

PARETO CRITERION (PARETO OPTIMUM): A requirement or status that an individual's welfare could not be further improved without making others in the society worse off.

PARETO EFFICIENCY (EDGEWORTH-PARETO POINT; EFFICIENT POINT; PARETO OPTIMALITY): A state of allocation of resources in which it is impossible to make any one individual better off without making at least one individual worse off.

PARIETAL: Of a structure, positioned along the edges or wall, other than on the axis, as shown in the illustration.

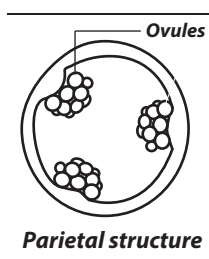
PARIETAL BONES: Either of two large, irregularly quadrilateral bones between the frontal and occipital bones that together form the sides and top of the skull. They are the largest bones of the skull and separated from each other by the sagittal suture, from the frontal bone by the coronal suture and from the occipital bone by the lambdoid suture. Together they form a large portion of the sides and roof of the cranium. Each bone has the shape of an irregular quadrilateral and has two surfaces, four borders and four angles.

PARIETAL LOBE: The lobe of the cerebral cortex that lies at the top of the brain above or superior to the occipital lobe and behind or posterior to the frontal lobe that integrates sensory information from different modalities, particularly determining spatial sense and navigation. This enables regions of the parietal cortex to map objects perceived visually into body coordinate positions.

PARIETAL PLACENTATION: Having ovules that are attached to the walls of the ovary.

PARIPINNATE: Pinnate leaflets that are symmetrically paired along the rachis, as shown in the illustration.

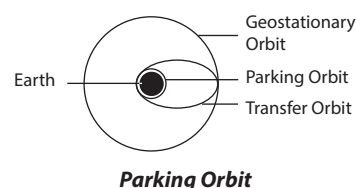
PARITY: 1. The property of a wave function that determines the behaviour of the wave function when all its spatial coordinates are reversed in direction, for example, when x , y and z are replaced by $-x$, $-y$ and $-z$. Its symbol is p . In mathematics, the parity of an object states whether it is even or odd. 2. In computing, it refers



to a basic method of checking data for errors during a transmission or on a data storage mechanism. It works by setting a parity bit to an even or odd number. For example, if the binary stream contains an even amount of 1's and the parity bit is odd, the computer would generate a parity error because the data is corrupt.

PARITY BITS (CHECK BITS): Bits of data that are used to check for errors in the transmission or storage of a unit of information before it is accepted. In the process, a 0 or 1 is added to each group of bits so that it will have either an odd number of 1's or an even number of 1's; for example, if the parity is odd then any group of bits that arrives with an even number of 1's must contain an error.

PARKING ORBIT: An orbit around the Earth with the same orbital period as that of the Earth's rotation period, which is 24 hours. A satellite in orbit having the same period of rotation as that of Earth as it turns about its axis is said to be in a parking orbit. Relay satellites are placed in a parking orbit, so that television programmes can be transmitted from one place to another. Transfer orbit is an elliptical trajectory by which a spacecraft can pass from one orbit to another at a higher altitude, especially a geostationary orbit.



PARKINSON'S DISEASE (PARKINSONISM; PARALYSIS AGITANS): A chronic and progressive movement disorder disease of the nervous system characterised by a progressive loss of mobility, muscular rigidity, tremor and speech difficulties. It is caused by the gradual breakdown or death of brain cells that make dopamine, a chemical messenger in the brain that helps neurons or nerve cells control movement. The condition affects mainly people over the age of 50. It was first described by British neurologist, James Parkinson (1755-1824) in 1817.

PARKLAND: A grazed grassland or heathland with single established trees or deliberately planted clumps, usually part of a designed and managed landscape. Parkland is economically important both for grazing and the production of substantial timber.

PARROT MOUTH: A deformity of the upper jaw of horses, preventing proper mastication. The condition prevents the horse from grazing.

PARSE: In computing, to break down a data stream into individual pieces of information that can be acted upon. On the World Wide Web, for example, data entered by a user can be sent to a database program for storage and later analysis; the ability to do this depends on being able to feed the right bit of data into the right record field. A parser in a program compiler is a program that takes each program statement that a developer has written and divides it into parts (for example, the main command, options, target objects, their attributes and so forth) that can then be used for developing further actions or for creating the instructions that form an executable program.

PARSEC: A unit of length used to express astronomical distance equal to 3.262 light-years. Light year is the distance travelled by light moving in a vacuum in one year, at its accepted speed of 299,792 km/second.

PARSEVAL'S THEOREM: In mathematics, the theorem of Fourier analysis stating that Bessel's inequality holds as equality for a square integrable function. In other words, it is a theorem that gives the integral of a product of two functions, $f(x)$ and $F(x)$, in terms of their respective Fourier coefficients.

PARSLEY: A common hardy biennial herb (*Petroselinum crispum*) used for garnishing and flavouring. It belongs to the Apiaceae or Umbelliferae and native to the Mediterranean region.

PARSLEY PIERT (LAMB'S FOOT): A common weed (*Aphanes arvensis*) affecting winter cereals.

PARSNIP: A plant (*Pastinaca sativa*) of the parsley family with a long white fleshy root and a distinctive, sweet flavour eaten as a vegetable.

PART: Distinct portion of a human or animal body or of a plant or a piece or component of a machine.

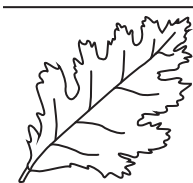
PART-TO-PART RATIO: A ratio that compares a part of a whole to another part of the same whole. For example, there are 8 unripe mangoes for every 12 ripe ones is a part-to-part ratio with a whole of 20 mangoes.

PART-TO-WHOLE RATIO: A ratio that compares a selected number of parts in a whole to the total of all parts in the entire whole. An example is 12 out of 20 mangoes are ripe and 8 out of 20 mangoes are not ripe.

PART-WHOLE FRACTION: A fraction that describes dividing an object or collection into equal parts. The object or collection is called the whole, or the ONE, and is the denominator of the fraction. The numerator is the number of parts of the whole.

PARTS-AND-TOTAL STORY: A number story in which a whole is made up of distinct parts. An example is "there are 15 girls and 12 boys in a class. How many students are there in all?" In other stories, the total and one or more parts may be known and the last part unknown.

PARTS-AND-TOTAL DIAGRAM: A diagram used to model problems in which two or more quantities (parts) are combined to get a total quantity.



Parted leaf

PARTED: Cleft deeply to reach more than half the distance to the base or midvein, as shown in the illustration.

PARTHENAIS: A breed of cattle originating from western France. It produces calves for a suckler herd and is known for easy calving and high growth rate.

PARTHENOCARPY: The production or formation of a fruit, either naturally or artificially without prior fertilisation of the ovules. The resulting fruits are seedless and, therefore, do not contribute to the reproduction of the plant. Examples are bananas and pineapples. In nature, it occurs as a result of mutation, but if it affects every flower, then the plant can no longer sexually reproduce but might be able to propagate by vegetative means. Plant hormones may have a role in this phenomenon, which can be induced by auxins in the commercial production of tomatoes and other fruits.

PARTHENOGENESIS: A form of reproduction in an organism in which an unfertilised egg or ovum develops into a new individual. It is the normal means of reproduction in some lower animals such as some fishes and especially in insects, for example, the aphid. In some aphid species, males are absent or very rare. The eggs formed by the females contain the full (diploid) number of chromosomes and are genetically identical. In many social insects, such as the honeybee and the ant, the unfertilised eggs give rise to the male drones and the fertilised eggs to the female workers and queens. It has also been observed in some snakes, fish and monitor lizards. Variation is limited in parthenogenetically reproduced species.

PARTI-COLOURED: Consisting of different colours.

PARTIAL: 1. In mathematics, concerning or relating to an operation in which only one of the independent variables is considered. 2. In mathematics, concerning or relating to a mapping, relation or predicate, which is defined only on the domain of the mapping. 3. In physics, a component vibration at a particular frequency in a compound tone. For example, when a musical instrument produces a note, say middle C, it will produce a complex tone in which the fundamental frequency is mixed with a number of partials. The component in the tone of a bell is described as partial.

PARTIAL AUTOCORRELATION: A correlation between two variables when the effects of one or more related variables are removed.

PARTIAL CHARGE TRANSFER REACTION: An ion/neutral species reaction, in which the charge on a multiply-charged reactant ion is reduced.

PARTIAL DECAY CONSTANT: For a radionuclide, it is the probability in unit time for the decay of one of its nuclei by one of several modes of decay.

PARTIAL DERIVATIVE (PARTIAL DIFFERENTIAL COEFFICIENT): The infinitesimal change in a function consisting of two or more variables when one of the variables changes and the

other remains constant. If $y = f(x, q)$, $\frac{\delta y}{\delta x}$ is the partial differential

of y with respect to x while q remains unchanged. Its symbol is ∂ and it is used in vector calculus and differential geometry.

PARTIAL DIFFERENTIAL EQUATION (PDE): A differential equation that involves partial derivatives of more than one variable. The equations are used to formulate and aid the solution of problems involving functions of several variables such as the propagation of sound or heat, electrostatics, electrodynamics, fluid flow and elasticity.

PARTIAL DIGESTION: In partial digestion and/or selective digestion procedures, only part or some of the analytes present are brought into solution. This may be preferred to total decomposition if relative concentrations of the analyte in the test samples provide sufficient information, for example, materials for geochemical exploration.

PARTIAL DOMINANCE: The reflection of both genes in the phenotype.

PARTIAL ECLIPSE: The partial obscuring of light from a celestial body as it passes behind or through the shadow of another body.

PARTIAL FRACTION: One of a set of fractions into which a more complex fraction can be resolved. For example, $\frac{3x}{x^2+7x+12} = \frac{A}{(x+3)} + \frac{B}{(x+4)}$, where A and B are constants to be determined. Two or more partial fractions can also be combined to form a single algebraic fraction.

PARTIAL ISOTHERM (INDIVIDUAL ISOTHERM): The function relating, at constant temperature and pressure, the amount of a particular component in the interfacial layer per unit area (or per unit mass of adsorbent) with its mole fraction (or concentration) in the liquid phase. This function can be evaluated only, when the location and the thickness of the interfacial layer have been defined.

PARTIAL KINETIC CURRENT: The current which would flow if mass transport were infinitely fast.

PARTIAL MASS DENSITY: Change in mass due to addition of a small amount of component to a system divided by the change in volume of the system. The systematic name is partial volumic mass.

PARTIAL MOLAL QUANTITY: The molal concentration of one component of a mixture of components as related to total molal concentration for all components in the mixture.

PARTIAL MOLAR VOLUME: That portion of the volume of a solution or mixture related to the molar content of one of the components within the solution or mixture. For example, one mole of water added to a large volume of water at 25 °C (298 K), increases the volume by 18 cm³, therefore making the molar volume of pure water 18 cm³ mol⁻¹. In contrast, the addition of one mole of water to a large volume of pure ethanol results in an increase in volume of 14 cm³. The reason is that the volume occupied by a given number of water molecules depends upon the identity of the surrounding molecules. Therefore, the value of 14 cm³ is the partial molar volume of water in ethanol.

PARTIAL ORDER (PARTIAL ORDERING): In mathematics, a relation ($\leq$) on a set S that is reflexive ($a \leq a$ for all a in S),

antisymmetric ($a \leq b$) and $b \leq a$ implies $a = b$ and transitive ($a \leq b$ and $b \leq c$ implies $a \leq c$). It generates chains of comparable elements, but unless it is a total ordering there will be pairs of elements in the set between which the relation does not hold in either order.

PARTIAL PIVOTING: In mathematics, the interchanging of rows in a pivoting system in a matrix in order to place a pivot element in a maximum diagonal position prior to a particular operation.

PARTIAL PRESSURE: The pressure exerted by one gas in a mixture of gases or the pressure which the gas would have if it alone occupied the volume. It is a measure of the thermodynamic activity of the gas's molecules. Gases dissolve, diffuse and react according to their partial pressures and not necessarily to their concentrations in a gas mixture.

PARTIAL PRODUCT: In mathematics, the product of an initial segment of an infinite product. It is formed by multiplying the multiplicand by one digit of the multiplier when the multiplier has more than one digit. For example, the product of 67 and 12 can be calculated as the sum of two partial products, 134 (67×2) + 670 (67×10), or 804. In this example 134 and 670 are partial products.

PARTIAL QUOTIENTS METHOD: An algorithm that uses repeated subtraction to solve division mathematical problems. When dividing a large number (dividend) by a small number (divisor), the following steps are followed: (i) subtract from the dividend an easy multiple (for example 100 $\times$, 10 $\times$, 5 $\times$, 2 $\times$, etc.) of the divisor; (ii) repeat the subtraction until the large number has been reduced to zero or the remainder is less than the divisor; (iii) add up the multiples of the divisor that were used in the repeated subtraction to find the answer of the division.

PARTIAL RATE FACTOR: The rate of substitution at one specific site in an aromatic compound relative to the rate of substitution at one position in benzene.

PARTIAL SPECIFIC VOLUME: Change in volume of a system when a small amount of component is added divided by the mass of the added component. The systematic name is partial massic volume. Its symbol is v_b .

PARTIAL STERILISATION/STERILIZATION: The elimination of a portion of a population of microorganisms, usually by heat or chemical treatment. The process is selective, and certain organisms or groups of organisms are destroyed to a greater extent than others. The process can be used to sterilise foods at a temperature that destroys harmful microorganisms without major changes in the chemistry of the food.

PARTIAL SUM: In mathematics, the sum of an initial segment of an infinite series or sequence. The first sum is the first element, the second sum is the first element added to the second element, the third sum is equal to the sum of the first three elements, etc. If a sequence is represented by $\langle a_n \rangle$, then its partial sum may be written $\sum_{i=1}^k a_i$ as the partial sum of first k terms.

PARTIAL-DIFFERENCES SUBTRACTION: A subtraction method in which separate differences are computed for each place value of the numbers and then added to get a final difference. This method is performed from left to right and can be used for adding any addition problems with 2 or more digit numbers, one place value at a time.

PARTIALLY DRAINING: An adjective describing a chain macromolecule that behaves in a hydrodynamic field as though the solvent within the domain of the macromolecule were progressively more immobilised with respect to the macromolecule in the direction from its outer fringes inward. A freely draining macromolecule and a non-draining macromolecule are two extremes of the concept of a partially draining macromolecule.

PARTIALLY ORDERED SET (POSET): A set endowed with a partial order. A partially ordered set formalises and generalises the intuitive concept of an ordering, sequencing or arrangement of the elements of a set. An example of a partially ordered set is the set of subsets of a given set, where $A \subset B$ for sets A and B to mean A is a proper subset of B .

PARTIALLY PERMEABLE MEMBRANE (SELECTIVELY-PERMEABLE MEMBRANE; PARTIALLY-PERMEABLE MEMBRANE; DIFFERENTIALLY-PERMEABLE MEMBRANE):

A membrane that is permeable to the small molecules or ions of water and certain solutes but does not allow the passage of large solute molecules by diffusion. The rate of passage depends on the pressure, concentration and temperature of the molecules or solutes on either side, as well as the permeability of the membrane to each solute. Examples of semi-permeable membranes are the lipid bilayer, on which is based the plasma membrane that surrounds all biological cells and the thin film on the inside of an egg.

PARTICLE: 1. Any tiny piece of matter that includes atoms, molecules, ions, alpha and beta particles. For example, electrons, protons and neutrons are fundamental particles. 2. A theoretical body that has mass but no physical extension. Because it has no volume and incapable of rotation, it can only have translational motion.

PARTICLE ACCELERATOR: A device that provides beams of high-energy particle, especially electrons, positrons and protons for scattering and collision experiments or for the generation of high-energy x-rays and gamma rays. Examples are cyclotrons and linear accelerators.

PARTICLE DENSITY: The number of particles (nuclear or elementary) divided by the containing volume. In soils, it is the mass per unit volume of the soil particles, usually expressed in grams per cubic centimetre.

PARTICLE DETECTOR (RADIATION DETECTOR): Any of a number of devices used to detect and measure subatomic particles emitted in nuclear processes such as those produced by nuclear decay, cosmic radiation or reactions in a particle accelerator. Almost all instruments used for detecting are based either on the ionisation of matter caused by radiation or on the luminescence it can cause in certain materials.

PARTICLE IN A BOX (INFINITE POTENTIAL WELL; INFINITE SQUARE WELL): A system in quantum mechanics used to illustrate important features of quantum mechanics, such as quantisation of energy levels and the existence of zero – point energy. A particle with a mass m is allowed to move between two walls having the coordinates $x = 0$ and $x = L$. The potential energy of the particles is taken to be zero between the walls and infinite outside the walls. The time-independent Schrodinger equation is exactly soluble for this problem, with the energies E_n being given by $n^2 h^2 / 8mL^2$, $n = 1, 2, \dots$ and the wave functions ψ_n being given by $\psi_n = (2/L)^{1/2} \sin(n\pi x/L)$, where n are the quantum numbers that label the energy levels and h is the Planck constant. The particles in a box can be used as a rough model for delocalised electrons in a molecule or solid. Thus, it can be used to explain the colour imparted to crystal by F – centres. The problem of a particle in a box can also be solved in two and three dimensions, enabling relations between symmetry and degenerate energy levels to be seen.

PARTICLE INDUCED X-RAY EMISSION ANALYSIS: A kind of analysis based on the measurement of the energies and intensities of characteristic X radiation emitted by a test portion during irradiation with charged particles other than electrons.

PARTICLE PHYSICS: The branch of physics that is concerned with the study of the properties and interactions of elementary particles that make up all atoms. This helps physicists to identify the simplest objects from which all matter is made of and to understand the forces which cause them to interact and combine to make more complex things. More than 300 subatomic particles have now been identified, categorised into several classes according to their mass, electric charge, spin, magnetic moment and interaction.

PARTICLE SCATTERING FUNCTION (PARTICLE SCATTERING FACTOR): The ratio of the intensity of radiation scattered at an angle of observation to the intensity of radiation scattered at an angle zero.

PARTICLE SIZE: The diameter or volume of the grains in samples such as a sediment or sedimentary rock, other solid materials,

suspensions, emulsions and even aerosols. To describe the size of liquid or solid particles (aerosol) the average or equivalent diameter is used. For non-spherical particles collected in an impactor, for example, the aerodynamic diameter of a particle of arbitrary shape and density refers to the size of a spherical particle of unit density that would deposit on a given impactor surface. **Particle size analysis** is the determination of the various amounts of the different separates in a given sample, such as solid materials, suspensions, emulsions and even aerosols. Some of the different methods employed to measure particle size include sedimentation, sieving, micrometry, laser diffraction or combinations of these methods.

PARTICLE SIZE DISTRIBUTION, PSD (GRAIN SIZE DISTRIBUTION): A way of measuring the composition of soil or particles dispersed in fluid. It defines the values or a mathematical function of the relative amounts of particles present, sorted according to size, important in understanding the physical and chemical properties of a medium, which can be used in planning irrigation and crop arrangement. It affects the reactivity of solids participating in chemical reactions and must be strictly monitored in many industrial processes such as the manufacture of printer toner and cosmetics. The size of the liquid or solid particles in the atmosphere usually extends from more than 0.01 to less than 100 μm in diameter. In the earth's atmosphere the distribution function which describes the number of particles as a function of diameter, mass or surface area of the aerosol can be determined reasonably well with modern instrumentation.

PARTICLE-BEAM EXPERIMENTS: Means of probing the structures of subatomic particles. There are essentially two arrangements for particle-beam experiments in elementary-particle physics. Before the early 1970s, fixed-target experiments were the only means of probing the structures of subatomic particles. In a fixed-target experiment a beam of accelerated particles is directed at a stationary target. An example of such an experimental arrangement is the Bevatron at Berkeley, built to produce antiprotons by striking a proton at rest with a high-energy proton: $p + p + p + p + p + p$. According to the mass-energy equation some of the kinetic energy of the accelerated proton becomes the mass energy of the proton-antiproton pair. Since this newly formed mass has only the mass equivalence of two proton rest masses, it is remarkable that in practice this fixed target experiment requires an incident kinetic energy equivalent to six times the proton rest mass to create a single proton-antiproton pair. This example illustrates the inefficiency of scattering using a stationary target; conservation of momentum means that the products of a fixed target experiment must possess some residual kinetic energy, which, therefore, cannot be involved in forming the mass-energy of collision products. This problem does not arise in colliding-beam experiments introduced in the early 1970s. In these, the beams of particles are caused to collide with their antiparticle counterparts moving in the opposite direction at the same speed. The total momentum of the colliding particles is, therefore, equal to zero and their total energy is available to form the mass energy of the collision products. Colliding-beam experiments are known to have lower collision rates than their fixed target counterparts. However, the likelihood of collision is increased by using finely focused beam and by allowing particles to accumulate in storage rings before they are made to collide. New machines in high-energy physics are, therefore, based on a colliding-beam arrangement.

PARTICULAR INTEGRAL: A function that satisfies a given differential equation, especially one that also satisfies certain given initial conditions or boundary conditions. It is the particular solution in a second order linear differential equation and is of the form $r \frac{d^2 y}{dx^2} + p \frac{dy}{dx} + qy = g(x)$, where r , p and q are constants and $g(x)$ is a function of x . A particular integral can be obtained by using a function which is similar to $g(x)$.

PARTICULAR PROPERTY: A property of a given object (phenomenon, body or substance). It includes the concept of particular quantity. The adjective 'particular' may be omitted, if no ambiguity is caused.

PARTICULAR SOLUTION: A solution of differential equation which is given by particular values of a constant that appears in the general solution of the differential equation.

PARTICULATE GEL: Gel in which the network component comprises solid particles.

PARTICULATE INHERITANCE: The transmission of genetic characteristics from generation to generation through discrete units or particles. It was proposed by Gregor Mendel and his theorists. Characteristics may be absent in the offspring of a cross in which only one parent possesses them, but reappears repeatedly in the progeny of subsequent crosses.

PARTICULATE MATTER (PARTICULATES; FINE PARTICLES): Tiny subdivisions of solid or liquid suspended in gas or liquid. In water, quality particulate matter, also called **suspended solids**, can be removed by settling or filtration with a particular filter, usually 0.45µm pore size. Sources of particulate matter can be man-made or natural. For example, air particulates can be classified as of natural origin, such as volcanoes, dust from storms, sea spray and forest fire. Man-made activities such as burning of fossil fuel in vehicles, power plant and various industrial processes produce particulates mixed with gas, known as **aerosol**. Particulates are classified according to their aerodynamic diameter. PM₁₀ are particulates less than 10µm. Such particulates are dangerous to health because they are not filtered out of the inhaled air. PM_{2.5} are fine particulates that are so small that they can only be detected using an electron microscope and generally emitted from activities such as industrial and residential combustion and from vehicle exhaust.

PARTICULATE SOL: Sol in which the dispersed phase consists of solid particles.

PARTITION: 1. In chemistry, it is the equilibration of a solute between two phases usually of differing composition. If a substance is in contact with two different phases then, in general, it will have a different affinity for each phase. Part of the substance will be absorbed or dissolved by one and part by the other, the relative amounts depends on the relative affinities. The substance is said to be partitioned between the two phases. For example, if two immiscible liquids are taken and a third compound is shaken up with them, then equilibrium is reached in which the concentration in one solvent differs from that in the other. The ratio of the concentrations is the **partition coefficient** of the system. The **partition law** states that this ratio is a constant for given liquids. Partitioning of a solute between two phases is the basis for extractions and chromatographic separations. 2. In mathematics, a partition of a positive integer n is a way of writing n as a sum of positive integers. Two sums that differ only in the order of their summands are considered to be the same partition. If order matters, then the sum becomes a composition. A summand in a partition is also called a part. The number of partitions of n is given by the partition function $p(n)$. For example, the partitions of 3 are: 3; 2 + 1 and 1 + 1 + 1. 3. In computer science, it is also known as **partition disk**, which is a reserved part of disk or memory that is set aside for some purpose. A very large drive is more efficient if partitioned because the file clusters are smaller because the computer does not have to search as large an area for data. Communication between partitions, however, is more cumbersome. On a computer, new hard disks must be partitioned before they can be formatted for the operating system. 4. In genetics, it is the events that lead to the separation of chromosomes or to the segregation of plasmids into daughter cells in prokaryotic organisms.

PARTITION CHROMATOGRAPHY: Chromatography in which separation is based mainly on differences between the solubility of the sample components in the stationary phase (gas chromatography) or on differences between the solubilities of the components in the mobile and stationary phases (liquid chromatography).

PARTITION CONSTANT (PARTITION RATIO; DISTRIBUTION RATIO): IUPAC names which have replaced partition coefficient. It is defined as the ratio of the concentration of a single solute in two immiscible solvent at equilibrium. It is independent of the actual concentration.

PARTITION FUNCTION (SUM OF STATES; SUM OVER STATES): The quantity Z defined by $Z = \sum \exp(-E_i/kT)$, where the sum is taken over all states i of the system and where E is the energy of the system, k is Boltzmann's constant and T is the temperature. From this function all the thermodynamic properties of the system can be derived. Z is a quantity of fundamental importance in equilibrium statistical mechanics. For a system in which there are nontrivial interactions, it is very difficult to calculate the partition function exactly. For such systems, it is necessary to use approximation techniques. The partition function links results at the atomic level to thermodynamics Z . It is related to the Helmholtz free energy (F) by $F = kT \ln Z$.

PARTITION ISOTHERM: Isotherm describing partition of the sample component between the bulk of a liquid stationary phase and a liquid, gaseous or supercritical mobile phase.

PARTITION OF UNITY: In mathematics, a family of non-negative continuous functions on a topological space that sums to unity and is locally finite in that all but a finite number of them vanish on some neighbourhood of each point. The partition is subordinate to a given covering if each function is zero outside some member of the covering.

PARTLY MIXED RATION: A winter feed for livestock, which combines total mixed ration and separate concentrate feeding, by mixing some concentrates with the roughage and keeping the rest aside as additional feed for higher-yielding animals.

PARTON: A point-like, almost free, particle postulated as a component of neutrons and protons by Richard Feynman in 1969 as a way to analyse high-energy hadron collisions. The parton model enabled the results of very high-energy experiments on nucleons to be understood.

PARTS PER MILLION (ppm): Dimensionless quantity and measurement unit that is used to express a very dilute concentration level of a solvent or particulate matter in a solution, such as when describing the level of pollutants in the air, water or other fluids. It can also be used to describe a concentration of something in soil.

PATTERN: A set of numbers, objects, or shapes that are arranged following a rule.

PARTURITION: The act of giving birth at the end of a gestation period. It begins by the secretion of ACTH by the foetal pituitary gland, which causes increased secretion of corticosteroid hormones by the foetal adrenal glands. The various stages are: (i) dilation, which causes uterine contractions resulting in severe labour; (ii) the widening of the opening to the cervix and the breaking of the amniotic sac accompanied by the expulsion stage in which the female pushes actively to aid the process; (iii) The expulsion of the placenta.

PARVOVIRUS: Any of a group of DNA viruses that have a single strand. They cause several diseases in animals. In humans, they cause diseases such as aplastic anaemia, arthritis, spontaneous abortion and erythema infectiosum, an acute disease usually affecting young children.

PASCAL: 1. The SI unit of pressure or stress, equal to one newton per square metre. Its symbol is Pa. 2. A high-level programming language designed to support structured programming, used in teaching, applications and systems programming.

PASCAL, BLAISE (1623-62): French mathematician, physicist and religious philosopher. A child prodigy, he wrote an essay on conic sections in 1640. In physics, he formulated Pascal's law concerning

fluid pressure and the principle behind the hydraulic press. In mathematics, he worked on the arithmetic triangle, inventing a calculating machine and contributing to the advance of differential calculus. He is regarded as the founder of the modern theory of probability for work done in his early years. The SI unit of pressure is named after him.

PASCAL'S DISTRIBUTION (NEGATIVE BINOMIAL DISTRIBUTION; PÓLYA DISTRIBUTION): A discrete probability distribution of the number of k^{th} successes in a sequence of Bernoulli trials before a specified (non-random) number x failures occurs. If the probability of success at each trial is p and the probability of failure is $q = 1 - p$, the distribution is given by:

$$P(X=x) = \binom{k+x-1}{x} p^k q^x, 0 \leq x \leq \infty,$$

$$\text{mean} = \frac{kq}{p}, \text{variance} = \frac{kq}{p^2}.$$

PASCAL'S PRINCIPLE (PASCAL'S LAW): A law of physics stating that in a confined fluid, externally applied pressure is transmitted uniformly in all directions. In a static fluid, force is transmitted at the velocity of sound throughout the fluid and acts at right angles to any surface in or bounding the fluid. The practical applications of the law are the hydraulic jack, balloon and pneumatic tyre.

PASCAL'S TRIANGLE: A triangle of numbers in which a row represents the coefficients of the terms in the expansion of $(a + b)^n$ where $n = 0, 1, 2, 3, \dots$. It was devised by the French mathematician, Blaise Pascal (1633–1662) and used to calculate probabilities. Each number is found by adding the two numbers above it and the sum of each row is a power of two.

				1					
			1		1				
		1		2		1			
	1		3		3		1		
1		4		6		4		1	
1	5		10		10		5		1
1	6	15		20		15	6		1

Pascal's Triangle

PASCHEN-BACK EFFECT: The splitting of a spectral line into two or more lines of different frequencies, which occurs when the light source is placed in a magnetic field. It has helped to identify the energy levels in atoms and also provides a means of studying atomic nuclei and electron paramagnetic resonance (magnetic resonance). It is also used in measuring the magnetic field of the Sun and other stars. It was discovered in 1896 by Pieter Zeeman (1865 - 1943).

PASSERINE: Any member of the order Passeriformes, with four suborders: Eurylaimi, Tyranni, Menurae and Oscines or Passeres. Passerines are characterised by three toes pointing forward and one backwards, to facilitate perching. Passeriformes are the largest order of birds and among the most diverse orders of terrestrial vertebrates, with more than 110 families and more than 5,000 identified species. Examples of some of the species are rooks, finches, sparrows, tits, warblers, robins, wrens, swallows, cuckoo-finches, and cowbirds.

PASSION FRUIT PLANT (*Passiflora edulis*): A flowering plant mostly vines, some being shrubs and a few being herbaceous in the family Passifloraceae and the genus *Passiflora* with about 500 species. The vine is a shallow-rooted, woody, perennial, climbing by means of tendrils. The alternate, evergreen leaves, deeply 3-lobed when mature, are finely toothed, 7.5-20 centimetres long, deep-green and glossy above, paler and dull beneath and, like the young stems and tendrils, tinged with red or purple, especially in the yellow form. A single, fragrant flower, 5-7.5 centimetres wide, is borne at each node on the new growth. It produces fruits which are round to oval, either yellow or dark purple with a soft to firm juicy interior containing many seeds. The passion plant is commercially cultivated in tropical and subtropical region such as South America, Central America, Africa,

Caribbean, United States and Hawaii. The fruits are used as a sauce in desserts and toppings for ice creams. It can also be used in some alcoholic cocktails.

PASSIVATION: The process of transition from the active to the passive state by formation of the passivating film. Passivation is achieved by an anodic current which at the respective electrode potential must be larger than the maximum current or by the presence of an oxidised substance in the neighbouring solution which passivates by being reduced (passivator).

PASSIVATION POTENTIAL: The most negative electrode potential at which the passivating film is formed. The passivation potential is equal to or more positive than the equilibrium potential of formation of the phase constituting the film. Usually, the corrosion current goes through a maximum at the passivation potential.

PASSIVE: 1. A term used to refer to a metal which possesses a surface layer of oxide which makes it unreactive except under special or extreme conditions. Examples are aluminium and iron dipped in concentrated nitric acid. 2. In electronics, a component such as a capacitor which exhibits no gain or contributing no energy and is capable of amplification.

PASSIVE BEAM: A type of chilled beam system that requires ventilation air to be delivered by a separate air-handling system and the warm air is drawn to the heat exchanger through the natural rise of warm air and does not need fans to push the warm air through.

PASSIVE CONTINENTAL MARGIN: Area along the coast that is not tectonically active where sediments are deposited. These areas form much of the Atlantic Ocean coastline.

PASSIVE DEVICE: 1. An electronic component such as capacitor or resistor that is capable of amplification and exhibiting no gain or contributing no energy. 2. An artificial satellite that reflects an incoming signal without amplification. 3. A solar powered device that makes use of an existing structure to collect and utilise solar energy without the use of pumps, fans, etc. 4. A radar device that provides information, guidance, surveillance, etc, by receiving the microwave radiation emitted by a warm body or reflected by a body from some other source. 5. A unit of a computer incapable of initiating a request for communication with another device itself, but which accepts such a request from another device.

PASSIVE IMMUNITY: A type of acquired or adaptive immunity resulting from the introduction of antibodies, for example, by a pregnant mother passing antibodies through the placenta to a foetus or through breast milk, called natural passive immunity; or by vaccination or the injection of antiserum, called artificial passive immunity.

PASSIVE INFRARED SENSOR (PIR SENSOR): An electronic device that measures infrared (IR) light radiating from objects in its field of view, sensing motion. PIRs are basically made of a pyroelectric sensor, which can detect levels of infrared radiation. Apparent motion is detected when an infrared source with one temperature, such as a human, passes in front of an infrared source with another temperature, such as a wall. They are small, inexpensive, low-power, easy to use and do not wear out. They are found in appliances and gadgets used in homes or businesses.

PASSIVE MATRIX OLED (PMOLED): A type of organic light-emitting diode (OLED display technology in which each row or line in the display is controlled sequentially (one at a time). PMOLED does not contain a storage capacitor and so the pixels in each line are actually off most of the time. To compensate for this more voltage is used to make them brighter. Another type of organic light-emitting diode (OLED) is the active-matrix organic light-emitting diode (AMOLED) that contains a storage capacitor that makes the line pixels light all the time allowing for higher resolution and larger display sizes and uses less power. PMOLED displays are usually small and are used in small gadgets and displays.

PASSIVE METAL: A metal corroding in the passive state.

PASSIVE SAMPLER: A device for preconcentration of trace substances from gaseous media based on molecular diffusion without

controlled conveyance of the gas to be investigated, for example, work place air.

PASSIVE STATE: The passive state is characterised by a contiguous 'passivating film' of solid corrosion products separating the metallic phase from the adjacent electrolyte. Corrosion in the passive state involves growth of the passivating film and/or transfer of metal ions through the film into the electrolyte.

PASSIVE TRANSPORT: Diffusion of biochemicals and atomic or molecular substances across a plasma membrane in which the cell expends no energy. The four main kinds of passive transport are diffusion, facilitated diffusion, filtration and osmosis.

PASSIVE-MATRIX: An LCD technology that uses a grid of vertical and horizontal wires to display an image on the screen. Each pixel is controlled by an intersection of two wires in the grid. By altering the electrical charge at a given intersection, the colour and brightness of the corresponding pixel can be changed.

PASSIVITY: The properties of a metal such as aluminium resisting corrosion because of a thin surface layer of oxide.

PASSWORD: A string of secret characters used for authenticating a user on a computer system. Passwords help ensure that computers or data can only be accessed by those who have been granted the right to view or access them. While usernames are generally public information, passwords are private to each user.

PASTE SPECIAL: A feature that gives users more control of how the content is displayed or functions when pasted from the clipboard. Paste special is typically found in Office Suites such as Microsoft Office and OpenOffice and is very commonly used in Word, Excel, Writer and Calc to provide special formatting or calculations when pasting content into a document.

PASTER: 1. The thin part of a horse's leg between the fetlock and the hoof, supported by proximal and middle phalanges. 2. The metacarpal region of dogs.

PASTEUR, LOUIS (1822-1895): French biologist and chemist, who was engaged in numerous scientific endeavours as dictated by the needs of industry and public health in his time; notably: (i) He explained the scientific events of fermentation. He clarified the causes of period spoilage in wines, particularly the conversion of wine to vinegar and the resultant sour flavour. The process he described is presently used in commercial vinegar production. (ii) He discovered another fundamental biological phenomenon – the existence of forms of life that can live only in the absence of free oxygen. He introduced the terms 'aerobic' and 'anaerobic' for, life in the presence of and, in the absence of oxygen, respectively. (iii) He invented the process named 'pasteurisation' to destroy contaminant microorganism in beverages. It is a mild-heating to 65 °C (338 K) for thirty minutes (Batch method) or 72 °C (345 K) for 15 seconds (Flash method). (iv) He completed definitive experiments that showed that spontaneous generation of organisms does not occur and that air contains organisms. He rounded up his study by showing that these living organisms are not evenly distributed through the atmosphere. (v) He contributed to the formulation of the 'Germ Theory of Disease' that diseases could arise in animal tissues when microbes entered the tissues and grew there. (vi) Together with some colleagues, he discovered a method for the production of anthrax vaccine. (vii) He discovered a special vaccine for the virus disease, rabies. A befitting building was erected in Paris to house the laboratories of Louis Pasteur and his co-workers. It was dedicated in 1888 and christened L' Institut Pasteur. Presently, the Pasteur Institute of Paris and its branches throughout the world carry out medical research and manufacture vaccines and various products including laboratory media.

PASTEUR PIPETTE (TEAT PIPETTE; EYE DROPPERS): A laboratory equipment that is used to transfer small amounts of measured volumes of liquid. It is a thin tube made of glass with a fine tip and rubber teat fitted to the other. It is not graduated and therefore needs a scale for precise measurements.

PASTEURELLOSIS: Various infectious diseases caused by the coccobacilli *Pasteurella multocida* and *P. haemolytica* mainly affecting young lambs, adult sheep and store lambs. They are transmitted by contaminated food or water. The symptoms include high temperature and difficulty in breathing.

PASTEURISATION (PASTEURIZATION): Treatment of food or beverages by mild heat irradiation or chemical agents to reduce the number of micro organisms it contains to protect consumers from diseases. For example, milk is heated to 65 °C (338 K) for 30 minutes or to 72 °C (345 K) for 15 minutes followed by rapid cooling to below 100 °C (373 K). Pasteurisation can destroy bacteria which cause diseases such as tuberculosis, diphtheria, typhoid and brucellosis.

PASTORAL FARMING: A type of non-nomadic farming in which the farmer raises livestock for by-products such as meat, cheese, milk, eggs, wool, feathers, etc. Pastoral farming is a sedentary process in which the animals are catered for without the need to move from one place to the other.

PASTURAGE: 1. The grazing of livestock or the right to graze livestock on a particular area of land. 2. The grass or other vegetation eaten by grazing animals.

PASTURE: The growing of plants and grass that are suitable food for livestock or grass-covered land used for grazing livestock. Land unsuited for cultivation such as hilly or stony land, may be used as pasture. Vegetation grown on pastures includes types of grass or clover.

PASTURE MATING: The uncontrolled and random mating in herd.

PASTURE TOPPER: An implement attached to the back of a tractor and used to keep grassy areas such as pasture free from weeds and coarse grasses.

PATAGIUM: In mammals, a fold of skin between the fore and hind limbs and sometimes the tail, used for gliding or flying. Examples are found in bats and flying squirrels.

PATCH: A small cultivated area with one type of plant growing in it, for example, a tomatoes patch or an onion patch.

PATCH CLAMP TECHNIQUE: A technique for studying the activation of voltage-dependent clamp trans membrane channels in neurons in which a small piece of membrane containing only a few channels is isolated and microelectrodes are sealed to both sides of the membrane.

PATCH PANEL: A panel consisting of group of sockets that allows multiple devices, usually of the same or similar type, to be connected and managed in a more efficient manner. They are used to manually interconnect incoming and outgoing lines in communications and electronic systems such as computer networks, sound studios, radio stations, television stations and a number of other electronic systems.

PATELLA (KNEE CAP): A thick, small, circular-triangular, movable bone that is situated in a tendon in front of the knee joint in most mammals. This bone covers the knee joint in most mammals and birds and articulates with the femur and protects the joints from injury. It is the largest sesamoid bone in the human body.

PATELLIFORM: Shaped like a kneecap.

PATH: 1. In mathematics, also known as **Hamilton walk**, it is a walk in which each vertex, except possibly the first, occurs only once; it is a closed path or cycle if its endpoint is identical with its initial point. 2. In computer science, it is a route that a computer operating system follows through the directories on a disk to locate a file or the sequence of keyed characters that identifies this route.

PATH DIFFERENCE: The distance between one wave motion relative to another. Path difference (x) and phase difference (ϕ) are related by the equation: $\phi = \frac{2\pi x}{\lambda}$, where λ is the wavelength.

PATH INTEGRAL FORMULATION: A formulation of quantum theory developed by Richard Feynman in 1948 in which all the possible paths followed by a particle in quantum mechanical system, weighted by

the probability of each path occurring, are added up. Path integrals are used in analysing the foundations of quantum mechanics and in solving certain types of problem. The theory replaces the classical notion of a single, unique trajectory for a system with a sum or functional integral, over infinity of possible trajectories to compute quantum amplitude.

PATHOGEN: Any disease-causing agent. Pathogens include bacteria, fungi, viruses, rickettsia, yeasts, helminths and protozoans.

PATHOGEN SUPPRESSIVE SOIL (SUPPRESSIVE SOIL): Soil in which a pathogen does not persist, or causes no damage. Suppressiveness can occur through substances such as certain chemicals present in the soil that inhibit or kill potential pathogens, competition for nutrients, parasitism and predation.

PATHOGENICITY: The ability of a parasitic or disease causing agent to infect or inflict damage on a host. Such agents include infectious bacteria, viruses, prions, fungi, viroids, etc.

PATHOGENESIS: The origin, production or development of a disease due to a series of changes in the structure and functions of a cell, tissue or organ caused by a microbial, chemical or physical agent.

PATHOGENIC MICROORGANISM: Microorganisms including viruses, bacteria, fungi and protozoa which cause diseases. These may be found outside the body on the skin surface and the mucosal surface or inside the body in the blood or tissues. Some bloodborne pathogens include the hepatitis B virus, hepatitis C virus and human immunodeficiency syndrome.

PATHOLOGY: The study of diseases and the modifications in cellular function and changes in cellular structure produced in organs and tissues that are caused by or give rise to disease. This involves the examination of tissue samples, x-ray photographs, testing body fluids and tissues for abnormal composition or metabolisms. The techniques used include electronmicroscopy, immunocytochemistry and molecular pathologies or other evidence taken from living patients or cadavers. Pathologists work mostly in the laboratory and consult with physicians after examining specimens. In experimental pathology, disease processes are studied using experimental animals, cell cultures or other means. **Pathological** means of or relating to pathology; alternatively, extreme in a way that is not normal or that shows an illness or mental problem. In mathematics, **pathological** defines a mathematical entity that has highly unexpected property or properties and therefore regarded as counterintuitive, bizarre or defective. Examples are the Weierstrass function that is everywhere continuous but nowhere differentiable and the Cantor set, a subset of the interval $[0,1]$ that has the pathological property that it is uncountable yet its measure is zero.

PATIENT: Anyone, who receives medical attention, care or treatment especially in a hospital most often because he/she is ill or injured and in need of treatment by a physician, a dentist, a registered nurse practitioner or other health care provider.

PATINA: The attractive thin film of corrosion products formed on metals by weathering, especially the green tarnish on copper or bronze.

PATRISTIC: Denoting similarity between organisms resulting from common ancestry.

PATTERN: A repetitive order or arrangement of numbers or shapes satisfying a given rule.

PATTERN FORMATION (PATTERNING): The genetically controlled mechanism by which basic body symmetries and repetition of parts are developed. It involves each cell in a field sensing and responding to its position along a morphogen gradient, followed by short distance cell-to-cell communication through cell signalling pathways to refine the initial pattern. In animal embryos, pattern formation defines the body axes, segmentation of animals, phyllotaxis and other basic reference points to produce the framework in which tissues and organs can develop in the correct place and orientation.

PATTERN RECOGNITION: A branch of statistics used to classify data based upon several variables. This type of statistical analysis is particularly useful in interpreting biological data that consists of multiple observations on large numbers of individuals.

PATTERSON FUNCTION: A function of three spatial coordinates, constructed in the Patterson-Harker method, which has peaks at all vectors between two atoms in a crystal, the heights of the peaks being approximately proportional to the product of the atomic numbers of the corresponding atoms. It was introduced in 1935 by Arthur Lindo Patterson (1902-1966) and used in the analysis of the results of x-ray crystallography. This function is plotted as a contour map with maxima of the contours at vector from the point distances between pairs of maxima in electron density. Thus, the Patterson function enables interatomic vectors, which give the lengths and directions between atomic centres, to be determined.

PATULOUS: Opening or spreading.

PAUCIDISPERSE SYSTEM: A colloidal system in which only a few particle-sizes occur.

PAULI EXCLUSION PRINCIPLE: A law of quantum physics formulated by the Austrian physicist, Wolfgang Pauli in 1925. It states that no two electrons can have the same four quantum numbers in an atom or no two electrons can simultaneously occupy the same quantum or energy state at the same time.

PAULI, WOLFGANG ERNST (1900-1958): Austrian-born physicist, who obtained his PhD at the University of Munich and studied with Neils Bohr in Copenhagen and Max Born in Göttingen and taught at Heidelberg and, finally in Zurich. His formulation in 1924 of the Pauli Exclusion Principle explained the electronic make-up of atoms, for which he was awarded the 1945 Nobel Prize in Physics. In 1930, he proposed that the energy and momentum apparently lost when an electron is emitted from an atomic nucleus in beta decay is carried away by an almost massless, uncharged and difficult-to-detect particle called the neutrino.

PAULING RULES: A set of five rules governing the number of ions of opposite charge in the neighbourhood of a given ion in an ionic crystal, in accordance with the requirement of local electrical neutrality of the structure. They work well for mainly ionic crystals, such as sulfides, in which the bonding is mainly covalent. These rules have the basis of empirical observation and calculations of crystal energies. They were published in 1929 by Linus Carl Pauling (1901 - 1994), an American chemist and biochemist.

Paullinia pinnata (SUPPLE JACK): A flowering plant in the genus Paullinia in the soapberry family (Sapindaceae) with a long flexible stem found in West Africa and South America. The stem is used to poison fish in shallow pools. The roots and leaves are used in traditional medicine for many purposes, especially for erectile dysfunction. The leaves also serve as food for many lepidopteran larvae.

PAWPAW (PAPAYA): A deciduous tree, *Carica papaya* or its fruit of the family Caricaceae, cultivated throughout the tropics and warm subtropics. It has purple flowers that bear usually green (yellow, when ripe) medium-sized oval sweet fleshy fruits, eaten fresh and also used in salads, pies, sherbets, juices and confections. The latex of the unripe fruit contains an enzyme that is useful in various remedies for indigestion and in meat tenderisers.

PAYLOAD: In internet terminology, it is the instructions a virus or worm executes in addition to copying itself when infected on the computer. For example, a payload of a computer virus may be to delete some or all of the files on the computer.

PDF: 1. An acronym for **portable document format**, a multi-platform file format developed by Adobe Systems. It enables users to capture document text, fonts, images and even formatting of documents from a variety of applications. PDF files are frequently used in legal documents such as tax papers, bank statements and

other forms that require the user to fill in data. Since PDFs contain colour-accurate information, they can also print the same way they look on the screen. 2. An acronym for **pair distribution function**, the probability of finding a second particle as a function of distance from an initial particle. This probability is found from a statistical average over the system concerned.

PEAK: In chromatography, it is the portion of a differential chromatogram recording the detector response when a single component is eluted from the column. If separation is incomplete, two or more components may be eluted as one unresolved peak.

PEAK ANALYSIS: The extraction of relevant peak parameters (i.e. position, area) from a measured spectrum.

PEAK AREA: In chromatography, it is the area enclosed between the peak and the baseline.

PEAK AREA METHOD: A kind of peak analysis in which a peak area is calculated by subtracting an estimate of the underlying continuum in a relevant part of a measured spectrum.

PEAK BASE: In chromatography, it is the interpolation, in a differential chromatogram, of the baseline between the extremities of the peak.

PEAK CONCENTRATION: The highest concentration of a given trace component which was measured with a continuous analyser during a specified sampling period.

PEAK CURRENT: In linear-sweep voltammetry, triangular-wave voltammetry, cyclic triangular-wave voltammetry and similar techniques, it is the maximum value of the faradaic current due to the reduction or oxidation of a substance B during a single sweep. This maximum value is attained after an interval during which the concentration of B at the electrode-solution interface decreases monotonically, while the faradaic current due to the reduction or oxidation of B increases monotonically, with time.

PEAK FITTING: A kind of peak analysis in which a relevant part of a spectrum is fitted with a theoretical response function.

PEAK MAXIMUM: In chromatography, it is the point on the peak at which the distance to the peak base, measured in a direction parallel to the axis representing the detector response, is a maximum.

PEAK POTENTIAL: In linear-sweep voltammetry, triangular-wave voltammetry, cyclic triangular-wave voltammetry and similar techniques, the potential of the indicator electrode at which the peak current is attained.

PEANO ARITHMETIC: The theory of the natural numbers as defined by the five Peano's axioms.

PEANO CURVE: A continuous curve that passes through every point of the unit square or which completely fills up higher dimensional spaces such as squares or cubes.

PEANO-JORDAN MEASURE (JORDAN CONTENT; JORDAN-MEASURE): In mathematics, a form of measure especially useful in surface integration. It is an extension of the notion of size (length, area, volume) to shapes more complicated than, for example, a triangle, disk or parallelepiped. It is to the Riemann integral what measure is to the Lebesgue integral and provides the most practical definition of area.

PEANO, GUIASEPPE (1858- 1932): Italian mathematician, recognised as founder of mathematical logic and set theory. He is best known for his work in the foundations of mathematics, where he attempted to derive all of mathematics from fundamental principles, using the notation which he invented and which was later adopted as standard. He also published important axioms named Peano's axioms, which defined the natural numbers in terms of sets and invented the artificial language, Interlingua. Some important mathematical terminologies such as Peano's axioms, Peano arithmetic and Peano curve have been named after him.

PEANO'S AXIOMS (DEDEKIND-PEANO AXIOMS): A set of axioms that describe the set of natural numbers, $\mathbb{N}$, including zero

by defining a first member and a unique successor to each member, by excluding loops and by providing for mathematical induction. The axioms include the following: (i) There exists a natural number zero (0). (ii) For each $n \in \mathbb{N}$, there exists a natural number that is the successor of n , denoted by n' . (iii) Zero is not the successor of any natural number: for each $n \in \mathbb{N}$, $n' \neq 0$. (iv) For all natural numbers n , $m \in \mathbb{N}$, if $n' = m'$, then $n = m$. (v) Given any predicate P on the natural numbers, if $P(0)$ is true and $P(k)$ implies $P(k')$ for any $k \in \mathbb{N}$, then $P(n)$ is true for all $n \in \mathbb{N}$. It is possible, therefore, to regard the natural numbers as excluding zero while the whole numbers include zero.

PEAR: Any of several species of trees of the genus *Pyrus*, especially *P. communis* with sweet juicy, yellow or green fruit with a rounded shape that becomes narrow towards the stalk or tree on which it grows. The fruit are commonly eaten fresh or used for fermented pear juice and also canned.

PEARL: 1. A shiny, hard, rounded abnormal growth composed of nacre, a chalky substance or a small lustrous sphere of aragonite or calcite that forms around a grain of sand in an ocean organism such as molluscs. It is used as ornaments or decorations. 2. Rounded masses of concentrically arranged squamous epithelial cells, seen in some carcinomas. 3. Mucous casts of the bronchi or bronchioles found in the sputum of asthmatic persons.

PEARSON SYMBOL (PEARSON NOTATION): A symbolic notation devised by W.B. Pearson for describing the structure of a crystal. It involves three steps characters made up of two letters followed by a number. An example is the diamond structure, $cF8$. The lower case letter denotes one of six crystal systems: a = anorthic (triclinic), m = monoclinic, o = orthorhombic, t = tetragonal, h = hexagonal or rhombohedral and c = cubic. The upper-case letter denotes the lattice type: P = primitive, I = body-centred, F = face-centred (full), C = face-centred (side) and R = rhombohedral and the number gives the number of atoms in a unit cell.

PEARSON, KARL (1857 - 1936): English mathematician, biometrician, lawyer and statistician, who was educated privately at the University College School and at King's College, Cambridge. He studied medieval and sixteenth century German literature at Berlin and Heidelberg Universities. He then read law at Lincoln's Inn and was called to the bar by the Inner Temple in 1881. He became Professor of Mathematics at King's College, London in 1881 and at University College, London in 1883. Pearson laid the foundations of modern statistics with papers on moments, correlation, maximum likelihood and the chi-squared goodness-of-fit test.

PEARSON'S CORRELATION COEFFICIENT (PEARSON'S PRODUCT MOMENT CORRELATION COEFFICIENT): In statistics, a measure of the linear correlation between two variables in a sample, giving a value between +1 and -1 inclusive, where 1 is total positive correlation, 0 is no correlation and -1 is total negative correlation. It is used as an estimate of the correlation, ρ , in the whole population.

PEARSON'S RESIDUAL: A relation that is used in the analysis of contingency table and logistic regression given by $r = \frac{O_i - E_i}{\sqrt{E_i}}$, where O_i represents the observed value and E_i represents the corresponding predicted value under some model.

PEAT: A mass of dark-brown or black fibrous plant debris produced by the partial decomposition and disintegration of vegetation such as mosses, sedges, trees and other plants in wet places. It may accumulate in depressions. When forest-type peat is subjected to burial and hence pressure and heat it may be converted to coal. Peat is used to improve soil and as fuel, especially in Ireland and Sweden. .

PEAT MOSS: Any species of plants that make up the bryophyte genus *Sphagnum*, which grow in dense clumps around ponds, in swamps and bogs, on moist, acid cliffs and on lakeshores from tropical to subpolar regions. As they die and are compressed, they form organic matter which is harvested and used as fuel, to improve garden soil or to prepare potting soil.

PEAUCLLIER'S CELL: A mechanical device that produces a straight-line motion as well as tracing the inversive image of a locus of points,

used for describing a circle of any radius including the infinite. It was invented in 1873 by French engineer and army officer, Captain Charles-Nicolas Peaucellier (1832–1913) .

PEBBLE: 1. A small stone made smooth and round by the action of water, found mostly in a stream or on the seashore. 2. Fragments of rock with a diameter between 4 mm – 64 mm. 3. A colourless type of quartz rock crystal used in making lenses.

PECKING ORDER: A term in animal behaviour, describing an order of dominance in a herd of animals or a flock of birds that gives priority in access to food, mates, etc, because of its success in previous aggressive encounter.

PECTEN: Any of various comb-like structures in animals. The pecten in the eyes of birds consisting essentially of a network of blood vessels attached to the optic nerve and projecting into the vitreous humour. It may be involved in the supply of nutrients and oxygen to the retina. In scorpion, it is located on the underside and used as a sense organ. It is also located on the sides of the bill of ducks serving both as food-strainer and as a comb for preening.

PECTIC SUBSTANCES (PECTIN): A group of polysaccharides made up primarily of sugar acids. They are important constituents of plant cell walls and the middle lamella between contiguous cell walls. They can be extracted with hot water, dilute acid or ammonium oxalate solutions. They are precipitated from aqueous solution by alcohol and are commercially used for their excellent gel-forming ability. The main sources of pectic substances are the skin of citrus fruits such as lemon, lime orange, grapefruit, apple pomace and sunflower heads. Pectin is widely used in the food industry, principally in the preparation of gels. It is used as a base for jelly and as a stabiliser in some dairy products and frozen desserts, such as sherbet and also as edible protective coatings for sausages, almonds, candied dried fruit and soft dates.

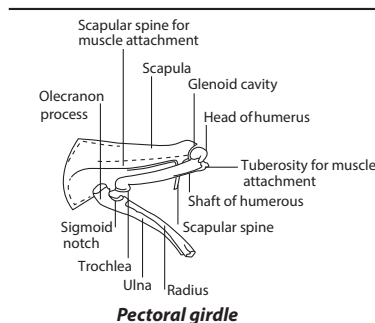
PECTIN: A colloidal mixture of large polysaccharides consisting chiefly of a galactose derivative called galacturonic acid. Pectin is an important constituent of the matrix of plant cell walls and of the middle lamella between adjacent cell walls and also found in certain plant juices. In fruits, pectin keeps the walls of adjacent cells joined together, helping them remain firm and hold their shape but breaks down to simple sugars that dissolve more readily, so that the fruits become soft and lose their shape. It is used in making jams and marmalades as it forms a gel with sucrose. It is also useful in the confectionery, pharmaceutical and textile industries.

PECTINATE: Having closely parallel projections that look like the teeth of a comb.

PECTORAL: Of or relating to the chest, breast or thorax. Examples are pectoral fin, pectoral muscles, etc.

PECTORAL FIN: Either of a pair of fins, situated just behind the head in fishes to the pectoral (shoulder) girdle, corresponding to the forelimbs of higher vertebrates that help to control the direction of movement during locomotion.

PECTORAL GIRDLE (SHOULDER GIRDLE): The bony or cartilaginous structure in vertebrates consisting of the right and left shoulder blades (scapulae) and collar bones (clavicles) to which the anterior limbs (pectoral fins, forelegs or arms) are attached. It consists of the clavicle and scapula in humans and, in those species with three bones in the pectoral girdle, the coracoid. Some mammals such as dog and horse have only the scapula and no joint exists between the forelimb and the thorax, the only attachment is muscles.



PECTORALIS MAJOR: A large, fan-shaped, powerful muscle of the upper chest wall that extends from the front of the thorax to the humerus and acts on the joint of the shoulder. It serves to flex, adduct and medially rotate the arm in the shoulder joint. **Pectoralis minor** is a triangular muscle that arises from the third, fourth and fifth ribs, close to their cartilages. Its fibres pass upward and laterally and its tendon of insertion is attached to the upper surface and the medial border of the coracoid process, near its tip. It functions to rotate the scapula, to draw it down and forward and to raise the third, the fourth and fifth ribs in forced inspiration.

PECULIAR GALAXY: A galaxy that has some abnormality of shape, size or content that sets it apart from the normal ellipticals, spirals and irregulars in the Hubble classification of galaxies.

PECULIAR MOTION: Motion of a galaxy that is a departure from the expansion described by Hubble's law. Peculiar motions occur because of the existence of large-scale structure in the universe.

PED: A unit of soil structure such as a prism, block, or granule, which is formed by natural processes and separated from each other by pores or void. This is in contrast with a clod, which is formed artificially.

PEDAL CURVE: The locus of the feet of the perpendiculars from a given point to the tangents to the given figure. The given point is known as the **pedal point**.

PEDAL TRIANGLE: The pedal triangle of a point P with respect to a triangle ABC is the triangle whose vertices are located at the feet of the perpendiculars from some given point to the sides of a specified triangle ABC .

PEDATE: 1. Resembling a foot. 2. Palmately divided into several lobes, as shown in the diagram. **Pedatifid** means pedately cleft.



Pedate leaves

PEDICEL: The stem that attaches a flower to the main floral axis, often developing in the axil of the bract.

PEDIGREE ANALYSIS: A type of genetic analysis in which a trait is traced through several generations of a family to determine the pattern and characteristics of the trait, including its mode of inheritance, age of onset and phenotypic variability. The information is displayed in a pedigree chart using standard symbols.

PEDIMENT: In geology, it is a broad, gently inclined rock surface formed at the base of a steeper slope of a mountain caused by erosion. Pediments are often covered with a thin layer of sediments, called alluvium, which have been eroded off the mountain.

PEDIPALP (PALPS; PALPI): Either member of the second pair of segmented appendages of the prosoma in the subphylum Chelicerata. Chelicerate pedipalps are appendages of six articles: the coxae, a single trochanter, the femur, a short patella, the tibia and the tarsus. In species with large chelicerae the pedipalps are typically adapted for walking or sensory functions. It is also variously adapted for defence, killing prey or digging. It is also used to manipulate food and transfer sperm to females during mating.

PEDOGENESIS (SOIL DEVELOPMENT; SOIL EVOLUTION; SOIL FORMATION; SOIL GENESIS): The process of soil formation as regulated by factors such as humification, weathering, leaching, and calcification. Biogeochemical processes act to both create and destroy order (anisotropy) within soils. These alterations lead to the development of layers, termed soil horizons, distinguished by differences in colour, structure, texture, and chemistry. These features occur in patterns of soil type distribution, forming in response to differences in soil forming factors.

PEDOLOGY: The scientific study of the properties and distribution patterns of soils worldwide, including their origin, characteristics, classification, habitats of living organisms and utilisation. It is one of the two main divisions of soil science, the other is edaphology.

PEDUNCLE: 1. The main stalk of a plant that bears the flowers which may be solitary or grouped in an inflorescence. 2. A stalk-like

structure in invertebrate animals, usually serving as an attachment for a larger part or structure. 3. A stalk-like bundle of nerve fibres connecting different parts of the brain. 4. The stalk-like base to which a polyp or tumour is attached.

PEEK: A feature introduced in Windows 7 that allows the user to view their desktop by placing (and not clicking) their mouse over the Show Desktop icon.

PEER: In computer networking, it is another computer on a network that has the same or similar rights as another computer.

PEER-TO-PEER COMMUNICATIONS (P2P, P-TO-P AND P2P COMMUNICATIONS): In a network, it refers to the transmission between two peer computers over a network. Files can be shared directly between systems on the network without the need of a central server. In other words, each computer on a P2P network becomes a file server as well as a client.

PEGMATITE: An extremely coarse-grained igneous rock composed chiefly of minerals found in ordinary igneous rocks. Pegmatites are widespread and very abundant, where they occur, especially in host rocks of Precambrian age, but their aggregate volume in the Earth's crust is small. Pegmatites have been economically valuable as sources of clays, feldspars, gem materials, industrial crystals, micas, silica and special fluxes, as well as beryllium, bismuth, lithium, molybdenum, rare-earth, tantalumnioibium, thorium, tin, tungsten and uranium minerals.

PEIERLS INSTABILITY: A periodic distortion in a one-dimensional metal (a chain of atoms with free electrons). A system of this type has a lower energy if there is a periodic lattice distortion, giving alternate longer and shorter bonds. It is observed in organic compounds containing alternating double and single bonds.

PEKING MAN: It is considered a subspecies of *Homo erectus* known from fossil remains of the Pleistocene Epoch found in the cave of Choukoutain in 1927 near Beijing (Peking). It is 500,000 to 750,000 years old. Peking man is characterised by a cranial capacity averaging about 1,000 cubic cm, though some individual skull capacities approached 1,300 cubic cm, nearly the size of modern man's. It had a skull that was flat in profile, with a small forehead, a keel along the top of the head for attachment of powerful jaw muscles, very thick skull bones, heavy browridges, an occipital torus, a large palate and a large, chinless jaw.

PELAGIC: A term describing organisms that swim or drift in a sea or a lake as distinct from those that live on the bottom (benthos). Pelagic organisms are divided into plankton and nekton.

PELAGIC ZONE: One of the two basic subdivisions of the marine biome describing any water in the sea that is not close to the bottom or near to the shore. It occupies 1,370 million cubic kilometres and has a vertical range of up to 11 kilometres. The other zone is the benthos, which is below the pelagic with increased pressure, lower temperature and less light.

PELL EQUATION: An equation of the form $y^2 = ax^2 + 1$, where a is any positive whole number except a square number. For example, the equation $y^2 = 92x^2 + 1$, has the solutions $x = 0, y = 1$; $x = 120, y = 1151$; $x = 276240, y = 2649601$; and so on. Each successive solution is about 2300 times the previous solution. In fact, the solutions are every eighth partial fraction (where x is the numerator and y is the denominator) of the continued fraction for $\sqrt{92}$. A Pell equation was used in finding the solution to Archimedes' cattle problem.

Archimedes' cattle problem, also known as the **problema bovinum** or **problema Archimedes** is a problem in Diophantine analysis, the study of polynomial equations with integer solutions.

PELL NUMBER: The n th term in the sequence 0, 1, 2, 5, 12 and 29. Defined by the recurrence $P_0 = 0, P_1 = 1$ and $P_n = 2P_{n-1} + P_{n-2}$. Pell numbers can be used to approximate the square root of two, to

find square triangular numbers, to construct integer approximations to the right isosceles triangle and to solve certain combinatorial enumeration problems.

PELL'S EQUATION: In mathematics, the Diophantine quadratic equation, of the form $X^2 - Dy^2 = \pm N$, where D and N are natural numbers usually with $N = 1$ and with D squarefree. In Cartesian coordinates, the equation has the form of a hyperbola; solutions occur wherever the curve passes through a point whose x and y coordinates are both integers, such as the trivial solution with $x = 1$ and $y = 0$.

PELLAGRA: A chronic disease of subtropical countries in which the staple food is maize or sorghum, with little meat or other vegetables. It is caused by the deficiency of niacin (vitamin B₃) and the acid tryptophan which are contained in protein foods such as beans, peas and yeast. The symptoms include digestive disorders, skin eruptions and mental disturbances.

PELLET: 1. A small ball or piece of material that has been pressed tightly together, for example, foodstuff produced from various raw materials and additives for animal feed. 2. A small bullet or ball of metal fired from a gun, especially an air gun or an imitation bullet for use in a toy gun or a ball, usually made of stone, formerly used as a cannonball or as a missile fired from a catapult. 3. A small round piece of the faeces of some animals such as sheep or rabbit. 4. An undigested mass of food, mostly bone and hair, which is regurgitated by owls and other birds of prey.

PELLICLE: The thin, outer covering, composed of protein that supports the cell membrane, giving protection and maintaining the shape of certain unicellular organisms such as Euglena. It is transparent and in ciliated organisms such as paramecium, contains small pores through which the cilia emerge. The pellicle consists of many bacteria that adhere to the surface by their attachment pili, allowing the organisms to remain in the broth, from which they take nutrients, while they congregate near air, where the oxygen concentration is greatest. Examples of protists with a pellicle are the euglenoids and the paramecium, a ciliate.

PELLUCID: Transparent; admitting light.

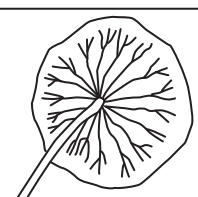
PELORIA: An abnormal regularity in the structure of a flower which occurs usually in irregular flowers. Such flowers are radially symmetrical instead of their normal bilateral symmetry. This is often a hereditary abnormality.

PELT: The skin of an animal with the hair attached.

PELTATE: 1. Having the shape of a shield. 2. A flat, circular structure, borne on a stem and attached near the centre of the lower surface, instead of at the base or margins. This is usually said of a leaf or other stalked structures or organs. **Peltafid** means shaped like a shield and divided into segments.

PELTIER EFFECT: A rise or drop in temperature produced when electric current flows through the junction of two dissimilar metals such as copper and bismuth. The rise or drop in temperature depends upon the direction of the current. For example, when a battery is connected by two pieces of copper wire to a length of bismuth wire, a temperature rise occurs at the junction where the current passes from copper to bismuth and a temperature drop occurs at the junction, where the current flows from bismuth to copper. It was discovered in 1834 by Jean Peltier (1785-1845), French physicist. The heat produced at the junction is proportional to the magnitude of the current, showing that an electromotive force known as Peltier's emf is present at the junction.

PELVIC FINS: Either of a pair of lateral hind fins of fishes, attached to the pelvic girdle and corresponding to the hind limbs of higher vertebrates.



Peltate leaf

PELVIC GIRDLE (PELVIS, HIP GIRDLE): The irregular ring-shaped bony or cartilaginous structure in vertebrates connecting the spine to the thigh bones (femurs). The pelvic girdle articulates dorsally with the backbone. It is made up of two halves, each produced by the fusion of the ilium, ischium and pubis.

PELVIC SPUR (CLOACAL SPUR; VESTIGIAL LIMBS): A claw in boas and pythons, located on each side of the cloaca, used by the male snake, while mating.

PELVIS: 1. The lower area of the abdomen featuring the bones and muscles, used to move the legs or hind limbs. In adults, the pelvis is composed of three large bones plus the coccyx totalling 3-5 bones. Before adolescence each hip bone consists of three separate bones yet to be fused, which are the ilium, ischium and pubis consisting of more than ten bones, depending on the composition of the coccyx. 2. A cup-shaped chamber in the kidney into which urine drains from the kidney tubules before passing to the ureter.

PEN DRIVE (JUMP DRIVE; USB FLASH DRIVE, DATA STICK; KEYCHAIN DRIVE; THUMB DRIVE): A portable drive, often the size of a thumb that connects to the computer USB port and used to transfer and store information. They are available in sizes such as 256MB, 512MB, 1GB, 5GB and 16GB. Because of the size it is conveniently carried on the body from place to place.

PENALTY FUNCTION: 1. In mathematics, the exterior penalty function for a set S is a non-negative continuous function P that is zero on S and strictly positive outside S . 2. **Interior penalty function** or **barrier function** for a set S with non-empty interior is a non-negative function B that is continuous over the interior of S and approaches infinity as the boundary is approached from within the set.

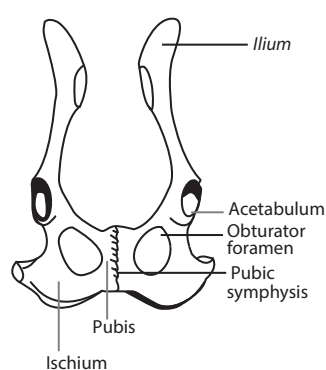
PENALTY FUNCTION METHODS: 1. In mathematics, methods for reducing constrained extremum problems to problems of unconstrained optimisation by solving instead a sequence of unconstrained problems constructed by adding internal or external penalty functions to the objective. 2. **Exact penalty function methods are methods**, where it is possible to obtain the solution to the original problem by solving the unconstrained problem for some fixed K .

PENCIL: 1. A set of lines or rays passing through a common vertex or a family of parallel lines. 2. A family of geometric figures with a common property, such as a pencil of circles that all intersect at the same pair of points; these are in fact lines in an abstract geometry.

PENDULOSE (PENDENT): Hanging loosely; suspended so as to swing freely.

PENDULUM: A weight hung from a pivot so that it can swing freely back and forth subject to a restoring force due to gravity that accelerates it back toward the equilibrium position. It has a weight, called a bob, swinging at the end of a rod or chord. One complete cycle of swing is called a period, which depends on the length of the pendulum. Pendulum was discovered by Galileo Galilei and was used for timekeeping and has found applications in pendulum clocks, accelerometers and seismometers.

PENETRANCE: In genetics, the frequency or rate of occurrence, typically in percentage, of a particular trait or condition such as disease, expressed among individuals carrying the same deleterious or disease-causing gene. There are two forms of penetrance: complete and incomplete. **Complete penetrance** occurs, when a deleterious or disease-causing gene is expressed in all individuals carrying the gene. **Incomplete penetrance**, also known as **reduced penetrance** occurs, when some individuals express the associated trait while others do not, although they carry the disease-causing gene.



A mammalian pelvic girdle

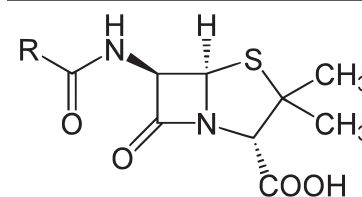
PENEPLAIN: A surface that was high at one time, but reduced by erosion to gently rolling surface resembling a plain, leaving so little gradient that essentially no more erosion could occur.

PENETRATION DEPTH: 1. The depth beneath the surface of a superconductor that a magnetic field can penetrate without being expelled if the field strength is not too high to destroy the superconducting state. This thin surface layer usually has a depth of about 5-10 nm and is temperature dependent. Its symbol is L . 2. In electricity, it is induction heating and defined as the thickness of a layer, extending inward from a conductor's surface, whose resistance to direct current equals the resistance of the whole conductor to alternating current of a given frequency. It is also the greatest depth in an ultrasonic test piece at which indications can be measured.

PENICIL: A small, brush-like tuft of hairs. **Penicillate** means having a brush-like tuft of hairs at the end.

PENICILLIN: An antibiotic belonging to a group originally derived from mould of the genus *Penicillium* or an antibiotic substance produced by a mould called *Penicillium notatum*. It is now produced synthetically and its formula

is $C_{16}H_{18}N_2O_4S$. Penicillin works by inhibiting the enzymes responsible for bacterial cell wall synthesis and so is not effective against microorganisms without cell walls or with certain variant cell walls such as the tuberculosis bacillus. It is used to treat infection caused by some types of bacteria.



Structural formula of Penicillin

It was the first antibiotic discovered by Sir Alexander Fleming in 1928 and used in medicine. The nature of R-group distinguishes one penicillin from the other.

PENINSULA: A piece of land surrounded on three sides by water, the fourth side being attached to a large landmass or a narrow piece of land that juts out from the mainland into water. Some examples of peninsulas are the Iberian Peninsula forming Portugal, Spain andorra and part of France and Italy; Malay Peninsula, which includes parts of Thailand, Malaysia and Singapore; and the Arabian Peninsula, which includes Saudi Arabia, United Arab Emirates (UAE), Oman, Yemen, etc.

PENIS: The male reproductive organ of mammals, some birds and reptiles (crocodiles). Other male reptiles possess a **hemipenis**, one half of paired erectile copulatory organs. In mammals the penis consists of three elongated masses of erectile tissue. The central corpus spongiosum (corpus urethrae) lies ventral to the paired corpora cavernosa. The urethra runs along the underside of the spongiosum and then normally rises to open at the expanded, cone-shaped tip, the glans penis, which fits like a cap over the end of the penis. Loose skin encloses the penis and also forms the retractable foreskin or prepuce. The penis becomes erect during the precopulatory activity, caused by nervous stimulation resulting in engorgement of the spiral helicine arteries and the plentiful venous sinuses of the organ making it possible to be inserted into the vagina or cloaca. Penis contains a duct through which the sperms pass into the female reproductive tract to ensure internal fertilisation or used to pass urine to the outside.

PENNATE (PINNATE): 1. With feathers or wings. 2. In botany, shaped like a feather.

PENNI-PARALLEL (PENNIVEINED): Having parts that arise parallel to one another from a main axis, as seen in feathers.

PENNING GAUGE: A type of cold-cathode ionisation gauge, invented in 1936 by the Dutch physicist Frans Penning (1894-1953). It consists of an anode and a cathode in which a discharge is maintained between the two electrodes with a potential difference of about 2.2 KV through current limiting resistors. An axial magnetic field is also applied to cause electrons to move in spiral paths and increase the ionisation current and sensitivity. This type of gauge is used in the range

$10^{-3} - 10^{-1}$ torr. The Penning trap can be used for storing thousands of particles or for bringing a single particle to rest.

PENNING IONISATION: A photochemical process that produces a positively charged ion. It was named after the Dutch physicist Frans Penning, who first reported it in 1927. It is the interaction between a gas-phase excited-state atom or molecule A^* and a target molecule B, resulting in the formation of a radical molecular cation B^+ , an electron e^- and a neutral gas molecule A, written as: $A^* + B \rightarrow A + B^+ + e^-$. An example of Penning ionisation is the ionisation of mercury by argon: $Ar^* + Hg \rightarrow Ar + Hg^+ + e^-$ which occurs because the energy of the metastable state of argon is higher than the ionisation energy of mercury.

PENROSE IMPOSSIBLE STAIRCASE (PENROSE STAIRWAY): A paradoxical impossible figure that appears to show a continuous four-sided staircase in which the stairs make four right angle turns as they ascend or descend yet form a continuous loop, so that person could climb them forever and never get any higher.

PENROSE PROCESS: A hypothetical process suggested by Sir Roger Penrose in 1969 in which the rotational energy of a rotating black hole can be extracted. A particle spirals into the ergosphere of the black hole in a direction opposite to the black hole's rotation and then splits into two particles, the one with negative energy, falls back into the black hole, causing the rotation rate to decrease. The other particle, with positive rotation and an energy greater than the energy of the original particle moves away. This results in the energy being extracted at the expense of the rotational energy of the black hole.

PENROSE TILING: A method of aperiodic tiling with a five-fold symmetry. A Penrose tiling may be constructed so as to exhibit both reflection symmetry and five-fold rotational symmetry. Since this type of symmetry is not allowed in crystallography, two types of tile are needed, which are called 'fat' and 'skinny'. Adjacent tiles have to obey certain matching rules.

PENROSE TRIANGLE: A figure that appears to represent a three-dimensional triangular solid, made of three straight beams of square cross-section which meet pairwise at right angles at the vertices of the triangle they form, but which is in fact impossible to construct.

PENROSE, SIR ROGER (1931-): English mathematical physicist, son of the geneticist Lionel Penrose, renowned for his work in mathematical physics and his contributions to general relativity and cosmology. He was educated at University College, London and obtained his PhD in 1957 from Cambridge University. In 1973 he was elected Emeritus Rouse Ball Professor of Mathematics at the Mathematical Institute, University of Oxford and Emeritus Fellow of Wadham College. He has received a number of prizes and awards, including the 1988 Wolf Prize for Physics, which he shared with Stephen Hawking for their contribution to the understanding of the universe. He is also a recreational mathematician and philosopher.

PENTA-: A prefix denoting five. For example, a pentagon is five sided polygon.

PENTACAMEROUS: Having five locules.

PENTACARPELLARY: Having five carpels.

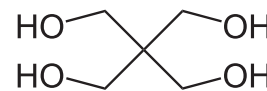
PENTACYCLIC: 1. In chemistry, it is having five usually fused rings as in many triterpenoids. 2. In botany, it refers to flowers that possess five whorls. Such a flower possesses a calyx, corolla, two whorls of stamens and a single whorl of carpel.

PENTADACTYL LIMB: A limb with five digits, as found in tetrapod vertebrates (amphibians, reptiles, birds and mammals) and having three parts: the upper arm, upper leg (thigh) containing one long bone, the forearm or shank containing two long bones and the hand or foot, which contains a number of small bones with modifications in the various species, based on the functions.

PENTADELPHOUS: Having stamens that are fused together partially by the filaments to form five separate groups or clusters. This is seen in the flowers of Linden.

PENTAERYTHRITOL (2,2-BIS(HYDROXYMETHYL)1,3-PROPANEDIOL):

A white crystalline compound, having the chemical formula $C_5H_{12}O_4$, density 1.396 g/cm^3 , melting point 260.5°C (533.65 K) and boiling point 276°C (549 K) at 30 mmHg. It is used in making the explosive pentaerythritol trinitrate and in producing resins and other organic products.



Chemical Structure of Pentaerythritol

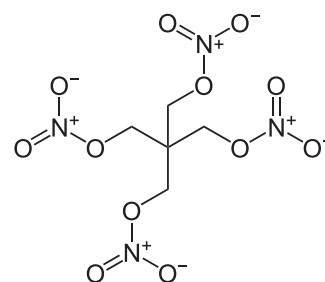
Derivatives of pentaerythritol are components of alkyd resins, varnishes, PVC stabilisers, oil esters and olefin antioxidants.

PENTAERYTHRITOL TETRANITRATE, PETN [(3-NITROOXY-2,2-BIS(NITROOXYMETHYL)PROPYL] NITRATE; PENTHRITE]:

The nitrate ester of pentaerythritol with the chemical formula

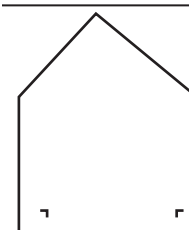
$C_5H_8N_4O_{12}$, density 1.77 g/cm^3 at 20°C , melting point 141.3°C (414 K) and boiling point 180°C (453 K). PETN refers to the five carbon atoms of the neopentane skeleton and known as one of the most powerful high explosives.

It is produced by reaction of pentaerythritol with concentrated nitric acid. PETN mixed with a plasticiser forms a plastic explosive. Semtex, another explosive is produced by mixing PETN with RDX (Research Department Explosive or cyclonite) and other minor additives. Medically, it is used as a vasodilator drug to treat heart conditions, such as angina.



Structural formula of Pentaerythritol Tetranitrate

PENTAGON: In geometry, it is a polygon with five sides and five angles. The sum of the internal angles in a simple pentagon is 540 degrees; and the sum of the external angles is 360 degrees. A regular pentagon is when all angles are equal and all sides are equal. An irregular one does not have all angles nor sides equal. **Pentagonal** is having five angles.



Pentagon

PENTAGONAL NUMBER: A number of

the form $n\left(\frac{3n-1}{2}\right)$, which is defined as the total number, $P(n)$, of dots marking off unit segments of the sides of a set of $n - 1$ nested pentagons, given by the formula $P(n) = n\left(\frac{3n-1}{2}\right)$.

PENTAGYNOUS: Having five styles or pistils. This relates to plants of the order Pentagyna.

PENTAGRAM (PENTANGLE; PENTACLE): A star-shaped figure formed by extending all the sides of a regular pentagon to meet in pairs at five points.

PENTAHEDRON: A five-sided polyhedron, for example, a pentagon-based pyramid.

PENTAHYDRATE: A chemical containing five molecules of water of crystallisation, for example, cupric sulfate pentahydrate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$).

PENTAMEROUS: With parts arranged in sets or multiples of five.

PENTANDROUS: Having five stamens. This relates to plants in the order Pentandria.

PENTANE: A hydrocarbon molecule that contains carbon-carbon single bonds with the general formula C_5H_{12} . It is a component of some fuels and used as special solvents in the laboratory.

PENTANEDIOIC ACID (GLUTARIC ACID; PROPANE-1,3-DICARBOXYLIC ACID; 1,3-PROPANEDICARBOXYLIC ACID): A simple dicarboxylic acid with the chemical formula $C_5H_8(COOH)_2$, molar mass 132.12 g/mol , melting point $95 - 98^\circ \text{C}$ ($368 - 371 \text{ K}$) and

boiling point 200 °C (473 K) at 20 mmHg. Pentanedioic acid can be prepared by the ring-opening of butyrolactone with potassium cyanide to give the mixed potassium carboxylate-nitrile that is hydrolysed to the diacid. It can also be prepared from reacting 1,3-dibromopropane with sodium or potassium cyanide to obtain the dinitrile, followed by hydrolysis. It is used in the perfume industry and also in the production of polymers such as polyester polyols, polyamides.

PENTANOIC ACID (VALERIC ACID; BUTANE-1-CARBOXYLIC ACID): Straight-chain alkylcarboxylic acid with a pungent odour and the chemical formula $C_5H_{10}O_2$, density 0.930 g/cm³ (liquid), melting point -34.5 °C (238.65) and boiling point 186 -187 °C (459 – 460 K). It occurs naturally in the perennial flowering plant valerian (*Valeriana officinalis*). It is primarily used in the synthesis of its esters. Volatile esters of valeric acid have pleasant odours and are used in perfumes and cosmetics.

PENTAPETALOUS: Having five petals.

PENTAPRISMO-: An affix used in names to denote ten atoms bound into a pentagonal prism.

PENTAPTEROUS: Having five wings.

PENTAQUARK: A hypothetical long-lived subatomic particle consisting of four quarks and one antiquark bound together having a mass of just over 1500 MeV. Originally hypothesised by Michał Praszalowicz in 1987, there is some evidence of the existence of this particle but, at present, it is not conclusive.

PENTARADIAL SYMMETRY (PENTAMERISM; PENTAGONAL SYMMETRY): The balanced distribution of duplicate body parts or shapes (sensory and feeding structures) in a five fold circular pattern around a central axis, 72° apart. Such organisms include echinoderms such as sea stars, sea urchins, and sea lilies are pentamerous as adults, with five arms arranged around the mouth; they are the only organisms having that symmetry.

PENTASEPALOUS: Having five sepals.

PENTASTICHOUS: Arranged in five vertical ranks or rows, as seen in the leaves of an apple or a cherry tree.

PENTAVALENT: Having a valency of five.

PENTIUM MICROPROCESSOR: A 32-bit microprocessor introduced by Intel in 1993. It contains 3.3 million transistors, nearly triple the number contained in its predecessor, the 80486 chip. Though still in production, the Pentium processor has been superseded by the Pentium Pro and Pentium II microprocessors. Since 1993, Intel has developed the Pentium III and more recently the Pentium 4 microprocessors.

PENTLANDITE: A yellowish-bronze, opaque mineral consisting of a mixed iron-nickel sulfide, crystallising in the cubic system. It is the main ore of nickel. Its chemical formula is $(Fe, Ni)_9S_8$ with a Mohs scale hardness of 3.5 -4 and specific gravity of 4.6 - 5.0. The largest deposit of this mineral is at Sudbury in Ontario, Canada and also mined in Manitoba, Canada; Russia and Western Australia.

PENTODE: A thermionic valve having five active electrodes. It consists of an anode, a cathode and three grids. It is used to suppress the loss of electrons from the anode as a result of secondary emission.

PENTOMINO: Any of the 12 plane figures that can be formed by joining five congruent squares along their sides.

PENTOSE PHOSPHATE PATHWAY (PHOSPHOGLUCONATE PATHWAY; HEXOSE MONOPHOSPHATE SHUNT): A pathway by which glucose is metabolised or transformed in plants and microorganisms into other molecules. Glucose-6-phosphate is oxidised to 6-phosphogluconic acid, which undergoes oxidative decarboxylation to form ribulose-5-phosphate and finally transformed into fructose-6-phosphate. In plants, the pentose phosphate pathway also plays a role in the synthesis of sugars from carbon dioxide. In animals, the pathway occurs at various sites, including the liver and adipose tissue; in plants, it takes place mainly in the plastids.

PENTYL GROUP (PENTYL RADIAL): Any of the organic group C_5H_{11} - derived from pentane.

PENUMBRA: 1. A partial outer shadow that is lighter than the darker inner shadow, umbra, found between regions of complete shadow and complete illumination in an eclipse. 2. A greyish area surrounding the dark centre of a sunspot.

PEPO: A modified berry that is fleshy, indehiscent and many seeded with a tough or hard rind. Examples are pumpkins, cucumbers, watermelon and squash.

PEPPER: 1. Any of the plants in the genus *Capsicum* of the nightshade family, notably *Capsicum annuum*, *Capsicum frutescens* and *Capsicum boccatum* cultivated for their edible, red, green or yellow berries; and native to Central and South America and cultivated extensively throughout tropical Asia and Africa. The berries are rich in vitamins A and C and used in flavouring food and as vegetable. Some varieties such as the red pepper can be very hot-tasting. Others have mild tastes. The pungency of hot peppers such as chili and cayenne peppers, comes from the compound capsaicin in the internal partitions of the fruit. 2. A species of plants in the genus *Piper* in the family, Piperaceae. They are tropical and subtropical plants that grow as vines, shrubs or small trees. An example is *Piper nigrum*, the source of black, white, red and green peppers, which is determined by the time of harvest and the method of processing. It is a perennial climbing vine, that can reach 10 metres and native to tropical South Asia, but grown in other tropical areas of the world. Its small berry-like fruit is called peppercorn, which is a hot and pungent spice, used for flavouring food. Black pepper is also used in traditional medicine, particularly for digestive ailments. *Piper amalago* or rough-leaved pepper, bears a pungent black berry, the size of a mustard seed, with the taste and flavour as that of black pepper and is used as a hot spice to flavour food; and also used medicinally.

PEPSIN: An enzyme found in gastric juice secreted by cells in the stomach of vertebrates that partially digest proteins into peptides. It catalyses the breakdown of proteins to polypeptides in vertebrate stomachs. Discovered in 1836 by Theodor Schwann, it is one of the three principal protein-degrading or proteolytic enzymes in the digestive system. The other two are chymotrypsin and trypsin. Pepsin functions only in the acidic (pH 1.5 – 2.5) environment of the stomach. It is secreted as an inactive precursor, pepsinogen. Pepsin is prepared commercially from the glandular layer of fresh hog stomachs. Its commercial uses include cheese making, in the leather industry to remove hair and residual tissue from hides and in the recovery of silver from discarded photographic films by digesting the gelatin layer that holds the silver.

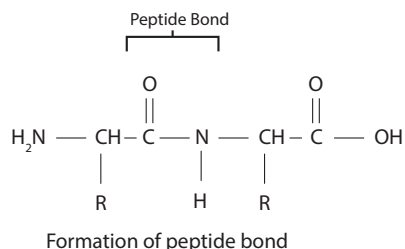
PEPSINOGEN: An inactive form of pepsin synthesised and stored in cells of the mucous membrane of the stomach into the lumen, where it is converted to pepsin by hydrochloric acid during digestion and also by the action of pepsin itself.

PEPTIC ULCER: Damage to the epithelial layer of the stomach lining generally caused by *Helicobacter pylori* bacteria infection and long-term use of aspirin and similar drugs. Ulcers can bleed, perforate the abdominal wall or block the gastrointestinal tract. It causes burning ache and hunger-like pain.

PEPTIDASE: A subclass of proteolytic enzymes often secreted in the body that catalyses the splitting of proteins into smaller peptide fractions and amino acids. It is the final process of digestion of proteins, a process known as **proteolysis**.

PEPTIDE: A linear molecule made up of two or more polymers of amino acids molecules, linked by the peptide bonds. Peptides are formed by the elimination of a molecule of H_2O from the amino group of one amino acid and the carboxyl group of the next amino acid. The number of amino acid molecules in the peptide is referred to as di-, tri- or polypeptide (two, three or many amino acids). Peptides include hormones, antibiotics and other compounds that participate in life processes.

PEPTIDE BOND: Covalent chemical bond formed between two molecules by the elimination of a molecule of H_2O from the amino group (NH_2) of one amino acid and the carboxyl group ($COOH$) of the next amino acid. In other words, a peptide bond is formed between two amino acids, when they are joined together by a condensation reaction (two molecules joining together and releasing a smaller molecule such as water as a by-product). A peptide bond can be broken by amide hydrolysis by adding water.



PEPTIDE MASS FINGERPRINTING (PMF; PROTEIN FINGERPRINTING): Developed in 1993, it is an analytical technique for identifying proteins using de Novo peptide sequencing and the information stored in a database. The partial hydrolysis of the unknown protein is cleaved into smaller peptides whose absolute masses are determined using a mass such as MALDI-TOF or ESI-TOF. The various masses obtained are compared to the database of protein sequences.

PEPTIDE NUCLEIC ACID (PNA): An artificially synthesised substance that contains the same bases as those found in biological nucleic acids but whose backbone is held together by amide bonds. DNA and RNA have a deoxyribose and ribose sugar backbone, respectively. PNA's backbone is composed of repeating N-(2-aminoethyl)-glycine units linked by peptide bonds. PNA is not known to occur naturally. The various purine and pyrimidine bases are linked to the backbone by methylene carbonyl bonds. PNAs are depicted like peptides, with the N-terminus at the first (left) position and the C-terminus at the right.

PEPTIDOGLYCAN: The main component of the bacterial cell wall, consisting of a two dimensional network of heteropolysaccharides running in one direction, cross linked with polypeptides running in perpendicular direction forming the cell wall. Peptidoglycan, gives structural strength to the bacterial cell wall and counteracts the osmotic pressure of the cytoplasm. Chains of amino sugars N-acetylglucosamine and N-acetylmuramic acid are linked to tripeptide of alanine, glutamic acids and lysine or diaminopimelic acids in a perpendicular direction. Archaeobacteria possess a similar polysaccharide, pseudopeptiglycan, with N-acetylalosaminuronic acid.

PEPTIZATION: Reversal of coagulation or flocculation, i.e., the dispersion of agglomerates to form a colloiddally stable suspension or emulsion.

PEPTONE: Any of a class of water-soluble and diffusible substance composed of a mixture of polypeptides and amino acids obtained by the action of the gastric and pancreatic juices. Peptones can also be obtained from albuminous matter by the action of boiling water and boiling dilute acids. It is used as a nutrient in culture media in bacteriology.

PER CAPITA: A figure that expresses an average of a particular datum for each person in the population being measured. It is used, for example, to compare the economic indicators of countries with different population sizes. The most commonly measured indicators are gross domestic product and income. It is Latin term for "by the head".

PER MIL (PER MILL; PERMILLAGE): A term used to denote a fraction, proportion, ratio or rate of a thousand, for example, 1/1000 denotes one part of 1000.

PER-: Prefix indicating that a chemical compound contains an excess of an element, for example, peroxide.

PER-UNIT RATE: A comparison of two measurements in which one of the terms has a value of 1. Per-unit rates tell how many of one

thing there are for 1 unit of another thing. Examples are kilometres per hour, cost per item, salary per month, etc.

PERAMORPHOSIS: Any change in which individuals of a species mature past adulthood and take on hitherto unseen traits. Peramorphic forms may arise either by acceleration or by hypermorphosis and conform to the theory of recapitulation.

PERCENT: A way of representing a number as a fraction of 100, for example, 50% equals 50/100. Its symbol is %. It is used to express a proportion of an amount stated in terms of one-hundredths that is calculated by multiplying a fraction by 100. A quantity can be increased by any percentage but cannot be decreased by more than 100 percent. For example, a disease such as smallpox that has been eradicated by 100 percent ceases to exist.

PERCENT CIRCLE: A tool that is used to measure and draw figures such as circle graphs that involve percentages.

PERCENT COMPOSITION OF MASS: The relative measure of the mass of each element in a compound.

PERCENT ERROR: Formula used to find the inaccuracy of a measurement. It is the difference between the estimated number and the actual number when compared to the actual number expressed in percent format. The formula is given by:

$$\% \text{ error} = \frac{\text{error}}{\text{actual value}} \times 100$$

PERCENT PURITY: The percent of a specified compound or element in an impure sample.

PERCENTAGE COMPOSITION: The proportion of an element in a given compound. This is usually expressed in percentage terms. Knowing the percentage composition of a compound, it is possible to work its empirical formula.

PERCENTAGE DECREASE: The percent decrease from an initial value y_1 to a final y_2 is $\frac{100(y_2 - y_1)}{y_1}$. For example, if a price falls from 20 to 16, it is $\frac{4}{20} = 0.2 = 20$ percent decrease.

PERCENTAGE EXPOSED: The accessibility of the atoms of metal in metallic catalysts, supported or unsupported, depends upon the percentage of the total atoms of metal which are surface atoms. It is recommended that the term percentage exposed be employed for this quantity rather than the term dispersion which has been frequently employed.

PERCENTAGE INCREASE: The percent increase from an initial value y_1 to a final y_2 is $\frac{100(y_2 - y_1)}{y_1}$. For example, if a price increases from

$$15 \text{ to } 20, \text{ it is } \frac{100(20 - 15)}{15} = \frac{100(5)}{15} = 33.3\% \text{ increase.}$$

PERCENTAGE IONISATION: The percentage of the weak electrolyte that ionises in a solution.

PERCENTILE (CENTILE): A term introduced by Galton in 1885, which denotes a value on a scale of one hundred that indicates whether a distribution is below or above it or the set of division that produces exactly 100 equal parts in continuous values. For example, the 75th percentile is the value of a variable that has 75% of the collection below that value. The 25th percentile is called the lower quartile, the 50th percentile is the median and the 75th percentile is the upper quartile. It can be used as a cut-off value to determine the performance of candidates in an examination.

PERCEPTION: Awareness or understanding of sensory information using both the raw data detected by the senses and previous experience.

PERCHED WATER TABLE: The upper surface of a local zone of soil water saturation on a relatively impermeable layer, usually clay, at some depth within the soil. The soil within or below the impermeable layer is not saturated with water.

PERCOLATION: 1. The movement and filtering of fluids or gas passing through a filter or porous substance. A typical application is coffee percolation, where the solvent is water, the permeable substance is the coffee grounds and the soluble constituents are the chemical compounds that give coffee its colour, taste and aroma. 2. Percolation of soil water is the process by which water with dissolved minerals move downwards slowly through the soil pores. The rate of percolation depends on the type of soil. Percolation rate helps in selection of suitable soil for crop growth.

PERCU-YERVICK APPROXIMATIONS (PERCUS-YEVICK EQUATION): An approximation used in statistical mechanics to calculate the radial distribution function of a system.

PERCUSSION: 1. A method of playing musical instruments such as cymbals, drums, triangles and xylophones by striking their surfaces with another object. It is used to accentuate and dramatise certain notes or rhythms. 2. A method of tapping on a surface to determine the underlying structure, used in clinical examinations to assess the condition of the thorax or abdomen. It is one of the four methods of clinical examination, together with inspection, palpation and auscultation. It is done with the middle finger of one hand tapping on the middle finger of the other hand, which is placed firmly on the body over tissue, not bone. There are two types of percussion: direct, which uses only one or two fingers and indirect, which uses the middle/flexor finger. There are four types of percussion sounds: resonant, hyper-resonant, stony dull or dull. A dull sound indicates the presence of a solid mass under the surface. A more resonant sound indicates hollow, air-containing structures.

PERENNATE: To renew or survive from one growing season to the next, usually with a period of reduced activity between seasons, as occurs in biennial plant or herbaceous perennial plants such as, cocoyam, carrot, iris, potato and dahlia. **Perennating organ** is the part of biennial or herbaceous perennial plants such as the stem, root, leaves or whole shoots that often develops underground and able to store food and allows it to survive adverse environmental conditions until it a favourable season, when it germinates.

PERENNATION: The survival of biennial or perennial plants from one year to the next by vegetative means. Examples are tubers like dahlia and cocoyam bulbs like onion, etc. In biennials and herbaceous perennials, the aerial parts of the plant die and the plants survive by means of underground storage organs. These perennating organs are also frequently responsible for vegetative propagation. Woody perennials also survive by reducing their metabolic activity, for example, leaf loss in deciduous trees and shrubs.

PERENNIAL: Plant species that live for more than two years. They include winter hardy herbaceous plants and woody plants like shrubs and trees. Woody perennials (trees and shrubs) have a permanent aerial form, which continues to grow year after year. Herbaceous perennials such as small flowering plants have aerial shoots that die down each autumn and are replaced in spring by new shoots from an underground structure.

PERFECT: In mathematics, precisely factorisable into equal integral or polynomial roots. For example, 36 is perfect square, because $6 \times 6 = 36$, 27 is a perfect cube, because $3^3 = 27$ and 32 is a perfect fifth power, because $2^5 = 32$.

PERFECT CUBE: An integer is a perfect cube, if it is of the form m^3 where m is an integer. For example, $8 = (2)^3$ and $64 = (4)^3$.

PERFECT FLOWER (BISEXUAL; HERMAPHRODITE; MONOCLINIC): A flower containing all the parts: stamen, pistil, petals and sepals.

PERFECT GAS (IDEAL GAS): A theoretical gas whose physical behaviour obeys all the gas laws exactly and for a given quantity of gas, the product of the volume V and pressure P is proportional to the absolute temperature T or $PV = kT$, where k is a constant. An ideal gas would consist of molecules that occupy negligible space and have negligible forces between them.

PERFECT MATCHING (1-FACTOR MATCHING; COMPLETE MATCHING): A matching of a graph in which all vertices are matched by an edge. In other words, every vertex of the graph is incident to exactly one edge of the matching.

PERFECT NUMBER (ARMSTRONG NUMBER; NARCISSISTIC NUMBER): An n -digit number, which is equal to the sum of its digits raised to the n th power. For instance, 371 is narcissistic because $3^3 + 7^3 + 1^3 = 371$ and 9474 is narcissistic because $9^4 + 4^4 + 7^4 + 4^4 = 9474$. The smallest narcissistic number of more than one digit is $153 = 1^3 + 5^3 + 3^3$. The largest narcissistic number (in base 10) is 11513221901876399256509559797 971522401, which is the sum of the 39th powers of its digits.

PERFECT PITCH (ABSOLUTE PITCH): 1. The position of a tone in the musical scale determined according to its number of vibrations per second, irrespective of other tones. 2. The ability of a person to identify and reproduce a note without reference to a tuned musical instrument.

PERFECT POWER: A positive integer of the form m^n where m and n are integers and $n > 1$ is called a perfect power.

PERFECT SET: In mathematics, it is a set that equals its derived set; a set that is closed with no isolated points; for example, Cantor set which is totally disconnected.

PERFECT SOLUTION: Perfect solutions occur when the intermolecular forces between molecules of the pure substance are similar to the forces between molecules of one and molecules of the other. The term refers to a mixture of liquids that obey Raoult's law over a whole range of concentrations.

PERFECT SQUARE (SQUARE NUMBER): A number or algebraic expression, which is equal to the square of another number or expression. Examples are $16 = 4^2$ and $(3x - 4)^2 = 9x^2 - 24x + 16$.

PERFLUOROCARBONS (PFCs): By-products of aluminum smelting and uranium enrichment composed only of the elements tetrafluoromethane (carbon tetrafluoride, CF_4) and hexafluoroethane (C_2F_6), which do not destroy the ozone layer in the stratosphere. They were introduced to replace chlorofluorocarbons (CFCs), as CFCs destroy the ozone layer. The Global Warming Potential of PFCs is 6,500–9,200 times that of carbon dioxide and is one of the six greenhouse gases to be banned under the Kyoto Protocol.

PERFOLIATE: Of a sessile leaf, having the margins entirely surrounding the stem, so that the stem appears to pass through the leaf.

PERFORATE: Having perforations or holes.

PERFORIN: A cytolytic protein occurring in the lytic granules of CD8 T-cells and NK cells. When released onto the surface of a target cell by an activated T cell, the perforin molecules polymerise to form a cylindrical pore through the target's plasma membrane, enabling entry of other cytotoxic molecules, such as granzymes and granulysin, which induce programmed cell death.

PERFUME: A blend of substances made from plant oils and synthetic materials which produce a pleasant odour. A main source of perfumes are essential oils extracted from different parts of plants such as the flowers of the rose, the leaves of lavender, cinnamon from bark and pine from wood. They are extracted by steam distillation; by using volatile solvents; by coating petals with fat or by soaking. Animal products, such as ambergris from the sperm whale, are used as fixatives to preserve fragrance.

PERFUSION TECHNIQUE: A process of maintaining a live organ in isolation by circulating nutritive delivery arterial blood through and around the organ. It is useful in studying the metabolism of drugs in isolation. Maintaining the circulation through an organ allows the delivery of nutrients to the tissues and the removal from them of toxins and waste products.



Perfoliate leaf

PERGULARIA: A small genus of the family Asclepiadaceae (Milkweed family) in the major group Angiosperms (flowering plants) containing 12 species. Pergularia provides highly toxic plants, ideally suitable as hunting poisons. *Pergularia daemia* Forsk is a foul-smelling perennial twinning herb with a hairy stem that exudes a milky juice when cut. It grows in damp savannah and forest margins of India and tropical and subtropical regions. The leaves are thin, broadly ovate, heart-shaped or nearly circular, hairless above, velvety beneath. The flowers are greenish yellow or dull white and sweet-scented and borne in lateral cymes which are at first corymb-like and become raceme-like. The fruit is a follicle, with soft spines all over and a long beak. The seeds are densely velvety on both sides. Its pharmacological activities include anti-inflammatory, liver disorders, anticancer, antidiabetic, antioxidant, antibacterial, antifungal, analgesic, anti-infertility and central nervous system depressant activities. Some species names are *Pergularia brunoniana* (Wight & Arn.) D.Dietr.; *Pergularia calesiana* (Wight) Buch.-Ham. ex Hook.f.; *Pergularia clausa* (R.Br.) Spreng; etc.

PERIANTH: A collective term for the outer structure of a flower, made up of the corolla, the calyx or both or a term for the outer whorls (sepals and petals that are indistinguishable) of the flower which protect the stamens and carpels during development. In monocotyledons, the two whorls are similar and often brightly coloured. In wind-pollinated flowers both whorls may be reduced or absent. In many horticultural varieties, the number of perianth parts is multiplied, but the resulting 'double' flowers are often sterile.

PERIAPSIS: The point in an orbit when two objects are closest together. Special names are given to this point for commonly used systems, including periastron, perihelion and perigee.

PERIASTRON: The point in the orbital motion of a binary star system when the two stars are closest together.

PERIBACTEROID MEMBRANE: A plant-derived membrane which surrounds rhizobia (nitrogen-fixing bacterium) in host cells of legume root nodules.

PERICARDIAL CAVITY: The cavity between the inner layer of the pericardium and the epicardium of the heart in vertebrates which contains a supply of serous fluid known as **pericardial fluid**.

PERICARDIUM: A tough double-layered fibrous membrane that contains the heart and the roots of the great vessels. The pericardium holds the heart in position, allowing it to relax and contract. It is made up of two main parts: **fibrous pericardium**, which is a tough outer fibrous layer and **serous pericardium**, which consists of a double layer of serous membranes, the inner layer being in close contact with the heart.

PERICARP (FRUIT WALL): The outer and often edible layer of a fruit, which is the tissue that develops from the ovary wall of the flower and surrounds the seeds. The pericarp is made up of three layers: the **exocarp**, which is the outer layer or peel; the **mesocarp**, which is the middle layer or pith; and the **endocarp**, which is the inner layer surrounding the seeds.

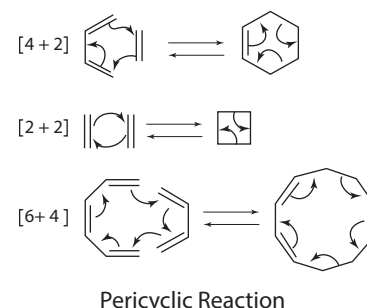
PERICHONDRIUM: A dense layer of fibrous connective tissue consisting of two separate layers: an outer fibrous layer and inner chondrogenic layer that surrounds the cartilage during bone formation and found around the perimeter of elastic cartilage and hyaline cartilage. Cells in the perichondrium lay down a peripheral layer that develops into compact bone. The **fibrous layer** contains fibroblasts, which produce collagenous fibres. The **chondrogenic layer** remains undifferentiated and can form chondroblasts or chondrocytes.

PERICLINAL: 1. Of, relating to or referring to a cell layer parallel to the surface of an organ or plant part. 2. Of, relating to or referring to strata and structures that dip radially outward from or inward toward, a centre, forming a dome or a basin.

PERICLINIUM: An involucre of bracts beneath or around an inflorescence which surrounds the common receptacle in composite flowers.

PERICYCLE: A cylinder of parenchyma cells lying immediately beneath the endodermis and phloem, forming the outermost part of the stele of plants. Pericycle produces and regulates the formation of lateral roots.

PERICYCLIC REACTION: A type of concerted (single-step) chemical reaction in which bond making and bond breaking occur simultaneously (but not necessarily synchronously) via a cyclic (closed-curve) transition state. Pericyclic reactions can be promoted either by heat or by light. The stereochemistry of the reaction is determined by the mode of activation employed and the number of electrons that are delocalised in the transition state. They can be understood using frontier orbital theory or the Woodward Hoffmann rules.



PERICYNTHION: The point in the orbit of the moon of a satellite launched from the Earth that is nearest to the moon.

PERIDERM: Impermeable outer layer that protects the inner stem tissues of woody plant from desiccation, freezing, heat injury, mechanical destruction and disease. It is made up of phellem, phellogen and phelloderm.

PERIDOTITE: A coarse-grained, heavy, igneous rock consisting largely of at least 10% of the mineral olivine. Pyroxene and other minerals may also be present.

PERIDROMA: 1. A genus of moths of the Notuidae family. 2. The rachis of a fern.

PERIGALACTICON: The point in a star's orbit around the Milky Way Galaxy, when the star lies closest to the galactic centre.

PERIGEE: The point at which an object travelling in an elliptical orbit of an object such as the Earth, Sun or planet is at its closest to object. The point at which it is furthest from the object is the **aphelion**.

PERIGON (ROUND ANGLE): An angle of 360°.

PERIGYNIUM (PERICHAETIUM): 1. A scale-like, tubular bract that encloses the pistil of the genus *Carex* such that the body is shaped like a flask. It may tightly envelope the achene or it may be inflated like a bladder. 2. A fleshy tube that surrounds the archegonium of various bryophytes, formed from the stem, apex or thallus.

PERIGYNY: A floral arrangement in which the ovary is situated in a cup-shaped receptacle without having attached to it or on flattened receptacle, from the margin of which the perianth and stamens arise. Examples are seen in cherry, rose and plum flowers.

PERIHELION: The point in the orbit of a planet, comet or artificial satellite in solar orbit at which it passes nearest to the Sun. The Earth is at perihelion on or about 3 January.

PERIKINETIC AGGREGATION: Collisions are caused by Brownian motion. The rate of aggregation is in general determined by the frequency of collisions and the probability of cohesion during collision.

PERILUNE: The point on an elliptical lunar orbit that is nearest to the Moon.

PERILYMPH: The extracellular fluid of the inner ear that fills the space between the bony labyrinth and the membranous labyrinth. The ionic composition of perilymph is comparable to that of plasma and cerebrospinal fluid. The major cation in **perilymph** is sodium. The values of sodium and potassium concentration in the perilymph are 138 mM (millimolar) and 6.9 mM (millimolar) respectively.

PERIMETER: The distance around a plane shape, especially one bounded by straight lines, such as a triangle, square or polygon. In the case of a curved shape, such as a circle or ellipse, the term circumference is more commonly used. The perimeter of a rectangle is given by $P = 2l + 2w$, where l is the length and w is the width.

PERINEUM: A region of the pelvis that lies between the anus and the vulva in the female and between the anus and scrotum in the male.

PERIOD: 1. The time taken for a moving object to return to its original position or the time taken for one complete cycle of a repeated sequence of events, for example, the time taken for a pendulum to swing from side and back again is the period of the pendulum. Its SI unit is the second (s) and the symbol is T . 2. In mathematics, it is an interval, generally the smallest, after which a periodic function takes the same values; a constant k such that $f(x) = f(x + k)$ for all x . For example, since $\sin\theta = \sin(\theta + 2n\pi)$, $2n\pi$ is a period of $\sin\theta$ for all integers n . 3. Of a point with respect to a function, f , the smallest positive integer n such that the n -fold composition $f^n(x) = x$. 4. In chemistry, the horizontal row of elements in the periodic table.

PERIOD DOUBLING: A mechanism for describing the transition process from regular motion to chaos in certain dynamical systems. If the force on a body produces a regular orbit with a specific period, a sudden increase in the force can suddenly double the period and the motion becomes more complex.

PERIOD OF A DECIMAL EXPANSION: The length of the smallest block of repeating digits in the decimal expansion of a rational number that does not terminate. For example, for a number $\frac{1}{3}$, its decimal expansion = 0.333333333333..., repeating block = 3 and period = 1; for $\frac{5}{7}$, the decimal expansion = 0.71428571428571..., repeating block = 714285 and period = 6; for $\frac{89}{26}$, the decimal expansion = 3.4230769230769..., repeating block = 230769 and period = 6.

PERIOD OF A FUNCTION: The least positive value of γ for which $f(x) = f(x + T)$. Example is the sine function $y = \sin\theta$ as shown in the diagram:

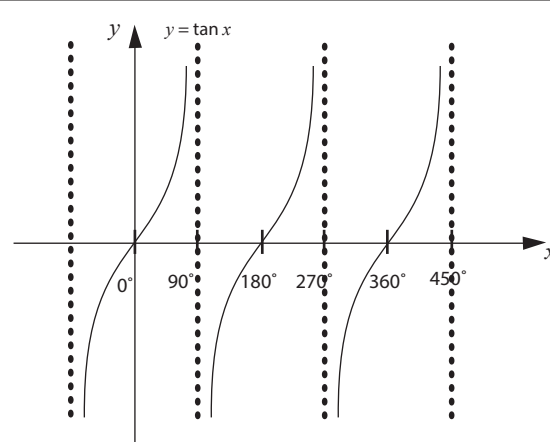
PERIOD OF ELEMENTS: A set of elements which have the same number of electron shells and whose atomic numbers increase by one unit from one atom to the next. An example is any horizontal row of the periodic table such as (lithium (Li), beryllium (Be), boron (B), carbon (C), nitrogen (N), oxygen (O), fluorine (F) and neon (Ne)).

PERIODIC ATTRACTOR (LIMIT CYCLE ATTRACTOR): An attractor that consists of a periodic movement back and forth between two or more values. The periodic attractor represents more possibilities for system behaviour than does the fixed-point attractor. An example of a period two attractor is the oscillating movement of a metronome or, in psychiatry, a bipolar disorder that causes a person's mood to shift back and forth from elation to depression.

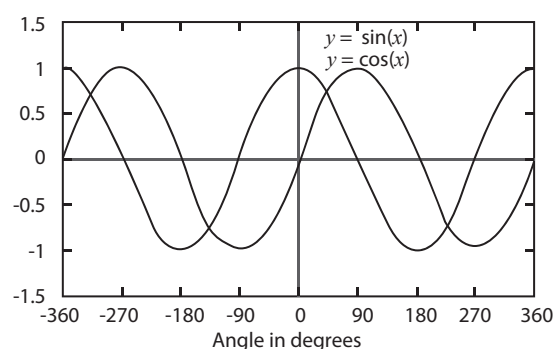
PERIODIC COPOLYMER: A copolymer consisting of macromolecules comprising more than two species of monomeric units in regular sequence.

PERIODIC FUNCTION: 1. A function $f(x)$ of a real or complex variable is periodic with period T if $f(x + T) = f(x)$ for every value of x . In other words, it is a function that repeats itself after every complete cycle. The most important examples are the trigonometric functions, which repeat over intervals of length 2π radians. Periodic functions are used to describe oscillations, waves and other phenomena that exhibit periodicity. Examples of periodic functions are the tangent, sine and cosine functions: $y = \tan x$, $y = \sin x$ and $y = \cos x$ as shown in the diagram. The tangent function has a period of π and the sine and cosine functions have a period of 2π . 2. A **doubly periodic function** is a complex function that is periodic in two directions: in the horizontal (real-axis) and vertical (imaginary-axis) directions. A

function $f(z)$ of a complex variable is doubly periodic if there exist two constants ω_1 and ω_2 such that $f(z) = f(z + \omega_1) = f(z + \omega_2)$ for all z .



(Fig i) Tangent Periodic Function



(Fig ii) Sine and Cosine Periodic Functions.

PERIODIC LAW: The principle of a periodic recurrence of the physical and chemical properties of elements when the elements are arranged in order of increasing atomic number. It can also be stated that, the properties of elements in the periodic table depends periodically on their atomic number.

PERIODIC MOTION: Any motion of a system that is continuously and identically repeated. The time that it takes to complete one cycle of an oscillation or wave motion is called the period, which is the reciprocal of the frequency. Examples of periodic motion include a rocking chair, a bouncing ball, a vibrating guitar string, a swinging pendulum and a water wave.

PERIODIC TABLE: A tabular display in which the elements are arranged in increasing order of their atomic number from left to right across the table or a table that lists the elements in order of increasing atomic numbers. A group or family is a vertical column in the periodic table. Groups are considered the most important method of classifying the elements. In some groups, the elements have very similar properties and exhibit a clear trend in properties down the group. Under the international naming system, the groups are numbered numerically 1 to 18 from the leftmost column (the alkali metals) to the rightmost column (the noble gases). Formerly the group consists of eight groups of elements listed in columns labelled I, II, III ... VIII and a block of transition metals. A period is a horizontal row in the periodic table, for example, Li, Be, B, C, N, O, F. Periods or horizontal rows are more significant than vertical group trends in classifying some elements such as the f-block, where the lanthanides and actinides form two substantial horizontal series of elements. Elements in the same period show trends in atomic radius, ionisation energy, electron affinity and electronegativity. Moving left to right across a period, atomic radius usually decreases. Russian

PERIODIC TABLE OF THE ELEMENTS

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1																		
	1 H 1.008																	
2	9 Li 6.941	4 Be 9.012																
3	11 Na 22.99	12 Mg 24.31																
4	19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
5	37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
6	55 Cs 132.9	56 Ba 137.3	57-71 La-Lu	72 Hf 178.5	73 Ta 180.9	74 W 183.9	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)
7	87 Fr (223)	88 Ra (226)	89-103 Ac-Lr	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (271)	111 Rg (272)	112 Cn (285)	113 Uut (284)	114 Fl (289)	115 Uup (288)	116 Lv (292)	117 Uus (294)	118 Uuo (298)
INNER TRANSITION ELEMENTS																		
	Lanthanides		57 La 138.9	58 Ce 140.1	59 Pr 137.3	60 Nd 144.2	61 Pm (145)	62 Sm 150.0	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0	
	Actinides		89 Ac (227)	90 Th 232.0	91 Pa (231)	92 U 238.0	93 Np (237)	94 Pu (242)	95 Am (243)	96 Cm (243)	97 Bk (247)	98 Cf (251)	99 Es (257)	100 Fm (252)	101 Md (258)	102 No (259)	103 Lr (260)	

chemist, Dmitri Mendeleev is generally credited with the present periodic table, which he developed in 1869.

PERIODIC VOLTAGE: A general term, which is applicable to square, triangular and other wave forms.

PERIODICITY: 1. The quality or a state of recurring at regular intervals. 2. In chemistry, it is the tendency of chemical elements to have similar properties when arranged according to atomic number in the periodic table or variation in properties of elements with their position in the periodic table.

PERIODONTAL MEMBRANE: The mode of attachment of the tooth to the alveolus by a membrane of connective tissues that surrounds the root of a tooth and anchors it to its socket in the jaw bone. Fibres of the periodontal membrane pass into the cement covering the root, which provides a firm attachment.

PERIODONTICS: The branch of dentistry concerned with the prevention, diagnosis and treatment of diseases that affect the gums and supporting structures of the teeth.

PERIOSTEUM: A fibrous membrane that covers bones and serves as the site of attachment for skeletal muscles except at the joints of long bones. Periosteum is divided into an outer fibrous layer, which contains fibroblasts and inner cambium layer or osteogenic layer, containing progenitor cells that develop into osteoblasts. In general, periosteum contains nerves, blood vessels and lymphatic vessels and a number of types of cell, including osteoclasts. The periosteum plays an important role in bone repair and growth.

PERIOSTRACUM: The external, outermost organic covering of the shell of some molluscs and brachiopods. It helps to protect the slender slimy inner portions from acids, as well as provide the shell with colour.

PERIPATRIC SPECIATION: Organisms that are closely related or sister species, whose ranges are closely adjacent but do not overlap, preventing them from exchanging genes. The isolated populations may undergo shifts in their gene frequencies under the influence of genetic drift. This is most likely to occur if new populations arise from a few founder individuals and no gene flow occurs between the isolates and the main population. Peripatric speciation was originally proposed by Ernst Mayr.

PERIPHERAL ARTERIAL DISEASE (PAD; PERIPHERAL VASCULAR DISEASE; ATHEROSCLEROTIC PERIPHERAL ARTERIAL DISEASE): A common circulatory problem in which narrowed arteries reduce blood flow to the limbs. PAD most commonly affects blood flow to the legs. It occurs when a fatty material called plaque builds up on the inside walls of the arteries that carry blood from the heart to the head, internal organs and limbs. Atherosclerosis causes the arteries to narrow or become blocked, which can reduce or block blood flow.

PERIPHERAL ATOM: A secondary reference (non-core) atom which at some point in the mechanistic sequence is situated in the molecule containing the core atom(s).

PERIPHERAL DEVICE: Any item of equipment attached internally or externally to a computer and controlled by the central processing unit of a computer. Examples are the keyboard, mouse, monitor, scanner and printer.

PERIPHERAL NERVOUS SYSTEM (PNS): All parts of the nervous system consisting of the cranial and spinal nerves and their branches, which link the receptors and effectors with the central nervous system but excluding the central nervous system. PNS is not protected by the bone of spine and skull or by the blood-brain barrier, leaving it exposed to toxins and mechanical injuries. The peripheral nervous system is divided into the somatic nervous system and the autonomic nervous system.

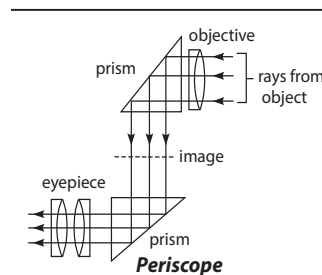
PERIPHYTON: Organisms such as algae and small crustaceans that are attached to the submerged surfaces of the stems and leaves of fresh water plants. Periphyton communities are composed of a complex mixture of algae, cyanobacteria, heterotrophic microbes and detritus forming an important food source for invertebrates, tadpoles and some fish. They are used in aquaculture food production systems

for the removal of solid and dissolved pollutants. It is being studied as aqua-cultural feed. It can also absorb contaminants, removing them from the water column and limiting their movement through the environment. The periphyton is also an important indicator of water quality.

PERIPLASM (PERIPLASMIC SPACE): The zone between the inner cytoplasmic membrane and the outer membrane in Gram-negative bacteria or the equivalent space outside the inner membrane of Gram-positive bacteria. It contains a thin layer of highly hydrated peptidoglycan with a gel-like consistency. It is concerned with the cell's metabolism.

PERIPTEROUS: Having a surrounding border or wing.

PERISCOPE: An optical device that enables an observer to see along a displaced or deflected axis, providing an observer with the view from a position, which may be inaccessible or dangerous. Periscopes range in complexity from the simple unit-power tank periscope to the complex multi element submarine periscope. It basically consists of the optical path, where light from the objective lens is directed 90°



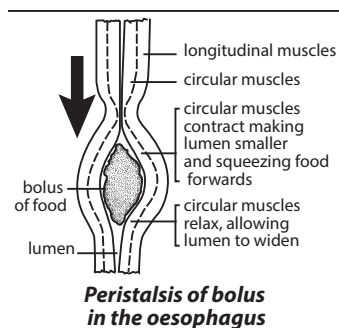
downwards by means of a prism. A relay lens inside the tube relays the image to a prism in the eye-piece, which redirects the path of the light back into the horizontal plane and into the eyes of the observer.

PERISPERM: A food storage tissue in some seeds that is derived from the nucellus of the ovule and surrounds the embryo of the seed. It consists mainly of thin-walled parenchymatous cells which are rich in starch or oil. Some seeds such as those of beets contain both the perisperm and endosperm.

PERISPORE: An outer membrane that covers the spore.

PERISSODACTYLA: An order of herbivorous, large and odd-toed ungulate mammals with the axis of symmetry of the foot running through the middle digit. Examples are zebras, asses, tapirs, horses and rhinoceros. They have large teeth specialised for grinding. Cellulose digestion occurs in the caecum and the large intestine. The perissodactyls are divided into three groups: the **Hippomorpha** which includes horses and their extinct relatives; the **Titanotheriomorpha** (the extinct brontotheres); and the **Moropomorpha** (tapirs, rhinoceroses and their extinct relatives). Fossils of the Eocene epoch, 60 million years ago, show that these animals were at that time already distinct from the cloven-hoofed Artiodactyla.

PERISTALSIS: A progressive wave-like involuntary muscular contraction and relaxation that passes along the alimentary canal including oesophagus, stomach and intestines forcing food contents along. The waves can be short, local reflexes or long continuous contractions along the length of the organ. It is brought about by the contraction of the circular muscles of the gut wall in sequence. Peristalsis in the large intestine pushes waste toward the anal canal and is important in removing gas and dislodging potential bacterial colonies.



PERISTOME: 1. A ring of tooth-like structures around the opening of a moss capsule. The teeth tend to bend and twist in dry weather, so opening the mouth of the capsule and allowing the spores to escape. In wet weather, they close over the opening of the capsule. 2. The area around the mouth in many invertebrates and some protists. It sometimes assists in food collection. Examples include the first

segment of the earthworm and the spirally ciliated groove around the mouth of some ciliated protozoans and the shells of gastropod molluscs such as snails.

PERITECTIC REACTION (INCONGRUENT REACTION): An isothermal, reversible reaction between two phases, a liquid and a solid, that results, on cooling of a binary, ternary, ..., n system in one, two, ... $(n - 1)$ new solid phases.

PERITECTOID REACTION (METATECTOID REACTION): An isothermal, reversible reaction in the solid state, that, on cooling of a binary, ternary, ..., n system, results in one, two, ... $(n - 1)$ new solid phases.

PERITECTOID TEMPERATURE: The maximum temperature at which a peritectoid reaction can occur.

PERITHECIUM: Flask-shaped ascocarp (fruiting body) that occurs in various ascomycetous fungi. It has a pore through which the spores are discharged.

PERITONEAL CAVITY: The space enclosed by the peritoneum. It is divided into two portions, the greater sac and the lesser sac or ommental bursa, which lies behind the stomach. The two sacs are connected by the foramen of Winslow or epiploic foramen. It is empty except for the film of lymph that keeps the surfaces moist.

PERITONEUM: The thin layer of the tissue that lines the abdominal cavity of vertebrates and covers the abdominal organs composing mainly of flattened epithelial cells that produce a small amount of watery or serous, fluid. **Peritonitis** is inflammation of the peritoneum, usually caused by bacterial infection or chemical irritation of the peritoneum, when internal organs become diseased or when contents of the gastrointestinal tract escape.

PERITUBULAR CAPILLARIES: In the vertebrate kidney, the capillaries located next to the nephrons that surround the renal tubule. Water and solutes are reabsorbed into the bloodstream through the peritubular capillaries and some substances are secreted from them into the renal tubule.

PERITRICHIOUS FLAGELLATION: Having multiple flagella (microscopic hair-like appendage) present all over the cell surface, as occurs in bacteria. This type of flagellation enables the bacteria to easily move in any direction.

PERKIN, SIR WILLIAM HENRY (1838-1907): A British chemist, who accidentally produced mauve, while still a student. It was the first aniline dyestuff to be synthesised. Perkin built a factory to produce it and made a fortune. His son, William Henry Perkin (1860-1927), also a chemist, synthesised numerous organic compounds, including camphor and several alkaloids.

PERL (PRACTICAL EXTRACTION AND REPORT LANGUAGE): A scripting language which uses a syntax similar to C/C++. It is commonly used by Web programmers to create scripts for Web servers. Perl is especially good at parsing text, so programmers often use it for reading and searching through text files.

PERLITE: Natural volcanic glass material having water holding capabilities typically formed by the hydration of obsidian. It has the unusual property of greatly expanding when heated sufficiently. It is an industrial mineral and a commercial product useful for its light weight, low density and relatively low price. It is used in lightweight plasters and mortars, insulation, ceiling tiles, in horticulture and as filter aid and others.

PERM ALLOYS: A group of alloys of high magnetic permeability consisting of iron and nickel (usually 40-50%) and often with small amounts (3-5%) of other elements such as molybdenum, copper, chromium or tungsten. Perm alloy has a high magnetic permeability, low coercivity, near zero magnetostriction and significant anisotropic magnetostriction. Perm alloy is used in transformer laminations and magnetic head sensors.

PERMAFROST: A thick subsurface to a depth of as much as 600 metres layer of soil that remains frozen throughout the year. It occurs in the treeless plains of Siberia and common through Arctic, sub-Arctic and Antarctic regions on Earth. While the top layer of soil

heats up in the summer, permafrost exists in deeper layers where the summer heating fails to melt the frozen ground. A continuous layer of permafrost is found in areas where the annual mean temperature is below about -5°C .

PERMANENT: In mathematics, the sum of all products of elements of a given square matrix, where each product contains exactly one element from each column and row. The permanent of a matrix is thus constructed in the same way as the determinant, except that the signs of the products do not alternate.

PERMANENT CHARGE: The net negative or positive charge of clay particles inherent in the crystal lattice of the particle. Permanent charges develop on the soil particle surface by isomorphic substitution. It is not affected by changes in pH or by ion exchange reactions.

PERMANENT CROPS: Crops not grown in rotation, but produced from plants occupying the soil and yielding harvests for several consecutive years, usually more than five. Permanent crops are usually perennial trees, bushes or crops like citrus, apples, blueberries, nuts or grapes. Permanent crops are usually intended for human consumption and generally yield a higher added value per hectare than annual crops.

PERMANENT CROSSLINK: Crosslink formed by covalent bonds, intermolecular or intramolecular interactions that are stable under the conditions of use of the material formed.

PERMANENT GAS: A gas that cannot be liquefied by pressure alone at temperatures below room temperature.

PERMANENT HARDNESS: Hardness of water that cannot be removed by boiling but may be removed by the addition of sodium carbonate and lime and filtration through natural or artificial zeolites, which absorb the hardness-producing metallic ions and release sodium ions to the water. Also sequestering agents in detergents inactivate the substances that make water hard.

PERMANENT LINK (PERMALINK): An URL that links to a specific news story or Web posting. Permalinks are most commonly used for blogs, which are frequently changed and updated. They give a specific Web address to each posting, allowing blog entries to be bookmarked by visitors or linked to from other websites.

PERMANENT MAGNET: A magnet that is surrounded by a steady magnetic field that does not rely on an electric current to maintain its magnetism. A permanent magnet loses its magnetism when heated or beaten.

PERMANENT MAGNET MOVING-COIL-INSTRUMENT: Electrical instrument in which the action depends upon a mechanical force exerted by a fixed permanent magnet on a movable coil carrying a current. This type of instrument is suitable for use only with direct current. It has two advantages: (i) relatively low power consumption on account of the high torque; and (ii) uniformly divided scale. It can be adapted for alternating-current measurements by means of rectifiers.

PERMANENT STORAGE: Computer storage that keeps the data or its contents even if the power is turned off or if the storage device is moved to another computer. The most commonly used one is computer hard drive.

PERMANENT TEETH: The second and final set of teeth that mammals produce after shedding the deciduous teeth. An adult human has thirty-two permanent teeth, consisting of six maxillary and

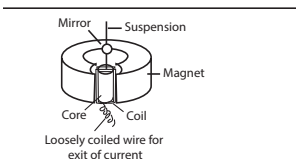


Fig. I: Moving coil in galvanometer

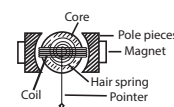


Fig. II: Plan of a type of moving-coil galvanometer

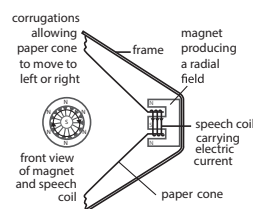


Fig. III: Moving coil in loudspeaker

Examples of permanent magnet moving-coil instrument

six mandibular molars, four maxillary and four mandibular premolars, two maxillary and two mandibular canines, four maxillary and four mandibular incisors, appearing between the ages of 6 and 21 years.

PERMANENT WILTING POINT, PWP (WILTING POINT): The volume of water remaining in soil when plants can no longer extract water, and the point at which a plant will not recover from wilt until water is added to soil. This point of soil water content is generally accepted as about 15 bars tension.

PERMEABILITY: 1. Also known as magnetic permeability, it is the measure or degree to which a material can support the formation of a magnetic field within itself. Its symbol is μ and refers to the ratio of the magnetic flux density, B , in a substance to the external field strength, H expressed as $\mu = B/H$. In SI units, permeability is measured in the Henry per metre (Hm^{-1}) or Newton per ampere squared (NA^{-2}).

Relative permeability also known as dielectric constant is the ratio of the permeability of a substance to the permeability of a vacuum at the same magnetic field strength. Its symbol is μ_r and expressed as $\mu_r = \mu/\mu_0$, where μ is the permeability of the medium and μ_0 is the permeability of free space or vacuum. 2. The rate at which a substance such as a liquid or a magnetic field passes through a membrane or other medium. 3. **Soil permeability** is the ease with which gases and liquids penetrate or pass through a bulk mass of soil or a layer of soil. It can also be defined as the property of a porous medium that relates to the ease with which gases or liquids can pass through it.

PERMEABLE: Describing a solid, which is usually porous allowing a fluid to pass through it. For example, sandstone and filter paper are permeable to water and to air.

PERMEASE: A cell-membrane protein that catalyses the transport of a specific small molecule in or out of the cell.

PERMEATE: A stream containing penetrants that leaves a membrane module.

PERMEATION CHROMATOGRAPHY: Separation based mainly on exclusion effects, such as differences in molecular size and/or shape, for example, molecular-sieve chromatography or on charge such as ion-exclusion chromatography. Gel-permeation chromatography is widely used for the process, when the stationary phase is a gel.

PERMEATION TUBE: A device used for dynamic preparation of test gas mixtures by means of controlled permeation of a gaseous analyte out of a container through polymer material into a carrier gas stream. These devices containing certain condensable gases, for example, nitrogen dioxide (NO_2) and sulfur dioxide (SO_2), when operated at closely controlled temperatures can be used as primary standards calibrated in terms of the weight loss per unit time.

PERMIAN: Of or belonging to the last geological period in the Palaeozoic era characterised by the formation of the supercontinent Pangaea, the rise of conifers and the diversification of reptiles and ending with the largest known mass extinction in the history of life. It extended from the end of the Carboniferous period, which is about 280 million years ago to the beginning of the Mesozoic era, which is about 225 million years ago.

PERMITTIVITY: The measure of the ability of a non-conducting material to retain electric energy when placed in an electric field. A high permittivity tends to reduce any electric field present. The capacitance of a capacitor can be increased by increasing the permittivity of the dielectric material. Its symbol is ϵ . **Relative permittivity**, also known as **dielectric constant**, with the symbol ϵ_r is the permittivity of a material given relative to that of free space or a vacuum or simply the ratio of the permittivity of a substance to that of free space. The

permittivity of free space, also known as **permittivity of vacuum** or **vacuum permittivity** with the symbol ϵ_0 , is a constant that defines the strength of the electric force between electric charges in vacuum. In other words, it defines how the electric field reacts to the presence of charges in empty space. Mathematically, it is expressed as $\epsilon_0 = \frac{1}{(\mu_0 c^2)}$, where μ_0 is the magnetic permeability of free space

and c is the speed of light in vacuum. Its value is approximately $8.854 \times 10^{-12} \text{ F m}^{-1}$ (farads per meter).

PERMSELECTIVITY: A term used to define the preferential permeation of certain ionic species through ion-exchange membranes.

PERMUTABLE (COMMUTATIVE): In mathematics, of a binary operation, having the property that one term operating on a second is equal to the second operating on the first, for example, $a \times b = b \times a$ and addition and multiplication are both commutative, but subtraction, division and composition of functions are not.

PERMUTATION (ORDERED ARRANGEMENT): A specified arrangement of a group of n objects or the arrangement of n distinct objects taken r of them at a time in all possible orders. It is given by ${}^nP_r = \frac{n!}{(n-r)!}$, where '!' stands for factorial. For example,

$${}^6P_2 = \frac{6!}{(6-2)!} = \frac{6 \times 5 \times 4!}{4!} = 30.$$

PERMUTATION GROUP (SUBSTITUTION GROUP): In mathematics, a group whose elements are permutations of a given set and whose group operation is the composition of permutations in the group and multiplication is defined as successive permutation. The group of all permutations of n objects has $n!$ elements and determines the full symmetric group and the group of all even permutations of n objects is called the alternating group.

PERMUTATION MATRIX: A square matrix with one unit element in each row and column and with all other entries zero. This corresponds to precisely one permutation of order n ; where the permutation takes the i^{th} element to the $\sigma(i)^{\text{th}}$, the matrix is zero except in the $[i, \sigma(i)]$ positions, where it is unity.

PERMUTATION REPRESENTATION: In mathematics, a group of permutations of a set that is isomorphic to a given group; the permuted set may be the given group itself.

PERMUTATIONS WITH REPEATITIONS: The number of arrangements of n objects of which r objects are alike and q are also alike, this can be calculated by using the expression $\frac{n!}{p!q!r!...}$, where

$n = p + q + r + ...$. If the number of arrangement of n objects of which r of them are taken at a time, when each of them can be used at any number of times up to r times in any arrangement is given by n^r .

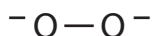
PERMUTE: In mathematics, it is to re-arrange a sequence of elements.

PERMUTIT: The trade name for a type of zeolite used in water softeners.

PERNICIOUS: Having a harmful or destructive effect.

PEROVSKITE STRUCTURE: A type of ion crystal structure shown by compounds of the types ABO_3 . The general chemical formula for perovskite compounds is ABX_3 , where A and B are two cations of very different sizes and X is an anion that bonds to both. There is a cubic arrangement with A atoms and B atoms surrounded by 12 O atoms and B atoms surrounded by 6 O atoms. Examples of the perovskite structure include CaTiO_3 , BaTiO_3 . It was first discovered in the Ural Mountains of Russia by Gustav Rose in 1839 and is named after Russian mineralogist, L. A. Perovski (1792–1856).

PEROXIDE: 1. A type of compound that contains two oxygen molecules joined together or a group of inorganic compounds that contain the O_2^{2-} ion. They are hypothetically derived from hydrogen peroxide. However, these ions do not exist in aqueous solution due to very rapid hydrolysis to OH^- . Most decompose giving off oxygen gas. 2. A group of organic compounds that contain the $-O_2-$ group. Organic peroxides are unstable and some are explosive and include peroxyacetic acid, dibenzoyl peroxide and cumene peroxide. They can be used to initiate free-radical reactions. Chemicals such as hydrogen peroxide are used as a bleaching and oxidising agent and also as a catalyst.

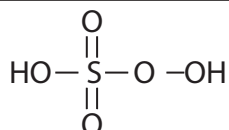


Chemical Structure of Peroxide Ion

PEROXISOMES: Subcellular organelles that contain flavin requiring oxidases and that regenerate oxidised flavin by reaction with oxygen. They are found in the cytoplasm of virtually all plant and animal cells except the archezoa (original lifeforms). Peroxisomes have enzymes that are involved in the oxidation processes, some of which generate the highly toxic compound hydrogen peroxidase (H_2O_2). Peroxisomes are important for lipid metabolism. In humans, the β -oxidation of fatty acids greater than 18 carbons in length occurs in peroxisomes. In yeast, all fatty acid β -oxidation occurs in peroxisomes. Peroxisomes contain the first two enzymes required for the synthesis of plasmalogens. They contain more than 50 characterised enzymes and perform many biochemical functions, including detoxification, especially in the cells of the liver parenchyma and proximal convoluted tubule of the kidney. In plants, peroxisomes detoxify certain by-products of photosynthesis and oxidise glycolate to glyoxylate which can then be salvaged by a series of reactions involving mitochondria and chloroplasts as well as peroxisomes.

PEROXYMONOSULFURIC ACID (PERSULFURIC ACID; PEROXSULFURIC ACID, CARO'S ACID):

A white crystalline compound having the chemical formula H_2SO_5 . It is a liquid at room temperature. It is synthesised by the combination of chlorosulfuric acid and hydrogen peroxide: $H_2O_2 + ClSO_3H \rightleftharpoons H_2SO_5 + HCl$. It is a powerful oxidation agent and highly explosive. It is used for a variety of disinfectant and cleaning applications such as swimming pool treatment and denture cleaning.

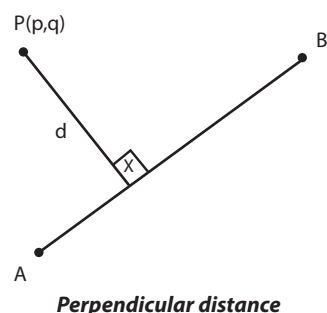


Chemical Structure of Peroxymonosulfuric Acid

PERPENDICULAR LINE: A line drawn at right angle (90°) to another or to a plane or two lines are said to be perpendicular if the gradient of one is a negative reciprocal of the gradient of the other. It is given by $m_1 = -\frac{1}{m_2}$ where m_1 and m_2 are the gradients of the lines. It is sometimes symbolised as $\perp$.

PERPENDICULAR BISECTOR: A line drawn at right angles to the midpoint of another line.

PERPENDICULAR DISTANCE: In mathematics (Euclidean geometry), the distance from a point to a line measured by the length of a perpendicular to the given line passing through the point, making it the shortest distance between the given point and a point on the line. Referring to the diagram, if the equation of the line AB is $ax + by + c = 0$, then the perpendicular distance



|PX| from $P(p,q)$ to line AB is given by $|PX| = \frac{|ap + bq + c|}{\sqrt{a^2 + b^2}}$, where X is the foot of the perpendicular.

PERPENDICULAR VECTOR: Two vectors **a** and **b** are perpendicular if their dot or scalar product is zero: $\mathbf{a} \cdot \mathbf{b} = 0$.

PERPETUAL CALENDAR: A table that can be used to determine the correct day of the week for any date in many years.

PERPETUAL MOTION: 1. The idea that a machine can be designed and constructed in such a way that, once started, it will continue in motion independently without requiring any further input or energy, would violate the laws of thermodynamics. 2. A machine that would produce more energy in the form of work than is supplied to it in the form of heat would violate the first law of thermodynamics, which is a special case of the law of conservation of energy and is known as a perpetual-motion machine of the first kind. 3. A machine that would completely convert heat from a warm body into work, without letting any heat flow into a cooler body, would violate the second law of thermodynamics, which is concerned with entropy changes and is known as a perpetual-motion machine of the second kind.

PERCEPTUAL SUBITIZING: In mathematics, recognising the number of objects without consciously using any mental or mathematical strategies.

PERRON-FROBENIUS THEOREM: In mathematics, any of various results generalising the theorem that a real square matrix, A, of strictly positive coefficients has a unique largest real positive eigenvector up to constant multiples; this has an eigenvalue equal to the spectral radius of A and all other eigenvalues have smaller modulus. The adjoint A^* has a positive eigenvector with the same latent root.

PERSICOLOR: Peach-coloured.

PERSISTENCE LENGTH: The average projection of the end-to-end vector on the tangent to the chain contour at a chain end in the limit of infinite chain length. The persistence length with the symbol a is the basic characteristic of the worm-like chain.

PERSISTENT: 1. Of or referring to a pesticide or other pollutant such as paraquat and DDT that remains in the environment for long periods, causing damage. 2. Referring to a part of a plant that does not fall off after maturation, for example, the calyx on an eggplant. 3. A body part that grows throughout life, examples are tooth or the persistent gills of fishes.

PERSISTENT ORGANIC POLLUTANT (POPs): An organic compound, usually considered toxic to the environment because they are not biodegradable through chemical, biological and photolytic processes and persists in the environment accumulating along the food chain, bioaccumulate in human and animal tissue and have potential significant impacts on human health and the environment. Examples of POPs are pesticides such as aldrin, DDT, dieldrin and toxaphene and organic compounds categorised as polychlorinated biphenyls, dioxanes and furans. Most POPs are created by humans in industrial processes, sometimes in the production of a range of products such as solvents, polyvinyl chloride and pharmaceuticals. Some POPs also occur naturally, examples are mycotoxins, marine biotoxins and cyanogenic. They are banned substances whose uses are not encouraged even under strict adherence to safety measures.

PERSISTENT VEGETATIVE STATE (PVS): A condition arising from a damage to the cerebral cortex in which the patient is unresponsive to mental and physical stimuli, though the eyes may be open and he/she may be capable of breathing. This condition may persist for a long time and the patient is usually kept alive only by medical procedures. Its causes include serious brain injuries, some serious brain disorders and severe abnormality of the nervous system.

PERSONAL COMPUTER (PC): A relatively inexpensive microcomputer that uses a microprocessor designed for individual use and used for

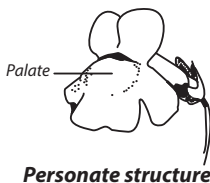
applications such as word processing, data management, desktop publishing, financial analysis and spreadsheet. Its physical components (hardware) include the monitor, mouse, keyboard, computer data storage, hard disk drive (HDD), system unit (graphic cards, sound cards, memory, motherboard and chips), etc.

PERSONAL DIGITAL ASSISTANT (PDA): Any small hand-held device with a touchscreen-based graphical user interface. It provides computing and information storage and retrieval capabilities. It is often used as a personal organiser, for example, as an address book, store personal information and to send and receive e-mail or faxes.

PERSONAL IDENTIFICATION NUMBER (PIN): A unique number used in conjunction with a magnetic card in security systems such as in banking as a password to establish an identity.

PERSONAL PRODUCTIVITY SOFTWARE (PRODUCTIVITY SOFTWARE): In computer science, it is a work-oriented software that is used to create or modify a document, image, audio or video clip, especially word processing, spreadsheets or presentation programs.

PERSONATE: In botany, it is an appendage that is raised on the lower lip of a bilabiate corolla, which partially or completely closes the throat or two-lipped with the upper arched and the lower protruding into corolla throat. This is usually seen in the snapdragon, which is a plant belonging to the genus *Antirrhinum*, of the figwort family, especially *Antirrhinum majus*, cultivated for its spikes of brightly coloured two-lobed flowers each having a corolla supposed to resemble the mouth of a dragon.



PERSPECTIVE: The representation of a three-dimensional object on a two dimensional plane surface such as paper by projection of the object onto the plane from a point with the characteristics that the features of the objects are smaller as their distance from the observer increases; and the object's dimensions along the line of sight are shorter than its dimensions across the line of sight. In other words, parallel lines are seen as converging in order to give the illusion of depth and distance. The perspective of two plane figures, have their points that can be put in one-to-one correspondence so that pairs of corresponding points lie on concurrent lines or dually, corresponding lines meet in collinear points. The two sets are perspective from a point, called the **centre of perspectivity**, if pairs of corresponding points are joined by lines through the centre. Two sets are perspective from a line, called the axis of **perspectivity**, if pairs of corresponding lines meet on the axis. A transformation in which two figures are perspective is called a **perspectivity**.

PERSPECTIVE DRAWING: A drawing that realistically represents a 3-dimensional object on a 2-dimensional surface.

PERSPECTIVE FORMULA: A geometric representation of stereochemical features of a molecule or model which appears as a view from an appropriate direction.

PERSPEX: A hard transparent brittle thermoplastic used as a light or shatter-resistant alternative to glass. Chemically, it is the synthetic polymer of methyl methacrylate. It was developed in 1928 in various laboratories and was first brought to the market in 1933 by Rohm and Haas Company, under the trademark Plexiglas. It is economical, light and easy to handle and does not contain any potentially harmful chemicals such as bisphenol-A subunits found in polycarbonate. Because of its brittleness, it is prone to scratching and cannot withstand much load.

PERSPICUOUS: 1. Transparent. 2. Clearly expressed or presented.

PERSPIRATION: Excretion of water and dissolved substances from the sweat glands of the skin of mammals without the active secretion of sweat. Perspiration has two main functions: body cooling by the evaporation of water from the skin surface and excretion of waste products such as salts. In the temperate zones, perspiration accounts for a loss of 300-400 ml daily in the average adult.

PERSTRACTION: A separation process in which membrane permeation and extraction phenomena occur by contacting the downstream with an extracting solvent.

PERTURBATION THEORY: 1. A method used in calculations in both classical physics and quantum mechanics in which the system is divided into a part that is exactly calculable and a small term, which prevents the whole system from being exactly calculable. 2. The theory of obtaining approximate solutions to the equations of motion of a physical system when these equations differ by a small amount from equations which can be solved exactly.

PERULA: 1. One of the scales of a leaf bud. 2. A portion of the perianth in certain orchids that looks like a pouch. **Perulate** means bearing scales.

PERVAPORATION: Membrane-based process for the separation of mixtures of liquids by partial vapourisation through a non-porous or porous membrane. The membrane acts as a selective barrier between the two phases: the liquid-phase feed and the vapour-phase permeate. The solution is placed in contact with a heated semipermeable membrane that allows the desired component(s) of the liquid feed to transfer through it by vapourisation. Separation of components is based on a difference in transport rate of individual components through the membrane.

PERVERSION (LATERAL PERVERSION): The effect produced by a mirror in reversing images from left to right.

PERVIOUS ROCKS: Another name for permeable rock or rock which allows water to pass through.

PEST: Any plant or animal which out of its feeding activities may cause harm or danger to other living organisms or things such as crops, livestock and food. Examples of pests are insects, fungus, bacteria, viruses, some protozoans, flatworms, nematodes, molluscs, arachnids and rodents. Most pests either compete with humans for natural resources or transmit disease to humans. Pests may be controlled by the use of pesticides and biological control methods.

PESTICIDE: Chemical substances that are useful for the mitigation, control or elimination of plants or animals detrimental to human health or economy. There are various types of insecticides designed for specific purposes. For example, algicides, defoliants, desiccants, herbicides, plant growth regulators and fungicides are used to regulate populations of undesirable plants which compete with or parasitise crop and other plants. Attractants, insecticides, miticides, acaricides, molluscicides, nematocides, repellants and rodenticides are used principally to reduce parasitism and disease transmission in domestic animals, the loss of crop plants, the destruction of processed food, textile and wood products and parasitism and disease transmission in humans. Most pesticides such as glyphosate, acephate, deet, dursban and the herbicide 2,4-D (2,4-Dichlorophenoxyacetic acid are obtained from chemicals. A few are also obtained from plants and minerals. Pesticides may be non-biodegradable and are non-specific and, therefore, harmful to other organisms not meant to be controlled. Organophosphorous insecticides such as malathion and parathion are biodegradable but can damage the respiratory and nervous systems in humans as well as killing useful insects such as bees.

PESTICIDE RESIDUE: Any substance or mixture of substances in food for man or animals resulting from the use of a pesticide including any specified derivatives, such as degradation and conversion products, metabolites, reaction products and impurities considered to be of toxicological significance.

PESTLE: A hand-held implement, consisting of a handle, which is blunt at the end used to grind things. It is usually made of materials such as wood, porcelain or ceramic. The shapes and sizes are varied to meet specific needs. Together with a mortar, they are used in grinding objects such as foodstuffs; in the chemistry laboratory to prepare reactants for a solid-state synthesis; and by pharmacists to grind substances for medical use.

PETA-: A prefix used to denote a multiple of 10^{15} in the SI system. Its symbol is P.

PETABIT (PB): 1. For processor storage, real and virtual storage and channel volume, it is 250 or 1 125 899 906 842 624 bits. 2. For disk storage capacity and communications volume, it is 1 000 000 000 000 000 (10¹⁵) bits.

PETABYTE (PB): 1. For processor storage, real and virtual storage and channel volume, it is 250 or 1 125 899 906 842 624 bytes. 2. For disk storage capacity and communications volume, it is 1 000 000 000 000 000 (10¹⁵) bytes.

PETAFLIPS (PFLOPS): A unit of measurement used for measuring the performance of a processor's floating point unit or FPU. One petaflops is equal to 1,000 teraflops or 1,0¹⁵ FLOPS.

PETAL: One of the four basic parts of a flower which is often brightly coloured, secrete perfume and nectar and functions to attract pollinators such as insects and birds. Petals make up the corolla of the flower. Epidermal hairs may be present and the cuticle is often covered by lines or dots known as honey guides, which direct the insects to the nectar.

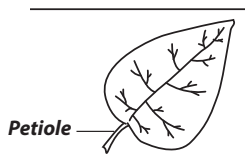
PETALANTHEROUS: A stamen that bears a distinctly petaloid filament, as seen in *Saxifraga*.

PETALODE: An organ, usually a stamen, which resembles a petal.

PETALOID: 1. Petal-like in appearance. 2. A petal-like stamen that does not possess a distinct anther and filament but with marginal microsporangia, as seen in *Magnolia nitida*. **Petaliferous** means producing petals; and **petaline** is anything which pertains to a petal.

PETALOSTEMONOUS: Having the filaments fused to the corolla and the anthers free.

PETIOLE: The generally non-leafy part of the leaf that attaches the leaf blade to the stem. Leaves without petioles are described as sessile and those possessing petiole are petiolate. The base of most petioles is enlarged into what is called **pulvinus**. Outgrowths appearing on each side of the petiole are called **stipules**. **Petiolate** refers to having a petiole. **Petiolar** means growing from a petiole. **Petioliform** means resembling a petiole.



PETRI DISH (PETRI PLATE; CELL CULTURE DISH): A shallow circular flat-bottomed dish made of glass or plastic and having a loose cover or a fitting lid that biologists use to culture cells. The common ones are 6 cm and 9 cm in diameter. It was named after German bacteriologist Julius Richard Petri, who invented it when working as an assistant to Robert Koch.

PETRIFICATION (PETRIFACTION; PETRIFICATION; SILICIFICATION): The process where the remains or traces of organic material that lived in geological past has been turned to stone or the process in which the porous structure of organic material such as bone, shell and wood is impregnated with salt-bearing ground water, which preserves the structure when it solidifies. The most common substances involved in petrifications are silica (SiO₂) and calcium carbonate (CaCO₃). Occasionally phosphate minerals, pyrite, hematite and other less common minerals make up all or part of the petrification matrix. For example, the Petrified Forest in Arizona contains whole tree trunks that have been turned into stone.

PETROCHEMICAL: A chemical derived from the processing of petroleum (crude oil) and natural gas. Examples include ethane and propane. These chemicals are used in the manufacture of other chemicals. Petrochemicals used as raw materials (feed stocks) include ethylene, propylene, butadiene, benzene, toluene, xylene and naphthalene. The numerous petrochemical products include plastics such as polyethylene, polypropylene, polystyrene; soaps and detergents, solvents, drugs, fertilisers, pesticides, explosives, synthetic fibres and rubbers, paints, epoxies, flooring and insulating materials, luggage and recording disks and tapes.

PETROL (GASOLINE): A colourless, volatile, inflammable liquid which is a mixture of hydrocarbons derived from petroleum. It is used mainly as a fuel for internal combustion engines and as a solvent for oils and fats.

PETROLEUM (CRUDE OIL): A natural mineral oil or thick greenish-brown flammable liquid derived from the geologic transformation and decomposition of plants and animals that lived hundreds of millions of years ago. Petroleum generally includes crude oil, from which petrol is extracted, natural gas, bitumen and asphalt, which are viscous or solid forms of hydrocarbons that occur in the Earth. Petroleum and its products are used as fuels for heating, for land, air and sea transport and for electric power generation and as petrochemical sources and lubricants.

PETROLEUM COKE: A carbonisation product of high-boiling hydrocarbon fractions obtained in petroleum processing (heavy residues). It is the general term for all special petroleum coke products such as green, calcined and needle petroleum coke. High-boiling hydrocarbon fractions (heavy residues) used as feedstock for petroleum coke are residues from distillation (atmospheric pressure, vacuum) or cracking, for example, thermal, catalytic, steam-based) processes. The nature of feedstock has a decisive influence on the graphitisability of the calcined coke.

PETROLEUM ETHER (BENZINE; PETROLEUM NAPHTHA; NAPHTHA ASTM; PETROLEUM SPIRITS): A colourless, volatile, flammable mixture of hydrocarbons consisting chiefly of pentanes and hexanes. It boils in the range 30 -70 °C (303-343 K) and used as a solvent. It has a specific gravity of between 0.6 and 0.8 depending on its composition. Petroleum ether is obtained from petroleum refineries as the portion of the distillate which is intermediate between the lighter naphtha and the heavier kerosene.

PETROLEUM JELLY (PETROLATUM; SOFT PARAFFIN): A tasteless, odourless, semi-solid mixture of hydrocarbons, obtained as the residue from petroleum after the more volatile and lighter components have been boiled off. It is used as protective covering, lubricant and ointment base.

PETROLEUM PITCH: A residue from heat treatment and distillation of petroleum fractions. It is solid at room temperature, consists of a complex mixture of numerous predominantly aromatic and alkyl-substituted aromatic hydrocarbons and exhibits a broad softening range instead of a defined melting temperature. The hydrogen aromaticity (ratio of aromatic to total hydrogen atoms) varies between 0.3 and 0.6. The aliphatic hydrogen atoms are typically present in alkyl groups substituted on aromatic rings or as naphthenic hydrogen.

PETROLEUM-BASED PROPELLANT: A type of liquid propellant in which the fuel is refined from crude oil and consists of a mixture of complex hydrocarbons (organic compounds containing only carbon and hydrogen). The petroleum used as rocket fuel is kerosene or a type of highly refined kerosene known as RP-1.

PETROLOGY: A branch of geology that studies the three major types of rocks: igneous, metamorphic and sedimentary. It is concerned with the composition, texture and structure, occurrence and distribution and conditions of origin.

PEWTER: Any of the various alloys consisting of about 91% tin, 7.5% antimony and 1.5% copper, used mainly for ornamental ware. Lead is now not included, making it safe to use for foods and beverages containers. It does not tarnish and maintains its colour of bluish white with either a bright finish or a soft, satin sheen.

PEYER'S PATCHES (AGGREGATED LYMPHOID NODULES): Named after the 17th-century Swiss anatomist Johann Conrad Peyer, they are lymphoid organs found in the wall of the small intestine, differentiating the ileum from the duodenum and jejunum. Humans have about 30, each consisting of a dome-shaped follicle containing chiefly B cells surrounded by smaller numbers of T-cells. Within the surface of the follicle exposed to the gastrointestinal tract lumen are specialised epithelial cells called M-cells (Multifingered cells). These have a microfolded luminal surface and take up particles from the gut by endocytosis. Peyer's patches play the important role of protecting the lumen of the gastrointestinal tract against infection and also facilitates the generation of the immune response within the mucosa.

PFUND SERIES: A series of lines in the infrared spectrum of atomic hydrogen, experimentally discovered in 1924 by August Herman

Pfund, which describes the emission spectrum of hydrogen when the electron is jumping to the fifth orbital. All of the lines are in the infrared portion of the spectrum.

PGML (PRECISION GRAPHICS MARKUP LANGUAGE): A hybrid of PDF and HTML under development by Adobe that is particularly suitable for use on electronic bookreaders and personal digital assistants. It is a 2D graphics language meant to satisfy both the Web's scalable lightweight vector graphics needs and the precision needs of graphic artists who want to ensure that their graphic designs appear on end user systems with precisely the correct fonts, colour, layout and compositing that they desire.

pH (POWER OF HYDROGEN): pH is an abbreviation of power of hydrogen, which is a scale from 0 to 14 used in measuring the acidity or alkalinity of a solution or the strength of an acid or an alkali. It is defined as the negative logarithm of the molar concentration of the hydroxonium or hydrogen ion. It is expressed as $\text{pH} = -\log[\text{H}^+]$ or $\text{pH} = -\log[\text{H}_3\text{O}^+]$. Acidic solutions have a pH less than 7, alkaline solutions have a pH greater than 7. A pH of 7.0 indicates neutrality.

pH GLASS ELECTRODE (GLASS pH ELECTRODE): A hydrogen-ion responsive electrode usually consisting of a bulb or other suitable form, of special glass attached to a stem of high resistance glass complete with internal reference electrode and internal filling solution system. Other geometrical forms may be appropriate for special applications, for example, capillary electrode for measurement of blood pH.

pH-DEPENDENT CATION EXCHANGE CAPACITY: The difference between the effective cation exchange capacity and the cation exchange capacity of a soil measured at a pH higher than that of its natural value.

PHAEANTHEROUS (PHANERANTHEROUS): Having stamens exerted from the corolla.

PHAEOCARPOUS: Having a carpel that is free from the surrounding floral parts.

PHAEOPHYTA (BROWN ALGAE): A division of algae of the kingdom Protista consisting of those organisms commonly called brown algae, in which the green chlorophyll pigments are usually masked by the brown pigment fucoxanthin. Brown algae are usually marine, found in abundance in cold water and many species such as the wracks (*fucus*), *Sargassum* (gulfweed) inhabit intertidal zones.

PHAGE (BACTERIOPHAGE): A virus that attacks and infects bacteria. The infection may or may not lead to the death of the bacterium, depending on the phage and sometimes on conditions. Each bacteriophage is specific to one form of bacterium. Bacteriophages make up a diverse group of viruses, some of which have complex structures, including double-stranded DNA. Since the first was discovered in 1915, bacteriophages have been important in the study of genetics.

PHAGOCYTES: A type of white blood cells that protect an animal against microbes that cause diseases by ingesting (phagocytosing) harmful foreign particles, bacteria and dead or dying cells. It is a form of endocytosis in which the plasma membrane of the phagocyte invaginates to capture the particle and then closes round it to form an acidified vesicle. This then coalesces with a lysosome, which contains enzymes that break down the particle, forming a phagolysosome. One litre of human blood contains about six billion phagocytes. It was first discovered in 1882 by Ilya Ilyich Mechnikov (1845-1916), Russian biologist and zoologist while he was studying starfish larvae, for which he was awarded the 1908 Nobel Prize in Medicine.

PHAGOCYTOSIS: The process by which foreign bodies invade the body or minute food particles are engulfed and broken down by certain animal cells known as **phagocytes**.

PHAGOTROPH: Any heterotrophic organism that ingests nutrients by ingesting organisms or organic particles, which are digested within its body.

PHALAENOPHILY: Pollinated by moths. Flowers pollinated by moths are usually white, night-opening, large and showy with tubular

corollas and a strong, sweet scent produced in the evening, night or early morning.

PHALANGES (PHALANX): The bones that make up the digits of the fingers and toes in vertebrates. They articulate with the metacarpals of the hand or with the metatarsals of the foot. Each toe and finger contains three bones, except the great toe (hallux) and thumb (pollex) which have two bones each.

PHANEROGAM: Any plant that produces seeds other than spores. On the other hand, **Cryptogam** is a plant that reproduces by spores and bears no flowers or seeds. The known groups of cryptogams are Algae, Lichens, Mosses and Ferns.

PHANEROPHYTE: A perennial tree or shrub with dormant buds borne on aerial shoots high above the ground, exposing them to drought stress and frost. Such plants occur in regions where frost and drought are uncommon, such as tropical forests.

PHANEROZOIC: Geological time since the end of the Precambrian era, represented by rock strata containing clearly recognisable fossils. It comprises the Palaeozoic, Mesozoic and Cenozoic eras and has extended for about 570 million years and goes back to the time when diverse hard-shelled animals first appeared.

PHANTOM CHAIN BEHAVIOUR: Hypothetical behaviour in which chains can move freely through one another when a network is deformed.

PHARATE: Inactive, non feeding stage in the life of insects that undergo complete metamorphosis, describing a larva or adult when inside of the cuticle of the previous developmental stage. For example, a newly metamorphosed adult insect may remain cloaked for some hours or even days within the pupal case controlled by hormones until larval structures break down and adult structures form. The adult emerges either by splitting the pupal skin and chewing its way out or by secreting a fluid that softens the cocoon. Pharate is also the stage prior to melanisation and hardening of the cuticle after metamorphosis.

PHARMACEUTICAL: 1. Of or relating to pharmacy or pharmacists
2. A substance that affects the function of living cells, used in medicine to cure, prevent, diagnose the occurrence of disorders and diseases and prolong the life of patients with incurable conditions.

PHARMACODYNAMICS: The study of biochemical and physiological effects of drugs on the body. It includes the physiological, therapeutic and toxicological responses to drugs with particular regard to the extent and time course of such effects.

PHARMACOGENOMICS (PHARMACOGENETICS): The study of how an individual's genetic inheritance affects the actions of drugs. Analysis of genetic data from individuals raises the prospect of drugs being tailor-made to suit the genetic make-up of particular patients or groups of patients. Environment, diet, age, lifestyle and state of health can influence a person's response to medicines, but understanding an individual's genetic makeup is thought to be the key to creating personalised drugs with greater efficacy and safety.

PHARMACOKINETICS (PK): 1. The movement of foreign substances, such as nutrients, metabolites, hormones, toxins, etc, but particularly drugs throughout the body of an animal. Processes such as uptake, distribution throughout the body tissues, the length of time the compound is active in the body and metabolism or excretion are taken into consideration and affect the development of new drugs.
2. The study of pharmacokinetics processes.

PHARMACOLOGY: The study of the origins, applications and effects of chemical substances on living organisms. Clinical pharmacology is concerned with the effect of drugs in treating diseases. It can also be defined as the science or art that deals with the study of drugs. It involves the history, physical and chemical properties, biochemical mechanisms of action, therapeutic effects and toxicity.

PHARMING: A web page that has been created to deceive visitors into believing that is another company or web page. While **phishing** attempts to capture personal information by getting users to visit a fake website, phishing redirects users to false websites without them even knowing it.

PHARYNGITIS: Inflammation of the pharynx (throat), the main symptom of which is a sore throat. Pharyngitis may be acute or chronic and have a number of different causes, bacterial, viral or other. The commonest cause of pharyngitis is a viral infection. Pharyngitis may also be caused by swallowing substances that scald, corrode or scratch the lining of the throat. Inflammation of the pharynx can be aggravated by smoking or excessive consumption of alcohol.

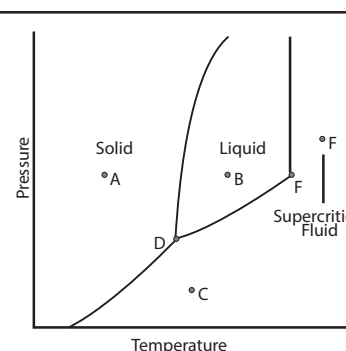
PHARYNX: 1. The interior of the throat or the cavity at the back of the mouth from the oral and nasal cavities to the trachea and oesophagus, which serves as a passage for food and respiratory gases. The pharynx has three connected sections, which are: the **nasopharynx**, at the back of the nasal cavity; the **oropharynx**, in the back of the oral cavity down to the epiglottis (flap of tissue that closes off the larynx during swallowing); and the **laryngopharynx**, from the epiglottis to the oesophagus. The presence of food in the pharynx stimulates swallowing. In fish and aquatic amphibians, the pharynx is perforated by gill slits. 2. The corresponding region in invertebrates.

PHASE: 1. In chemistry, chemically and physically uniform quantity of matter that can be separated mechanically from a non-homogeneous mixture. It may consist of a single substance or of a mixture of substances. The three basic phases of matter are solid, liquid and gas. Other phases that are considered to exist include crystalline, colloidal, glass, amorphous and plasma. The different phases of a pure substance are related to each other in terms of temperature and pressure. For example, if the temperature of a solid is raised enough or the pressure is reduced enough, it will become a liquid. 2. In mathematics, also known as **argument**, it is an angle θ from the positive real axis to the vector representing z in the complex plane, such that $r(\cos\theta + i\sin\theta)$ is equal to a given complex number $z = x + iy$. The numeric value is given by the angle in radians and is positive if measured counterclockwise. The principal phase is that value of θ in the half open interval $(-\pi, \pi)$ radians. Geometrically, in the complex plane, as the angle ϕ from the positive real axis to the vector representing z . 3. In physics, it is a particular point in the cycle of a waveform, measured as an angle in degrees. Also, it is a stage in an oscillatory motion such as a wave motion. Two waves are in phase when their peaks and their troughs coincide. In a wave motion, the fraction of the time required to complete a full cycle that a point completes after last passing through the reference position. Two periodic motions are said to be in phase when corresponding points of each reach maximum or minimum displacements at the same time. If the crests of two waves pass the same point at the same time, they are in phase for that position. If the crest of one and the trough of the other pass the same point at the same time, the phase angles differ by 180° and the waves are said to be of opposite phase. Phase differences are important in alternating electric current technology. 4. In astronomy, it is the apparent shape of the Moon or a planet when all or part of its illuminated hemisphere is facing the Earth. The monthly phases of the Moon are a common example. When the Sun is approximately on the far side of the Moon as seen from Earth called conjunction, the dark side of the Moon faces the Earth and there is a new moon. The phase waxes, beginning with crescent phases, as an increasing fraction of the illuminated face of the Moon is seen. At quadrature, when half the visible face of the Moon is illuminated, the phase is called the **first-quarter moon**, since the Moon is now one-quarter of the way through its cycle of phases. The waxing moon continues through its gibbous phases until it is in opposition. If the entire visible face of the Moon is illuminated, it is **the full moon**. During the full moon, the Moon and the Sun are on opposite sides of the Earth, a configuration known as a **syzygy**. Then the Moon wanes, going through waning gibbous, third-quarter and waning crescent phases until it is new again. The cycle of Moon phases takes approximately 29.53 days and explains the origin of the word month. 5. In electricity, one of the circuits in an electrical system or a device in which there are two or more alternating currents that are not in phase with each other. In a three-phase system, the displacement of the currents from one another is a third of a period.

PHASE ANGLE: 1. An angle between the sight lines to the Sun and the Earth measured at a remote location such as other celestial bodies. 2. The number of electrical degrees between the time an alternating current (AC) voltage passes through zero moving in the positive direction and the time the current passes through zero going in the same direction. 3. The phase difference between two sets of periodic phenomena expressed in an angular measure.

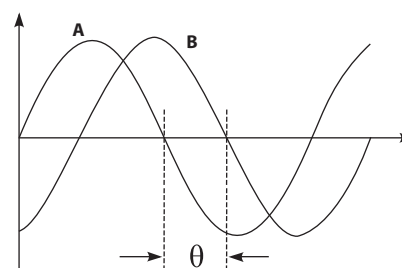
PHASE CHANGE: Metamorphosis of a substance from one state to another, as from gas to liquid or from solid to gas.

PHASE DIAGRAM: 1. A graph showing the relationship between equilibrium relationships between phases such as solid, liquid and gaseous phases over a range of conditions, for example, temperature and pressure. 2. A graph showing the pressures at which phase transitions between different states of a pure compound occur, as a function of temperature. 3. A graph showing the temperatures at which transitions between different phases of a binary system occur, as a function of the relative concentrations of its components.



Phase Diagram - Phases of Matter and Phase Transitions

PHASE DIFFERENCE: The angle in radians, which shows the extent to which two sinusoidally varying quantities in motion at the same frequency and wavelength are out of phase. In other words, it is the fraction of a complete cycle in radians, by which one wave motion is retarded or advanced to be in phase with another of the same frequency and wavelength. In the diagram shown, θ is the phase difference between waveforms A and B.



Phase difference of two waves

PHASE DOMAIN: A region of a material that is uniform in chemical composition and physical state.

PHASE EQUILIBRIA: In thermodynamics, equilibria between substances in different phases. Phase diagram, basic to engineering, metallurgy and mineralogy, are empirical graphs showing what phases exist at different pressures and temperatures. The phase rule, deduced by Josiah Gibbs, states that, for a closed system, $F = C - P + 2$, where P is the number of phases, C the number of independent chemical components and F the number of degrees of freedom, i.e., the number of variables (pressure, temperature, composition) whose values must be specified to define the system.

PHASE I METABOLISM: The first stage in metabolism of foreign compounds, such as a drug or toxin, in a form that can be eliminated by the body. Common reactions during this phase are oxidation, reduction and hydrolysis. The resulting metabolites are chemically more reactive than the parent compound, enabling them to undergo the reactions of the second stage.

PHASE II METABOLISM: The second stage in adapting foreign compounds, for elimination from the body, in which the activated

derivatives formed in phase I metabolism are conjugated with amino acids, glucuronic acid or glutathione, to yield water-soluble derivatives that are excreted in urine or bile.

PHASE ALTERNATION BY LINE; PAL: The coding system for colour television broadcasting developed in Germany and adopted by the United Kingdom, the Netherlands and Switzerland in 1967. It broadcasts 25 interlaced frames per second (50 half frames per second) at 625 lines of resolution. Brazil uses PAL M, which broadcasts 30 frames per second (fps). PAL is used throughout Europe and China as well as in various African, South American and Middle Eastern countries. PAL's colour signals are maintained automatically and the TV set does not have a user-adjustable hue control.

PHASE INTERACTION: Molecular interaction between the components present in the interphases of a multiphase mixture. The interphase elasticity is the capability of a deformed interphase to return to its original dimensions after the force causing the deformation has been removed.

PHASE OF THE MOON: The appearance of the illuminated portion of the moon by the sun as seen by an observer on earth. This is caused by the angle from which an observer on earth can see the moon illuminated by the sun as it orbits the earth. The phase changes as a result of the relative positions of the Earth, Sun and Moon. The eight phases of the moon are: (i) New Moon when the moon is not visible in the sky except during a solar eclipse that occurs when the moon is between earth and the sun, called conjunction; (ii) Waxing Crescent when the moon is less than one-half illuminated by the sun but less than one quarter illuminated that occurs when the moon's illumination is increasing; (iii) First Quarter when the moon is one-half illuminated by the sun that occurs when the moon's illumination is increasing; (iv) Waxing Gibbous when the moon is more than one-half illuminated by the sun but not completely illuminated that occurs when the moon's illumination is increasing; (v) Full Moon when the moon is totally visible in the sky except during a lunar eclipse; (vi) Waning Gibbous when the moon is more than one-half illuminated by the sun but not completely illuminated that occurs when the moon's illumination is decreasing; (vii) Third Quarter when the moon is one-half illuminated by the sun that occurs when the moon's illumination is decreasing; (viii) Waning Crescent when the moon is less than one-half illuminated by the sun but less than one quarter illuminated that occurs when the moon's illumination is decreasing. The cycle of moon phases takes approximately 29.53 days and explains the origin of the word month.

PHASE PORTRAIT: A picture which plots the evolution of possible paths in phase space that a point in that space can have, starting from a certain set of initial conditions. In the case of simple harmonic motion the phase portrait consists of a set of ellipses which all have the same centre. In the case of non-linear oscillation more complicated patterns appear in the phase portrait. The phase portrait is a convenient way of showing the complicated behaviour that can occur in dynamical systems, including the presence of attractors, chaotic behaviour and limit cycles.

PHASE RATIO: 1. In chromatography, it is the ratio of the volume of the mobile phase to that of the stationary phase in a column. 2. In liquid-liquid distribution, it is the ratio of the quantity of the solvent to that of the other phase.

PHASE RULE: The number of degrees of freedom, F , (the number of intensive properties such as temperature or pressure which are independent of other intensive variables) of a heterogeneous system is related to the number of components (C) such as one-component system like pure chemical or a two component system such as mixtures of water and ethanol and phase (P) such as solids, liquids and gases, present at equilibrium. The equation is given by $P + F = C + 2$.

PHASE SEPARATION: The process by which a single solid (liquid) phase separates into two or more new phases.

PHASE SHIFT: The phase shift of the function of the form, $y = a \sin(k\theta + c)$, is $-c/k$. If $c > 0$, the shift is to the left and if $c < 0$ the shift is to the right. This applies to all trigonometric functions.

PHASE SPACE: 1. A space in which all possible states of a system are represented, with each possible state of the system corresponding

to one unique point in the phase space. In other words, it is the union of the coordinate space and momentum space of the system in which the rectangular coordinates represent the position and momentum of the points of the system. It was introduced by Willard Gibbs in 1901. Each point represents a state of the system. In a gas of N point particles, each particle has three positional coordinates and three corresponding momentum coordinates so that the phase space has $6N$ -dimensions. If the particles have internal degrees of freedom, such as the vibration and rotations of molecules, then these must be included in the phase space, which is consequently of higher dimension than that for point particles. As the system changes with time the representative points trace out a curve in phase space known as a trajectory. 2. In differential equations, the space with coordinates $y, \dot{y}, \ddot{y}$, etc., that are the successive derivatives of the dependent variable.

PHASE SPEED (PHASE VELOCITY): The speed of propagation of pure sine wave $V_p = \lambda f$, where V_p is the symbol for the phase speed, λ is the wavelength and f is the frequency. The value of the phase speed depends on the medium through which it is travelling and may also depend on the mode of propagation.

PHASE TRANSITION (STATE CHANGE): A change in the physical state (solid, liquid, gas) or the phase transition of a material. For example, melting, boiling, evaporation, solidification and condensation are changes of state or phase transition. Phase transition is brought about by a change in the temperature of the system. The temperature at which the change of state occurs is called **transition temperature**, denoted by T_c . Examples of phase transitions are the **gas-liquid transition** called **condensation**, the **liquid-solid transition** called **freezing**, the **normal-to-superconducting transition** in electrical conductors, the **paramagnet-to-ferromagnet transition** in magnetic materials and the **superfluid transition** in liquid helium. It also includes transitions involving amorphous or glassy structures, spin glasses, charge-density waves and spin-density waves.

PHASE VARIATION: A form of antigenic method in which certain bacteria can change their surface antigens to evade the adaptive immune response of their host. For example, Salmonella uses this technique to switch between different types of the protein flagellin. As a result, flagella with different structures are assembled. The mechanism involves the inversion of segments of the bacterial genome that encode antigenic proteins, with different proteins expressed depending on the orientation of the segment relative to its promoter. Such inversions are heritable and reversible, depending on trigger stimuli encountered by successive generations.

PHASE-CONTRAST MICROSCOPE: A type of microscope that is widely used in examining cells and tissues. It makes visible the differences in phase or optical path in transparent or reflecting specimens. In passing through an object, the light is slowed down and becomes out of phase with the original light. With transparent specimens, having some structure, diffraction occurs, causing a larger phase change in light outside the central maximum of the pattern. The phase-contrast microscope provides a means of combining this light with that of the central maximum by means of an annular diaphragm and a phase-contrast plate, which produces matching phase change in the light of the central maximum only. This gives greater contrast to the final image, due to constructive interference between the two sets of light waves.

PHASE-SHIFT OSCILLATOR: A linear transistor oscillator (a sine wave generator) circuit that is used to produce a sine wave output for generating signals over a wide frequency range. It has an inverting amplifier element such as a transistor or operational amplifier with its output fed back to its input through a phase-shift network consisting of resistors and capacitors in a ladder network. The frequency may be varied from a few Hz to 200 Hz by employing one set of resistors with three capacitors arranged together to vary over a capacitance range in the 1:10 ratio.

PHASE-SPACE THEORY: A theory applied to unimolecular or bimolecular reactions proceeding through long-lived complexes. The probability of reaction is assumed to be proportional to the number of states available to a particular product channel divided by the

number of states corresponding to all product channels. The theory is used to predict rates, product energy distributions, product velocity distributions and product angular momentum distributions.

PHASE-TRANSFER CATALYSIS: The phenomenon of rate enhancement of a reaction between chemical species located in different phases (immiscible liquids or solid and liquid) by addition of a small quantity of an agent called the phase-transfer catalyst that extracts one of the reactants, most commonly an anion, across the interface into the other phase so that reaction can proceed. These catalysts are salts of onium ions, for example, tetraalkylammonium salts or agents that complex inorganic cations, for example, crown ethers. The catalyst cation is not consumed in the reaction although an anion exchange does occur.

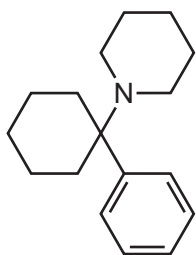
PHASOR: 1. A rotating vector used to represent a sinusoidally varying quantity. Its length, representing the value of the quantity is shown by its projection upon a fixed axis. 2. Any quantity such as impedance or admittance which is a complex number. 3. A low-energy collective excitation of the conduction electrons in a metal, corresponding to a slowly varying phase modulation of a charge-density wave.

PHELLEM: A protective water-proof layer of plant tissue produced by the cork cambium. It develops in plants undergoing secondary growth and replaces the epidermis. The cells are dead and the walls contain suberin.

PHELLODERM: The inner secondary tissue produced by the division of the cork cambium.

PHELLOGEN (BARK CAMBIUM; PERICAMBIUM; PHELLOGEN): A lateral meristem forming the periderm, outer protective tissue common in stems and roots of dicotyledonous plants and gymnosperms. It produces phellan (cork) towards the surface of the organ and phelloderm toward the inside. The periderm, therefore, consists of three different layers: phelloderm, phellogen (cork cambium) and phellem. It is found in woody and many herbaceous dicots, gymnosperms and some monocots, which usually lack secondary growth.

PHENCYCLIDINE [1-(1-PHENYLCYCLOHEXYL) PIPERIDINE; PCP; ANGEL DUST]: A recreational, hallucinogenic and neurotoxic drug originally used as a veterinary anaesthetic known as 'angel dust'. PCP works primarily as an N-methyl-D-aspartate receptor (NMDA receptor) antagonist, which blocks the activity of the NMDA receptor and, like most antilutamatergic hallucinogens, is significantly more dangerous than other categories of hallucinogens. Although the primary psychoactive effect of the drug lasts for a few hours, the total elimination rate from the body typically extends eight days or longer.



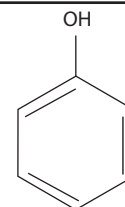
Chemical Structure of Phencyclidine

PHENETICS (TAXIMETRICS): A term describing an attempt at system classification of organisms based on similarities and differences in as many observable characteristics as possible.

PHENOCOPY: An environmentally induced, non hereditary variation in an organism, closely resembling a genetically determined trait. It occurs most commonly when only environmental influences evoke a developmental trait that has a very similar genetic counterpart. For example, dietary deficiency of vitamin D causes the bone disease rickets, which cannot be readily distinguished from genetically determined forms of the disease due to malabsorption or excessive loss of bone minerals. Another example of a phenocopy is the Vanessa genus of butterflies, which can change phenotype based on the local temperature. If introduced to tropical forests, for example, they mimic

butterflies localised to this area and if again sent to the grassland region, they mimic butterflies of that area. These phenotypes are not inherited and are solely due to the environment.

PHENOLS (PHENOLICS): Any of a class of organic compounds that contain a hydroxyl group (-OH) bound directly to a carbon atom in a benzene ring. Unlike normal alcohols, phenols are acidic because of the influence of the aromatic ring. Phenol (C_6H_5OH) ionises in water by the reaction: $C_6H_5OH \rightarrow C_6H_5O^- + H^+$. Phenols can either be found in nature (produced by plants and microorganisms) or synthesised. Phenols are made by fusing a sulfonic acid salt with sodium hydroxide to



Chemical Structure of phenol or phenolic acid, the simplest of phenols

form the sodium salt of the phenol. The free phenol is liberated by adding sulfuric acid. Examples of naturally-occurring phenols are cannabinoids (found in cannabis); capsaicin (the pungent compound of chili peppers); cresol (found in coal tar and creosote); eugenol (the main constituent of the essential oil of clove); gallic acid (found in galls); and salicylic acid, (a precursor compound to aspirin). Some synthetic phenols include phenol, also known as phenolic acid (formerly called carbolic acid, with the chemical formula C_6H_5OH); picric acid also called trinitrophenol, which is an explosive material); and xlenol (used in antiseptics and disinfectants). Phenols can be classified as either simple or polyphenols based on the number of phenol units in the molecule. Polyphenols contain large multiples of phenol structural units; an example is tannic acid. The simplest of the phenols is phenol (phenolic acid, formerly called carbolic acid). Its most important reaction is with formaldehyde, with which it forms widely used polymers called phenol-formaldehyde resins. It is used to manufacture dyes, pharmaceuticals, antiseptics and disinfectants.

PHENOL-FORMALDEHYDE RESIN (PF): A class of resins produced by the reaction of phenols with formaldehyde (methanal) using acid or basic catalysts, for example, bakelite. Phenolic resins are mainly used in the production of circuit boards, moulded products such as pool balls, laboratory countertops and as coatings and adhesives.

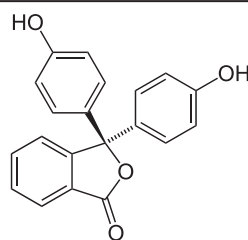
PHENOLATES (PHENOXIDES): Salts or analogous metal derivatives of phenols.

PHENOLIC RESINS: Synthetic resins made by copolymerising phenols with aldehydes such as formaldehyde and furfural. Phenol-formaldehyde resins are the commonest type. Phenolic resins can be cast from syrupy intermediates or moulded from B-stage solid resins. Laminated products can be produced by impregnating fibre, cloth, wood and other materials with the resin. An important type of phenolic resin product is rigid foam. Hardened phenolic plastics are rigid, hard and resistant to chemicals except strong alkali and to heat. Phenolic resins are used for making precision moulded articles such as telephone parts, for manufacturing strong and durable laminated boards or for impregnating fabrics, wood or paper. They are also used for adhesives such as the binder for grinding wheels, as thermal insulation panels, as ion-exchange resins and in paints and varnishes.

PHENOLOGY: The study of the periodic plant and animal life cycle events such as flowering in plants and breeding and migration in animals and how they are affected by climate and other environmental factors. Examples include when leaves and flowers appear, the first flight of organisms such as butterflies and the first appearance of migratory birds.

PHENOLPHTHALEIN (3,3-BIS(4-HYDROXYPHENYL) ISOBENZOFURAN-1(3H)-ONE):

A colourless or yellowish dye having the chemical formula $C_{20}H_{14}O_4$, density 1.277 g cm^{-3} at 32°C (305.15 K) and melting point 260°C (533.15 K). It is colourless below (acidic solutions) pH 8 and pink in basic solutions (above pH 9.6). In strongly basic solutions, phenolphthalein's pink colour undergoes a rather slow fading reaction and becomes colourless again. It is used in titrating weak acids against strong bases and also used as laxative.

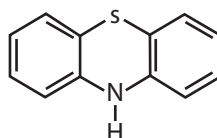


Chemical Structure of Phenolphthalein

PHENOMENON: 1. An observable event that happens. An example is earthquake. 2. Any state or process known through the senses rather than by thought or intuition. 3. A fact or event of scientific interest susceptible to scientific description and explanation.

PHENONIUM IONS: The traditional generic name for those cyclohexadienyl cations that are spiro-annulated with a cyclopropane unit. Phenonium ions constitute a subclass of arenium ions.

PHENOTHIAZINE (10H-PHENOTHIAZINE): A greenish-yellow crystalline compound with the chemical formula $C_{12}H_9NS$, melting point 185°C (458 K) and boiling point 371°C (644 K). It is used to disinfect threadworms and roundworms of animals, as tranquilisers in the treatment of psychiatric disorders such as schizophrenia and as a source of dyes.



Chemical Structure of Phenothiazine

PHENOTYPE: An expressed characteristic of an organism that may be visible such as brown eyes, colour or curly hair or may not be outwardly visible such as blood group. The visible characteristic results from the interaction of its genotype with environmental factors such as light, soil moisture, and temperature. The phenotype may change throughout the life of an organism because of factors like environmental changes and aging.

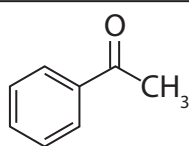
PHENOTYPIC PLASTICITY: The ability of an organism to adapt to its phenotype in response to changes in the environment by modifying its own development, form, behaviour or other trait. It depends on the capacity of a given genotype to respond to environmental factors and to adjust the phenotype accordingly. Social insects such as bees with the same genotype undergo different developmental pathways to become members of different castes, depending on diet and incubation temperature.

PHENOXY RESINS: Thermoplastic materials made by condensation of phenols, used for injection moulding, extrusion, coatings and adhesives.

PHENYL AMMONIUM ION: The ion $C_6H_5NH_3^+$ derived from phenylamine.

PHENYL GROUP: The organic group C_6H_5- present in benzene, where the six carbon atoms are arranged in a cyclic ring structure. One of the simplest phenyl containing compounds is phenol (C_6H_5OH).

PHENYL METHYL KETONE (ACETOPHENONE): A colourless, crystalline ketone with an odour of bitter almonds and chemical formula $C_6H_5COCH_3$, melting point 19.6°C (292 K) and a specific gravity of 1.028. It is made by treating benzene with ethanoyl chloride in the presence of aluminium chloride. It is used as a solvent and in making perfumes.

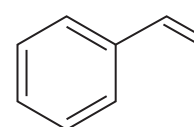


Chemical Structure of Phenyl Methyl Ketone

PHENYLALANINE: An essential amino acid found in most protein-rich foods, with the chemical formula $HO_2CCH(NH_2)CH_2C_6H_5$. It is one of the twenty common amino acids used by terrestrial living things to form proteins. The hereditary condition known as phenylketonuria (PKU) stems from an inability to metabolise phenylalanine normally.

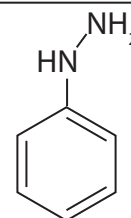
PHENYLAMINE (ANILINE; AMINO BENZENE): A colourless, oily, liquid aromatic amine having the chemical formula $C_6H_5NH_2$, whose molecule is formed by replacing one hydrogen atom of a benzene molecule with an amino group. It boils at 184°C (457 K) and melts at -6°C (267 K). The compound turns brown on exposure to sunlight. It is basic, forming the phenylaminium or anilinium ion ($C_6H_5NH_3^+$) with strong acids. It is prepared by the reduction of nitrobenzene under acidic conditions or by the addition of ammonia to chlorobenzene using a copper(II) salt catalyst at 200°C (473 K). The compound is used extensively in the rubber industry and in the manufacture of drugs and dyes.

PHENYLETHENE (STYRENE): A liquid hydrocarbon, with the chemical formula $C_6H_5CHCH_2$, boiling point 145.2°C (418 K) and freezing point -30.6°C (243 K). The ethylenic linkage of styrene readily undergoes addition reactions and under the influence of light, heat or catalysts undergoes self-addition or polymerisation to yield polystyrene. It can be made by dehydrogenating ethylbenzene. It is used in making polystyrene, thermoplastic and elastomers or synthetic rubbers.



Chemical Structure of Phenylethene

PHENYLHYDRAZINE (HYDRAZINOBENZENE): A toxic colourless dense liquid with the chemical formula $C_6H_5N_2H$, density 1.0978 g/cm^3 , melting point 19.5°C (292.65 K), 24°C (297 K) for hemihydrate and decomposes at a boiling point of 243.5°C (516.65 K). It forms monoclinic prisms that melt to an oil around room temperature. Its colour changes to yellow to dark-red on exposure to air. It is slightly soluble in water and miscible with ethanol, diethyl ether, chloroform and benzene. It is prepared by oxidising aniline with sodium nitrite in the presence of hydrogen chloride to form the diazonium salt, which is subsequently reduced using sodium sulfite in the presence of sodium hydroxide to form the final product. It is a powerful reducing agent. It is used in analytical chemistry to detect sugars and aldehydes and as a chemical intermediate. It is also used to make dyes and derivatives of indole.



Chemical Structure of Phenylhydrazine

PHENYLKETONURIA: A human disease caused by a genetic deficiency in the enzyme that converts phenylalanine to tyrosine. The immediate cause of the disease is inability to metabolise phenylalanine due to the absence of the enzyme phenylalanine hydroxylase resulting in an excess of phenylalanine in all body fluids and low melanin levels, with light hair, eye and skin colour. The disease occurs in individuals who are homozygous for the defective recessive allele at the PAH gene on chromosome 12. Both parents of such individuals are thus heterozygous carriers of the allele. Phenylalanine is present in foods such as meat, dairy and high-protein foods and aspartame and avoiding them restores normalcy.

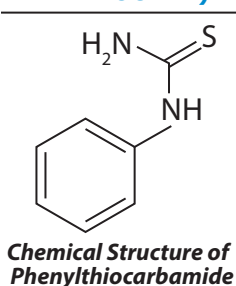
PHENYLMETHANE (METHYLBENZENE; TOLUENE): A colourless liquid aromatic hydrocarbon, derived from benzene by replacement of hydrogen atom by a methyl group. Its chemical formula is $CH_3C_6H_5$, melting point -95°C (178 K) and boiling point 110.8°C (394 K). It is insoluble in water but highly soluble in most organic solvents. It can be obtained from coal tar or made from methylcyclohexane (extracted from crude oil) by catalytic dehydrogenation. Its main uses are as a solvent and as a raw material for producing trinitrotoluene (TNT). It is also used in aviation fuel and other high-octane fuels, in dyestuffs, explosives and as a solvent for gums and lacquers.

PHENYLMETHANOL (BENZYL ALCOHOL; PHENYLCARBINOL):

A colourless liquid aromatic alcohol having the chemical formula $C_6H_5CH_2OH$, density 1.044 g/cm^3 , melting point -15°C (258 K) and boiling point 205°C (478 K). It is partially soluble in water and miscible in alcohols and diethyl ether. It is used mainly as a solvent.

PHENYLTHIOCARBAMIDE (PTC; PHENYLTHIOUREA):

A crystalline compound having the chemical formula $C_6H_5NHCSNH_2$. It is an organosulfur thiourea containing a phenyl ring. It tastes extremely bitter to people who possess a specific dominant gene or is tasteless. Non-smokers and those who are not habitual coffee or tea drinkers are more likely to taste PTC than others. It is used for testing the PTC gene.



PHEOPHYTIN: The first electron acceptor in the light-dependent reactions of photosynthesis. It is the first electron carrier intermediate in the electron transfer pathway of photosystem II in plants and the photosynthetic reaction centre RC P870 found in purple bacteria. It is passed on via another acceptor to plastoquinone. Pheophytin is a form of chlorophyll molecule lacking a central Mg^{2+} ion (the magnesium ion is replaced by two hydrogen ions). It participates in the crucial step of converting light energy to chemical energy.

PHEROMONE (ECTO HORMONE; FEROMONE): A chemical substance secreted by an organism in minute amounts into the environment as a specific signal to another organism, usually of the same species. Pheromones play an important role in the social behaviour of certain animals, especially insects and mammals. It is present in some fungi, slime moulds and algae. They are used to attract mates, signal danger, mark trails and to promote social cohesion and coordination. Pheromones may be secreted by special glands or contained into other substances such as urine (for example dog), shed freely or deposited in selected locations.

PHI-COEFFICIENT: The measure of correlation between two variables forming two-by-two contingency table. This is given by

$$\phi = \sqrt{\frac{\chi^2}{n}}$$

where χ^2 is the chi-squared statistic for the independence

of the two variables and N is the sample size. The coefficient has a maximum value of 1. The closer the value of the coefficient is to 1, the stronger the correlation between the two variables.

PHILLIPS PROCESS: A process for making high-density polyethylene by polymerising ethene at high pressure (30 atmospheres) and a temperature of 1500°C (1073 K). The catalyst is chromium(III) oxide supported on silica and alumina.

PHILOPATRY: 1. The tendency of an animal to stay in or return to its home area. 2. The behaviour of elder offspring sharing the parental burden in the upbringing of their siblings. 3. A dispersal method in which reproductive particles remain near their point of origin.

PHILOSOPHICAL LOGIC: The branch of philosophy that studies the relationship between formal logic and ordinary language and especially the extent to which the former can be held accurately to represent the latter or, equivalently, to which the latter is an adequate model for the former.

PHISHING: A term used to describe a malicious individual or group of individuals that attempt to get an individual to provide personal information such as banking details, credit card details and passwords, which can be used to access the individual's account or for other identity theft purposes.

PHLOEM: The principal conducting tissue of vascular plants that transports food substances and other organic materials such as hormones from their sites of synthesis to all parts of the plant. It contains basically three living cell types: the sieve tube elements, companion cells and typical parenchyma cells. The substances travel along the phloem in the sieve tube elements. They are wide elongated

cells joined end to end with large pores in the end plates to facilitate fast movement of large amounts of substances.

PHLOEM LOADING: The processes by which sugars and other substances enter the phloem driven by positive hydrostatic pressures to other tissues within the plant. The carbon fixed by photosynthesis is transported in phloem primarily as sucrose, formed by combination of glucose and fructose in the cytosol of photosynthetic cells (the source). From here, the sucrose travels the short distance to the vicinity of the nearest phloem sieve tube elements which have perforated areas in their walls, providing the main channels in which food substances travel, mainly by diffusing from cell to cell via the plasmodesmata.

PHLOEM PROTEIN (P-PROTEIN): A protein found in large amounts in the sieve elements tube of phloem tissue in plants. It takes various forms in the mature sieve tube element, depending on plant species, ranging from a network of filaments to discrete crystalline bodies. One general property is its ability to form a gel and functions as a puncture repair substance, forming a plug at any site of damage in the sieve tube element, thus preventing loss of food materials being translocated by the phloem. In an intact functioning sieve tube element, the P-protein is mainly found lining the interior wall.

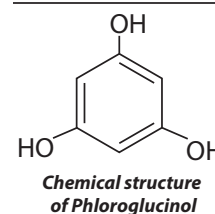
PHLOEM TISSUE: Plant conductive tissue responsible for transport of carbohydrates and other organic materials.

PHLOEM UNLOADING: The process by which sugars and other substances leave the sieve tube elements of the phloem and are delivered to their destination in the plant by diffusion via the plasmodesmata and an apoplastic pathway, via intercellular space.

PHLOGISTON THEORY: A former hypothesis of combustion in which all flammable objects were supposed to contain a substance called phlogiston, which was released when the object burned. Phlogiston was supposed to be a substance without colour, odour, taste or weight that is given off in burning. The existence of this hypothetical substance was proposed in 1669 by Johann Becher, who called it 'combustible earth' (*terra pinguis*: 'fat earth'). For example, according to Becher, the conversion of wood to ashes by burning was explained on the assumption that the original wood consisted of ash and *terra pinguis*, which was released on burning. In the early 18th century, Georg Stahl renamed the substance phlogiston (from the Greek 'burned') and extended the theory to include the calcination and corrosion of metals. Thus, metals were powdery residue and phlogiston. When a metal was heated, phlogiston was set free and the calx remained. The process could be charcoal (a substance believed to be rich in phlogiston, because combustion almost totally consumed it). The calx would absorb the phlogiston released by the burning charcoal and become metallic again. The theory was finally demolished by Antoine Lavoisier, who showed by careful experiments with reactions in closed containers that there was no absolute gain in mass of the air used in combustion. After experiments with Priestley's dephlogisticated air, Lavoisier realised that this gas, which he named oxygen, was taken up to form a calx (now called an oxide). The role of oxygen in the new theory was almost exactly the opposite of phlogiston's role in the old. In combustion and corrosion phlogiston was released; in the modern theory, oxygen is taken up to form an oxide.

PHLOGOPITE: A range of magnesium-rich varieties of mica, grading into biotite.

PHLOROGLUCINOL: 1. A red dye usually acidified with hydrochloric acid, with the chemical formula $C_6H_6O_3$ that stains lignin in plant cells red. 2. A colourless crystalline solid used in printing, photography and in the manufacture of pharmaceuticals.



PHOBIA: A neurosis characterised by exaggerated anxiety on confrontation with a specific object or situation or the anxiety itself. Phobia is sometimes linked with obsessional neurosis, sometimes with hysteria; in each case the object of phobia is usually symbolic. Example of phobias are **agoraphobia** (the fear and often

avoidance of places perceived as dangerous or uncomfortable) and **claustrophobia** (the fear of being in closed or small spaces or rooms and having no escape).

PHOBOS: The larger and inner of the two small moons of Mars. Its distinguishing feature is the crater Stickney, some 9 km across, which forms about 40% of the diameter of the moon.

PHOENICEOUS (PHENICIOUS): Purple-red; red colour with a slight mixture of grey.

PHON: A unit of loudness of sound that measures the intensity of a sound relative to a reference tone of defined intensity and frequency.

PHONG SHADING: In computer science, it is a type of shading developed by Phong Bui-Tuong used in animation and 3-D graphics, based on the colour and illumination at each pixel.

PHONOCHEMISTRY: The use of ultrasound, with a frequency greater than about 20 kHz to induce or accelerate certain types of chemical reaction.

PHONON: A quantum of crystal lattice vibrational or acoustic energy having a magnitude hf , where h is the Planck constant and f is the frequency of the vibration. In analogy to a photon, a phonon is viewed as a wave packet with particle-like properties. Phonons are useful in the treatment of the thermal conductivity of nonmetallic solids and, through consideration of electron-phonon interactions, the temperature dependence of the electrical conductivity of metals and understanding superconductivity in certain metals.

PHONOTAXIS: The movement of an organism in relation to a sound source. For example, females are often attracted by the courtship song of a potential mate called **positive phonotaxis** or animals, may flee from the sound of a predator called **negative phonotaxis**.

PHORANTHIUM: The receptacle of the head of composite flowers.

PHORESY: A relationship between two different species of organisms especially among arthropods, insects and some fishes in which the larger or host organism transports a smaller organism, the guest to a new site, where it releases itself, with little or no harm to the carrier. Animals use this when searching for new food sources; and parasites use it, when seeking a new host. The tropical botfly, *Dermatobia hominis*, which is parasitic on humans and other mammals, captures a suitable carrier, such as a blood sucking mosquito, to which it attaches 30 or so of its eggs. When the carrier insect reaches a human or bovine host, the botfly eggs quickly hatch and the larvae release themselves from the carrier and invade the hair follicle of their host, where they develop.

PHOSGENE (CARBONYL CHLORIDE; CARBON OXYCHLORIDE; CHLOROFORMYL CHLORIDE; DICHLOROFORMALDEHYDE; DICHLOROMETHANONE): A colourless highly toxic gas with penetrating and suffocating smell. Its chemical formula is COCl_2 , melting point -118°C (155 K) and boiling point 8.3°C (281 K). It is manufactured by the reaction of carbon monoxide and chlorine in the presence of a catalyst. If inhaled it reacts with water in the lungs to form hydrochloric acid and carbon monoxide, which causes pulmonary oedema and may also cause bronchial pneumonia and lung abscesses. It has been employed as a war gas and is now used in organic synthesis.

PHOSPHAGEN: High-energy phosphate compounds that provide a reserve of chemical energy in the form of high-energy phosphate compounds found mainly in muscular tissue in animals. As muscle tissues can have sudden demands for lots of energy, these compounds can maintain a reserve of high-energy phosphates that can be used as needed, to provide the energy that could not be immediately supplied by glycolysis or oxidative phosphorylation.

PHOSPHANES: The saturated hydrides of tervalent phosphorus having the general formula P_nH_{n+2} . Individual members having an unbranched phosphorus chain are named phosphane, diphosphane, triphosphane, etc. The name of a saturated hydride of phosphorus wherein one or more phosphorus atoms have a bonding number of 5 is formed by prefixing locants and symbols to the name of the

corresponding phosphane. Hydrocarbyl derivatives of PH_3 belong to the class phosphines.

PHOSPHATASE: An enzyme that catalyses the hydrolysis of a phosphate group in an organic compound important in the absorption and metabolism of carbohydrates, nucleotides and phospholipids and in the calcification of bone.

PHOSPHATE: A salt or ester of phosphoric acid (H_3PO_4). It is formed by the reaction of a metal, alcohol or other radical with phosphoric acid. Phosphate esters are organic compounds in which the hydrogens of phosphoric acid are replaced by organic groups such as methyl, ethyl, phenyl, with one of their carbon atoms bonding to an oxygen atom in the phosphate group. Examples of substances containing phosphate are nucleic acids and adenosine triphosphate (ATP), bones and teeth. Phosphates are used in fertilisers to replace the phosphorous containing compounds in the soil and in baking powder and toothpaste.

PHOSPHATIDE (PHOSPHOGLYCERIDE): A lipid containing charged hydrophilic phosphate groups. It is a member of a class of complex lipids that are the main component of biological membranes. The molecular structure is like the triglycerides, except for the replacement of a fatty acid with a phosphate-containing polar group. They consist of molecules containing a phosphoric (V) acid ester of glycerol, with the remaining hydroxyl groups of the glycerol being esterified by fatty acids. Examples of phosphatide are choline, ethanolamine, serine or inositol.

PHOSPHATIDIC ACIDS: Derivatives of glycerol in which one hydroxy group, commonly but not necessarily primary, is esterified with phosphoric acid and the other two are esterified with fatty acids.

PHOSPHAZENES: Compounds containing a phosphorus–nitrogen double bond, i.e. derivatives of $\text{H}_3\text{P}=\text{NH}$ and $\text{HP}=\text{NH}$.

PHOSPHIDE: A compound of phosphorus with a less electronegative element or elements. Typically there are a range of stoichiometries with each element. For example, with potassium there are nine potassium phosphides: K_3P , K_4P_3 , K_5P_4 , KP , K_4P_6 , K_3P_7 , K_5P_{11} , $\text{KP}_{10.3}$, KP_{15} . Nickel has eight compounds. One way to classify them is on the basis of structure and reactivity such as: (i) **Principally ionic** with P^{3-} ions, for example, Na_2P , Ca_3P_2 . (ii) **Polyphosphides** with P_2^{4-} ions, cluster ion P_{11}^{3-} , polymeric chain anions, for example, the helical $(\text{P})_n$. (iii) Compounds with individual p atoms in a metal lattice that can be semiconducting, for example, GaP.

PHOSPHINE (PHOSPHORUS; TRIHYDRITE PHOSPHORUS(III) HYDRIDE): A colourless, flammable, toxic gas with the chemical formula PH_3 , density 1.379 g/l (gas at 25°C), melting point -132.8°C (140 K) and boiling point -87.7°C (185 K). Phosphines are also a group of organophosphorus compounds with the formula R_3P , which are important in catalysis where they complex to various metal ions. Complexes derived from a chiral phosphine can catalyse reactions to give chiral products.

PHOSPHINIC ACID (HYPOPHOSPHORUS ACID): Organic derivative of hypophosphorous acid which is a white crystalline solid with the chemical formula H_3PO_2 , density 1.493 g/cm³, melting point 26.5°C (300 K) and decomposes above 130°C (400 K). It is soluble in water, ethanol and ethoxyethane. Salts of phosphinic acid may be prepared by boiling white phosphorus with the hydroxides of alkali and alkaline earth metals. The free acid is made by the oxidation of phosphine with iodine. It is a weak monobasic acid in which the -O-H group is ionised to give the ion H_2PO_2^- . The acid and its salts readily oxidise to the orthophosphate and consequently are good reducing agents.

PHOSPHOBACTERIUM: Bacteria that can dissolve insoluble inorganic phosphate from the soil.

PHOSPHODIESTER: A molecule containing two alcohols esterified to a single molecule of phosphate. For example, the backbone of

nucleic acids is connected by 3' carbonatom of one sugar molecule and the 5' carbon of another deoxyribose in DNA and ribose in RNA.

PHOSPHODIESTER BOND: Strong covalent bonds that link a phosphate group and a sugar group, by means of oxygen bridge, in the sugar phosphate backbone of a nucleic acid molecule. Phosphodiester bonds are important to all life on Earth, as they make up the backbone of the strands of DNA. In DNA and RNA, the phosphodiester bond is the linkage between the 3' carbonatom of one sugar molecule and the 5' carbon of another, deoxyribose in DNA and ribose in RNA.

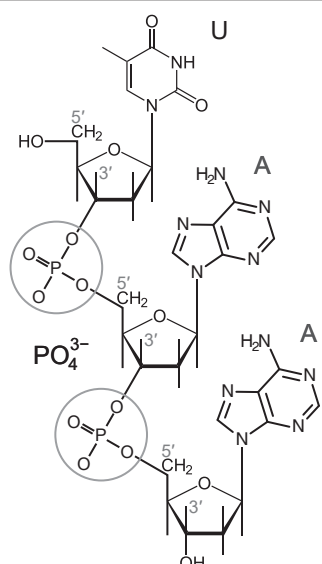
PHOSPHOENOLPYRUVATE (PHOSPHOENOLPYRUVIC ACID; PEP): A three carbon compound that is the substrate for carbon dioxide fixation during photosynthesis and also involved in the biosynthesis of various aromatic compounds in C4 plants. It is also an intermediate in glycolysis and gluconeogenesis. It is also used as the source of energy for the phosphotransferase system in bacteria.

PHOSPHOGLYCERATE (PGA): A three-carbon molecule formed when carbon dioxide is added to ribulose biphosphate (RuBP) during the dark reaction of photosynthesis (Calvin or Calvin-Benson Cycle). PGA is converted to PGAL, using adenosine triphosphate (ATP) and NADPH.

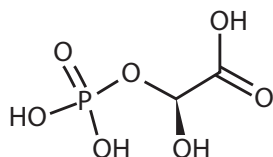
PHOSPHOGLYCERIDES: Phosphoric diesters, esters of phosphatidic acids, generally having a polar head group (OH or NH₂) on the esterified alcohol which typically is 2-aminoethanol, choline, glycerol, inositol, serine. The term includes lecithins, cephalins.

PHOSPHOLIPID (GLYCEROL PHOSPHOLIPASE): Any of various enzymes that cleave particular bonds in the polar phosphate 'heads' of glycerophospholipids. Phospholipase C (PLC) comes in multiple forms and plays a key role in the signal transduction process for many receptors. For example, it cleaves the phosphate-glycerol bond to liberate the second messengers inositol 1, 4, 5-triphosphate (IP₃) and diacylglycerol (DG) from phosphatidylinositol diphosphate in cell plasma membranes. DG is necessary for further activation of Protein Kinase C while IP₃ leads to the release of intracellular calcium. Phospholipase C plays a central role in signal transduction. It is activated by protein tyrosinase kinases, which themselves are activated by ligand binding to cell surface receptors.

PHOSPHONIC ACID (PHOSPHOROUS ACID; PHOSPHONATES; ORTHOPHOSPHORUS ACID): A colourless to pale-yellowish deliquescent crystalline solid with a garlic like odour and the chemical formula H₃PO₃, melting point 73 °C (346 K) and decomposes at 200 °C (473 K). It is very soluble in water and alcohol. The structure of this material is unusual in that it contains one direct P–H bond and is more correctly written as (HO)₂HPO. The acid is dibasic, giving rise to the ion H₂PO₃[−] and HPO₃^{2−} (phoshonates, formerly phosphites) and on heating gives phosphine and phosphoric(V) acid. It is prepared by hydrolysis of phosphorus trichloride or tetraphosphorus hexaoxide with alcohols or phenols. Phosphorous acid esters are called phoshonates



Phosphodiester Bond



Chemical Structure of Phosphoglycerate

or phosphites with the chemical formula (RO)₃P. Phosphorous acid and phosphonates (phosphite) are used as reducing agents in the chemical industry because of easy oxidation property to phosphoric acid, used as antioxidant, stabiliser and chelating agent in plastic system, as solvent in paint and as flame retardant on fibres. They are also used as chemical intermediates in the production of pharmaceutical ingredients, pesticides, optical brighteners and in lubricant additives and adhesives.

PHOSPHONITRILES: Compounds of stoichiometric composition [X₂PN]_n, in which X is alkoxy, halogen or other electronegative group and n is a variable integer, the value of which may not be known.

PHOSPHONIUM ION: The ion PH₄⁺ or the corresponding organic derivatives of the type R₃PH⁺, RPH₃⁺. The phosphonium ion PH₄⁺ is formally analogous to the ammonium ion NH₄⁺ but PH₃ has a much lower proton affinity than NH₃ and reaction of PH₃ with acids is necessary for the production of phosphonium salts.

PHOSPHOR: Any substance, which is a transition metal compound or a rare earth compound of various types that gives out visible light when it is illuminated by a beam of electrons or ultraviolet light or when irradiated with particles of electromagnetic radiation. They are most commonly used in CRT displays and fluorescent lights.

PHOSPHOR BRONZE: An alloy of copper containing 3.5 to 10% of tin and 0.05%-1% of phosphorous, which is added as a deoxidising agent during melting. It is noted for toughness, strength, low coefficient of friction and fine grained thus improving the fluidity of the molten metal and its casting abilities. It also improves mechanical properties by cleaning up the grain boundaries. Phosphor bronze is used for springs, bolts and ship's propellers and also used in some dental bridges.

PHOSPHORAMIDES: Compounds in which one or more of the OH groups of phosphoric acid have been replaced with an amino or substituted amino group; commonly confined to the phosphoric triamides, P(=O)(NR₂)₃, since replacement of one or two OH groups produces phosphoramidic acids P(=O)(OH)(NR₂)₂, P(=O)(OH)₂(NR₂).

PHOSPHORESCENCE: The emission of light by certain substances after they have absorbed energy, whether from visible light or other electromagnetic radiation such as ultraviolet rays, x-rays or cathode rays. But unlike fluorescence, a phosphorescent material does not immediately re-emit the radiation it absorbs due to the "forbidden" energy state transitions in quantum mechanics.

PHOSPHORESCENCE LIFETIME: A parameter describing the time evolution of the decay of the phosphorescence radiant intensity.

PHOSPHORIC(V) ACID (TETRAOXO-PHOSPHORIC(V) ACID; PHOSPHORIC ACID; ORTHOPHOSPHORIC ACID): A tribasic acid derived from phosphorous and oxygen with the chemical formula H₃PO₄, relative density 1.834 and melting point 42.35°C (315.5 K). It is very soluble in water and soluble in ethanol. It is manufactured by two methods: the wet process, in which the phosphate rock, Ca₃(PO₄)₂, is treated with sulfuric acid and the calcium sulfate removed either as gypsum or the hemihydrate. In the thermal process, molten phosphorus is sprayed and burned in a mixture of air and steam. The wet process product contains some of the impurities originally present in the rock and is used in the fertiliser industry. The thermal process produces a much purer product suitable for the foodstuffs and detergent industries. Phosphoric acid is also used in rust removers and for rust-proofing iron and steel.

PHOSPHOROLYSIS: Phosphate-induced cleavage of a molecule in which inorganic phosphate is the attacking group. An example of this is glycogen breakdown by glycogen phosphorylase, which catalyses attack by inorganic phosphate on the terminal glycosyl residue at the non reducing end of a glycogen molecule. The result is glucose 1-phosphate and glycogen glucose units.

PHOSPHOROSCOPE: A mechanical device used to separate phosphorescence from fluorescence.

PHOSPHORUS: A highly reactive, non-metallic element with the symbol P and atomic number 15, atomic weight 30.9738, melting point (white) 44.1 °C (317.25 K), boiling point (white) 280.5 °C (553 K) and specific gravity (white) 1.82. Phosphorus exhibits allotropy. The white allotrope, called white phosphorus glows in the dark (phosphorescence) and combusts spontaneously in air, producing dense white fumes of the oxide phosphorus pentoxide (P_4O_{10}). When white phosphorus is heated to about 250 °C (523 K) in the absence of air, it changes into the more stable red phosphorus. Black phosphorus is made by heating white phosphorus under pressure. Phosphorus is commonly found in inorganic phosphate rocks. It is a component of DNA, RNA, adenosine triphosphate (ATP) and also the phospholipids that form all cell membranes, essential for plant and animal life. It is an important constituent of bones and teeth tissue and of cells. It is required for the formation of nucleic acids and energy carrying molecules, for example, ATP and is also involved in various metabolic reactions. Its compounds are used in fertilisers, various organic compounds, toothpaste and detergents, for matches, explosives and fireworks, semiconductors, in glass and steel.

PHOSPHORUS CYCLE: The process involving the transfer of inorganic phosphates PO_4^{3-} , HPO_4^{2-} and $H_2PO_4^-$ between the biotic and abiotic components of the environment. Phosphorus is absorbed by plants from the soil and bodies of water and eventually passes into animals through food chains. It is used to build nucleic acids and energy systems and released upon death of the organisms and decay into the environment. Through a gradual process of erosion, these phosphates are returned to the soil, seas, rivers and lakes. Rocks containing phosphorus are mined for the manufacture of fertilisers, which provide an additional supply of inorganic phosphate to the abiotic environment.

PHOSPHORUS OXYGEN RATIO (P/O RATIO): A measure of oxidative phosphorylation showing the number of inorganic phosphate molecules used for adenosine triphosphate (ATP) synthesis for every oxygen consumed. It may also indicate the number of ATPs produced for every pair of electrons travelling through the electron transport system.

PHOSPHORUS(III) BROMIDE (PHOSPHORUS TRIBROMIDE; PHOSPHOROUS BROMIDE; TRIBROMOPHOSPHINE): A colourless, corrosive, fuming liquid with the chemical formula PBr_3 , density 2.852 g/cm³, melting point -41.5 °C (231.7 K) and boiling point 173.2 °C (446.4 K). It is prepared by treating red phosphorus with bromine. It is soluble in acetone, alcohol, carbon disulfide and hydrogen sulfide and decomposes in water. It is used as an analytical reagent to test for sugar and oxygen.

PHOSPHORUS(III) CHLORIDE (PHOSPHORUS TRICHLORIDE; PHOSPHOROUS CHLORIDE): A colourless fuming liquid with the chemical formula PCl_3 , density 1.574 g/cm³, melting point -93.6 °C (179.6 K) and boiling point 76 °C (349 K). It decomposes rapidly in moist air and water. It is soluble in ether, benzene, carbon disulfide and carbon tetrachloride. It may be prepared by the reaction of phosphorus(III) chloride with oxygen or by the reaction of phosphorus(V) oxide with phosphorous(V) chloride. Its reactions are very similar to those of phosphorus(III) chloride. Hydrolysis with water gives phosphoric(V) acid. Phosphorus(III) chloride oxide has a distorted tetrahedral shape and can act as a donor towards metal ions, thus giving rise to a series of complexes. It is used as a chlorinating agent, phosphorus solvent and in the manufacture of saccharin.

PHOSPHORUS(III) OXIDE (PHOSPHORUS TRIOXIDE): A white waxy solid with the chemical formula P_4O_6 , relative density 2.13, melting point 24 °C (297 K) and a boiling point of 174 °C (447 K). It is soluble in ether, chloroform and benzene but reacts with cold water to give phosphonic acid (H_3PO_3) and with hot water to give phosphine and phosphoric(V) acid. It is formed when phosphorus is burned in an oxygen-deficient atmosphere (about 50% yield).

PHOSPHORUS(V) BROMIDE (PHOSPHORUS PENTABROMIDE; PHOSPHORUS PENTABROMIDE): A yellow readily sublimable solid with the chemical formula PBr_5 , density 3.61 g/cm³, which decomposes at 106 °C (379 K). It is soluble in benzene and carbon tetrachloride (tetrachloromethane). It has the structure $PBr_4^+Br^-$ in the solid state but in the vapour phase is completely dissociated to PBr_3 and Br_2 . Rapid cooling of this phase to 258 °C (15 K) leads to formation of an ionic species $[PBr_4]^+[Br_3]^-$. It may be prepared by the reaction of phosphorus(III) bromide with bromine or the direct reaction of phosphorus with excess bromine. It is very readily hydrolysed to give hydrogen bromide and phosphoric(V) acid. It is used in organic chemistry as a brominating agent.

PHOSPHORUS(V) CHLORIDE (PHOSPHORUS PENTACHLORIDE; PENTACHLOROPHOSPHORANE): A toxic yellowish-white crystalline solid with the chemical formula PCl_5 , density 2.1 g/cm³, melting point 166.8°C (440 K), which sublimates at its boiling point of 160.5°C (434 K). It is prepared by the chlorination of PCl_3 and reacts upon contact with water to release hydrogen chloride and gives phosphorus oxides. It is soluble in carbon disulfide. It is used as a chlorinating agent and as a catalyst and chlorinating agent.

PHOSPHORUS(V) OXIDE (PHOSPHORUS PENTOXIDE): A white crystalline solid chemical compound and the anhydride of phosphoric acid with the chemical formula P_4O_{10} , density 2.39 g/cm³, melting point 340 °C (613 K) under pressure and sublimates at 360 °C (633 K). It reacts violently with water to give phosphoric(V) acid. It is prepared by burning elemental phosphorus in a plentiful supply of oxygen and purified by sublimation. It is used as a desiccant.

PHOSPHORYLATION: The addition of a phosphate (PO_4) or a phosphate group to an organic molecule in order to produce an organic phosphate. The first stage in many biochemical reactions is phosphorylation. The conversion of adenosine monophosphate (AMP) and adenosine diphosphate (ADP) to adenosine triphosphate (ATP) occurs by phosphorylation reactions in two main metabolic pathways: oxidative phosphorylation and photophosphorylation. This is normally controlled by a phosphorylase enzyme. Phosphorylation activates or deactivates many protein enzymes, causing or preventing the mechanisms of diseases such as cancer and diabetes.

PHOT: The centimetre-gram-second (cgs) unit of illuminance, which is equal to 10⁴ lux or one lumen per square centimetre. The International System of Units (SI) is the currently used system of units, and the lux (lx) has replaced the phot (ph).

PHOTIC ZONE (EUPHOTIC ZONE): The upper layer of a sea or lake, in which there is sufficient light for photosynthesis to occur. It is often affected by seasonal turbidity. Euphotic depths vary from only a few centimetres in highly turbid eutrophic lakes, to around 200 metres in the open ocean.

PHOTINO: A hypothetical counterpart of the photon, postulated to exist in supersymmetry theories. It relates the known fermions to hypothetical bosons and the known bosons to hypothetical fermions.

PHOTO NEGATIVE: 1. A term describing organisms that move away from a source of light. 2. A term describing a conductive material whose electrical conductivity decreases in response to increasing illumination.

PHOTOACOUSTIC DETECTOR: A pressure-sensitive detector used to detect intermittent radiation absorbed in a black body or in the sample concerned. The resulting rapid temperature change produces a transient pressure oscillation that is observed with the help of a microphone or a piezoelectric device.

PHOTOACOUSTIC EFFECT: Generation of heat after absorption of radiation, due to radiation less deactivation or chemical reaction.

PHOTOACOUSTIC SPECTROSCOPY: A spectroscopic technique in which spectra of opaque materials such as powders are obtained by exposing the material to light that is modulated at acoustic frequencies. This produces a photoacoustic signal, the magnitude of

which can be related to the ultraviolet or infrared absorption coefficient of the material. This technique was first used by Alexander Graham Bell in 1880, when he demonstrated that thin discs emitted sound when exposed to a beam of sunlight that was rapidly interrupted with a rotating slotted disk.

PHOTOADSORPTION: Irradiation by light (usually visible or ultraviolet) may affect adsorption. In a system containing adsorptive and adsorbent exposure to light may lead to increased adsorption (photoadsorption) or it may lead to desorption of an adsorbate (photodesorption).

PHOTOAFFINITY LABELLING (PAL): A technique in which a photochemically reactive molecular entity, specifically associated with a biomolecule, is photo-excited in order to covalently attach a label to the biomolecule, usually via intermediates. Identification of the resulting photo-crosslinked product provides structural information on the protein's binding site. The technique is widely used for identifying in general ligand-receptor interactions in biological macromolecules and in particular the binding site within proteins. Typical examples are nitrenes, carbenes, ketone excited states, cations and radicals.

PHOTOASSISTED CATALYSIS: Catalytic reaction involving production of a catalyst by absorption of light.

PHOTOAUTOTROPH: An organism such as a green plant or a photosynthetic bacterium, that synthesises its own food from inorganic components using energy derived from the Sun in the process of photosynthesis.

PHOTOMULTIPLIER TUBE (PHOTOMULTIPLIER): A vacuum tube that produces electrical signals from faint pulses of light, using the photoelectric effect to produce a few electrons from incident light, then multiplying the number of electrons by a cascade process to produce enough electrons to generate a useful current.

PHOTOPHOBIA: Abnormal sensitivity to or intolerance of light. It may be caused by eye inflammation, lack of pigmentation in the iris or various diseases, such as migraine.

PHOTOBIOLOGY: A branch of biology concerned with the biological and bioenvironmental effects of ultraviolet, visible or infrared radiation.

PHOTOBLASTIC: Of, relating to or referring to a seed whose germination is influenced by light. It is a period of temporary suspension of growth activity called **dormancy**, until conditions, in this case light favour germination. Photoblastic is classified into three kinds: positive photoblastic, negative photoblastic and non-photoblastic types. Seeds that are stimulated to germinate by light are described as **positively photoblastic**. When seeds are exposed to one or two cycles of intense source of white light, they germinate. Seeds whose germination is inhibited by light are said to be **negatively photoblastic**, they do not respond to white light treatment, but they germinate if they are maintained in complete dark conditions. The non photoblastic seeds are insensitive to light and they germinate irrespective of the presence or absence of light. The response to light is apparently mediated by phytochrome-regulated production of plant hormone gibberellins.

PHOTOCATALYSIS: A change in the rate of a chemical reaction or its initiation under the action of ultraviolet, visible or infrared radiation in the presence of a substance, the photocatalyst that absorbs light and is involved in the chemical transformation of the reaction partners.

PHOTOCATALYST: A catalyst that is able to produce, upon absorption of light, chemical transformations of the reaction partners. The excited state of the photocatalyst repeatedly interacts with the reaction partners forming reaction intermediates and regenerates itself after each cycle of such interactions.

PHOTOCATHODE: A negatively charged electrode such as a photomultiplier or phototube coated with a photosensitive compound that emits electrons when light falls upon it, as a result of the photoelectric effect.

PHOTOCELL (PHOTOELECTRIC CELL): Any of several solid-state devices producing an electric signal in response to exposure to

electromagnetic radiation. It has a photosensitive cathode that emits electrons when illuminated and an anode for collecting the emitted electrons. A positive potential on the anode enables a current to flow through an external circuit, the current being proportional to the intensity of the illumination on the cathode. Photocells are used in control systems, where interrupting a beam of light opens a circuit, actuating a relay that supplies power to a mechanism to bring about a desired operation, such as opening a door or setting off a burglar alarm. It is also used in photometry and spectroscopy.

PHOTOCHEMICAL EQUIVALENCE: According to the principle of photochemical equivalence, there is a one-to-one relationship between the number of photons absorbed by a system and the number of excited species produced. The principle often fails for light of high intensity such as laser beams, when molecules may absorb more than one photon.

PHOTOCHEMICAL FUNNEL: Molecular structure where the excited-state reactant or intermediate is delivered to the ground state to initiate product formation. For a broad class of organic reactions, the structure of the funnel can take the form of a conical intersection or a singlet-triplet crossing.

PHOTOCHEMICAL REACTION (PHOTOREACTION): A chemical reaction that is initiated by the absorption of light, ultraviolet or infrared radiation. The most important phenomenon of this type is photosynthesis. It is also the basis of photography. Other photochemical reactions are bleaching of laundry, tanning of human skin, formation of Earth's ozone layer, etc. Vision depends on photochemical reactions that occur in the eye. In photographic film and paper, light activates their silver salts to a state that is easy to reduce to metallic silver grains during development.

PHOTOCHEMICAL REACTION PATH: A sequence of geometries adopted by a molecule after initial electronic excitation, combined with a specification of the electronic state at each geometry, from radiation absorption to product formation. Theoretically, this usually involves the computation of the MERP connecting the Franck-Condon point, located on the potential-energy surface of the spectroscopic state, to the final photoproducts located on the ground-state potential energy surface. Experimentally, a photochemical reaction path can be probed by detecting transient species or reaction intermediates.

PHOTOCHEMICAL SMOG: Air pollution produced by the chemical reaction of sunlight, nitrogen oxides and volatile organic compounds (VOCs) in the atmosphere. The nitrogen oxides are released from the exhaust of fossil fuel-burning engines and power plants, the VOCs are released from anthropogenic sources such as gasoline, paints, solvents and pesticides. The resulting mixture of air pollutants includes nitrogen oxides, tropospheric ozone, VOCs, peroxyacetyl nitrate (PAN), aldehydes, etc. These are all highly reactive and oxidising and especially harmful to the elderly and children.

PHOTOCHEMISTRY: The branch of chemistry concerned with chemical reactions of molecules in electronically excited states produced by the absorption of infrared, visible, ultraviolet or vacuum ultraviolet (100-200 nm) light.

PHOTOCHROMISM: The reversible transformation of a chemical species between two forms by the absorption of electromagnetic radiation, where the two forms have different absorption spectra or the change of colour occurring in certain substances when exposed to light. Photochromic materials are used in sunglasses that darken sunlight. This was discovered in the late 1880s by German organic chemist, Dr. Willi Marckwald (1864-1950).

PHOTOCONDUCTIVE EFFECT (PHOTOELECTRIC PROCESS): The emission of electrons from matter (metals and non-metallic solids, liquids or gases) by absorption of energy from electromagnetic radiation of very short wavelength such as visible or ultraviolet light. It was first observed by Heinrich Hertz in 1887. Electrons emitted in this manner may be referred to as **photoelectrons**. The process

includes photoemission, photoionisation, photoconduction, the photovoltaic effect and the Auger effect.

PHOTOCONDUCTIVITY: Electrical conductivity resulting from photoproduction of charge carriers.

PHOTOCOPIER (COPIER; COPY MACHINE; PHOTOCOPYING MACHINE): An electrically-operated machine for reproducing duplicates of written, drawn, printed or graphic material using light, heat, chemicals or electrostatic charges. The technology used in some photocopiers is called xerography, a photographic method that works on the basis of electrostatic charges: the electrical resistance of a photoconductive substance decreases when light falls on it, forming an image. Some also use technologies such as inkjet printing, which sprays quick-drying ink onto paper to create a digital image. Some modern photocopiers can interface with computer network and used as a printer (including printing on both sides of the paper, printing booklets), scanner, fax server, etc.

PHOTOCROSSLINKING: The formation of a covalent linkage between two macromolecules or between two different parts of one macromolecule.

PHOTOCURRENT YIELD: The quantum efficiency of electron transport between the two electrodes of a photovoltaic cell or photoelectrochemical cell.

PHOTOCYCLIZATION: Intramolecular photochemical process leading to carbo- or hetero- cyclic ring systems by formation of one new single bond, either by a concerted process, for example, electrocyclization or by multi-step processes such as Norrish–Yang reaction, photoinduced electron transfer processes, Yang photocyclization and others.

PHOTOCYCLOADDITION: Intermolecular photochemical process leading to carbo- or hetero- cyclic ring systems by formation of two new single bonds, either by a concerted process, for example, a concerted addition or by multi-step processes, for example, the Paternò–Büchi reaction. Triplet state photocycloadditions are two-step processes involving the formation of diradical intermediates.

PHOTODECARBONYLATION: Photoinduced extrusion of carbon monoxide (CO).

PHOTODECARBOXYLATION: Photoinduced extrusion of carbon dioxide (CO₂). Typical examples are the CO₂ extrusion upon excitation of anti-inflammatory 2-arylpropionic acids, such as ketoprofen.

PHOTODECOMPOSITION: The chemical break down of a substance by the removal of one or more atoms from a molecule by the absorption of a quantum of electromagnetic energy, especially sunlight.

PHOTODECONJUGATION: Photoinduced reaction of conjugated compounds, for example, enones and carboxylic acid derivatives leading to loss of conjugation. This reaction has important synthetic applications for the synthesis of frameworks of numerous natural products, such as β,γ -unsaturated carboxylic acids from photoexcitation of α,β -unsaturated carboxylic acid derivatives, excluding the use of strong bases, as required for similar isomerisations performed in the ground state.

PHOTODEGRADATION: The photochemical transformation of a molecule into lower molecular weight fragments, usually in an oxidation process. This term is widely used in the destruction (oxidation) of pollutants by UV-based processes.

PHOTODETACHMENT: Ejection of an electron from a negative ion upon photoexcitation.

PHOTODIODE (PHOTOCONDUCTOR DIODE): A semiconductor diode in which the reverse current varies with illumination. It is used to detect the presence of light or to measure its intensity, for example, alloy-junction photocell and the grown-junction photocell.

PHOTODIODE ARRAY: An arrangement of a number of photodiodes on a single chip.

PHOTODISINTEGRATION: The decay of a nuclide as a result of the absorption of high-energy radiation such as gamma ray photon.

PHOTODYNAMIC EFFECT: In photobiology, it is photoinduced damage requiring the simultaneous presence of light, photosensitiser and molecular oxygen. A sensitised photooxidation which involves molecular oxygen.

PHOTOELASTICITY: An effect in which certain materials exhibit double refraction when subjected to stress. It is used for detecting strains to determine stress concentration factors in irregular geometries and transparent materials such as perspex, celluloid and glass. When polarised white light is passed through a stressed sample, the birefringence causes coloured patterns to be seen on the viewing screen of a suitable polariscope. If monochromatic polarised light is used, a complex pattern of light and dark fringes is produced.

PHOTOELECTRIC ATTENUATION COEFFICIENT: The attenuation coefficient when only the photoelectric process is taken into account.

PHOTOELECTRIC CELL (PHOTOCELL): Any of several solid-state devices producing an electric signal in response to exposure to electromagnetic radiation. It has a photosensitive cathode that emits electrons when illuminated and an anode for collecting the emitted electrons. A positive potential on the anode enables a current to flow through an external circuit, the current being proportional to the intensity of the illumination on the cathode. Photocells are used in control systems, where interrupting a beam of light opens a circuit, actuating a relay that supplies power to a mechanism to bring about a desired operation, such as opening a door or setting off a burglar alarm. It is also used in photometry and spectroscopy.

PHOTOELECTRIC EFFECT (PHOTOELECTRICAL EFFECT; PHOTOCONDUCTIVE EFFECT; PHOTOVOLTAIC EFFECT): The emission of electrons from the surface of a metal, when it is incident by photons of appropriate frequency called threshold frequency (f_0). Electrons can only be removed from the metal surface if the frequency of the incident photon or radiation is equal or greater than the threshold frequency (f_0). There would be no emission of electrons if the frequency of the incident photon or radiation is less than the threshold frequency (f_0). The effect was discovered by H. R. Hertz in 1887.

PHOTOELECTRIC PEAK (PHOTOPEAK): Of a spectrum of γ -radiation, the part of the spectral response curve corresponding to the absorption in the radiation detector by the photoelectrical effect of the detected γ -energy. In most cases the peak also contains events caused by multiple processes and the use of the expression "total absorption peak" or "full energy peak" is to be preferred.

PHOTOELECTROCHEMICAL ETCHING: The dissolution of a semiconductor in an electrolytic solution upon exposure to light. It is used in the photopatterning of semiconductor surfaces.

PHOTOELECTROCHEMISTRY: A term applied to a hybrid field of chemistry employing techniques which combine photochemical and electrochemical methods for the study of the oxidation-reduction chemistry of the ground or excited states of molecules or ions. In general, it is the chemistry resulting from the interaction of light with electrochemical systems.

PHOTOELECTROLYTIC CELL: A cell in which radiant energy causes a net chemical conversion in the cell, for example, so as to produce hydrogen as a useful fuel. These cells can be classified as photosynthetic or photocatalytic. In the former case, radiant energy provides Gibbs energy to drive a reaction such as $\text{H}_2\text{O} \rightarrow \text{H}_2 + 1/2\text{O}_2$ and electrical or thermal energy may be later recovered by allowing the reverse, spontaneous reaction to proceed. In a photocatalytic cell the photon absorption promotes a reaction with $\Delta G < 0$ so there is no net storage of chemical energy, but the radiant energy speeds up a slow reaction.

PHOTOELECTRON: An electron emitted from a substance by irradiation as a result of the photoelectric effect or photo ionisation.

PHOTOELECTRON SPECTROSCOPY, PES (PHOTOELECTRON SPECTROSCOPY): A technique using the photoelectric effect, in order to determine the binding energies of electrons in a substance. The techniques commonly used are: (i) **X-ray photoelectron spectroscopy (XPS)** developed by Kai Siegbahn. It is used to study the energy levels of atomic core electrons, primarily in solids. (ii) The **ultra-violet photoelectron spectroscopy (UPS)** developed originally for gas-phase molecules in 1962 by David W. Turner. In UPS, the sample is a gas or vapour irradiated with a narrow beam of ultraviolet radiation, usually from a helium source at 58.4 nm, 21.21 eV photon energy. (iii) **X-ray photoelectron spectroscopy (XPS)** also known as **electron spectroscopy for chemical analysis (ESCA)**, is a similar analytical technique in which a beam of X-rays is used. In this case, the electrons ejected are from the inner shells of the atoms. Peaks in the electron spectrum for a particular element show characteristic chemical shifts, which depend on the presence of other atoms in the molecule.

PHOTOELECTRON YIELD: The number of photoelectrons emitted by the sample per incident photon.

PHOTOEMISSION: The process in which electrons are emitted by a substance such as a solid or less commonly by a liquid as a result of incident electromagnetic radiation. The visible and ultraviolet regions of the electromagnetic spectrum are most often involved, although the infrared and x-ray regions are also of interest.

PHOTOEMISSIVE DETECTOR: A detector in which a photon interacts with a solid surface, which is called the photocathode or a gas, releasing a photoelectron. This process is called the external photoelectric effect. The photoelectrons are collected by an electrode at positive electric potential (the anode).

PHOTOEXCITATION: The production of an excited state by the absorption of ultraviolet, visible or infrared radiation.

PHYTOEXTRACTION: A type of phytoremediation in which plants or algae are used to remove contaminants from soil, sediments, or water, and turning them into harvestable plant biomass.

PHOTOFISSION: A nuclear fission that is caused by absorption by the nucleus of a gamma-ray photon.

PHOTO GALVANIC CELL: An electrochemical cell in which current or voltage changes result from photochemically generated changes in the relative concentrations of reactants in a solution phase oxidation-reduction couple.

PHOTOGRAM: A picture produced on photographic material by placing objects directly onto the surface of a photo-sensitive material such as photographic paper and exposing it to light but without using a camera. This results in a negative on which portions of the paper that have received no light appear white; those exposed through transparent or semi-transparent objects appear grey.

PHOTOGRAPHIC DENSITY (OPTICAL DENSITY): A measure of the opacity of a specific part of a negative or transparency, which is the reciprocal of its transmittance.

PHOTOGRAPHIC EMULSION: A layer of light-sensitive material coated on a substrate. For example, in silver-gelatin photography, the emulsion consists of silver halide crystals suspended in gelatin and the substrate may be glass, plastic, paper or fabric.

PHOTOGRAPHY: The process of reproducing still or moving pictures by recording radiant energy, including visible light, ultraviolet, infrared, x-rays, atomic radiation and electron beams on a radiation-sensitive medium such as a photographic film or an electronic sensor.

PHOTOHETEROTROPH: Any organism that uses energy derived from the Sun to manufacture organic compounds from organic precursors in photosynthesis. For example, under certain conditions, some purple sulfur bacteria use organic acids rather than hydrogen sulfide as a source of hydrogen. Examples of such organisms are purple non-sulfur bacteria, green non-sulfur bacteria and heliobacteria.

PHOTOHYDRATION: A photochemical process leading to the addition of water across a C–C double (or triple) bond or to a carbonyl (or carbonyl analogue) group.

PHOTOINDUCED ELECTRON TRANSFER: An electron transfer resulting from an electronic state produced by the resonant interaction of electromagnetic radiation with matter.

PHOTOINDUCED POLYMERISATION: The polymerisation of a monomer by a free radical or ionic chain reaction initiated by photoexcitation.

PHOTOINHIBITION: The reduction in photosynthesis capacity of a plant, alga or a cyanobacterium caused by exposure to abnormally high intensities of visible or ultraviolet light energy trapped by the chloroplast. Some degree of light induced damage occurs normally in all photosynthesising organisms. They are repaired by mechanisms such as degradation and synthesis of the D1 protein of the photosynthetic reaction centre of Photosystem II (PSII).

PHOTOINITIATION: The photo production of a free radical or ion capable of initiating a chain reaction such as a polymerisation.

PHOTOIONISATION (ATOMIC PHOTOELECTRIC EFFECT): The ejection of one or more electrons from an atom, molecule or positive ion following the absorption of one or more photons such as light or other electromagnetic radiation. For photoionisation to occur the incident photon of the radiation must have energy in excess of the ionisation potential of the species being irradiated.

PHOTOIONISATION DETECTOR: A detector, in which radiation from an ultraviolet lamp ionises certain species in the carrier gas. A potential difference is applied and the resulting ionisation current is detected. The detector is only useful for substances with ionisation potentials below about 11 eV. This makes it quite useful for detecting one component of a combined eluent when the other component, for example, nitrogen, has a high ionisation potential. The detector has a small linear dynamic range and is capable of detecting substances below 1 ppmv.

PHOTOISOMERISATION: Photochemical process leading to an isomerisation of the substrate, either by bond rotation, skeletal rearrangement or atom- or group- transfer. Typical examples are *cis-trans* photosomerisation of alkenes, polyenes and phototautomerisation.

PHOTOKERATITIS (SNOW BLINDNESS): A type of keratitis that occurs in the cornea of the eye due to exposure to ultraviolet light (UVL) reflected off snow or ice, welder's arc, photographic flood lamps, lightning, carbon arcs, photographic flood lamps, lightning, electric sparks and halogen desk lamps. Its signs and symptoms include red and painful eye, excess tears or other discharge from the eye, blurred and decreased vision, sensitivity to light and difficulty in opening the eye. It is prevented by sun-glasses or protective goggles.

PHOTOLITHOGRAPHY: A technique used in the manufacture of semiconductor components, integrated circuits, etc, to selectively remove parts of a thin film or the bulk of a substrate. It uses light to transfer a geometric pattern from a photo mask to a light-sensitive chemical photo resist. A series of chemical treatments then engraves the exposure pattern into the material underneath the photo resist. In complex integrated circuits, for example a modern CMOS, a wafer will go through the photolithographic cycle up to 50 times.

PHOTOLUMINESCENCE (PL): A process, in which a substance absorbs photons, stores the photons leading to the excitation of electron(s) to higher energy state in the material and then upon de-excitation of the electron(s), re-radiates them as it returns to a lower energy state. It is also referred to as a glow in the dark. The period between absorption and emission is typically extremely short, in the order of 10 nanoseconds. Under special circumstances, however, this period can be extended into minutes or hours. Photoluminescence is generally limited to cases in which the incident radiation is in the ultraviolet, visible or infrared regions of the electromagnetic spectrum. Photoluminescence may be either fluorescence or phosphorescence

or both. Energy can be stored in certain luminescent materials by subjecting them to light or some other exciting agent and can be released by subsequent illumination of the material with light of certain wavelengths.

PHOTOLUMINESCENT POLYMER: Polymer that exhibits luminescence (fluorescence or phosphorescence arising from photoexcitation).

PHOTOLYSIS: A chemical reaction in which a chemical compound is broken down due to absorption of incident light. Photolysis is not limited to visible light. Any photon with sufficient energy, such as UV and x-ray, can affect the chemical bonds in a compound. Photolytic reactions often involve free radicals, the first step being homolytic fission of a chemical bond. Many metal complexes such as azides, nitrides and sulfides and most organometallic compounds undergo decomposition when exposed to light, often with concomitant evolution of a gaseous product. In the presence of water and oxygen, illumination of many semiconductors results in their corrosion.

PHOTOMAP: Aerial photographs showing physical and cultural features upon which a grid and data pertinent to maps have been added.

PHOTOMETER: An instrument used to measure luminous intensity or electromagnetic radiation, in the visible range, usually by comparing relative intensities from different sources. Photometers can be classified into two divisions: **laboratory photometers**, which are usually fixed in position and yield results of high accuracy; and **portable photometers**, which are used in the field or outside the laboratory and yield results of lower accuracy.

PHOTOMICROGRAPHY: A photograph taken of anything that can be viewed under magnification through a microscope. Photomicrographs are usually taken with cameras fitted via an attachment to the normal eyepiece of the microscope or, more commonly, through attachment to a special facility in the image path through the instrument.

PHOTOMORPHOGENESIS: The development of plants involving growth, differentiation of cells, tissues and organs under the influence of light. Photomorphogenic influences of light on plant form are quite different from light effects that nourish the plant through photosynthesis, since the former usually occur at much lower energy levels than are necessary for photosynthesis and determine the nature and direction of a plant's growth and its ecological adaptations to various environmental changes. The light mediated responses are mediated by various photoreceptor systems, including phytochrome, which responds to red and far-red light and cytochrome, which is sensitive to blue and green light. These photoreceptor molecules interact with cell signalling networks to regulate the expression of genes involved in development and also influence the production of plant hormones, notably gibberellins and brassins.

PHOTOMULTIPLIER: A sensitive vacuum-tube detector of light or radiant flux that detects low levels of electromagnetic radiation (visible light or infrared radiation) and amplifies it to produce a detectable signal. It consists of a photocathode which converts the light to photoelectrons; one or more secondary-electron-emitting electrodes or dynodes which amplify the number of photoelectrons; and an output electrode or anode which collects the secondary electrons and provides the electrical output signal.

PHOTOMULTIPLIER TUBE: A vacuum phototube with additional amplification by electron multiplication. It consists of a photocathode, a series of dynodes, called a dynode chain on which a secondary-electron multiplication process occurs and an anode.

PHOTON: An elementary particle or quantum of electromagnetic radiation which demonstrates both particle and wave properties. It has neither rest mass nor electric charge but possesses energy, which is the product of Planck's constant and the frequency of the electromagnetic field and momentum and angular momentum corresponding to a spin of unity. The effects of this force are easily observable at both the macroscopic and microscopic level, because the

photon has no rest mass. Photons also exhibit wave-particle dualism. The interaction of radiation with matter involves the absorption, scattering and emission of photons.

PHOTON COUNTING: Recording of sequential single photons counted by way of recording and counting sequential electron pulses at the anode of the photomultiplier.

PHOTON SPLITTING: A physical process in which a very-high-energy gamma-ray photon can borrow energy from an external source, such as a very strong magnetic field and split into two lesser energy through coupling to the external field.

PHOTONEUTRON: A neutron emitted by an atomic nucleus undergoing a photonuclear reaction.

PHOTONIC CRYSTAL: A macroscopic, periodic dielectric structure that has spectral gaps, which restrict the movement of electromagnetic waves. It increases the efficiency in optical fibres such as optical semiconductors. It is used in microscopic lasers.

PHOTONICS: The study of devices covering all technical applications of light in the whole spectrum from ultraviolet to all aspects of infrared (750nm to 1nm). It includes the generation, emission, transmission, modulation, signal processing, switching, amplification, detection and sensing of light. It is analogous to electronics but with the electrons replaced by photons.

PHOTONUCLEAR REACTION: A nuclear reaction that is initiated from the collision of a photon with a nucleus.

PHOTOOXIDATION: Oxidation reactions induced by light. Common processes are: (i) the loss of one or more electrons from a chemical species as a result of photoexcitation of that species; (ii) the reaction of a substance with oxygen under the influence of light. When oxygen remains in the product this latter process is called photooxygenation. Reactions in which neither the substrate nor the oxygen are electronically excited are sometimes called photoinitiated oxidations.

PHOTOXYGENATION: The incorporation of molecular oxygen into a molecular entity. There are three common mechanisms: Type I in which the reaction of triplet molecular oxygen with radicals formed photochemically. In Type II, the reaction of photochemically produced singlet molecular oxygen with molecular entities to give rise to oxygen containing molecular entities. The third mechanism proceeds by electron transfer producing superoxide anion as the reactive species.

PHOTOPERIODISM: The response of an organism to changes in daily, seasonal or yearly cycles of light and darkness. Many plant responses such as seed setting, germination, flowering and fruiting are controlled by day length. In mammals, such activities as sleep, migration, reproduction and the changing of coats or plumage are controlled considerably by day length. Such responses enable organisms to exploit favourable conditions associated with seasonal changes in climate and vegetation. Examples are producing the young, when food is plentiful or producing flowers when pollinating insects are abundant. Plants have various photoreceptor molecules that are sensitive to light, including phytochromes and cytochromes. In livestock such as in poultry keeping, photoperiodism is commonly induced by artificial lighting to maximise egg laying and body weight.

PHOTOPHORE: A gland or organ that is specialised for the production of light. It appears as luminous spots on various marine animals, including fish and cephalopods. They contain the light producing chemicals or symbiotic bacteria responsible for the bioluminescence; from specialised mitochondrial cells in the organism, called photocytes or from compounds during the digestion of prey. The cells of many different tissues can be modified to form a photophore, including mucous glands in various fish and even the suckers of some deep-sea octopuses.

PHOTOPHORESIS: In aerosol physics, the motion of particles due to the influence of light. In many cases, this amounts to a special form of thermophoresis due to the heating of the particles by the light.

PHOTOPHOSPHORYLATION: The transformation of light (solar) energy for the reduction of CO₂ and the generation of adenosine5'-triphosphate, adenosine triphosphate (ATP).

PHOTOPHYSICAL PROCESSES: Photoexcitation and subsequent events which lead from one to another state of a molecular entity through radiation and radiationless transitions. No chemical change results.

PHOTOPIC VISION: The type of vision that occurs when the cones in the eye are the principal receptors for colour perception. The human eye uses three types of cones to sense light in three respective bands of colour. The biological pigments of the cones have maximum absorption values at wavelengths of about 420 nm (blue), 534 nm (bluish-green) and 564 nm (yellowish-green). Their sensitivity ranges overlap to provide vision throughout the visible spectrum. The maximum efficacy is 683 lumens/W at a wavelength of 555 nm (green).

PHOTOPOLYMERISATION: Polymerisation processes requiring a photon for the propagation step.

PHOTOPROTECTION: Protection of the human body and a plant's photosynthetic apparatus from the harmful effects of UV radiation. In humans, the damage mostly occurs in the skin, but the rest of the body especially the testicles can be affected by the oxidative stress caused by UV light. In plants, during periods of peak light intensity, they utilise less than half the incoming energy. The surplus energy may cause photooxidation and the formation of highly reactive superoxide radicals that can destroy the cells' chlorophyll and many other cellular components. Much of the excess energy is trapped and dissipated as heat by carotenoids of the xanthophylls cycle. Also, chloroplasts contain the enzyme superoxide dismutase which scavenges superoxide radicals. Photoprotection of the human skin is achieved by internal conversion of DNA, proteins and melanin, which is a photochemical process that converts the energy of the UV photon into small amounts of heat. This small amount of heat is harmless. If the energy of the UV photon were not transformed into heat, then it would lead to the generation of free radicals or other harmful reactive chemical species such as singlet oxygen or hydroxyl radical.

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PHOTOREACTIVATION: DNA repair enzymes in which the damaged region caused by exposure to ultraviolet light is repaired with the help of visible light, preferably from the violet/blue end of the spectrum. The damaged part is repaired without excision from the DNA. It usually contains a pigment that undergoes a chemical change when light is absorbed, thus stimulating a nerve.

PHOTOREARRANGEMENT: Photoisomerisation involving changes in the arrangement of the atoms within the molecular species. It may lead to unstable isomers that react further by, for example, dehydrogenation, deprotonation or other reactions. Examples are photorearrangements of alkenes (other than *cis-trans* photoisomerisation), of cyclohex-2-enones, of endoperoxides, of thioarenes, the di- π -methane rearrangement and others.

PHOTORECEPTOR (PHOTORECEPTOR CELL): A specialised type of neuron (nerve cell) found in the eye's retina that is capable of phototransduction. The great biological importance of photoreceptors is that as cells they convert light (electromagnetic radiation) into the beginning of a chain of biological processes. The photoreceptor absorbs photons from the field of view and through a specific and complex biochemical pathway, signals this information through a change in its membrane potential. Examples are rhodopsin in the photoreceptor cells of the vertebrate retina, phytochrome in plants and bacteriorhodopsin and bacteriophytochromes in some bacteria. They mediate light responses as varied as visual perception, phototropism and phototaxis, as well as responses to light-dark cycles such as circadian rhythm and other photoperiodisms including control of flowering times in plants and mating seasons in animals.

PHOTOREDUCTION: Reduction reactions induced by light. Common processes are: (i) addition of one or more electrons to a photoexcited species; (ii) the photochemical hydrogenation of a substance. Reactions in which the substrate is not electronically excited are sometimes called **photoinitiated reductions**.

PHOTORESIST: A photoimaging material, generally applied as a thin film, whose local solubility properties can be altered photochemically. A subsequent development step produces an image which is useful for the fabrication of microelectronic devices, for example, integrated circuits.

PHOTORESISTOR (LIGHT-DEPENDENT RESISTOR, LDR):

A high resistance electronic component, whose resistance varies according to the intensity of light it is exposed to, usually ranging as much as several megohms in the dark and as low as a few hundred ohms when exposed to light. For example, when exposed to a light level of 1000 lux, the resistance is just 400 ohms; but when exposed to a light level of 10 lux, the resistance rises to 10.43 megohms (10,430,000 ohms). It is made of a high resistance semiconductor that absorbs photons and based on the quantity and frequency of the absorbed photons the semiconductor material gives bound electrons enough energy to jump into the conduction band. The resulting free electrons conduct electricity resulting in lowering the resistance of the photoresistor. LDRs are classified as intrinsic or extrinsic based on the materials used. Intrinsic photo resistors are semiconductors that are not doped, consisting of pure semiconductor materials such as silicon or germanium. Extrinsic photo resistors are made of doped semiconductor materials. LDRs are often used as light sensor devices such as camera light meters, clock radios, alarm clock, burglar alarm circuits, in street light, etc.

PHOTORESPIRATION: A reaction that occurs in plants, alongside photosynthesis, by the fixation of oxygen instead of carbon dioxide during photosynthesis. This oxygenation reaction forms phosphoglycolate, which represents carbon lost from the photosynthetic pathway. Phosphoglycolate also inhibits photosynthesis if it is allowed to accumulate in the plant. Most of the fixed carbon represented by the glycolate can be salvaged by a series of reactions in the glycolate pathway, involving the peroxisomes and mitochondria and returned to the chloroplasts. However, some of the carbon is lost as carbon dioxide.

PHOTOSELECTION: Selection of an aligned subset of sample molecules by an exciting light beam. The selected subset will have an orientation distribution that is determined by the electric vector of the beam. If the beam is linearly polarised a uniaxial sample with a known orientation distribution will be produced. Also, the set of remaining molecules will be aligned with a known orientation distribution (residual anisotropy).

PHOTOSENSITIVE POLYMER: Polymer that responds to ultraviolet or visible light by exhibiting a change in its physical properties or its chemical constitution.

PHOTOSENSITIVE SUBSTANCE: Any substance when exposed to electromagnetic radiation produces a photoconductive, photoelectric or photovoltaic effect or any substance such as the emulsion of a photographic film, in which electromagnetic radiation produces a chemical change.

PHOTOSENSITISATION: The process by which a photochemical or photophysical alteration occurs in one molecular entity as a result of initial absorption of radiation by another molecular entity called a photosensitiser. In mechanistic photochemistry, the term is limited to cases in which the photosensitiser is not consumed in the reaction.

PHOTOSPHERE: The visible surface of the Sun, about 400 km thick and the source of its light that reaches Earth directly. It is a gaseous layer with an average temperature of 5507°C (5,780 K).

PHOTOSTATIONARY STATE: A steady state reached by a reacting chemical system, when light has been absorbed by at least one of the components. At this state the rates of formation and disappearance are equal for each of the transient molecular entities formed.

PHOTOSYNTHESIS: The process occurring in the chloroplasts by which green plants and some other organisms trap light energy and use it to drive a series of chemical reactions leading to the formation of carbohydrates from carbon dioxide and water. The two main reactions are light dependent reaction and the light independent reaction or

the dark reaction. **Light dependent reaction** requires the presence of light energy from sunlight which is absorbed by the chlorophyll to excite some electrons in the photosynthetic pigments and used to bring about the photolysis of water. The electrons released by this reaction pass along a series of electron carrier molecules, as they do so they lose their energy, which is used to convert adenosine diphosphate (ADP) to adenosine triphosphate (ATP) in the process of photophosphorylation. The light reaction also produces NADPH which is used to reduce 1,3-bisphosphoglycerate to glyceraldehyde-3-phosphate in the dark reaction. The **light independent reaction** or the **dark reaction** is sustained by the ATP generated during the light independent reactions. At this phase, carbon dioxide is reduced to carbohydrate in a metabolic pathway known as the Calvin Cycle.

PHOTOSYNTHETIC BACTERIA: Bacteria, for example, cyanobacteria (blue-green algae) that make photosynthesis just like green plants and algae. It is light-dependent, the bacteria use chlorophyll-type molecules, called bacteriochlorophyll, to collect sunlight as an energy source to convert carbon dioxide to glucose, which is used for energy production and biosynthesis. Some other bacteria also use bacteriochlorophyll to collect sunlight to make photosynthesis, but unlike cyanobacteria, do not generate oxygen from the oxidation of water. These bacteria are called **anoxygenic photosynthetic bacteria** which include the purple phototrophic bacteria, green sulfur bacteria, green filamentous bacteria, the heliobacteria and the cyanobacteria.

PHOTOSYNTHETIC GAS EXCHANGER: A device that utilises plants and light energy to convert back into oxygen the carbon dioxide generated by man or animals through breathing oxygen.

PHOTOSYNTHETIC NITROGEN USE EFFICIENCY: The ratio of a plant's rate of carbon assimilation to the concentration of organic nitrogen in its photosynthesis organs, which is a measure of the efficiency with which the plant allocates nitrogen to its photosynthetic apparatus.

PHOTOSYNTHETIC PIGMENTS (ACCESSORY PIGMENT; CHLOROPLAST PIGMENT; ANTENNA PIGMENT): The plant pigments responsible for the capture of light energy during the light reactions of photosynthesis. Chlorophyll is the principal light receptor. The carotenoids and various other pigments are the accessory pigments, absorbing light energy and passing this on to the chlorophyll as molecules. The six related photosynthetic pigments in plants are: Carotene (an orange pigment), xanthophyll (a yellow pigment), phaeophytin *a* (a grey-brown pigment), phaeophytin *b* (a yellow-brown pigment), chlorophyll *a* (a blue-green pigment) and chlorophyll *b* (a yellow-green pigment).

PHOTOSYSTEM I PRIMARY DONOR (P700): A special form of chlorophyll of photosystem I that is most reactive and best in absorbing light at a wavelength of 700 nm. It is found within the photosystem I light-gathering centre of photosynthesis. Other ordinary chlorophyll *a* molecules are also present and these absorb light energy and pass it to P700, creating a charged form, P700⁺. Electron transfer occurs from photosystem II, electrons being passed on ultimately to produce NADPH, the reduced form of NADP.

PHOTOSYSTEM II PRIMARY DONOR (P680): A special form of chlorophyll that is most reactive and best in absorbing light at a wavelength of 680 nm. It is one of the constituents of the photosystem II light-gathering centre of photosynthesis. The absorption of light energy by the P680 molecules causes, among other things, the formation of a strong oxidant that can then oxidise water. This results in the evolution of molecular oxygen and initiates electron flow, which continues, via photosystem I, through to the production of reduced NADP.

PHOTOSYSTEMS I/II: The two systems of photosynthetic pigments found in the thylakoid membranes of plants, algae and cyanobacteria. In plants and algae they are located in the chloroplasts or in the cytoplasmic membrane of photosynthetic bacteria. They are

involved in the light dependent reactions of photosynthesis. Each photosystem is an enzyme containing about 300 chlorophyll molecules that trap light energy which is then passed onto a reaction centre, comprising chlorophyll and a molecule, known as **P680** found in chloroplasts and in non-sulfur purple bacteria, which utilises light of wavelength 680 nm. In **photosystem I** the chlorophyll molecule, known as **P700** found in chloroplasts and in green-sulfur bacteria absorbs light at a wavelength of 700 nm. Light energy is used in each reaction centre to raise electrons to higher energy levels to enable them to be taken up by electron acceptors. This causes the P680 and P700 to become positively charged or oxidised.

PHOTOTAXIS: The movement of a cell such as gamete or a unicellular organism in response to the stimulus of light. It may be either towards or away from a source of light. Phototaxis is beneficial to phototrophic organisms as they can orient themselves most efficiently to receive light for photosynthesis. For example, some algae, such as *Chlamydomonas* can perceive light by a sensitive eyespot and move to regions of higher light concentration to enhance photosynthesis. Phototaxis is called positive if the movement is in the direction of increasing light intensity and negative if the direction is opposite.

PHOTOTHERMAL EFFECT: An effect produced by photoexcitation resulting partially or totally in the production of heat.

PHOTOTHERMOGRAPHY: A process utilising both light and heat, simultaneously or sequentially, for image recording.

PHOTOTRANSISTOR: A junction transistor that is photosensitive and uses light rather than electricity to cause an electrical current to flow from one side to the other. When radiation falls on the emitter-base junction, new free charge carriers are created in the base region and the collector current is increased.

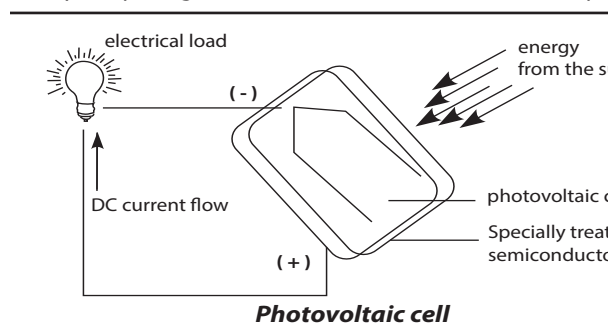
PHOTOTROPH: An organism that obtains its energy from sunlight to manufacture organic compounds by photosynthesis. They use the energy from sunlight to convert carbon dioxide and water into organic materials to be utilised in cellular functions such as biosynthesis and respiration. The process of obtaining energy from light or any organism that obtains sunlight to manufacture organic compounds by photosynthesis is described as **phototrophic**.

PHOTOTROPIN: Any of a class of photosensitive flavoproteins sensitive to blue light and trigger reorientating growth of a plant towards a light source. They also cause the opening of stomata and the movement of chloroplasts inside leaves. Light activates phototropins by inducing autophosphorylation.

PHOTOTROPISM: The directional growth or movement observed in plants, which also occurs in other organisms such as fungi towards or away from the stimuli of light. Shoots are positively phototropic and roots are negatively phototropic. The phototropic response is triggered by photosensitive molecules such as phototropins and brought about by the differential distribution of the plant hormone auxin in the illuminated organ, which in turn causes differential growth of the shoot or root. Growth towards a light source as exhibited by shoots is a **positive phototropism**, while growth away from light exhibited by roots is called **negative phototropism** or **skototropism**.

PHOTOVOLTAIC CELL (SOLAR CELL): A solid state device, usually a semiconductor such as silicon, which absorbs photons with energies higher than or equal to the bandgap energy and simultaneously produces electric power. In other words, it obtains energy from solar radiation and converts it into electrical energy. It consists of a thin semiconductor such as silicon, which is specially treated to form an electric field, positive on one side and negative on the other. Any electron generated by sunlight in the depletion region may pass from the p-side to the n-side of the junction and constitutes current flow. If a wire or any load is connected between the ends of the n-type and p-type region with metal contacts, electron will flow

to the p-type through an external load. The electricity generated by photovoltaic panels is direct current DC and an inverter is used to convert it to alternating current (AC). A group of cells are connected electrically and packaged into a frame called a module or solar panel.



PHRAGMOPLAST: An array of short microtubules observed in bryophytes and vascular plants and a few algae that organise the formation of the cell plate in dividing plant cells. It is a complex assembly of microtubules (MTs), microfilaments (MFs) and endoplasmic reticulum (ER) elements that assemble in two opposing sets perpendicular to the plane of the future cell plate during anaphase and telophase. The phragmoplast is orientated parallel to the mitotic spindle and accumulates vesicles in the plane of metaphase equator, midway between the daughter nuclei. The vesicles fuse and their components form the new plasma membranes and cell walls of the daughter cells.

PHREATIC ERUPTION: A steam-driven explosion that occurs when water beneath the ground or on the surface is heated by magma, lava, hot rocks or new volcanic deposits, for example, tephra and pyroclastic-flow deposits. The intense heat of such material (as high as 1,170°C for basaltic lava) may cause water to boil and flash to steam, thereby generating an explosion of steam, water, ash, blocks and bombs.

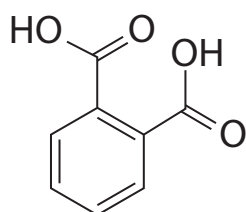
PHREATOPHYTE: A deep rooted plant with its roots typically in soil saturated with water. They have deep roots which are constantly in touch with moisture. They can mostly be found along streams and areas where the water table is near the surface. Many tamarisk species are phreatophytes.

PHRENIC NERVE: The nerve originating mainly from the 4th cervical nerve that forms a network in the diaphragm. It is involved in the control of ventilation. The phrenic nerves contain motor, sensory and sympathetic nerve fibres. These nerves provide the only motor supply to the diaphragm as well as sensation to the central tendon. In the thorax, each phrenic nerve supplies the mediastinal pleura and pericardium.

PHRENOLOGY: A detailed study of the shape and protuberances of the cranium. This was based on the theory of the Austrian physicist, Franz Josef C. Gall (1758-1828) that such features revealed measurable intellectual aptitudes and character traits.

PHTHALEINS: Triphenylmethane derivatives obtained by the action of phenols upon phthalic benzene 1,2- dicarboxylic anhydride. An example is phenolphthalein.

PHTHALIC ACID: A white crystalline solid with the chemical formula $C_6H_4(CO_2H)_2$. It is alcohol-soluble, slightly soluble in water and ether and decomposes at 191 °C (464 K). On heating converts to its anhydride. It is used in organic synthesis and to make polyester resins dyes, medicine and synthetic perfumes.

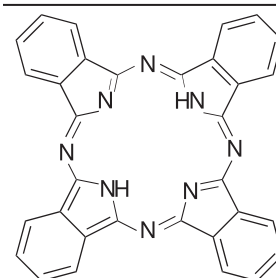


PHTHALIC ESTER: An ester of phthalic acid made by reacting phthalic anhydride with an alcohol in the presence of sulfuric acid. They are high boiling-point liquids used as plasticisers and in the manufacture of

polymers and alkyl resins. The dimethyl ester is used as an insert repellent.

PTHALIDES: 3,3-Di(hydrocarbonyl) or 3-hydrocarbonylidene-2-benzofuran-1(3H) ones.

PTHALOCYANINE: A member of an important class of synthetic organic dyes and pigments with the chemical formula $C_{32}H_{18}N_8$. Phthalocyanines form coordination complexes with most elements of the periodic table. These complexes are also intensely coloured and also used as dyes. They are blue to green and used in chemicals, plastics, linoleum, inks, wallpapers, etc.



PHYCOBILIN: Water-soluble pigments (mostly red, orange, and blue) that is seen in cyanobacteria and red algae that enable deep water organisms to collect light through a special form of chlorophyll that is most reactive and best in absorbing light at a wavelength of 680 nm to enable them make photosynthesis.

PHYCOBILIPROTEIN: Any one of water-soluble proteins present in the blue-green bacteria, red algae and photosynthetic cryptophytes that act as accessory pigments in photosynthesis, being attracted to thylakoid membranes. Each comprises a protein bound to a phycobilin, a coloured tetrapyrrole prosthetic group that acts as chromophores (the light-capturing part). The main phycobiliproteins are phycocyanin, phycoerythrin, allophycocyanin and phycoerythrocyanin. Within cells, phycobiliproteins form complex clusters called phycobilisomes.

PHYCOBIONT: A photosynthetic partner of a symbiotic pair such as the algal component of the fungal-algal association in lichens.

PHYCOCYANIN: An accessory pigment from the light-harvesting phycobiliprotein family, along with allophycocyanin and phycoerythrin found in cyanobacteria and the chloroplasts of red algae. It is a phycobiliprotein in which the pigmented prosthetic group is phycocyanobilin, which gives phycocyanin its blue colour. Phycocyanin absorbs orange and red light, particularly near 620 nm (depending on which specific type it is) and emits fluorescence at about 650 nm. It is used in the food and beverage industry as a natural colouring agent and found in sweets and ice cream.

PHYCOERYTHRIN: An accessory pigment found in cyanobacteria and the chloroplasts of red algae. It is a phycobiliprotein in which the pigmented prosthetic group is phycoerythrobilin, which gives phycoerythrin its red colour. Phycoerythrin is composed of a protein part organised in a hexameric structure of alpha and beta chains, covalently binding chromophores called phycobilins.

PHYCOMYCETES: A class of the lower fungi, many of which are found in water or damp areas. Many are unicellular but those that form mycelia generally have hyphae-lacking cross walls, which distinguish them from the Ascomycota and Basidiomycota. They include the Zygomycota. They lack regularly spaced septa in the actively growing portions of the plant body and have sporangiospore, produced in the sporangium by cleavage, as the fundamental, asexual reproductive unit.

PHYCOPLAST: An array of microtubules observed during cytokinesis in members of the Chlorophyceae. The phycoplast consists of microtubules arranged perpendicular to the axis of the mitotic spindle and parallel to the lane of formation of the new cell wall. The new wall is formed by deposition of a cell plate, which eventually disrupts the telophase spindle.

PHYLETIC CHANGE: The changes taking place in a single lineage of organisms over a long period of time; one of the principal patterns of evolutionary change.

PHYLETIC SERIES: A sequence of fossil forms that represent the succession of species in the evolution of a particular lineage.

Often there are gaps in the fossil record and phyletic series are incomplete. However, in some cases, the fossil record is good and more complete phyletic series can be constructed. For example, most of the intermediate forms between the modern horse (*Equus*) and its dog-sized ancestor, *Hyracotherium* (*Eohippus*) of the Eocene epoch, have been charted.

PHYLLARY: An involucre bract of a composite flower head or capitulum.

PHYLLITE: Fine-grained metamorphic rock produced under increasing temperature and pressure in which nice crystals are aligned so that the rock splits along their plane of orientation.

PHYLLOCLADE: A flattened green stem that is leaf-like in appearance and functions as the main photosynthetic organs of the plant. These flattened green stems may also be called **cladodes**, are not phylloclades. In some cases, the cladode so resembles a leaf that it is difficult to make a distinction. A phylloclade is an aerial stem or branch modification; the stem is modified into a flat or cylindrical, succulent and green structure; it arises in the axil of the leaf which falls off, leaving a scar; and it bears succession of nodes and internodes at long or short intervals. Examples of plants with phylloclade are of the genera *Opuntia* and *Casuarina*. Examples of plants having cladodes are some cacti and asparagus.

PHYLLODE: A flattened and widened petiole that resembles and functions as a leaf and may or may not be combined with an actual lamina. In plants with phyllodes the true leaves are often reduced or do not develop. Australian and Pacific species of *Acacia* typically have phyllodes, whose vertical orientation is thought to protect the photosynthetic apparatus from excessive light.

PHYLOGENETIC CLASSIFICATION: Classification of organisms based on their assumed evolutionary ancestry and relationships.

PHYLLOME: A leaf, a structure corresponding to a plant leaf or a plant part that evolved from a leaf.

PHYLLPODIC: Having well-developed leafy bases that are not reduced to scales. On the other hand, **aphyllopodic** means having the lowermost leaves reduced to small scales.

PHYLLOQUINONE (PHYTOMENADIONE): The major form of vitamin K, with the chemical formula $C_{31}H_{46}O_2$ that is involved in the production of prothrombin, which is essential in blood clotting. It is a fat-soluble chemical found in green vegetables such as alfalfa; synthesised constantly by intestinal bacteria in mammals; and is also prepared synthetically. Because of its blood clotting properties, it is used to prevent haemorrhages.

PHYLLOSILICATES: Silicate structures in which the SiO_4 tetrahedra are linked together in infinite two-dimensional sheets and are condensed with layers of AlO_4 or MgO octahedra in the ratio 2:1 or 1:1. Isomorphous substitution of certain elements often occurs. Most clays belong to phyllosilicates, which have formed from the breakdown products of other minerals.

PHYLLOTAXIS (PHYLLOTAXY): The arrangement of leaves on a plant stem. The basic patterns are alternate, opposite, whorled or spiral. In the majority of plants, the leaves are inserted singly either in a spiral arrangement up the stem or on alternate sides. When arranged in pairs, the two leaves arise on opposite sides of the stem and are usually at right angles to the leaf pairs above and below them. Phyllotaxis generally results in the minimum of shading of leaves by those above them.

PHYLLOXERA: 1. An aphid which attacks vine. 2. Any of numerous, insect species, genus *Phylloxera* order Homoptera, many of which are serious pests of plants.

PHYLOGENETIC (PHYLETIC): A system of classification of organisms' evolutionary relatedness, for example, species or populations, which is discovered through molecular sequencing data and morphological data matrices.

PHYLOGENETIC SPECIES CONCEPT: The concept of a species of an irreducible group whose members are descended from a common ancestor and who all possess a combination of certain defining or derived traits. It permits successive species to be defined even if they have evolved in an unbroken line of descent, with continuity of sexual fertility.

PHYLOGENOMICS: The field of bioinformatics that integrates knowledge about the evolutionary history of organisms with structural and functional analysis of their genomes (genomics) and proteins (proteomics). It refers to various types of analysis involving genome data and evolutionary reconstructions, especially phylogenetics. A basic supposition is that, well-characterised genes (or proteins) in one organism provide valuable insights into the likely occurrence and function of homologous genes or proteins in related organisms. On a molecular level, vast amounts of data from genomics projects are used to construct phylogenetic data bases for genes and proteins. Sequence data is correlated with experimental findings about protein structure and function and subject to statistical analysis.

PHYLOGENY: The historical sequence of changes that occur in a given species during the course of its evolution. Phylogenetic relationships are depicted by branching diagrams called **cladograms** or **phylogenetic trees**. Cladograms show relative affinities of groups of organisms called taxa. Such groups of organisms have some genealogical unity and are given a taxonomic rank such as species, genera, families or orders. For example, two species of cats, the lion (*Panthera leo*) and the tiger (*Panthera tigris*) are more closely related to each other than either is to the grey wolf (*Canis latrans*). The family including all cats, Felidae, is more closely related to the family including all dogs, Canidae, than either is to the family that includes giraffes, Giraffidae. The lion and tiger and the Felidae and Canidae, are called sister taxa because of their close relationship relative to the grey wolf or to the Giraffidae, respectively.

PHYLOGRAM: A diagram representing the evolutionary interrelations and distances between different groups or species of organisms. The relationships are depicted by dichotomously branching lines to form a tree like pattern, with the lengths of the branches proportional to distance of the relationship in evolutionary terms. The ancestor is in the tree trunk organisms that have arisen from it are placed at the ends of tree branches. The distance of one group from the other groups indicates the degree of relationship. Closely related groups are located on branches close to one another.

PHYLUM: Phyla is the plural of Phylum. A taxonomic grouping of animals or plants with similar characteristics in biological classification. For example, lions are classified in the phylum Chordata (chordates), which includes all animals with a backbone. Other examples of phyla are: Rhodophyta, Ascomycota, Bryophyta, Annelida and Chordata. Phylum is a taxonomic rank below Kingdom and above Class and equivalent to the botanical term division.

PHYSICAL: Of, referring to or concerning material things or of the body or according to the laws of nature.

PHYSICAL ADDRESS: 1. In computers or computer memory, it is the computer memory address of a physical hardware device. 2. In a network address, physical address is sometimes used to describe the MAC address.

PHYSICAL CHANGE: A type of change that does not produce a new chemical substance, does not involve large energy changes and that can be easily reversed. In other words, it is a change in a substance that does not affect its chemical composition. Examples of physical properties include texture, shape, size, colour, volume, mass, weight and density.

PHYSICAL CHEMISTRY: The branch of chemistry concerned with interactions and transformations of materials, examining the relationships between the chemical composition of substances and the physical properties that they display.

PHYSICAL EQUILIBRIUM: An equilibrium in which only physical properties change.

PHYSICAL HALF-LIFE, T_P $\left(T_{\frac{1}{2}}\right)$: 1. The time taken for half the

number of radioactive nuclei in sample to decay. 2. The time taken for the activity of a sample to decrease to half its initial value.

PHYSICAL LAYER: In the Open System Interconnection Model (OSI Model), it is the lowest or first layer (layer 1) of a seven layer model. It is made up of the fundamental network components and defines how bits are transmitted. It encompasses details such as electrical and mechanical connections to the network, transmission of binary data as changing voltage levels on wires or similar concepts on other connectors and data rates.

PHYSICAL MAP: 1. Any map that shows the arrangement of the material (nucleoprotein) making up the chromosome or segments of a genome. There are several types of physical map differing vastly in scale and detail. The coarsest physical maps are the ones depicting chromosome banding patterns, which are dark and light transverse bands obtained by staining entire chromosomes in mitosis. These cytological maps enable characterisation of individual chromosomes and can reveal gross anomalies, such as missing or duplicating segments. 2. A map showing a country's borders, major cities and significant bodies of water and the location of landforms like deserts, mountains and plains.

PHYSICAL MEMORY (REAL STORAGE; PHYSICAL STORAGE): It defines the total amount of memory installed in the computer. For example, if the computer has two 64MB memory modules installed, it has a total of 128MB of physical memory.

PHYSICAL MODEL: Representation of an object, usually on a reduced scale.

PHYSICAL NETWORK: In computing, it describes the physical topology of the network, i.e., the placement of the various components.

PHYSICAL PROPERTY: A description of the appearance of a substance or of a physical change that can happen to it without involving a change in chemical composition. Examples of physical properties of water are being liquid at room temperature, melting and boiling point.

PHYSICAL QUANTITY: A measured quantity that uses numbers and units. It is an attribute of a phenomenon, body or substance that may be distinguished qualitatively and determined quantitatively that can be used in the mathematical equations of science and technology. The value of a physical quantity is usually expressed as the product of a number and a unit. For example, in expressing the mass of a substance as 100 kg, the number is 100, and unit is kg. The unit represents a specific example or prototype of the quantity concerned, which is used as a point of reference. The number represents the ratio of the value of the quantity to the unit. All physical quantities have units which can be built from seven base units, which are as follows: length - metre (*m*), mass - kilogram (*kg*), time - second (*s*), electric current - ampere (*A*), temperature - kelvin (*K*), amount of substance - mole (*mol*), luminous intensity - candela (*cd*).

A few examples of physical quantities are as follows:

Physical Quantity	Symbol	Unit	Unit Symbol
length	<i>l</i>	metre	<i>m</i>
area	<i>A</i>	square metre	<i>m</i> ²
volume	<i>V</i>	cubic metre	<i>m</i> ³
mass	<i>m</i>	kilogram	<i>kg</i>

density	ρ	kilogram/cubic metre	kg m^{-3}
time	<i>t</i>	second	<i>s</i>
speed	<i>v</i>	metre/second	m s^{-1}
temperature	<i>t</i>	kelvin or degree celsius	<i>K</i> or $^{\circ}\text{C}$
force	<i>F</i>	newton	N m^{-2}
pressure	<i>P</i> or <i>p</i>	newton/metre squared	
electric current	<i>I</i>	ampere	<i>I</i>
amount of substance		mole	<i>mol</i>
luminous intensity		candela	<i>cd</i>
energy	<i>E</i>	joule	<i>J</i>

PHYSICAL TOPOLOGY: The way in which the physical devices are actually interconnected with wires and cables and the type of cabling used. Types of physical topologies include: linear bus topology, star topology and tree (extended star) topology, ring topology, hierarchical topology and mesh topology.

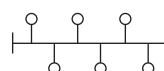


Fig I: **Bus Topology**



Fig II: **Ring Topology**



Fig III: **Star Topology**

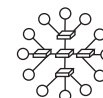


Fig IV: **Tree (Extended Star)**



Fig V: **Hierarchical Topology**



Fig VI: **Mesh Topology**

Types of physical topologies

PHYSICAL WEATHERING (MECHANICAL WEATHERING):

The form of weathering responsible for the mechanical breakdown of rocks which involves no chemical change. Processes include exfoliation, frost shattering, granular disintegration, salt weathering, spalling and thermal weathering.

PHYSICIAN: A doctor, especially one that specialised in areas of treatment other than surgery and concerned with maintaining or restoring human health through the study, diagnosis and treatment of disease or injury. It requires a detailed knowledge of the academic disciplines such as anatomy and physiology underlying diseases and their treatment, the science of medicine and also competence in the practice of medicine.

PHYSICS: The scientific study of the properties of matter and energy and the interactions between the fundamental constituents of the observable universe, for example, heat, light, sound, magnetism, gravity and the relationship between them. The goal of physics is to formulate comprehensive principles that bring together and explain all discernible phenomena.

PHYSIOGNOMY: The general appearance of a person, object or terrain, without reference to its implied characteristics. Plant physiognomy can be broadly correlated with environmental conditions, so that regions of the world with similar climates tend to have a dominant vegetation of similar life forms.

PHYSIOLOGICAL SALINE: A sterile solution of sodium chloride that is isotonic to and having the same pH as the body fluids of the animal. Animal tissues may be kept alive for a few hours during experiments without pathological changes or distortion of the cells taking place. A common example is Ringer's Solution, which is a mixture of sodium chloride, calcium chloride, sodium bicarbonate and potassium chloride solutions. It is also a solvent for drugs administered parenterally (drugs administered through injection, infusion or implantation).

PHYSIOLOGICAL SPECIALISATION: The occurrence within a species of several forms that are identical in appearance but differ in physiology, termed physiological races. Many pathogenic fungi develop new physiological races in response to the strong selection pressure exerted when disease-resistant crop varieties are grown over large areas.

PHYSIOLOGY: The branch of biology concerned with the vital functions of living organisms or their constituent tissues or cells.

PHYSIOME: A complete description of the functional behaviour and all processes that sustain the life of an organism or species and how these change in time and space. It integrates all aspects of an organism's biology, from its genes and proteins, through cellular metabolism and biochemistry, to anatomy.

PHYSIOTHERAPY (PHYSICAL THERAPY): The medical treatment of diseases, injuries or deformities by physical methods such as massage, heat application and exercise rather than by drugs or surgery. These diseases, bodily defects or weaknesses often have to do with the muscles, nerves, joints or other bodily parts.

PHYSISORPTION: 1. The taking up of a gas by a solid or liquid or of a liquid by a solid. 2. The conversion of the energy of electromagnetic radiation, sound, streams of particles, etc, into other forms of energy on passing through a medium.

PHYTOALEXIN: An antimicrobial substance that is produced by a plant in response to infection by fungi or bacteria. It helps to defend the plant by inhibiting the growth of the invading microbes. When these barriers are breached, the plant must actively protect itself by employing a variety of strategies. Plant cell walls are strengthened and special cell layers are produced to block further penetration of the pathogen. Phytoalexins vary in their chemical nature. For example, in legumes, they are mainly isoflavonoids, whereas in the potato family, they are predominantly terpenes. Substances such as small polysaccharides and proteins that are produced by the invading pathogen act as elicitors, triggering the manufacture of phytoalexins in the tissues surrounding the infection site. The production of phytoalexins is one component of a broader defensive reaction, known as **hypersensitivity**.

PHYTOCHROME: A blue protein pigment that absorbs red (666nm) and far red light with a molecular mass of about 125kDa (kilodaltons). It is present in small quantities in many plant organs and controls most photomorphogenic responses in higher plants. It enables the plant to sense not only the presence of light but also its intensity, direction, duration and spectral quality. It is important in photomorphogenesis of plants such as seed germination, growth direction, growth rate, chloroplast development, pigmentation, flowering and senescence.

PHYTOHAEMAGGLUTININ (RPHA): Lectins found in plants, especially legumes such as white kidney beans, green beans and broad beans such as fava beans. They consist of two closely related proteins, called Leucoagglutinin (PHA-L) and PHA-E and have carbohydrate-binding specificity for a complex oligosaccharide containing galactose,

N-acetylglucosamine and mannose. They have various physiological effects, one of which is causing mitogenic effects, especially on T lymphocytes. It is used in medical research. If taken in high doses it can be toxic.

PHYTOHORMONE (PLANT HORMONE): A chemical substance produced in extremely low concentrations. Plant hormones shape the plant, affect seed growth, time of flowering, the sex of flowers, senescence of leaves and fruits. It also affects tissues which grow upward and downward, leaf formation and stem growth, plant longevity and even plant death. When the hormones are chemically synthesised, they are called **growth regulators**. Examples are auxin, gibberellin, ethylene and cytokinin.

PHYTOMENADIONE (PHYLLOQUINONE): The major form of vitamin K, with the chemical formula $C_{31}H_{46}O_2$ that is involved in the production of prothrombin, which is essential in blood clotting. It is a fat-soluble chemical found in green vegetables such as alfalfa; synthesised constantly by intestinal bacteria in mammals; and is also prepared synthetically. Because of its blood clotting properties, it is used to prevent haemorrhages.

PHYTOMERE: A segment of the shoot of a grass that includes an internode, the leaf and a portion of the node at the top and bottom of the internode.

PHYTOMETER: A plant or group of plants usually under controlled conditions and used to measure the physical factors of the habitat in terms of physiological activities or responses to various environmental factors.

PHYTOMORPHIC: Having or represented with the attributes of a plant. **Phytomorphic soils** are well-drained soils that have developed under the dominant influence of the natural vegetation characteristic of a region.

PHYTONUTRIENT: Substances such as vitamins in plants that are beneficial to humans. They are naturally occurring chemicals and provide flavour, colour, texture and smell. Some plants which produce phytonutrients are beans, lentils, peanuts, broccoli, cauliflower, cabbage, celery, parsley, parsnips berries, citrus fruits, whole grain, etc. Phytonutrients have potential health effects, as they may boost enzyme production or activity, which may, in turn, block carcinogens, suppress malignant cells or interfere with processes that can cause heart disease and stroke.

PHYTOPATHOLOGY (PLANT PATHOLOGY): The scientific field that studies diseases of plants caused by pathogens and environmental factors. It encompasses studies in botany, crop science, soil science, genetics, microbiology, physiology and molecular biology.

PHYTOPLANKTON: The plant component of plankton, consisting of groups with silica-containing skeletons, such as diatoms and dinoflagellates. Members of phytoplankton use carbon dioxide, release oxygen and convert minerals to a form animals can use and form the basis of food for all other forms of aquatic organisms.

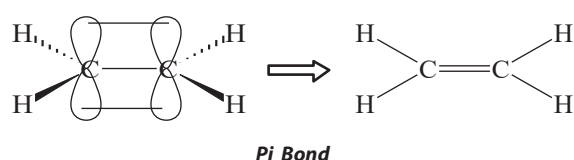
PHYTOREMEDIATION: The use of plants to decontaminate environmental problems such as polluted land, water or air without the need to excavate the contaminant material and dispose of it elsewhere. Plant species that grow on contaminated sites have evolved various ways of containing, degrading and eliminating high concentrations of heavy metals, oil, solvents or other toxic substances. The various ways of achieving this are: (i) **Phytotransformation**, in which the plant takes up the toxin and degrades it by enzyme activity, thereby eliminating its toxic nature altogether; (ii) **Phytostabilization**, in which the plant's root exudates or associated microorganisms precipitate the toxins and make them harmless to other organisms; (iii) **phytoaccumulation or phytoextraction** in which a plant takes up the contaminants and stores them in its tissues, the plant can then be harvested, thus removing the toxin with the plant.

PHYTOTELM: An accumulation of water found on a plant. Examples include the contents of a pitcher plant or the small pool within a hollow in a tree. Phytotelms may contain organisms which can be used in ecological studies.

PHYTOTOXICANT: An agent which produces a toxic effect in vegetation.

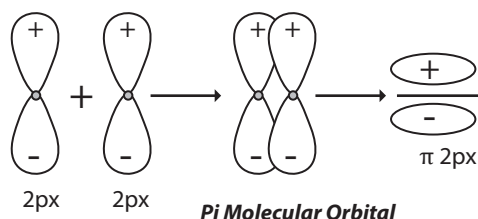
pi: 1. An irrational number with the symbol π denoting the ratio of the circumference of a circle to its diameter. The value of pi is 3.14215926 (22/7) correct to seven decimal places. It is generally used in calculations involving lengths, areas and volumes of circles, spheres, cylinders and cones as well as motion of pendulums and alternating electric currents. The number of radians in a complete circle is 2π and in circular measure, $\pi/2$ radians is a right angle. 2. The symbol, Π that denotes a product, for example the product of the elements $x_a, \dots, x_b$ is written $\prod_{i=a}^b x_i$. 3. The symbol, π for a permutation. 4. The arithmetic function, $\pi(n)$, used to denote the number of primes not exceeding n .

Pi BOND (π BOND): A covalent chemical bond formed by the side-by-side or lateral overlapping of two lobes of p-orbitals of adjacent atoms running parallel to each other. It can also be defined as a covalent bond formed as a result of the collateral overlap of two parallel p-orbitals of two bonding atoms.



PI ELECTRON: An electron which participates in a pi-orbital.

PI MOLECULAR ORBITAL: A molecular orbital in which the electron density is concentrated above and below the line joining the two nuclei of the bonding atoms.



PIA MATER: The delicate innermost layer of the three membranes (meninges) that surround the brain and spinal cord of vertebrates. It is immediately adjacent to the central nervous system and the choroid plexus. It supplies blood to the superficial areas of the cortex and supports larger blood vessels passing over the surface of the brain.

PICARD'S METHOD: The iterative solution of an ordinary differential equation, by a sequence of functions which converge to the solution.

PICARD'S THEOREMS: Picard's two theorems; the first theorem, known as **little Picard theorem** states that a non-constant entire function assumes every value in the whole complex plane or the plane minus a single point save at most one. The second theorem, also known as **great Picard's theorem** states that in every neighbourhood of an essential isolated singularity an analytic function takes all possible complex values, except possibly for one. An example is $\sin(1/z)$.

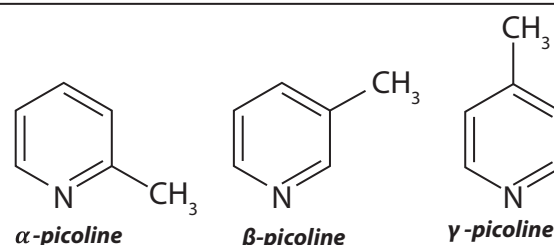
PICK-UP REEL: A part of a combine harvester with spring tines, used to improve cutting efficiency in tangled crops.

PICKLING: 1. A technique for treating metals with acid in order to remove the surface layer of oxide so that they can be painted or treated in some way. 2. A method of preparing hides for tanning by

immersion in a salt solution with a pH of 2.5 or less. 3. Preserving foods using salt, sugar, spices and vinegar (acetic acid).

PICO-: A prefix, with the symbol p used in the metric system to denote 10^{-12} . For example, 10^{-12} farad = 1picofarad (pF).

PICOLINE (METHYLPYRIDINE): A heterocyclic colourless liquid at room temperature and pressure with the chemical formula $C_5H_4N(CH_3)$. It is miscible with water and most organic solvents. It occurs in coal-tar and bone oil and is used in organic synthesis. It exists in three methylpyridine isomeric forms: **2-Methylpyridine (α -picoline)** with a melting point of -66.7°C (206.45 K) and a boiling point of 129.4°C (402.55 K); **3-Methylpyridine (β -picoline)** has a melting point of -18°C (255 K) and a boiling point of 141°C (414 K); and **4-Methylpyridine (γ -picoline)** that has a melting point of 3.6°C (276.6 K) and a boiling point of 145.4°C (418.55 K).



PICONET: A network that is created using a wireless Bluetooth connection. Examples of piconets are cell phone and a computer; a laptop and a Bluetooth-enabled digital camera; or several Personal Digital Assistant (PDAs) that are connected to each other. A PDA is a device similar to a computer that fits in the palm of and allows the user to collect such information as contacts, appointments, files, programs and more.

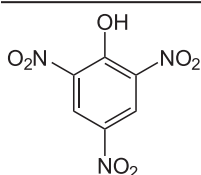
PICORNAVIRUS: Any of a group of the smallest known spheroidal RNA-containing viruses commonly present in the alimentary and respiratory tracts of vertebrates. They cause infections in the tracts. The group also includes the poliovirus, which attack the central nervous system causing poliomyelitis, viruses that infect tissues in the vertebrate nose (rhinoviruses) and the aphtho viruses responsible for foot and mouth disease in cattle, sheep and pigs.

PICRALIMA NITIDA: A tree in the family Apocynaceae, reaching up to 25 m high and a stem diameter of up to 60 cm. It occurs from Cote d'Ivoire eastwards to Uganda and southwards to Democratic Republic of Congo and Angola. The stem is cylindrical, erect, glabrous and much ramified. The bark is 1 cm thick, hard and fibrous, white-grey, ochre or black. The wood is pale yellow and very hard. White latex is found in the fruit (pericarp) but not in the leaves and branches. All plant parts taste very bitter and have an anaesthetic effect. The leaves are opposite, large, green to glossy dark and glabrous. The flowers are bisexual and in terminal, with a white or yellow corolla. The fruits consist of two large obovoid to ellipsoid follicles touching each other but free. Its colour is yellow to orange when ripe and finely flecked. It is fleshy and very rich in milky latex. Akuammine, an indole alkaloid, also known as vincamajoridine, is the most abundant active alkaloid found in the seeds. The roots, bark and seeds are used extensively in traditional medicine for the treatment of malaria, fever, chest problems, pneumonia, intestinal worms and also for pain relief. Its hard timber is used to make things such as paddles, spoons, mortars, pestles, bows and arrows, combs and shuttles used in weaving.

PICRATE: 1. Salt or an ester of picric acid. 2. Compound formed between picric acid and an aromatic compound, aromatic amines, aliphatic amines, alkalines and other compounds. Picrates are frequently used to identify these classes of compound. Some picrates are explosives, for example ammonium picrate known as Dunnite. Some are used as primary explosives, for example, lead picrate

or potassium picrate is used as primers for cartridge ammunition. Ferrous picrate is used in some applications as a diesel fuel additive to achieve better mileage.

PICRIC ACID (2,4,6-TRINITROPHENOL; CARBAZOTIC ACID; PHENOL TRINITRATE; PICRONITRIC ACID; TRINITROPHENOL; 2,4,6-TRINITRO-1-PHENOL): A toxic yellow crystalline solid, derived from phenol with the chemical formula $(\text{NO}_2)_3\text{C}_6\text{H}_2\text{OH}$, density 1.763 g/cm^3 (solid), melting point 122.5°C (395.65 K) and exploding at boiling point of $>300^\circ\text{C}$ ($>573 \text{ K}$). It is soluble in most organic solvents. It reacts with metals to form metal picrates, which like picric acid itself are highly sensitive explosives that can be detonated by heat, flame, shock or friction. Although picric acid can be synthesised by nitration of phenol, higher yields are obtained if chlorobenzene is used as a starting material; the latter method involves several steps and the formation of several intermediate products. In addition to its use in explosives, picric acid has been used as a yellow dye, as an antiseptic and in the synthesis of chloropicrin or nitrotrichloromethane (CCl_3NO_2) a powerful insecticide.



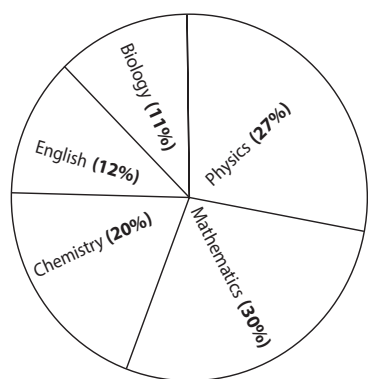
Chemical Structure of Picric Acid

PICTOGRAM (PICTOGRAPH; PICTOGRAMME): A type of graphic symbol that conveys its meaning through its pictorial resemblance to a physical object.

PICTORIAL PATTERN: A set of pictures, drawings, or geometric shapes that are arranged following a rule.

PICTURE GRAPH: A graph that uses symbols, icons, and pictures to represent data.

PIE CHART (CIRCLE GRAPH): A method of displaying statistical information where a circle is divided into different-sized sectors by radial lines. The angle of each sector is directly proportional to the the figure it represents and it is expressed as a fraction of the figures in the group data:



Pie Chart

$$\frac{\text{Angle of sector}}{360^\circ} = \frac{\text{Figure sector represents}}{\text{Sum of figures in data}}$$

PIECEWISE (SECTIONALLY): A function which is defined by a specified property, as smoothness or continuity, on each of a finite number of pieces into which its domain is divided. A function is piecewise continuous if it can be broken into different segments such that it is continuous in each segment.

PIEZOCHEMISTRY: The study of the effect that high pressure has on chemical reactions.

PIEZOELECTRIC CRYSTALS: Quartz crystals that change shape when an electric current flows through them and can be used with AC voltage to produce and detect ultrasound.

PIEZOELECTRIC EFFECT: The production of potential difference across the opposite faces of certain non-conducting crystals, called piezoelectric crystals by applying a mechanical stress between these faces. Pressure polarises some crystals, such as quartz, by slightly separating the centres of positive and negative charge, which is detected as voltage. Using this effect, a high-frequency alternating electric current can be converted to an ultrasonic wave of the same frequency, while a mechanical vibration, such as sound, can be converted into a corresponding electrical signal. Piezoelectricity is utilised in microphones, phonograph pickups and telephone communications systems.

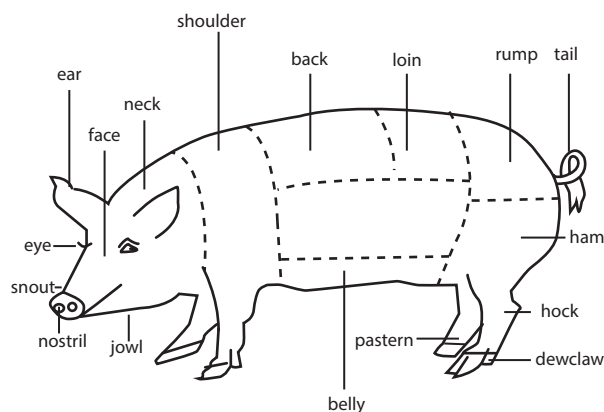
PIEZOELECTRIC POLYMER: 1. A polymer that exhibits a change in dielectric properties on application of pressure. 2. A polymer that shows a change in its dimensions when subjected to an electric field.

PIEZOLUMINESCENCE: Luminescence observed when certain solids are subjected to a change in pressure.

PIEZOMAGNETISM: The magnetic field produced by some crystals and ceramic materials when subjected to mechanical deformation, characterised by a linear coupling between the system's magnetic polarisation and mechanical strain. A spontaneous magnetic moment may be induced by applying physical stress or a physical deformation by applying a magnetic field.

PIEZOMETER: An instrument for measuring the pressure of a liquid or gas. Piezometers are often placed in boreholes to monitor the pressure or depth of groundwater.

PIG: An omnivorous, wild or domesticated mammal, also called a swine, which is usually black or pink, with a curved tail and short legs. It is hoofed, with sparse bristly hair and a long snout which is flat at the end and used for rooting in the soil. It is kept for its meat, which is known as bacon, pork, ham or gammon.



External features of a pig

PIG IRON (CAST IRON): The impure form of iron produced by a blast furnace which is cast into pigs (blocks) for converting it later into cast iron, steel, etc. It contains around 4% carbon plus some other impurities.

PIGEON: Monogamous birds of the family Columbidae, found almost worldwide and recognisable by their head-bobbing strut. Unlike other birds, it provides the young with regurgitated milk. The 175 species of true pigeons include the Old and New World *Columba* species and the Old World *Streptopelia* species; all eat seeds and fruit. Common street pigeons or rock doves, are descendants of the Eurasian rock dove (*Columba livia*). About 115 species of fruit pigeons occur in Africa, southern Asia, Australia and the Pacific islands. Pigeons have been trained since ancient times to carry messages over long distances.

PIGEON PEA (ANGOLA PEA; DHAL; CONGO PEA; NO-EYE PEA; *Cajanus cajan*; *Cajanus indicus*; *Cajanus flavus*; *Cytisus cajanus*): An annual or short-lived perennial leguminous shrub in the family, Fabaceae, whose seeds have become a common food grain. The plant is normally cultivated in the tropics and subtropics and is resistant to dry conditions, but can also be grown in some temperate areas. It thrives very well in well-drained soils and does not survive waterlogged conditions. It is a shrub that grows averagely between 1-2 metres tall, but can also reach about 3-5 metres in height. It has pubescent, upright, ribbed stem that is woody at the base, with a deep and quick growing tap root. The oblong, lanceolate leaflets are stalked and pubescent, about 5-10 centimetres long and 2-4 centimetres wide, with minute resinous glands underneath. Its yellow

or yellow and red bi-laterally symmetrical, bell-shaped blooms may occur in pairs and grow in the angle between the stem and leaf on an unbranched inflorescence. The upper two lobes are united and two-toothed and the lower lip is smaller and three-toothed. The mottled red seedpods are two-valved and pointed, borne in clusters, about 5-9 centimetres long and 12 millimetres wide, flat, covered with soft hairs and taper to a sharp point. The pod contains 2-9 light beige to dark brown round or oval seeds. Pigeon peas are eaten as dried peas, flour or green vegetable peas; and planted as a forage or cover crop. Being a legume, it enriches the soil by fixing nitrogen compounds in the soil.

PIGEON-HOLE PRINCIPLE (DIRICHLET'S PRINCIPLE; DRAWER PRINCIPLE; LETTER-BOX PRINCIPLE): It states that if n items are put into m containers, with $n > m$, then at least one container must contain more than one item. It was introduced in 1834 by German mathematician, Peter Gustav Lejeune Dirichlet (1805-1859).

PIGMENT: 1. A substance that changes the colour of reflected or transmitted light as the result of wavelength-selective absorption. A pigment must have a high tinting strength relative to the materials it colours. It must be stable in solid form at ambient temperatures. Pigments are used for colouring paint, ink, plastic, fabric, cosmetics, food and other materials. 2. A natural substance in animal or plant tissue that gives its colour initiated through exposure to light.

PILE-UP: The processing by a radiation spectrometer of pulses resulting from the simultaneous adsorption of independent particles or photons in a radiation detector. As a result they are counted as one single particle or photon with energy between the individual energies and the sum of these energies.

PILEATE: 1. Having a crest extending from the bill to the nape, as seen in some birds. 2. Having a pileus or cap.

PILEUS: The umbrella-shaped cap of mushrooms.

PILIFORM: Having the form of a hair.

PILL: 1. A small ball or pellet of medicine which is usually swallowed whole. 2. A popular term used for contraceptive tablet.

PILLIFEROUS LAYER: The absorbing part of the root epidermis that bears root hairs. It extends over a region about 4-10mm behind the root tip. In young plants this region is responsible for most of the plant's water uptake and in older plants and in conditions of water stress a greater proportion of water moves in across the suberised regions of the root further back from the root tip.

PILOGLANDULOSE: Having glandular hairs.

PILOMOTOR: Describing erector muscles of hairs in the skin and the postganglionic sympathetic nerve fibres that innervate them.

PILOSE: Covered with hairs, especially long, soft and straight ones.

PILOSULOSE: Covered with minute, long, soft and straight hairs.

PILOT: 1. A person who is responsible for the safe navigation of an airplane. Such a person is normally in charge of the aircraft. 2. A person with special knowledge of a canal, the entrance to a harbour, etc, who is licensed to guide ships through them.

PILOT STUDY: A small feasibility investigation carried out before the main study, in order to gain information and to identify problems necessary to the actual study.

PILOT SURVEY: A small scale of investigation carried out before the main survey primarily to gain information and to identify problems relevant to the survey proper.

PILTDOWN MAN: Fossil remains which were claimed to have been found by Charles Dawson (1846-1916) during excavations at Piltdown, Sussex, in 1912 that were named *Eoanthropus dawsoni*. They were described as a representative of the true ancestors of modern humans. The skull resembled that of a human, but the jaw was apelike. In 1953, fluorine tests showed the specimen to be a fraud.

PILUS: A fine, filamentous structure found on the surface of many gram-negative bacteria. It is shorter, thinner and straighter than flagella. There are two kinds of pili (plural of pilus): a larger form

that is hollow and found on male bacterial cells only and used in bacterial cell conjugation. The smaller form is of major significance in adherence of bacterial cells to epithelial surfaces such as the intestinal mucosa or mammary gland epithelium. Antipilus antibody can provide protection against disease.

PIN: 1. In electricity, it is a thin metal terminal extending from an electrical or electronic device such as a plug or a vacuum tube, used to connect the device by socket to other circuitry. 2. In mechanical engineering, it is the part of a key that inserts into a lock. 3. In security systems, it is an acronym for **Personal Identification Number** is a unique number used in conjunction with a magnetic card in security systems such as in banking as a password to establish an identity. 4. A heterostylic flower that possesses a fairly long style and short stamen. In this type of flower, the anthers are halfway down the central flower tube, in a ring around the style. An example is Primrose (*Primula vulgaris*). On the other hand, a **thrum** is a flower whose style reaches only half way up the flower tube so that the stigma is also positioned halfway up inside the tube.

PIN SEMICONDUCTOR DETECTOR: A semiconductor detector consisting of a compensated region between a p and a n region. The compensated region is often referred to as intrinsic.

PINACOL REARRANGEMENT (PINACOL-PINACOLONE REARRANGEMENT): A reaction in which the diol pinacol, $(CH_3)_2COH(CH_3)_2COH$, converts to the ketone pinacolone, $CH_3COC(CH_3)_3$. This rearrangement takes place under acidic conditions, with loss of a molecule of water. A methyl group moves from one carbon atom to an adjacent one in order to stabilise an intermediate carbocation. The reaction, also gives its name to a class of similar rearrangements.

PINATIFID: A term used of a leaf, which is cleft nearly to the midrib in broad divisions not separated into distinct leaflets or pinnately divided into lobes reaching more than halfway to the midrib.

PINCER: 1. Either of the pair of curved claws of arthropods such as scorpions, lobsters, crabs, etc, that enabling it to carry loads, to defend itself from other creatures or to attack prey. The pincer may also be used as an offensive tool through which venom or acid can be injected into an enemy. 2. Pincers is a hand tool made of metal consisting of two handles and a set of jaws that are operated with the use of a pivot. It is used for holding or gripping objects so that they may be handled more easily, cutting or pulling out objects such as nails out of wood. A mechanical advantage is required to grip, cut or pull an object. Pincers are first-class levers, but differ from pliers in that the concentration of force is either to a point or to an edge perpendicular to the length of the tool. This allows pincers to be brought close to a surface, as is often required when working with nails. If the pincers have perpendicular cutting edges, the pincers are often called end-nippers or end-cutters.

PINCH EFFECT: 1. The constriction of a liquid conductor of electricity such as mercury or molten metal, which occurs when a substantial current is passed through it. 2. The constriction of plasma due to the magnetic field of a high current within the plasma.

PINE: An evergreen coniferous tree which has clusters of long needle-shaped leaves. The term applies to any conifer in the genus of the family pinaceae.

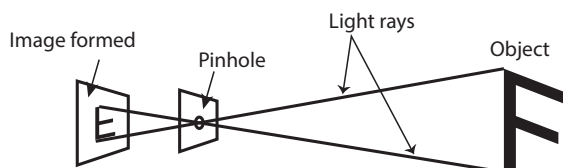
PINEAL BODY (PINEAL GLAND; EPIPHYSIS; PINEAL ORGAN): A cone-shaped outgrowth about the size of a pea in the forebrain of vertebrates. In humans, its functions are obscure but in other vertebrates it is sensitive to different levels of light and plays major roles in sexual maturation, circadian rhythm and sleep induction and seasonal affective disorder (SAD) and depression.

PINEAL EYE (PARIETAL ORGAN; THIRD-EYE): An eye-like structure, with a lens and retina, found on top of the head of some lizards, frogs, lampreys, tuna, pelagic sharks and hagfishes. The eye may be photoreceptive and is usually associated with the pineal gland, regulating circadian rhythmicity and hormone production for thermoregulation.

PINEAPPLE (*Ananas comosus*): A tropical plant or its fruit, with a short thick stem and long spiny succulent leaves. It is cultivated for its large edible multiple fruit, also called pineapple. The fruit has a yellowish juicy flesh. It is typically covered with floral bracts and topped with short spiny leaves. A determinate inflorescence forms 15-20 months after planting. After fertilisation, the many lavender flowers fuse and become fleshy to form the fruit, which ripens in 5-6 months. The plant does not contain any viable seeds and the crowns from the top of the fruit are usually used for propagation; also slips (tiny plantlets that grow at the base of the fruit on the fruit stalk) as well as the suckers (little plantlets that grow between the leaves of the mature pineapple) can be used. Plants propagated by the slips and suckers mature more quickly than those propagated by the crowns. Matured or ripe fruits can be peeled and eaten raw and can be processed into a syrupy juice. It can be used to make wine and the leaves are used to produce. Medicinally, the plant has been used traditionally to induce menstruation and abortion, kill parasitic amoebas and expel worms.

PING (PACKET INTERNET GROPER): In computer science, it is computer network administration software that is built into Windows and UNIX operating systems that can send short messages over a network by one computer to another. It is used by system administrators to test network response time and performance and generally to diagnose problems.

PINHOLE CAMERA: The simplest type of camera, in which a pinhole rather than a lens is used to form an image. Light from a scene passes through this single point and projects an inverted image on the opposite side of the box. The human eye in bright light acts similarly, as do cameras using small apertures. A common use of the pinhole camera is to capture the movement of the sun.



Pinhole Camera

PINING: A disease of sheep caused by the lack of cobalt. It causes the animals to lose condition and have poor fleece, dull eyes and general weakness.

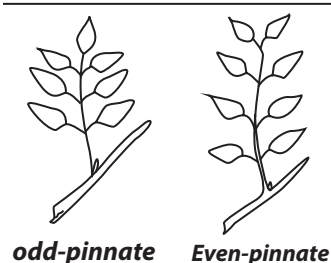
PINK: 1. A pale red colour. 2. A garden plant with sweet smelling pink, crimson or variegated flowers.

PINK-EYE: A disease causing inflammation of the surface of the eye (the conjunctiva) caused by infection, chemical burn, physical injury or allergy.

PINNA: 1. The external part of the ear or the visible part of the outer ear that resides outside of the head, present in some mammals which collect sound waves into the auditory canal. 2. The primary division of the pinnate compound leaf.

PINNATE COMPOUND

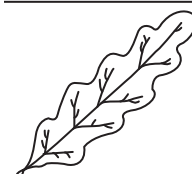
LEAF: A compound leaf with a feather-like or multi-divided leaflets arranged in two rows, one along either side of the rachis. Each leaflet is known as pinna. An **odd-pinnate leaf** is a pinnate leaf with a single



leaflet at the apex. An **even-pinnate leaf** is a pinnate leaf with a pair of leaflets at the apex.

PINNATIFID: Pinnately cleft or divided into lobes to reach halfway or more to the midrib.

PINNATILOBATE: Having pinnately arranged lobes.



Pinnatilobate

PINNATION: Pinnate condition or development.

PINNATIPARTITE: Having lobes that extend more than half-way toward the midrib.

PINNATISECT: Incisions reaching nearly to the midrib.

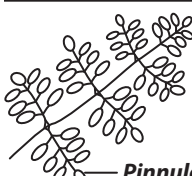
PINNIPALMATE: Intermediate between pinnate and palmate, usually in a leaf with the lowermost pair of veins slightly larger and more distinctive than the others.



Pinnipalmate

PINNIVEINED: Pinnately veined.

PINNULE: 1. Any of the smaller leaflets into which each leaflet of a bipinnately compound leaf is subdivided. 2. A feather-like or plume-like organ such as a small fin or one of the appendages of a crinoid.



Pinnule

PINOCYTOSIS: The process by which small particles such as the already dissolved or broken-down food are brought into the cell suspended within small vesicles which subsequently fuse with lysosomes to hydrolyse or to break down the particles.

PINT: A unit measure of liquid capacity in the United States and formerly in the United Kingdom. It is equal to 0.473 litre in the United States and 0.568 litre in the United Kingdom.

PINTEREST: A social media arena platform consisting of digital bulletin boards where businesses can pin their content mainly using images, GIFs and videos. It was launched in March 2010. By August 2018, it has 200 million monthly active users.

PION (PI-MESON): An elementary particle classified as a meson and the lightest in the family. It was first detected in cosmic rays by C. F. Powell in 1947. It exists in three forms: **neutral π^0** with a mass of about 264 times that of the electron; and **the pions positively charged (π^+)** and **negatively charged (π^-)**, which have a mass of about 273 times each that of the electron.

PION DECAY EXPERIMENT: An experiment involving the decay of a fast-moving pion into two gamma ray photons which indicates that the speed of light in a vacuum is independent of its source. Pion was the first meson to be discovered that has a mass approximately 270 times that of an electron.

PIONEER CROP: A crop grown to improve general soil fertility, prior to the sowing of another more valuable crop.

PIONEER SPECIES: A species that is the first to colonise and thrive in a new area, where nothing is growing or in an area that has been devastated by fire, flood, ploughing, etc. These species are usually annuals, disappearing after the second year when perennials take over.

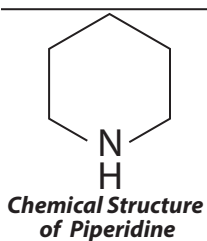
PIP: The small seed of a fleshy fruit such as an apple or orange.

PIPE: A tube or hollow cylinder through which liquids or gases or very fine particles can flow. Examples are water pipe and gas pipe.

PIPELINE (PIPLINING): In computer processors, it is an advanced micro-processing technique that allows multiple instructions to be processed at the same time. While one stage of an instruction is being processed, other instructions may be undergoing processing at

a different stage. Without a pipeline, each instruction would have to wait for the previous one to finish before it could even be accessed.

PIPERIDINE: A colourless, fuming liquid with an odour like ammonia having the chemical formula $C_5H_{11}N$, density 0.862 g/ml (liquid), melting point $-7^\circ C$ (266 K) and boiling point $106^\circ C$ (379 K). It is a saturated heterocyclic compound having a nitrogen atom in a six-membered ring, containing five methylene units and one nitrogen atom. The structure is present in many alkaloids. It is a widely used building block and chemical reagent in the synthesis of organic compounds, including pharmaceuticals.



PIPETTE (PIPETTOR; CHEMICAL DROPPER): A laboratory device used to transport an accurate measurement of a known volume of liquid, to a graduated tube. It is a glass tube often with an enlarged bulb, which is calibrated in one or more positions or it may be a plastic device with an adjustable plunger fitted with one or more disposable plastic tips. Many pipette types work by creating a partial vacuum above the liquid-holding chamber and selectively releasing this vacuum to draw up and dispense. It is usually used in chemistry and biology laboratories.

PIPIDAE: A family of exclusively aquatic tongueless frogs with no vocal chords comprising 30 species found in tropical South America and sub-Saharan Africa with no vocal chords. They have dorso-ventrally compressed bodies and tiny eyes; and invisible eardrums on the sides of their heads (unlike many other frogs). They have a lateral line system evident by stitch-like marks running down each side of the body from the head to the rump. These marks allow the frog to feel the movements made by other animals in the water. They have long and muscular hind limbs splayed to the side. The feet are webbed, with three inner toes having keratinous claw-like tips. The forelimbs are relatively short, with small long unwebbed fingers.

PIRACY (PIRATING; SOFTWARE PIRACY): In computer science, it is a term used to describe the act of illegally using, copying or distributing software without purchasing the software or having the legal rights.

PIRANI GAUGE: Invented in 1906 by Marcello Pirani, it is an instrument used to measure low pressures in the region of between 0.5 torr to 10^{-4} torr. It consists of an electrically heated filament, usually platinum, which is suspended in a tube which is connected to the system whose gas pressure is to be measured. If the gas is at high pressure, gas molecules collide frequently with the filament and absorb energy from the filament which results in cooling of the filament. As the pressure of the gas molecules decreases the number of gas molecules inside the chamber also goes down resulting in fewer collisions with the filament. As a result, the temperature of the filament increases because of decreased cooling. Electrical resistance of a wire varies with temperature. Hence by studying the variation of the resistance of the wire the vacuum surrounding the wire can be predicted.

PIRIFORM: Shaped like a pear.

PIRSSONITE: A mineral consisting of a hydrated mixed carbonate of sodium and calcium with the chemical formula $Na_2CO_3 \cdot CaCO_3 \cdot 2H_2O$, specific gravity 2.35 and Moh's scale hardness 3. It is one of several carbonate minerals that form in non-marine evaporite deposits. Evaporite minerals are geologically important because they clearly are related to the environmental conditions that existed at the time of their deposition, namely arid. They also can be easily recrystallised in laboratories in order to confirm their specific characteristics of formation.

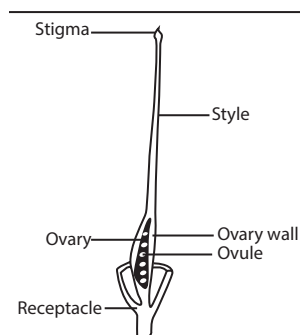
PISACEOUS: Pea-green.

PISCES (THE FISHES): A group of cold-blooded aquatic vertebrates that have gills and fins and includes the Agnatha or jawless fishes, the most primitive; the Placodermi or armoured fishes, known only as Paleozoic fossils; the Chondrichthyes or cartilaginous fishes; and the Osteichthyes or bony fishes.

PISCIVOROUS: A fish-eating animal or bird.

PISIFEROUS: Bearing peas. **Pisiform** means shaped like a pear.

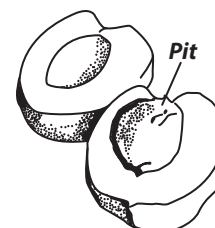
PISTIL: A general term for the female reproductive part of a flower consisting of either a single carpel, known as a **simple pistil** or a group of several fused carpels called a **compound pistil**. The pistil has a swollen base called the **ovary**, which contains the structures that develop into a seed when fertilised called **ovules**. The stalk (style) arises from the ovary and has a pollen-receptive tip, the stigma, which is variously shaped and often sticky. There may be a single pistil, as in the lily or several as in the buttercup. **Pistillate** means possessing a pistil or pistils but lacking stamens.



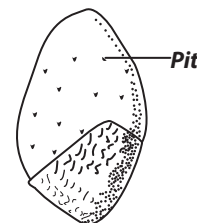
**A section of the
pistil of a flower**

PISTON: 1. A component of reciprocating engines, pumps and gas compressors consisting of a metal cylinder that slides up and down a tubular housing, known as a cylinder. It receives pressure from or exerting pressure on a fluid, especially one of several in an internal-combustion engine. Its function is to transfer force from expanding gas in the cylinder to the crankshaft via a piston rod and/or connecting rod to create motion in a vehicle, such as a motor car. In a pump, the force is transferred from the crankshaft to the piston for the purpose of compressing or ejecting the fluid in the cylinder. 2. The valve mechanism in a brass musical instrument that is used to alter its pitch.

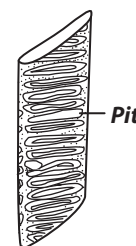
PIT: 1. A large usually deep hollow or opening in the ground. 2. A natural hollow in the surface of a plant or an animal's body such as the pit of the stomach or a hollow scar left on the skin, after small pox. 3. A small hollow on a surface especially of metal or glass. 4. The hardened stone of the endocarp of a drupe. Fruits with pits usually develop from a single carpel and from flowers with superior ovaries. Some flowering plants that produce drupe with pit are coffee, mango, coconut, oil palms, almond, apricot, cherry and peach. 5. The small openings in the walls of tracheids and vessel elements which allow water to move from cell to cell. The pit membranes are highly permeable to water and dissolved solutes. Pits are found in the xylem. **Pitted** means having small pits or depressions. 6. In optical recording, a microscopic hole in the information layer of a videodisc surface made by the recording laser beam. Recorded information is contained in the pits.



Pit of a fruit or drupe



Pit



Pit

PIT CRATER: A circular-shaped crater formed by the sinking or collapse of the ground. Fissures may erupt from the walls or base of a pit crater, but pit craters are not constructional features built by eruptions of lava or tephra. Pit craters may also partially fill with lava to form a

lava lake. They are common along rift zones of shield volcanoes, for example, Mauna Loa and Kilauea volcanoes in Hawaii.

PIT ORGAN (HEAT SENSITIVE PIT): An organ located on each side of the head, between the nostril and the eye of some snakes, especially the vipers. These organs are very sensitive to small changes in temperature and are therefore used in detecting preys such as birds and rodents that maintain a constant body temperature.

PITCH: 1. A black sticky substance, hard when cold but liquid when hot, obtained from the distillation residue of coal tar, wood tar or petroleum and used for waterproofing, roofing, caulking and paving. 2. The distance between the adjacent threads of a screw or bolt. When a screw is turned through one full turn it moves up or down a distance equal to the pitch of its thread. 3. The property of sound that characterises its highness or lowness to an observer. It is related to but not identical to frequency.

PITCH-BASED CARBON FIBRES: Carbon fibres obtained from pitch precursor fibres after stabilisation treatment, carbonisation and final heat treatment. The term pitch-based carbon fibres comprises the isotropic pitch-based carbon fibres as well as the anisotropic mesophase pitch-based carbon fibres (MPP-based carbon fibres). The isotropic type belongs to the carbon fibre type LM (low modulus) and is mainly used as filler in polymers and insulation materials and for similar applications. The anisotropic type (MPP-based carbon fibres) belongs to the carbon fibres type HM and is used mainly for reinforcement purposes due to its high Young's modulus value.

PITCHBLEND: A dark-coloured lustrous, heavy mineral, which is the oxide ore of uranium and radium. It consists mainly of natural uranium oxides, triuranium octaoxide (U_3O_8) and uranium dioxide (UO_2) and also some lead and variable amounts of thorium and rare-earth elements. The major sources in the world are the Great Lakes region of Canada, Democratic Republic of the Congo (DRC), the Czech Republic, the Colorado Plateau, Australia and South Africa.

PITFALL TRAP: A trap used to collect organisms such as insects, amphibians and reptiles living on or just below the soil surface and from leaf litter for ecology studies and ecologic pest control.

PITH: 1. The central cylinder of parenchyma tissue found in plant stems surrounded by the vascular bundles or the central parenchymatous tissue of the stem of a vascular plant. Pith is composed usually of parenchyma cells often arranged in longitudinal files. This arrangement results from predominantly transverse division of pith mother cells near the apical meristem. 2. The soft whitish fibrous tissue that lies under the outer rind of a citrus fruit. 3. The soft spongy inner tissue of a part of the body such as a hair shaft or bone. 4. To cut or destroy the spinal cord of a vertebrate as part of a laboratory experiment. 5. To kill animals, especially cattle, by cutting through the spinal cord.

PITOT TUBE: An instrument that measures the stagnation pressure due to isentropic deceleration of fluid (gas and liquid) flow or the speed of a fluid. It consists of two tubes, one with an opening facing the moving fluid and the other with an opening at 90° to the direction of the flow. Water rises in the tube at a distance, h , above the surface and if friction losses are negligible, the velocity of the stream, V , is approximately $2gh$, where g is the acceleration of gravity.

PITTING CORROSION: A special type of non-uniform corrosion of passive metals, resulting in the formation of pits, which usually occurs in the presence of certain anions at electrode potentials positive to a critical pitting potential.

PITUITARY GLAND: A small oval endocrine gland of vertebrates situated in the centre or at the base of the brain that plays a major role in regulating the endocrine system. It consists of two lobes: the **anterior lobe** secretes six hormones such as growth hormone, prolactin and thyroid stimulating hormone. The **posterior lobe** stores and releases two hormones, oxytocin and vasopressin, from nerve cells in specialised regions of the hypothalamus that control

pituitary function. Generally the pituitary gland produces hormones that control other glands and influences growth of the bone structure, sexual maturing and general metabolism.

PIVOT: 1. A turning point in a machine on which a pin, shaft, rod or a related part rotates or swings. 2. A property used to effect a smooth transition from one key to another and which has different harmonic or melodic functions in two different keys.

PIVOT JOINT: A type of synovial joint in which there is a central bony point surrounded by a bony-ligamentous ring. In this type of joint, rotation is the only movement possible. The atlanto-axial and superior radioulnar joints are good examples.

PIVOTAL QUANTITY: A function of data, derived from an experiment and the parameter of interest, regarded as a random variable whose probability distribution does not depend on the unknown parameters.

PIXEL: One of the smallest addressable unit dots or spots on a screen of a computer's visual display unit (VDU) or bitmapped images. Screens are rated by their number of horizontal and vertical pixels. For example, 1024 x 768 means 1024 pixels are displayed in each row and there are 768 rows. With colour systems, each pixel contains red, green and blue subpixels and the subpixel is actually the smallest addressable unit.

PIXELS PER INCH (PPI): A measure of the pixel density of an electronic image on a display screen of a device such as a computer or television. The pixels per inch for a given picture resolution will differ based on the overall screen size since the same number of pixels are being spread out over a different space. A greater density of pixels reproduces the overall image more accurately on a screen. The term PPI can also describe the resolution, in pixels, of an image file.

PK VALUE: Negative logarithm of the equilibrium constant for the dissociation of a weak electrolyte in aqueous solution. For an acid, the value, denoted pK_a , is given by $\log_{10}(1/K_a)$, where K_a is the acid dissociation constant. For a base, pK_b is given by $\log_{10}(1/K_b)$. pK_a . Values are often used to compare the strengths of different acids.

PLACE VALUE: The value of the place that a digit occupies in a numeral, for example, in the number 6,745, the place value of 7 is hundred, hence 7 is 700.

PLACE VALUE NOTATION (POSITIONAL NOTATION): A system of arranging or encoding numbers such that their position in relation to each other gives each number a particular value. With the use of a radix point, the notation can be extended to include fractions and the numeric expansions of real numbers.

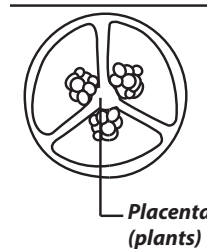
PLACE-VALUE SYSTEM: A number system in which the value of a number symbol depends not only on the symbol itself but also on the position it occurs.

PLACEBO: Any inert substance that has no active ingredient, but which may bring about improvement in a patient's condition psychologically rather than for any physiological effect. It is also used as a control in an experiment or test to determine the effectiveness of a medicinal drug.

PLACEMENT DRILL: A machine which drills seeds and fertilisers at the same time, placing the fertiliser close to the side of and below the rows of seeds.

PLACENTA: 1. An organ that attaches the developing embryo or foetus to the uterus in some mammals through which the developing embryo is nourished in its mother's uterus. The placenta is expelled at parturition. 2. The section of the ovary that bears ovules in plants.

PLACENTAL MAMMAL (EUTHERIAN MAMMALS): A member of a group of mammals, including humans, whose young



complete their embryonic development in the uterus, joined to the mother by a placenta and umbilical cord.

PLACENTALIA: A subclass of mammals such as cows, bats, mice and humans in which the embryos are retained in the uterus of the mother's body and nourished during gestation through a placenta. Unlike young marsupials, young placental mammals spend a relatively long time developing inside their mother's body before birth.

PLACENTATION: 1. The fusion of a tissue or organ at the embryonic stage of an animal to its parent for physiological exchange to promote the growth and development of the young. This makes the foetus to respire, acquire nourishment and eliminate wastes by bringing the bloodstreams of mother and young into close association but never into direct connection. 2. In plants, it is the part within the ovary of a flowering plant to which the ovules are attached. Some of the most common kinds of placentation are marginal as found in cowpea, axile in tomato or parietal in pawpaw.

PLACODE: An area of thickening in the ectoderm of an early embryo where some organ or structure such as cranial placodes, peripheral nervous system structures associated with the special senses and cranial ganglia develops. For example, the inner ear develops from the otic placodes, located on either side of the hindbrain and the hair follicles develop from individual epithelial placodes.

PLACODERM: A member of an extinct class of fish-like vertebrates of the Devonian period that were enclosed in a tough, outer armour and had jaws.

PLACOID SCALE (DENTICLE): A small, tooth-like scale with a projecting spine and a flattened base embedded in the skin. It is typical of bony and cartilaginous fish. Fish scales are produced from the mesoderm layer of the dermis.

PLACOZOA: A phylum of simple aquatic animals generally classified as a single species, *Trichoplax adhaerens*, first discovered in 1883. They are the simplest in structure of all non-parasitic multicellular animals. They are covered in cilia, which they use to crawl over surfaces and feed by secreting enzymes from their ventral surfaces, part of which they may invaginate to form a temporary stomach.

PLAGIARISM: The act of copying and republishing another individual's work such as text, media and even ideas without an appropriate reference or citation.

PLAGIOTROPISM: The tendency of roots, stems or branches to be oriented at an oblique or horizontal angle to the line of action of the stimulus concerned. For example, the growth of lateral branches and lateral roots is at an oblique angle to the stimulus of gravity.

PLAGUE: An infectious disease caused by the bacterium *Yersinia pestis*, carried by the rat flea that occurs in epidemics, killing many organisms. There are three types: bubonic, pneumonic and septicemic plague.

PLAIN (GRASSLAND): A land which is usually flat and predominated by grass, but also trees. It covers large areas of the Earth's surface especially between the deserts and the tropics except Antarctica.

PLAIN TEXT (PLAINTEXT): A text file or document that contains only text and is capable of being viewed in any text editor. It supports standard ASCII characters, including numbers, symbols and spaces, but does not support any type of text formatting. An example of a plain text editor and viewer is Microsoft Notepad, although Microsoft WordPad and Word could also be used to view plaintext. Unlike a rich-text document, a plaintext document cannot have bold text, fonts, larger font sizes or any other special text formatting.

PLAIT: Intertwined strands of textiles or hair; a pleat or fold.

PLAN: Projection of an object on a horizontal plane, usually drawn to scale showing the relative position of objects.

PLANAR: 1. A molecule which has all its atoms in the same plane and flat. All linear and triatomic molecules must be planar. 2. Relating to, involving or typical to a geometric plane. 3. Having only two dimensions or lying in a single geometric plane or flat.

PLANAR CHROMATOGRAPHY: A separation technique in which the stationary phase is present as or on a plane. The plane can be a paper, serving as such or impregnated by a substrate as the stationary bed (paper chromatography, PC) or a layer of solid particles spread on a support, for example, a glass plate (thin layer chromatography, TLC). Sometimes planar chromatography is also termed open-bed chromatography.

PLANAR FILM: A film whose boundaries are represented by ideal parallel planes. The planar film thickness is defined as the distance between these planes. The terms nanometre thick film, micrometre thick film and millimetre thick film refer specifically to film thickness in the ranges 0.1 – 100 nm, 0.1 – 100 μ m and 0.1 – 100 mm, respectively.

PLANARIANS: A class of free living flatworms, of the family Planariidae and related families, which occur in wet soils, fresh water and marine environments. The soft, ciliated body is leaf-shaped when elongated. The spade-shaped head has two eyes and sometimes tentacles and the under surface is covered with cilia, used for gliding over stones and weeds.

PLANCK CONSTANT: The fundamental constant, which is equal to the ratio of the energy of a photon and the frequency of its associated electromagnetic wave having a value $6.626\ 0755(40) \times 10^{-34}$ Js. It is named after Max Planck (1858–1947), German physicist. In quantum-mechanical calculation, rationalised Planck constant or Dirac constant $\hbar = h/2\pi = 1.054\ 589 \times 10^{-34}$ Js is frequently used.

PLANCK LENGTH: The length scale at which a classical description of gravity ceases to be valid and quantum mechanics must be taken into account. It is given by, $\sqrt{(\hbar G/2\pi c^3)}$, where G is the gravitational constant, $\hbar$ is the rationalised Planck constant and c is the speed of light. The value is equal to 1.6162×10^{-35} m (twenty orders of magnitude smaller than the size of a proton, 10^{-15} m).

PLANCK MASS: The mass, $\sqrt{(\hbar c/2\pi G)}$, of a particle whose Compton wavelength is equal to the Planck length, where $\hbar$ is the rationalised Planck constant, c is the speed of light and G is the gravitational constant. Planck mass is equal to 21.764 micrograms or 1.2209×10^{19} GeV/c.

PLANCK TIME: The time taken for light to travel in a vacuum, a distance equal to the Planck length. It is given by $t_p = (\hbar G/c^5)$, where G is the gravitational constant and $\hbar$ is the rationalised Planck constant. The value of the Planck time is of order 10^{-43} s. In the big-bang cosmology, up until a time t_p after the initial instant, it is necessary to use a quantum theory of gravity to describe the evolution of the universe. The unit is named after Max Planck, who was the first to propose it.

PLANCK UNITS: A system of units of measurement, used principally in discussion of quantum theories of gravity, in which length, mass and time are expressed as multiples of the Planck length, mass and time respectively. This is equivalent to setting the gravitational constant, the speed of light and the reduced Planck constant all equal to unity. All quantities that ordinarily have dimensions involving length, mass and time become dimensionless in Planck units. Since, in the subject area where Planck units are used, it is normal to employ Gaussian units or Heaviside-Lorentz units for electromagnetic quantities, these then become dimensionless. Planck units have profound significance for theoretical physics since they simplify several recurring algebraic expressions of physical law by not having dimensions. They are particularly relevant in research on unified theories such as quantum gravity.

PLANCK, MAX KARL ERNST LUDWIG: A German physicist, who studied at the Universities of Munich and Kiel, then became professor of theoretical physics at the University of Berlin (1889 – 1928). His work on the second law of thermodynamics and blackbody radiation led him to formulate quantum theory of radiation. One of the most important scientific discoveries of the century, this theory earned him the 1918 Nobel Prize for Physics. He also discovered the quantum of action, now known as Planck's constant, $\hbar$.

PLANCK'S RADIATION LAW: A law stated by Max Planck (1889 – 1928) which gives the spectral energy distribution of the heat radiation

emitted from a so-called blackbody at any temperature. The law laid the foundation for the advent of the quantum theory because it was the first physical law to postulate that electromagnetic energy exists in discrete bundles or quanta.

PLANE: 1. A flat surface with the condition that a straight line that lies entirely in the surface joins any two points in the same plane. In Cartesian three-dimensional space, the equation for a plane is $ax + by + cz = d$, where the vector (a, b, c) is the normal to the plane. 2. Any subgeometry of dimension 2 of an algebraic geometry.

PLANE ANGLE: An angle formed by two straight lines meeting in the same geometric plane.

PLANE FIGURE: A two dimensional figure such as triangles, squares, pentagons, circles, and parabolas that is entirely contained in a single plane.

PLANE GEOMETRY: The branch of geometry that studies two-dimensional figures.

PLANE GRAPH: In mathematics, it is a graph drawn in a plane with the edges meeting only at the vertices.

PLANE OF SYMMETRY (PLANE SYMMETRY): Imaginary lines traced on polyhedrons such as cubes and octahedrons depicting a point, which if rotated 180° from that line will yield the same shape. In other words, it is a solid object which has a reflective symmetry.

PLANE TREE: A common temperate deciduous hardwood tree, frequently grown in towns because of its resistance to air pollution.

PLANE WAVE: A wave which does not spread (diverge) or converge

PLANE-POLARISED LIGHT: A display in which sets of parallel conductors at right angles to each other are deposited on glass plates, with the very small space between the plates filled with a gas. Each intersection of two conductors defines a single cell that can be energised to produce a gas discharge forming one element of a dot-matrix display.

PLANET: A large body of nonluminous ball of rock, metal or gas in orbit around a star or a body that revolves around a central astronomical body, especially one of the eight bodies: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune. Pluto, long regarded after its discovery in 1930 as the ninth planet, was gradually recognised as a Kuiper belt or transneptunian object and in 2006 was reclassified by astronomers as a dwarf planet.

PLANETARIUM: An optical projection device of the stars, Sun, Moon, planets and other celestial objects upon a large hemispherical dome, showing their motions as viewed from the Earth or space near the Earth. There are over 100 major planetariums around the world with domes over 15 m in diameter and over 1000 smaller planetariums.

PLANETARY MOTION: The orbital motion of planets around the Sun. Johannes Kepler (1571-1630) developed three laws, which apply to the motion of the planets across the sky. They are: (i) The Law of Orbits, which states that all planets move in elliptical orbits, with the Sun as one of the foci. (ii) The law of areas, which states that a line that connects a planet to the Sun sweeps out equal areas in equal times. (iii) The law of periods, which states that the square of the period of any planet is proportional to the cube of the semi major axis of its orbit.

PLANETARY SYSTEM: A collection of planets and other celestial bodies such as moons, asteroids and comets, which are bound to a star by gravity and orbits around it. Planetary systems are believed to originate from rotating clouds of dust and gas that encircle stars shortly after their birth.

PLANETESIMAL: A small body formed from collisions with other objects in the solar system from which a planet originated.

PLANETOID (ASTEROID; MINOR PLANET): Any small planet with a diameter of 10 metres up to 1,000 km that orbits the Sun or another star (usually between Mars and Jupiter). The largest ones are spherical in shape, and the smaller ones are irregular.

PLANIMETER: A simple integrating instrument for measuring the area of an arbitrary two-dimensional shape or an irregular plane figure,

such as the area under a curve. **Planimetry** is the measurement of plane areas.

PLANISPHERE: 1. A circular star map drawn for a given latitude, having a rotary overlay that shows which constellations are visible at any time and date in the year. 2. A map in which the celestial sphere is projected onto a plane surface, especially in a polar projection and to which a grid system is added to help locate celestial objects.

PLANKTON: Tiny marine and freshwater organisms that float or drift about in the water. Plankton includes algae (phytoplankton) and the animal-like community (zooplankton). They form the productive base of both marine and freshwater ecosystems and provide food for larger marine organisms, such as fish.

PLANO-CONCAVE LENS: A diverging spherical lens with one plane face and one concave face.

PLANO-CONVEX LENS: A converging spherical lens with one plane face and one convex face.

PLANT: 1. An organism in the kingdom Plantae, consisting of multicellular, eukaryotic life forms that use photosynthesis to make its own food, has cellulose cell wall, absence of sensory and nervous systems, unlimited growth at meristems and complex cells and is immobile. They are distinguished from other multicellular organisms by their life cycles in which a haploid generation (gametophyte) alternates with a diploid generation (spermatophyte). The radiant energy for photosynthesis is absorbed by chlorophyll, which is a complex pigment not found in animals. They respond slowly to external stimuli, often grow in soil or water and usually have green leaves. 2. A factory, power station or other large industrial complex where something is manufactured or the equipment and machinery necessary for carrying on an industrial or engineering activity.

PLANT BREEDING: The practical application of genetic principles to the development of improved strains of horticultural and agricultural crops in order to increase their yield.

PLANT ECOLOGY: The study of the distribution and abundance of plants, the interactions among and between members of plant species and with their environment. It includes plant ecophysiology, plant population ecology, community ecology, ecosystem ecology, landscape ecology and global ecology.

PLANT GEOGRAPHY (PHYTOGEOGRAPHY; PHYTOCHOROLOGY; GEOBOTANY; GEOGRAPHICAL BOTANY): The study of the spatial distribution of the world vegetation with particular emphasis on the influence of the environmental factors that determine this distribution.

PLANT GROWTH REGULATOR: A synthetic plant substance which is used to control the growth and regulation of plants. An example is 2,4-dichlorophenoxy acetic acid (2,4-D) and kinetin. It is used, for example, to control the growth of vegetation in areas such as road sides.

PLANT HEALTH: The areas of study or operation related to the prevention of pests and diseases affecting plants and plant produce including the control of imports and exports.

PLANT HORMONE (PHYTOHORMONE): A chemical substance produced in extremely low concentrations. Plant hormones shape the plant, affect seed growth, time of flowering, the sex of flowers, senescence of leaves and fruits. It also affects tissues which grow upward and downward, leaf formation and stem growth, plant longevity and even plant death. When the hormones are chemically synthesised, they are called **growth regulators**. Examples are auxin, gibberellin, ethylene and cytokinin.

PLANT INTRODUCTION (PI): The transfer of germplasm or a group of genotype of crop plants from another part of the world, where they are cultivated to a new place for breeding. For safety reasons the material is quarantined and inspected at the country of entry for pests and diseases before being released.

PLANT NUTRITION: The provision of nourishments to plants necessary for their growth. There are 17 essential plant nutrients.

Carbon and oxygen are absorbed from the air, while other nutrients including water are obtained from the soil. Those obtained from the soil are: (i) the three primary macronutrients: nitrogen, phosphorus and potassium; (ii) the secondary macronutrients are calcium (Ca), sulfur (S) and magnesium (Mg); (iii) the macronutrient is silicon (Si); (iv) the micronutrients or trace minerals are boron (B), chlorine (Cl), manganese (Mn), iron (Fe), zinc (Zn), copper (Cu), molybdenum (Mo), nickel (Ni), selenium (Se) and sodium (Na). The macronutrients are consumed in larger quantities and are present in plant tissue in quantities from 0.2% to 4.0% (on a dry matter weight basis). Micronutrients are present in plant tissue in quantities measured in parts per million, ranging from 5 to 200 ppm or less than 0.02% dry weight.

PLANT NURSERY: A carefully selected site where seedlings are raised on seedbeds, seedboxes or in containers.

PLANT PATHOGEN: Any of the organisms that infects plants and cause injury or disease. Examples of such organisms are fungi, oomycetes, protists, nematodes, bacteria, or viruses. Plant pathogens affect agriculture production causing hunger and starvation as well as harming the economy. For example, *Phytophthora* attack cocoa causing black pod disease.

PLANT PATHOLOGY (PHYTOPATHOLOGY): The scientific study of diseases in plants caused by pathogens and environmental factors. It encompasses studies in botany, crop science, soil science, genetics, microbiology, physiology and molecular biology.

PLANT PROPAGATION: Growing new plants from seeds or from parts of existing plants. Plant propagation occurs in nature to ensure survival and spread of species. Plant propagation is done by two means: asexual (vegetative) and sexual propagation. Asexual (vegetative) propagation is achieved through the growing of new plants from a leaf, stem or root of a single parent plant. Sexual propagation occurs by the union of sperm and egg to form seeds. Plant propagation transmits genetic information between plants of the same species. Genetic engineering transfers genes from one organism to another that may or may not be of the same species to introduce desirable traits into an organism.

PLANT PROTECTION: The process of protecting plants from diseases by bio-control, cultivation practices and especially by the application of pesticides.

PLANT RESIDUE COVER: Plant material or biomass remaining on the soil surface after crop harvest.

PLANT SENESCENCE: The final stage in the life cycle of a plant, leading to the death of part or all of the plant.

PLANTAE: The plant kingdom, which accounts for the largest proportion of the Earth's biomass with approximately 250,000 species of mosses, liverworts, ferns, flowers, bushes, vines, trees and others. They are immobile, multicellular, autotrophic eukaryotes with cellulose in their cell walls and starch as a carbohydrate storage product, with chlorophylls *a* and *b* as photosynthesis pigments.

PLANTAIN: 1. A herbaceous annual or perennial plant, *Musa paradisiaca* of the genus *Musa* or its starchy fruit, which is similar to banana. It is cultivated in tropical and subtropical regions. It is a tall herb, which can grow up to about 7.5 m high. The stem is cylindrical, succulent and very juicy. The leaves are broad and flat, up to 3 m long and up to 1 m broad. The blades are simple with a prominent midrib and numerous penni-parallel lateral veins. The inflorescence is pendulous reaching 1.5 m long. The inflorescence axis arises from the corm and grows upward through the channel formed by the overlapping leaf bases and produces a terminal series of large overlapping bracts, each of which subtends and hides a cymose cluster of flowers. As each cyme reaches anthesis, the subtending bract reflexes to expose the flowers and eventually abscises from the inflorescence axis. The flowers are zygomorphic and functionally unisexual, the proximal ones are female and the distal ones are male. The cylindrical fruits are berries occurring in several clusters. They are green in colour, but when ripe, are golden yellow or yellowish green. 2. Also known as **broad-leaved plantain**, **snakeweed**, **common**

plantain or **ripple grass**, it is a perennial weed, *Plantago major* of the family Plantaginaceae, found in the temperate regions of North America and Europe. It spreads by seeds and grows from a very short rhizome, which bears below a great number of long, straight, yellowish roots and above, a large, radial rosette of leaves and a few long, slender, densely-flowered spikes. It is a low-growing, green plant with oval, ribbed short-stemmed leaves. The leaves form basal rosettes which tend to hug the ground. It is edible and also used medicinally.

PLANTATION: A large piece of land, especially in a tropical country where tea, cotton, sugar, tobacco, etc. are grown; or an area of land planted with trees or group of trees.

PLANTIGRADE: One of three forms of locomotion of many mammals, including humans, in which the podials and metatarsals are flat on the ground. The leg of a plantigrade mammal includes the bones of the upper leg called femur or humerus and lower leg (tibia and fibula/radius and ulna). In humans the podials and metatarsals constitute the sole of the foot. The other two ways of locomotion adopted by mammals are **digitigrade**, walking on the toes with the heel and wrist permanently raised and **unguligrade**, walking on the nail or nails of the toes (the hoof) with the heel/wrist and the digits permanently raised.

PLANULA: The flattened, ciliated, free swimming larva of many cnidarians, consisting of a solid mass of cells. It eventually settles on a suitable surface and develops into a polyp. Planulae of the subphylum Medusozoa have no mouth or digestive tract and are unable to feed themselves, while those of Anthozoa can feed.

PLAQUE: 1. A thin layer of organic material composed of bacteria, mucus and food particles that forms on the surfaces of teeth, which contributes to tooth decay and gum disease. 2. A clear area in a bacterial culture grown on an agar plate due to lysis of the bacteria by a bacteriophage. 3. A combination of cholesterol and lipids that can accumulate on the inside of arteries, causing atherosclerosis.

PLASMA: 1. The liquid part of blood including dissolved chemicals but not the blood cells and platelets. It consists of 90% of water and about 7% proteins and 3% of other substances. Plasma proteins include albumin, coagulation factors and globulins, including gamma globulin and a hormone that stimulates erythrocyte formation. It serves as the blood's transport medium, helps maintain blood pressure, distributes body heat and maintains the pH balance in the bloodstream and body. It can be prepared in the laboratory by removing the cells and corpuscles from donated sterile blood and freezing the resulting fluid until it is needed. 2. A substance similar to gas and considered as one of the four states of matter (solid, liquid, plasma and gas) in which a certain portion of the particles are ionised to release electrons making it electrically conductive. Because it contains positive and negative charge carriers in approximately equal numbers it responds strongly to electromagnetic fields. It occurs naturally on the Sun and other stars, but can be artificially produced in fluorescent lights and plasma displays by electrically charging a gas in order to release ultraviolet light.

PLASMA CELLS (PLASMA B CELLS; PLASMOCYTES; EFFECTOR B CELLS): White blood cells with small, eccentrically placed nucleus that produce antibodies. They are derived from a B cell. They develop in the lymph nodes, spleen and bone marrow and are transported by the blood plasma and the lymphatic system.

PLASMA DESORPTION IONISATION: The ionisation of any species by interaction with heavy particles (which may be ions or neutral atoms) formed as a result of the fission of a suitable nuclide adjacent to a target supporting the sample.

PLASMA DETECTOR: A device for measuring the density, composition, temperature, velocity and three-dimensional distribution of plasmas that exist in interplanetary regions and within planetary magnetospheres.

PLASMA DISPLAY PANEL (PDP): A form of display common to large television (80 cm or larger), in which light output is produced from the interaction between an electric current and an ionised

inert gas such as neon. A panel typically has millions of tiny cells in compartmentalised space between two panels of glass. The display consists of matrix of individual cells, one per pixel. The pixels rely on plasma cells or what are in essence chambers more commonly known as fluorescent lamps.

PLASMA FREQUENCY: The maximum frequency of internal oscillation of plasma. The plasma frequency is proportional to the square root of the electron density.

PLASMA OSCILLATION (LANGMUIR WAVES): A collective rapid oscillations of the electron density that occur in plasmas or metals. To ensure that the plasma is electrically neutral there has to be a positive background to the free electrons. They were discovered by American physicists Irving Langmuir and Lewi Tonks in the 1920s.

PLASMA PROTEIN: Proteins dissolved in the plasma of vertebrate blood. Examples are antibodies, fibrinogen and some hormones. They help to regulate the body's osmotic pressure, which keeps the body's systems working properly. They also transport various compounds needed by the body, in addition to playing a role in the immune system function and blood clotting. An imbalance of plasma proteins can lead a patient to abnormally dilated blood vessels and a weakened immune system.

PLASMA SHEET: Low energy plasma, largely concentrated within a few planetary radii of the equatorial plane, distributed throughout the magnetosphere throughout which concentrated electric currents flow.

PLASMA TV: A television display technology in which each pixel on the screen is illuminated by a tiny bit of plasma (charged gas). The plasma is encased between two thin sheets of glass. Within each cell two glass panels are separated by a narrow gap in which neon-xenon gas is injected and sealed in plasma form. Plasma displays are generally considered to offer better dark-room viewing and wider viewing angles than LCD.

PLASMA WAVE DETECTOR: A device for measuring the electrostatic and electromagnetic components of local plasma waves in three dimensions.

PLASMAGEL: The specialised outer jellylike state of cytoplasm of living cells that move by extruding part of the cell (known as a pseudopodium) in the direction of motion, characteristically occurring in the pseudopod of the amoeba. A conversion of plasma gel to plasmasol is believed to be responsible for the continuous forward flow of cytoplasm necessary for forming a pseudopodium.

PLASMALEMMA (CELL MEMBRANE; PLASMA MEMBRANE): The semi permeable membrane forming the outer covering of the cell of animals. It separates an animal cell from its environment or a plant cell from its cell wall and consists mostly of protein and lipid. By osmosis processes, it allows nutrients to enter the cell and waste products to leave.

PLASMASOL: The specialised inner sol-like cytoplasm of living cells that move by producing pseudopodia or a state of cytoplasm that is more liquid than plasmagel.

PLASMID: Genetic element not contained within a chromosome that can exist and replicate independently of the chromosomes. Some plasmids determine the production of proteins that can kill other bacteria; others make bacteria resistant to antibiotics. Bacteria plasmids are used to produce recombinant DNA in techniques of genetic engineering techniques.

PLASMIN (FIBRINASE; FIBRINOLYSIN): A proteolytic enzyme, present in blood plasma that breaks down a blood clot by destroying the fibrin threads of the clot and by inactivating factors involved in blood clotting, such as prothrombin and the clotting factors. It is derived from an inactive precursor, plasminogen and can act on many protein substrates. It liquefies coagulated blood by digesting fibrin (fibrinolysis), the insoluble meshwork of the clot; it also digests fibrinogen, the precursor of fibrin, rendering it incoagulable. Plasmin activator is produced in blood vessel walls, from which it is released following vascular injury.

PLASMODESMATA: Microscopic pores in plants that traverse the cell walls of plant cells and some algal cells and plasma membranes, allowing direct communication between the cytoplasm of adjacent cells. Species that have plasmodesmata include members of the Charophyceae, Charales and Coleochaetales (which are all algae), as well as all embryophytes, better known as land plants. There are two forms of plasmodesmata: the primary ones are formed during cell division and the secondary ones can form between mature cells.

PLASMODIUM: 1. A parasitic protozoan, that causes malaria. The method of transmission is through the bite of the female anopheles mosquito carrying the parasite. The genus *Plasmodium* was created in 1885 by Marchiafava and Celli. There are over 300 known species of *Plasmodium* and at least 11 species infect humans. 2. A mass of protoplasm containing many nuclei, which is a stage in the life cycle of some organisms, especially slime moulds.

PLASMOGAMY: Fusion of the cytoplasm of two or more cells in the sexual reproduction of fungi. It precedes the formation of the secondary mycelium, which consists of dikaryotic cells, one nucleus from each of the parent mycelia.

PLASMOLYSIS: The loss of water through osmosis in plant cells when a cell is surrounded by a solution more concentrated than the concentration of its cell sap to the extent that the protoplasm shrinks away from the cell wall.

PLASMON: A collective excitation for quantised oscillations of the free electron gas density, for example, at optical frequencies in a metal. Plasmons can couple with a photon to create another quasiparticle called a plasma polariton.

PLASMONICS: The design and study of devices that make use of plasmons. A plasmon is a quantum of plasma oscillation. Metals provide the best evidence of plasmons, because they have a high density of electrons free to move. Plasmonics is to plasmons what electronics is to electrons and photonics to photons. It has many appealing possibilities in technology but has the practical difficulty that plasmons are short-lived excitations.

PLASTER: 1. A powder made by roasting and grinding gypsum with some limestone and clay which sets into a hard, coherent solid when mixed with water. It is used to coat walls to produce a smooth surface which can be painted or papered. 2. A mixture of lime and sand and sometimes Portland cement which is mixed with water just before it is used to coat walls to produce a smooth surface.

PLASTER OF PARIS (P.O.P.): A white powder, calcium sulfate dihydrate (CaSO_4), composed of $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$, made by calcining the mineral gypsum, at temperatures of up to 250°C (523 K), which when mixed with water forms a quick-hardening paste. It is used in making casts and moulds and in medicine for moulding casts around broken limbs.

PLASTIC: 1. An extremely versatile synthetic material of high molecular mass, made from the polymerisation of organic compounds. Monomers of plastic are either natural or synthetic organic compounds. It can be moulded into shapes or fabricated in many different forms for use in commerce and industry. 2. Capable of adapting to conditions during growth or development.

PLASTIC BEHAVIOUR: The permanent change of shape or size of some solids when stress is applied beyond a certain value (the yield point) without rupturing. It is caused by the slip between layers of atoms when highly stressed. This leads to a form of flow in solids. Plastic deformation occurs in many metal-forming processes such as rolling, pressing, forging, wire drawing and in geologic processes such as rock folding and rock flow within the Earth under extremely high pressures and at elevated temperatures.

PLASTIC DEFORMATION: The change in shape or size of a solid material, when stress is applied to it close to the breaking point. The energy stored in the material is converted into heat energy and it does not regain its original size and shape.

PLASTIC FLOW: Steady flow occurring only above a certain finite stress.

PLASTIC SOIL: A soil capable of being moulded or deformed continuously and permanently into various shapes by moderate pressure.

PLASTIC TRANSITION: A first-order transition that occurs when a low-temperature phase transforms to an orientationally disordered high-temperature phase with a strong variation of enthalpy.

PLASTICINE: A substance similar to clay but which does not harden like clay. It is made from calcium salts, petroleum jelly and aliphatic acids. It is used for modelling structures.

PLASTICISER: A substance which, when added to another substance increases the plasticity or fluidity of the material, making it more flexible.

PLASTICITY: The property of solids, which causes them to change permanently in size and shape as a result of the application of a stress that exceeds a certain value, called yield point. Plastic deformation occurs in most materials including metals, soils, rocks, concrete, foams, bone and skin. In brittle materials such as rock, concrete and bone, plasticity is caused predominantly by slip at microcracks.

PLASTICITY NUMBER: The numerical difference between the liquid limit and the plastic limit, or between the lower plastic limit and the upper plastic limit.

PLASTICITY RANGE: The range of moisture weight percentage within which a small sample of soil exhibits plastic properties.

PLASTIDS: Major organelles found in the cells of plants that are enclosed by double membranes and contain pigments used in photosynthesis. The types of pigments present can change or determine the cell's colour.

PLASTOCYANIN: A blue copper containing protein that is found in chloroplasts. It acts as an electron carrier molecule between cytochrome f of the cytochrome b_6f complex from photosystem II and P700+ from photosystem I in the light-dependent reactions of photosynthesis. Plastocyanin consists of monomeric, with a molecular weight around 10,500 Daltons and 99 amino acids in most vascular plants. It is a member of the plastocyanin family of copper binding proteins.

PLASTOGLOBULUS: A densely staining droplet, 10-500 nm in diameter found singly or in groups in the stroma of chloroplasts. Plastoglobuli consists of lipid and lipophilic compounds such as galactolipids, quinones and polyisoprenols, but no chlorophylls or carotenoids and are especially prominent in coloured plastids (chromoplasts), for example in ripening fruits.

PLASTOME: All the genetic material that is contained in cells' cytoplasmic organelles, such as mitochondria or chloroplasts. It is part of the entire genome of photosynthetic organisms.

PLASTOQUINONE: A quinone molecule, found in chloroplasts, that functions as one of the carrier molecules of the electron transport chain in the light-dependent reactions of photosynthesis.

PLASTRON: The ventral part of the shell of tortoises and turtles formed of bony plates divided into plates called **scutes** and an underlying layer of interlocking bones. It is flatter but similar to the carapace.

PLATE: 1. One of the number of slabs of solid rocks about a hundred kilometres thick and often several thousands of kilometres across, making up the Earth's surface. 2. A metal other than silver or gold that has been covered with a thin coating of silver or gold. 3. A thin flat sheet of metal, glass, etc, used in the construction industry. 4. A sheet of metal, plastic, rubber, etc, treated so that words of picture can be printed from it. 5. A sheet of special glass coated with a film sensitive to light.

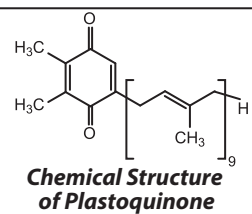


PLATE COUNT: An estimate of the number of viable cells in a culture. A plate count of a bacterial culture is made by inoculating the culture plate of nutrient medium with sample microorganisms and counting the number of cells that appear in the resultant culture after incubation at a specific temperature and time.

PLATE MARGIN (PLATE BOUNDARY): The point at which a plate meets another plate. Plates make up the upper layer or lithosphere and cover the surface of the Earth. There are four types of plate margin: destructive, constructive, collision and conservative. A volcano may be found along two of the types of the plate margin and an earthquake may occur at all four plate margins.

PLATE TECTONICS: The theory formulated to explain the phenomena of continental drift and seafloor spreading and the formation of the major physical features of the Earth's surface or the theory that the lithosphere (the crust and upper portion of the mantle) is divided into about twelve large plates and several small ones that float on and travel independently over the asthenosphere.

PLATEAU: 1. An extensive elevated area of fairly flat land or a mountainous region in which the peaks are at the same height or an area of high ground with a fairly level surface. Plateaus are one of the four major landforms, together with mountains, plains and hills; occupying about a third of the Earth's surface. There are two kinds of plateaus: **dissected plateaus** that form as a result of upward movement in the Earth's crust; and **volcanic plateaus** that are formed by volcanic eruptions that slowly build up over time. Together with enclosed basins they cover about 45% of the Earth's land surface, often creating their own local climate such as arid and semiarid conditions. 2. A phase in mental or physical development during which little headway is made or a period or phase in something when there is little increase or decrease.

PLATELET: A membrane bound cell fragment that buds off from large cells in the bone marrow and functions in blood clotting. They are produced in the bone marrow and stored in the spleen and accumulate to block a cut in a blood vessel and provide a surface for fibrin strands to adhere to, contract to pull the strands together and take part in the conversion sequence of coagulation factors. They also store and transport several chemicals.

PLATELET ACTIVATING FACTOR (PAF): A potent phospholipid activator and mediator of many leukocyte functions, including platelet aggregation, inflammation and anaphylaxis produced by a variety of cell types including neutrophils, basophils, platelets and endothelial cells. It contributes to the increased permeability of local blood vessels, helping to attract immune cells to the infection site. Originally named after its ability to activate and aggregate platelets as part of the blood-clotting mechanism, it is now known to have a broad spectrum of biological effects. PAF is one of the front-line agents produced by activated macrophages in the initial phase of an innate immune response to infection. Its release from mast cells during an allergic reaction contributes to constriction of the airways in asthma and also to the reduction in blood pressure during anaphylaxis.

PLATFORM: 1. In computing, it is a set of standards of a computer's system on which all application software run, enabling the development of software. Examples are Windows 2000 that runs on a Windows platform and the Mac OS X, which runs on a Macintosh platform. 2. A flat elevated portion of ground. 3. The stable, interior layer of a craton composed of igneous or metamorphic rocks covered by a thin layer of sedimentary rock.

PLATEAU'S PROBLEM: The problem of finding a minimal surface with a given twisted curve as its boundary. This is often solved experimentally by means of a soap-film stretched on a wire frame work.

PLATINUM: A very unreactive, heavy, soft, silvery-white, malleable and ductile transition metal element with the symbol Pt, atomic number 78, melting point 1,769 °C (2,043 K), boiling point 3,827 °C (4100 K) and specific gravity 21.45. Platinum is found usually as alloys of 80-90% purity in placer deposits or more rarely combined

with arsenic or sulfur. It is used in jewellery and alloys and for electrodes in the electrolysis of laboratory apparatus, in dental alloys and surgical pins and as a catalyst during the manufacture of nitric acid from ammonia.

PLATINUM BLACK (PT BLACK): Finely divided form of platinum, with good catalytic properties produced by vacuum evaporation. It is used as absorbent and a catalyst and also as a thin film covering solid platinum metal, forming platinum electrodes for applications in electrochemistry.

PLATINUM METALS: The three members of the second and third transitions series immediately preceding silver and gold: ruthenium (Ru), rhodium (Rh) and palladium (Pd), osmium (Os), iridium (Ir) and platinum (Pt). They have similar physical and chemical properties and tend to occur together in the same mineral deposits. The platinum group metals are relatively hard and resistant to corrosion and resistant to chemical attack, excellent high-temperature characteristics and stable electrical properties. All these properties have been exploited for industrial applications. In solution they form a vast range of complex ions. They also form coordination compounds with carbon monoxide and other pi-bonding ligands. The metals and their organic compounds have considerable catalytic activity.

PLATONIC HYDROCARBONS: Hydrocarbons that have the same molecular geometry as the five platonic solids (the tetrahedron, the cube, the octahedron, the dodecahedron and the icosahedrons) with vertices replaced by carbon atoms and with edges replaced by chemical bonds. There are only two known platonic hydrocarbons: cubane (C_8H_8) and dodecahedrane ($C_{20}H_{20}$). The hydrocarbon based on a tetrahedron (C_4H_4) does not exist since 5 edges meet at each vertex. Angle strain is also the reason for the nonexistence of octahedrane (C_6H_6) and icosahdrane ($C_{12}H_{12}$) since if 4 edges meet at each corner, there would be no hydrogen atoms, thus this hypothetical octahedron molecule would be an allotrope C_6 of elemental carbon, not a hydrocarbon.

PLATONIC SOLID (REGULAR SOLID): Any of the five regular polyhedrons, which are solids with regular polygon faces and the same number of faces meeting at each corner that are possible in three dimensions. They are the tetrahedron (a pyramid with triangular faces), the octahedron (an eight-sided figure with triangular faces), the dodecahedron (a 12-sided figure with pentagonal faces), the icosahedron (a 20-sided figure with triangular faces) and the hexahedron or cube. A regular solid with hexagonal faces cannot exist because if it did, the sum of the angles of any three hexagonal corners that met would already equal 360° , so such an object would be planar (of, relating to, or lying in a plane).

PLATONISM (MATHEMATICAL PLATONISM): The philosophical theory that mathematical objects such as numbers and sets exist and are independent of us and our language, thought and practices and therefore that mathematical truth does not consist in, but is the aim of, the construction of proofs.

PLATY: In soils, consisting of aggregates that have developed predominantly along the horizontal axes and are flat and plate-like. Platy structures are usually found in subsurface soils that have been subject to leaching or compaction by animals or machinery.

PLATYCERIUM: The scientific name of about 18 fern species in the polypod family, Polypodiaceae. Ferns in this genus are widely known as staghorn or elkhorn ferns due to their uniquely-shaped fronds. They are epiphytic fern that obtain moisture chiefly from the air and are native to Australia and tropical Asia. The sporophytes have tufted roots growing from a short rhizome that bears two types of fronds, basal and fertile fronds. Basal fronds are sterile, shield or kidney shaped and laminated against the tree and protects the fern's roots from damage and desiccation.

PLATYHELMINTHES: A phylum of invertebrates consisting of three classes: the Turbellaria, mainly free-living, predacious worms; the Trematoda or flukes, holozoic ecto- or endoparasites; and the Cestoda or tapeworms, saprozoic endoparasites in the enteron of vertebrates, whose larvae are found in the tissues of invertebrates or vertebrates.

They are characterised by a flattened unsegmented body, lack coelom, anus, circulatory and respiratory systems and exo- or endoskeleton. They possess a protonephridial excretory system, a complicated hermaphroditic reproductive system and a solid mesenchyme which fills the interior of the body.

PLATYKURTIC: In statistics, a type of distribution in which the points along the x -axis are highly dispersed, thus its lower peak or kurtosis is less heavily concentrated about the mean than a normal distribution.

PLATYZOA: A clade of protostome animals based on molecular systematic. It includes the flatworms, rotifers and certain other phyla.

PLAYSTATION: A home entertainment video game console containing a processor and a DVD reader. It has a port for controllers, a slot for memory card and a video output for it to be connected to a Television set. The original PlayStation was followed by Playstation 2 (PS2) in 2000 and Playstation 3 (PS3) in 2006.

PLAYFAIR'S AXIOM: In mathematics, the axiom that through a point not on a given line there is at most one line parallel to the given line that can be drawn through the point. This axiom is equivalent to the parallel postulate (Euclid's fifth postulate).

PLEATED SHEET: One form of the secondary structure of proteins in which the polypeptide chain folds back and forth or where two regions of the chain lie parallel to each other and are held together by hydrogen bonds.

PLEATED SURFACE: A surface in Euclidean space or hyperbolic geometry that resembles a polyhedron in the sense that it has flat faces that meet along edges. Unlike a polyhedron, a pleated surface has no corners, but it may have infinitely many edges that form a lamination.

PLEIOCHASMIUM: A cyme with three or more branches from the main axis.

PLEIOMERY: A condition whereby a flower possesses more than the usual number of parts or floral whorl.

PLEIOMORPHISM: The existence in distinctly different forms during the life cycle of certain individual animals. An example is the caterpillar, pupa and winged adult of a butterfly.

PLEIOPETALOUS: Having many petals.

PLEIOSEPALOUS: Having many sepals.

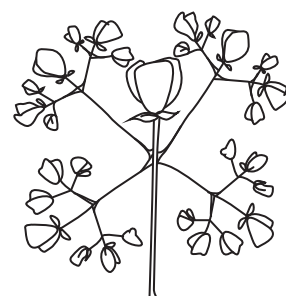
PLEIOSPERMOUS: Having many seeds.

PLEIOTROPIC: Of, relating to or referring to an allele that has multiple effects in an organism. For example, the allele that causes the erythrocytes to have a distorted form in sickle-cell anaemia also causes these blood vessels to rupture easily, thereby inducing anaemia.

PLEIOTROPY: The process in which a single gene influences multiple different observed characteristics of an organism. This can become a problem when selection on one trait favours one specific version of the gene (allele), while the selection on the other trait favours another allele.

PLEISTOCENE: The first epoch of the Quaternary period. It extended from the end of the Pliocene which is about 2,000,000 years ago to the beginning of the Holocene, which is about 10,000 years ago. The Pleistocene is commonly characterised as an epoch when the Earth entered its most recent phase of widespread glaciation.

PLEOCHROIC: Referring to a crystal that appears to be of different colours, depending on the direction from which it is viewed. It is caused by polarisation of light as it passes through an anisotropic medium. In coloured transparent tourmaline the effect may be so strong that one polarised component of a light beam is wholly absorbed and the crystal can be used as a polariser.



Pleiochasium cyme

PLESIOMORPHY (ANCESTRAL TRAIT; SYMPLESIOMORPHY; SYMPLESIOMORPHIC CHARACTER): An evolutionary trait which is shared between two or more but is not unique to members of that group and, therefore, cannot be used as a diagnostic or defining character for the group. For example, vertebrae are found in zebras, cheetahs and orang-utans, but the common ancestor in whom this trait first evolved is so distant that the trait is shared by many other animals.

PLESIOSAUR: A prehistoric carnivorous marine reptile of the Jurassic and Cretaceous periods, which reached a length of 12 metres and had a long neck and paddle-like limbs.

PLEURA: The double membranes of the thoracic cavity covering the exterior surface of the lungs. It consists of the parietal pleura that lines the walls of the chest cage and covers the upper surface of the diaphragm and the pulmonary pleura or visceral layer, which tightly covers the surface of the lungs.

PLEURAL CAVITY (PLEURAL SAC; PLEURAL SPACE): The thin space formed between the layer lining the chest wall and the pleural layer surrounding the lungs in humans and other mammals. The pleuron is a serous membrane, which folds back onto itself to form a two-layered, membrane structure.

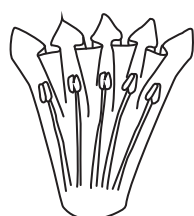
PLEURAL EFFUSION: A disorder of the pleura, the large, thin layers of tissue that wraps around the outside of the lungs and lines the inside of the chest cavity. Pleural effusion occurs when too much fluid collects in the pleural space (the space between the two layers of the pleura). It is characterised by shortness of breath, chest pain, gastric discomfort (dyspepsia) and cough.

PLEURISY: Inflammation of the pleura (the double membrane of the thoracic cavity covering the exterior surface of the lungs). Pleurisy occurs when the two layers of the pleura become red and inflamed. Because they rub against each other every time the lungs expand to breathe in air, it causes sharp pain with breathing. Infections like pneumonia are the most common causes of swelling or inflammation, of the pleura and pleurisy.

PLEURODONT: Having the teeth of an animal attached to the inner surface of the jawbone without roots, as in some lizards.

PLEXUS: The five major branching networks of nerve fibres consisting of the brachial, cervical, coccygeal, lumbar and sacral plexuses. For example, the brachial plexus is a network of spinal nerves that supply branches to the fore limbs in vertebrates. Some of their fibres carry sensory messages to the central nervous system (CNS) and others carry motor impulses away from the CNS to the effector organs such as muscles.

PLICATE: Arranged in folds or plaits like those of a fan, as seen in some leaves or in some young flowers.



Plicate flower

PLIERS: A hand tool consisting of a pair of metal first-class levers joined at a fulcrum positioned close to one end of the levers, creating short jaws on one side of the fulcrum and longer handles on the other side. This creates a mechanical advantage, allowing the force of the hand's grip to be amplified and focused on an object with precision. It has long jaws used for holding, bending, twisting or cutting wire.

PLIESTESIAL: Monocarpic; living several years before flowering and dying promptly thereafter. An example is *Agave*.

PLIOCENE: The fifth and last epoch of the Tertiary period of geological time, about 5.2 to 1.6 million years ago following the Miocene Epoch but preceding the Pleistocene Epoch of the Quaternary Period. During this period, the earliest hominid, the humanlike ape 'australopithecines' evolved in Africa and mammals grew larger than those of earlier epochs.

PLIOSAUR: The prehistoric carnivorous reptiles from the Jurassic and Cretaceous Periods. They descended from the plesiosaurs but have shorter necks and longer heads and jaws.

PLOCOCARPIUM: A fruit which consists of follicles positioned around an axis.

PLOIDY: A term indicating the number of sets of chromosomes present in an organism, e.g. haploid (one) or diploid (two).

PLOT: To locate or mark exact position of points on a coordinate system, using x and y coordinates; or to draw a curve through points on a graph.

PLOTTER (GRAPH PLOTTER): A graphics printer that draws pictures or diagrams under computer control. It is used to draw business charts, architectural plans, engineering drawings, etc.

PLOUGH: An agricultural implement used for tilling the soil.

PLOUGH BODY: The main part of the plough consisting of the frog, mouldboard, share and landside.

PLUG: 1. In electricity, it is a device used to connect an appliance to a power supply. It has a cable connection of two or three metal pins/bars that fit into a socket. 2. Spark plug is a device designed to fit in each cylinder of a petrol engine such as a car to produce electric spark for igniting the mixture of petrol and air in an internal combustion engine. 3. A piece of wood or other material shaped to fit a hole or aperture tightly, and used to block it. 4. In geology, it is a mass of igneous rock filling the vent of a volcano.

PLUG-AND-PLAY (PNP): An ability of a computer to detect and configure a new piece of hardware automatically, without the requirement of the user to physically configure the hardware device with jumpers or dip switches.

PLUG-FLOW: A packed bed flow reactor is commonly called a fixed bed reactor and the term plug-flow is also used to indicate that no attempt is made to back-mix the reaction mixture as it passes through the catalyst bed.

PLUG-IN (ADD-IN; ADD-ON): A software utility or other program that can be added to a primary or existing program to increase memory or add graphics or extend the capabilities of larger programs in a computer. They can come in the form of chips, circuit boards, cartridges, video adapters, graphics accelerators, sound cards, accelerator boards and internal modems. For example, Excel add-ins complement the basic functionality offered by Excel, Microsoft Bookshelf is an add-in for Word and Analysis Toolpak is an add-in for Microsoft Excel. Also, Mozilla Firefox, a popular Web browser, supports add-ons such as the Google toolbar, ad blockers and Web developer tools. Some computer games support add-ons that provide extra maps, new characters or give the player game-editing capabilities.

PLUM BOX: A viral disease affecting plants, damsons and peaches. The fruit has dark blotches and ripens prematurely and is often sour.

PLUMAGE: The name given collectively to the feathers covering the body of a bird.

PLUMBAGO: 1. Also known as **graphite**, it is a mineral allotrope of carbon and one of the few non-metals that can conduct electricity. 2. An evergreen flowering shrub or climber of the genus *Plumbago*, which is widely distributed in warm regions and grown elsewhere as a greenhouse or indoor plant.

PLUMBANE (LEAD(IV) HYDRIDE): An extremely unstable gas with the chemical formula PbH_4 , molar mass 211.23 g/mol and boiling point -13°C (260 K). It is formed by the action of acids on magnesium-lead alloys. It was first reported in 1924, although doubts have since been expressed about the existence of the compound. Derivatives of plumbane include lead tetrafluoride (PbF_4) and tetraethyl lead ($(\text{CH}_3\text{CH}_2)_4\text{Pb}$).

PLUMBATE: A compound formed by the reaction of lead(IV) oxide (PbO_2) with alkali. The oxides of lead are amphoteric and react to give plumbate ions. Plumbate salts contain either the hydrated plumbate anion $\text{Pb}(\text{OH})_6^{2-}$ or the anhydrous anions, PbOH_3^{2-} (meta-plumbate).

or ortho-plumbate (PbO_4^{4-}). For example, dissolving PbO_2 in a hot, concentrated aqueous solution of potassium hydroxide forms the potassium salt $\text{K}_2\text{Pb}(\text{OH})_6$. The anhydrous salts may be synthesised by heating metal oxides or hydroxides with PbO_2 . All plumbate(IV) salts are very strong oxidising agents. Some hydrated plumbate(IV) salts decompose upon dehydration. They are also decomposed by carbon dioxide.

PLUMBEOUS: Having the colour of lead.

PLUMBIC COMPOUNDS: Compounds of leads in higher (+4) oxidation state. An example of plumbic oxide is lead(IV) oxide (PbO_2).

PLUMBOUS COMPOUNDS: Compounds of lead in lower (+2) oxidation state. An example of plumbous oxide is lead(II) oxide (PbO).

PLUMBOUS NITRATE (LEAD(II) NITRATE; LEAD NITRATE; LEAD DINITRATE): A white, colourless crystal or powder with the molecular formula $\text{Pb}(\text{NO}_3)_2$, molar mass 331.2g/mol, density 4.53 g/cm³ at 20 °C and a melting point of 270 °C (543 K) at which it decomposes. Together with lead ethanoate (lead acetate), they are the only water-soluble compounds of lead. It can be made by two methods: dissolving metallic lead in aqueous nitric acid:

$\text{PbO} + 2\text{HNO}_3 \rightarrow \text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{O}$, or by dissolving lead(II) oxide in nitric acid: $\text{PbO} + 2\text{HNO}_3 \rightarrow \text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{O}$. When heated, it decomposes readily into oxygen, nitrogen(IV) oxide gas and lead(II) oxide: $2\text{Pb}(\text{NO}_3)_2 \Rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$. It is a hazardous substance, which is injurious to health if inhaled or ingested. It is not combustible but it accelerates the burning of combustible materials and is sometimes used in pyrotechnics such as fireworks.

PLUMBOUS OXIDE (LEAD(II) OXIDE; LEAD MONOXIDE): Lead(II) oxide is the International Union of Pure and Applied Chemistry's (IUPAC) name for lead monoxide or litharge. It is a yellow crystalline inorganic compound with the chemical formula PbO . It exists in two polymorphs, red, having a tetragonal crystal structure and yellow, having an orthorhombic crystal structure. Both forms occur naturally as rare minerals. The red form is known as **litharge** and the yellow form is known as **massicot**. It can be prepared by heating the nitrate and is manufactured by heating molten lead in air at approximately 600 °C (873 K). If the temperature used is lower than the melting point of the oxide, the product is massicot; above this, litharge is formed. Variations in the temperature and in the rate of cooling give rise to crystal vacancies and red orange and brown forms of litharge can be produced. The oxide is amphoteric, dissolving in acids to give lead(II) salts and in alkalis to give plumbates. It is used extensively in manufacturing of lead glasses and ceramic glazes as well as in fine dinnerware.

PLUME: 1. A long, soft feather or arrangement of feathers used by a bird for display or worn by a person for ornament. 2. The gaseous and aerosol effluents which are emitted from a chimney and the volume of space which they occupy (often visible). The shape of the plume and the concentrations of pollutants in it at various points along the path of the plume are sensitive functions of the meteorology, local topography and the chemistry which occurs in the plume. Urban plumes are observed over many urban areas and downwind of these areas in which a combination of man sources of pollution are concentrated.

PLUMERIA (FRANGIPANI): A plant genus made up of about 29 species in the Apocynaceae (dogbane) family containing primarily deciduous shrubs and small trees. It is native to Central America, South America and the Caribbean and can be cultivated in tropical and sub-tropical regions. It has large, oblong, leathery, green leaves that are arranged alternately on squat branches and cluster at the branch ends. It has waxy flowers that consist of five petals arranged in a tubular funnel shape, the colours varying from species to species. When a limb is broken or the bark is cut, a white sap comes out, which like Oleander and Euphorbia, causes irritation to the eye and skin. Examples of the species include *Plumeria rubra*, a tree that grows up to 7.6 metres in height with large leaves of about 51 centimetres long and pink to red flowers 10 cm in diameter; *Plumeria alba* grows

to about 12.2 metres high, with leaves about 0.3 metres long and white flowers with a yellow centre; and *Plumeria obtusa* grows to about 7.6 metres with leaves 20 centimetres long and flowers (white with yellow centres) 7.6 centimetres in diameter. Plumeria species are used as garden plants because of their beautiful flowers and fragrance.

PLUMOSE: Having many fine filaments or hairs arranged on both sides of the axis, as seen in a feather.

PLUMULE: 1. A down feather, which are fine feathers found under the tougher exterior feathers of birds. 2. Part of a seed embryo situated at the end of the hypocotyl, consisting of the epicotyl that develops into the shoot bearing the first true leaves of the plant.

PLURALITY: 1. A number greater than one. 2. The qualification of a sentence by expressions such as 'many', 'most', 'few' and 'a few'. Plural logics study such concepts by analogy with the standard quantifiers of predicate calculus.

PLURI-: A prefix meaning many or several.

PLURICELLULAR (PLURILOCULAR): Having many cells.

PLURICIPITAL: Having more than one head, as in a highly branched caudex.

PLURIPOTENT STEM CELL: A cell within bone marrow that is a progenitor for any kind of blood cell. Pluripotent stem cells are master cells, which can make cells from all three basic body layers, potentially producing any cell or tissue the body needs to repair itself. Types of pluripotent stem cells include the following: induced pluripotent cell (iPS cells), true embryonic stem cell (ES cells) derived from embryos, embryonic stem cells made by somatic cell nuclear transfer (ntES cells) and embryonic stem cells from unfertilised eggs (parthenogenesis embryonic stem cells or pES cells).

PLURISERIATE: Arranged in many rows or series.

PLUS: 1. Increased by the addition of a quantity or value; symbolised by '+', for example, 2 plus 3 is written as 2+3. The plus symbol, +, written before a number indicates the number has a positive value and is greater than zero, for example +2 is read as 'plus two'. 2. Operated upon by any defined addition, such as the symmetric difference. For example, the first operator in both $a + (-b)$ and $A \oplus (A \cup B)$ is read as 'plus'.

PLUTO: The smallest and usually outermost planet of the solar system, which inclines at 17.14°. Its mean distance from the Sun is 5.9×10^9 km. Pluto has a diameter of about 2,340 km, less than 1% that of Earth's mass and only about 6% of Earth's surface gravity. Its estimated average surface temperature is about -235 °C (38 K). Its thin atmosphere contains nitrogen, methane and perhaps other heavier gases. It was discovered in 1930 by Clyde W. Tombaugh (1906-97) and named after the Greek god of the underworld. Pluto used to be the smallest planet of the nine planets until reclassified as a dwarf planet in August 2006. A **dwarf planet** is defined as a spherical celestial body revolving about the Sun, similar to a planet but not large enough to gravitationally clear its orbital region of most or all other celestial bodies.

PLUTONIC ROCK: Igneous rock such as granites and gabbro, derived from magma that has cooled and solidified slowly at great depth in the crust of the Earth.

PLUTONIUM: A silvery, poisonous metal, with six allotropic forms with the symbol Pu, atomic number 94, mass number of most stable isotope 244, melting point 64 °C (914 K), boiling point 3,232 °C (3,505 K) and specific gravity 19.84 at 20 °C (293 K). It is very radioactive and tarnishes in air, taking on a yellow cast when oxidised. It dissolves in hydrochloric, hydriodic and perchloric acids and reacts with the halogens, carbon, nitrogen and silicon. Plutonium is found naturally in very small quantities in association with uranium ores. One of its isotopes, Pu -239, is used in atomic weapons and in some nuclear reactors.

PLUVIOMETER (RAIN GAUGE): An instrument for measuring precipitations such as rain, snow, hail or sleet over a set period of

time. It consists of a cylinder that catches rain covered by a funnel-like lid. If 10 cm of rain collects in the cylinder, it means 10 cm of rain has fallen.

PLYWOOD: A manufactured panel of wood noted for its strength, durability, lightness, rigidity and resistance to splitting and warping. It is composed of an odd number of thin sheets of wood glued together under pressure with grains of the successive layers at right angles. The glues commonly used are starch pastes, animal glues or casein and waterproof synthetic resins. It is widely used in building construction.

PM₁₀: Particulate matter that is less than 10 micrometre (10µm) in diameter, consisting of clusters of small particles, such as carbon particles in the air that come mostly from vehicle exhausts. Such particulates are dangerous to health because they are not filtered out of the inhaled air.

PM_{2.5}: Fine particulates that are so small that they can only be detected using an electron microscope and generally emitted from activities such as industrial and residential combustion and from vehicle exhaust. They are also formed in the atmosphere when gases such as sulfur dioxide, nitrogen oxides and volatile organic compounds, emitted by combustion activities, are transformed by chemical reactions in the air. Chemical composition of particulates varies widely, depending on location and time of year.

PMOLED (PASSIVE MATRIX OLED): A type of organic light-emitting diode (OLED display technology in which each row or line in the display is controlled sequentially (one at a time). PMOLED does not contain a storage capacitor and so the pixels in each line are actually off most of the time. To compensate for this more voltage is used to make them brighter. Another type of organic light-emitting diode (OLED) is the active-matrix organic light-emitting diode (AMOLED) that contains has a storage capacitor that makes the line pixels light all the time allowing for higher resolution and larger display sizes and uses less power. PMOLED displays are usually small and are used in small gadgets and displays.

PNEUMATIC: 1. Referring to a machine that moves mechanical energy from one place to another by means of compressed air or an object containing compressed air such as pneumatic pump and pneumatic car tyre. 2. In zoology, it is having cavities filled with air such as the bones of certain birds.

PNEUMATIC DETECTOR: A pressure-sensitive detector based on the pressure increase of a gas. A special type is the Golay cell where the pressure change is detected by observing the deflection off one of the chamber walls.

PNEUMATIC DISTRIBUTOR: A machine which conveys fertiliser from a hopper to nozzles for spreading by a stream of air.

PNEUMATIC DRILL: A drill operated by compressed air. It is used in mining and tunnelling, for drilling short holes and in road repairs for breaking up pavements.

PNEUMATIC GRAIN DRILL: A machine which sows grain, the seed being moved from a hopper down the drill pipe by compressed air.

PNEUMATOPHORE (AEROPHORE, BREATHING ROOT, RESPIRATORY ROOT): An erect root that protrudes some distance above soil level. Pneumatophores are formed in large numbers by certain plants, for example, *Sonneratia* and some mangrove species, growing in areas with waterlogged badly aerated soils. The surface of the pneumatophore is perforated by numerous lenticels that promote gaseous exchange.

PNEUMOCONIOSIS: A restrictive disease of the lungs caused by inhalation of various kinds of dust, especially mineral or metallic dust, during years of exposure, often in mines extracting minerals such as coal, asbestos, iron, tin, bauxite and tin. Prolonged inhalation of cotton fibre particles can also cause pneumoconiosis.

PNEUMOCYTE: One of the two epithelial cells lining the alveoli in the lung. They are the type 1 and type 2 pneumocytes: Type 1 pneumocyte cell is involved in the process of gas exchange between the alveoli and the capillaries. Type 2 pneumocyte, the smaller of the two, is responsible for the secretion of pulmonary surfactant, which reduces surface tension in the alveoli.

PNEUMONIA: A serious illness with inflammation of the lungs causing difficulty in breathing. It is usually caused by many factors, including bacterial infections, viral infections, fungal infections and the inhalation of chemical irritants.

PNEUMOTHORAX: Accumulation of air in the pleural cavity, between the lungs and the chest wall causing it to expand and thus compress the underlying lung, which may then collapse. Symptoms include a sharp pain in one side of the chest and shortness of breath.

PNG (PORTABLE NETWORK GRAPHICS): An open source raster graphics file format in which graphical images are stored as a bit map (arrangement of dots) and transmitted.

PNICTOGENS: The elements of group 15 (VA) of the Periodic Table, which includes nitrogen (N), phosphorus (P), arsenic (As), antimony (Sb) and bismuth (Bi). Each member of this family contains five valence electrons. Due to these valence electrons, the members of group 15 are able to form double and triple bonds. Elemental nitrogen is a diatomic molecule held together by a triple bond. The elements arsenic (As), antimony (Sb) and bismuth (Bi) form a range of solid-state ternary compounds with interesting electrical and magnetic properties. Compounds of this type are known as pnictides.

POACEAE: The grasses, the Gramineae or Poaceae, which is a very large family of plants including bamboo and cereals such as wheat, barley, oats and sorghum and maize.

POCHHAMMER SYMBOL: The symbol $(a)_n$, used to represent either the rising factorial or the falling factorial. For rising factorial, it is written as $(a)_n = a(a+1) \dots (a+n-1) = \frac{\Gamma(a+n)}{\Gamma(a)}$.

POCKELS CELLS: An electro-optic device which changes the state of polarisation of light beam passing through it by producing birefringence in the optical medium as a result of the application of a constant or varying electric field. Pockels cells contain crystals that lack inversion symmetry, such as lithium niobate or gallium arsenide and in other noncentrosymmetric media such as electric-field poled polymers or glasses. Pockels cells are used in conjunction with polarisers in varying applications. They can be used as Q-switches, preventing feedback in a laser cavity and for quantum key distribution.

POD: 1. A dehiscent fruit of a member of the family Leguminosae (pulse family). Examples are legumes such as beans, lentils, peas and soybeans. At maturity the pod splits along its two seams and releases the enclosed seeds. 2. A protective covering that encases the eggs of some insects and fish.

PODOCARP: 1. A fruit borne on a stipe. 2. An evergreen shrub or tree that belongs to the family Podocarpaceae. They are mostly conifers.

PODCAST (AUDIOBLOG): Often abbreviated as **pod**, it is a multi-media digital file distributed over the Internet that a user can download and listen to on a computer or any portable audio device such as iPod capable of playing MP3 files.

PODOCYTE (VISCERAL EPITHELIAL CELLS): A type of epithelial cell found in Bowman's capsule in the kidneys that wrap around the capillaries of the glomerulus. Podocytes possess major footlike processes, each supporting a series of minor processes. These minor processes are interwoven with those from other podocytes to form a number of slits, through which filtration can occur. These slits (approximately 0.1µm wide) permit the passage of all plasma constituents but act as a barrier to blood cells.

PODOGYNE: Species of flowering plants in the mint family.

PODSOL (PODZOL): A type of zonal acid infertile soil (a pedalter) found in regions of heavy rainfall and long cool or cold winters. The soils of the northern coniferous forest zone of North America and Eurasia are typical examples. The soil is heavily leached by the sudden snow-melt in the spring that washes out lime and iron compounds. Thus beneath the humus layer in the A horizon there is a bleached horizon composed of quartz sand. Iron compounds accumulate in the B horizon, which stains brown and has a clayey texture. These compounds may form an impermeable hard pan that, if developed sufficiently, can stop water percolating from above, resulting in the A horizon becoming waterlogged.

POIKILOTHERM (COLD-BLOODED ANIMAL): An organism such as fish or reptile whose internal temperature varies with the ambient environmental temperature. Poikilothermy is characteristic of all animals except birds and mammals which maintain their body temperatures by homeothermy: they are warm-blooded.

POINCARÉ GROUP (POINCARÉ TRANSFORMATIONS): In physics and mathematics, it is the group of isometries of Minkowski space-time.

POINCARÉ SPHERE: A geometrical representation of the state of polarisation of a wave of monochromatic radiation predicting how any given retarder will change the polarisation form. This was conceived by Henri Poincaré in about 1892. The upper and lower poles represent left and right-circularly polarised light. Points on the equator indicate linear polarisation. Other points on the sphere represent elliptical polarisation. An arbitrarily chosen point H on the equator designates horizontal polarisation and the diametrically opposite point V designates vertical polarisation. A general point P on the surface of the (unit radius) Poincaré sphere is specific in terms of the longitude and the latitude.

POINCARÉ STRESSES: Non-electric forces postulated to give stability to a model of the electron.

POINCARÉ, JULES HENRI (1854- 1912): French mathematician, physicist, engineer astronomer and philosopher who made major contributions to physics and mathematics. He formulated Poincaré's conjecture and originated the study of automorphic functions. He also made major contributions in many fields of mathematics, celestial mechanics, fluid mechanics, etc. Some specific topics include algebraic topology, the theory of Abelian functions, algebraic geometry, hyperbolic geometry, number theory, theory of electromagnetism, etc.

POINCARÉ'S CONJECTURE: A conjecture in topology that a simply connected three-dimensional compact manifold is topologically equivalent to a three-dimensional sphere precisely when a certain algebraic condition is satisfied.

POINCARÉ'S LEMMA: The result that every closed differential form defined on a simply connected region is exact.

POINT: A dimensionless geometric object having no properties other than location or place; also the intersection of two straight lines or of a straight line and a plane. More generally, a point is an element in a geometrically described set.

POINT AND CLICK: In computing, it is an interface for reacting with a computer in which the user points at an object using a cursor and a mouse and clicking to activate it.

POINT AT INFINITY: 1. An ideal element in affine geometry. 2. The point added in the one-point compactification of the complex plane. The extended plane may then be identified with a sphere of which it is the conformal image under stereographic projection. The point at infinity then corresponds to the pole of the projection.

POINT CHARGE: A charge that has a very small physical size compared to its distance from other objects. While an electron can for many purposes be considered a point charge, its size can be characterised by length scale known as the electron radius.

POINT DEFECT: An imperfection in the lattice structure of a crystal that affects only one or two lattice sites. It consists either of a missing atom or ion, creating a vacancy in the lattice.

POINT DISCHARGE: A silent, nonluminous, gaseous electrical discharge from a pointed conductor maintained at a potential that differs from that of the surrounding gas. It occurs when one of two electrodes in a gas has a shape causing the electric field at its surface to be significantly greater than that between the electrodes.

POINT ESTIMATION: Point estimation of a population parameter allows obtaining a unique value calculated from a sample. This value will be considered the estimate of the unknown parameter.

POINT EVALUATION: A linear functional that assigns to each member of a space of function its value at a given point, so that $\delta_f(t) = f(t)$.

POINT GROUP: A group consisting of the elements that leave a molecule or a point unchanged from the previous when applied to a molecule. This has been used in classifying compounds into various point shapes.

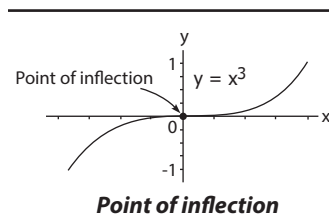
POINT MASS: In mathematics, a measure whose support is singleton.

POINT MEASURE: In mathematics, μ , for which there is a point, p , such that for every measurable set E , $\mu(E) = 1$, if p is in E and $\mu(0) = 0$ if not; that is, $\mu(E) \dots \chi_E(p)$, the characteristic function of E .

POINT MUTATION (SUBSTITUTION MUTATION): A replacement of a single base nucleotide with another nucleotide of the genetic material in DNA or RNA. A gene in which such a change has occurred is known as a **mutant gene** or **allele**. The DNA sequence can be altered in several ways such as by insertion, substitution deletion and inversion. Point mutations can result in a misreading of the genetic code during the translation phase of protein synthesis and can change the order of amino acids making up a protein, which may or may not affect the function of that protein. Types of point mutations include the following: transitions, transversions, nonsense mutations, silent mutations, conservative mutations and non conservative mutations. Gene mutations are most commonly caused by environmental factors such as chemicals, radiation and ultraviolet light from the sun can cause mutations. They are also caused by errors made during mitosis and meiosis. Point mutations can cause several genetic disorders such as sickle cell anaemia, cystic fibrosis, Tay-Sachs disease, Huntington disease, haemophilia and some cancers.

POINT OF CONTACT (TANGENCY POINT): The point at which a tangent line or plane is tangent to a given curve or surface.

POINT OF INFLECTION: A point on a curved line in which the tangent changes direction of movement. Moving from one side of the point of inflection, the slope of the tangent to the curve increases. Moving away from it on the other side, it decreases. For example, the curve $y = x^3$ has a point of inflection at $x = 0$. If the second derivative of a function at the point x is zero then, the point x is a point of inflection.



POINT OF LAY: A term referring to pullets (young hen, less than 1 year old) that are approaching the point when they will lay their first eggs.

POINT OF PRESENCE (POINT-OF-PRESENCE, PoP): In telecommunications, it is a point of access provided by an Internet service provider (ISP) that enables users to establish a connection with the Internet through communication devices such as a telephone.

It consists of servers, modems, routers, ATM switches and other communication equipment that are permanently connected to the network. An Internet access provider may have several PoPs distributed at different locations in their area of operation to enable remote users to connect to the Internet.

POINT OF SALE TERMINAL (POS TERMINAL): A computer terminal at a point of sale that processes data such as to record and track customers orders, process credit and debit cards, print out a fully itemised receipt for the customer, connect to other systems in a network and manage inventory.

POINT OF ZERO CHARGE (P.Z.C.): A surface charge is at its point of zero charge when the surface charge density is zero. It is a value of the negative logarithm of the activity in the bulk of the charge-determining ions.

POINT PROCESS: In statistics, a sequence of isolated events, occurring in time or some space, where the interval between any pair of consecutive events follows a common probability distribution. An example is emissions from a radioactive source.

POINT SOURCE: A source whose angular extent is so small that it cannot be measured with existing equipment.

POINT SOURCE POLLUTION: Pollution resulting from any single identifiable source, such as a pipe, ditch, tunnel, well, wastewater treatment plants, concentrated animal-feeding operation, wastes from industries, etc.

POINT SYMMETRY: A figure has point symmetry if it is a reflection image of itself through a centre of symmetry. It occurs when there exists a position or a central point on an object with the following conditions: the central point splits the object or shape into two parts; every part on each has a matching part on the other that is the same distance from the central point; and both parts face different directions.

POINT TO POINT PROTOCOL (PPP): A communication protocol that enables a user to utilise their dialup connection (commonly a modem) to connect to other network protocols like TCP/IP, IPX, etc. It is a set of rules that defines how a modem exchanges packets of data with other systems on the Internet.

POINT-CONTACT TRANSISTOR: A transistor having a base electrode and two or more point contacts located near each other on the surface of an n -type semiconductor. It was the first type of solid-state electronic transistor ever constructed. It was made by researchers John Bardeen and Walter Houser Brattain at Bell Laboratories in December 1947.

POINT-SET TOPOLOGY (GENERAL TOPOLOGY): A branch of topology concerned with how to put a structure on a set in such a way as to generalise the idea of continuity for maps from the real numbers to itself. A topology on a set X is a certain set of so-called open subsets of the set X which satisfy various axioms. The set X together with this topology is called a topological space.

POINT-SLOPE FORM: A form of equation of a line where $m = \frac{y - y_1}{x - x_1}$, where m is the gradient or slope of the line.

POINT-TO-POINT MAPPING (CARRIER; MULTIVALUED FUNCTION; MULTIFUNCTION; SET-VALUED FUNCTION): A mapping that associates at least one input element with multiple output elements. It is a mapping from a set into the power set of another set. In mathematics a mapping that associates a number of different elements of the output with the same element of the input; a mapping from a set into the power set of another set. The inverses of trigonometric, hyperbolic, exponential and integer power functions are multivalued. For example, the function z^2 maps each complex number z to a well-defined number z^2 , while its inverse function $\sqrt{z}$ maps, for example, the value $z = 1$ to $\sqrt{1} = \pm 1$.

POINTER: 1. In general, a pointer is a term used to describe a suggestion, tip, or a hint about a particular product or subject. 2.

In computer science, a programming language object, whose value refers to another value stored elsewhere in the computer memory using its memory address. A pointer contains the address of a location in memory. The location is the starting point of an allocated object, such as an object or value type, or the element of an array. Obtaining the value stored at that location is known as **dereferencing the pointer**. 3. Also known as **pointing device**, it is an input hardware device used to move the pointer or cursor in a computer program or Graphic User Interface (GUI) operating system.. The major pointing device is the mouse for the desktop computer and the touchpad for the laptop. It is used to select commands and options or to manipulate any object or text on the screen.

POINTWISE CONVERGENCE: In mathematics, it is the convergence of a sequence of functions to a particular function in terms of the convergence of their values at each point of their domain, so that a pointwise convergent sequence is not necessarily uniformly convergent.

POISE: A centimetre–gram–second (cgs) unit of dynamic viscosity with symbol P. It is equal to the tangential force in dynes per square centimetre required to maintain a difference in velocity of one centimetre per second between two parallel planes of a fluid separated by one centimetre. One poise is equal to 10^{-1} Nsm^{-2} . It is named after Jean Louis Marie Poiseuille. The International System of Units (SI) is the currently used system of units, and the pascal second (Pa·s) has replaced the poise.

POISEUILLE'S EQUATION (HAGEN-POISEUILLE LAW; POISEUILLE LAW): An equation relating the volume flow of an incompressible fluid through a circular tube, V , to the pressure difference p , between the ends of the tube. The volume flowing per second, $\frac{dv}{dt}$, is given by: $\frac{dv}{dt} = \frac{\pi p r^2}{8l\eta}$, where r is the radius and l is the length of the tube and η is the viscosity of the fluid.

POISON: 1. A substance which, if absorbed by an organism at a certain dosage causes harm and death. Poisoning is the act of an organism being poisoned using a poisonous substance 2. A substance that absorbs neutrons in a nuclear reactor and therefore slows down the reaction. 3. A substance that prevents the activity of a catalyst.

POISONOUS GAS (POISON GAS): Any toxic chemical agent in the form of a gas or vapour-forming liquid or solid that may be used in chemical warfare to kill or cause injuries through inhalation or contact. Examples of such toxic gases are carbon monoxide, chlorine, nitrogen dioxide and phosgene.

POISSON DISTRIBUTION: A discrete probability distribution that expresses the number of events that occur randomly in a given time at an average rate and independently of the time since the last event, written as $Po(l)$. The distribution was first introduced by Siméon-Denis Poisson (1781–1840), a French physicist and mathematician.

POISSON EQUATION (POISSON'S DIFFERENTIAL EQUATION): A generalisation of the Laplace equation in which there is a function $f(x, y, z)$ on the right hand side of the equation. This implies that the Poisson equation is of the form $\nabla^2 \phi = f(x, y, z)$ where ∇^2 is the three-dimensional Laplacian.

POISSON, SIMÉON DENIS (1781 -1840): French mathematician and physicist, who studied under Laplace and Lagrange. He worked on the mathematical theory of electricity and magnetism, integrals, differential equations and Fourier analysis and corrected Laplace's second order partial differential equation for potential. He contributed to celestial mechanics by extending the work of Lagrange and Laplace on the stability of planetary orbits and his gravitational equations are now used in interpreting satellite orbits. His approximation to the binomial distribution is used to model traffic flow and radioactive decay.

POISSON'S INTEGRAL: In mathematics, the integral $U^*(r, \theta)$ defined as $\frac{1}{2\pi} \int_0^{2\pi} \frac{u(\phi)[a^2 - r^2]}{a^2 - 2ar \cos(\theta - \phi) + r^2} d\phi$, where $U(\Phi)$ is continuous

on the boundary of a disk of radius a in the complex plane and the coefficient of $U(\Phi)$ in the integrand is the Poisson kernel. from the x -axis at which a point in the xy -plane lies.

POISSON'S RATIO: The ratio of a lateral strain to the longitudinal strain when a rod is stretched by forces which are applied at its ends and which are parallel to the rod's axis.

POISSON'S SPOT (ARAGO'S SPOT; FRESNEL BRIGHT SPOT): A bright spot at the centre of the shadow of a circular opaque obstacle occurring as a consequence of the wave theory of light.

POLAR (POLARITY): Having parts or areas with opposed or contrasting properties, such as positive and negative charges, head and tail.

POLAR ANGLE: The angle, measured counterclockwise, between the polar axis (x -axis) of a system of polar coordinates and the line joining the origin to any given point.

POLAR AXIS: The fixed axis in a system of polar coordinates from which the polar angle is measured anticlockwise.

POLAR BODY: A structure produced and ultimately discarded in the development of an oocyte, containing little or no cytoplasm but having one of the nuclei derived from the first or second meiotic division.

POLAR COMPOUND: A compound such as water (H_2O) in which the electric charge is not symmetrically distributed resulting in partial or complete separation of charges and formation of definite positive and negative poles; or a compound with a permanent dipole. Compounds in which charge separation is incomplete is said to be a polar covalent compound whereas compounds in which there is complete separation of charges is called an ionic compound. Hydrogen chloride (HCl) is a polar covalent compound while sodium chloride ($NaCl$) is an ionic compound.

POLAR COORDINATES: A pair of coordinates, denoted by (r, θ) that locate a point in a plane by the length of its radius vector, r , and the angle, θ , this vector makes with the polar axis. The polar coordinates of the point whose Cartesian coordinates are (x, y) are $(r, \theta) = (\sqrt{x^2 + y^2}, \arctan(x/y))$, so that in the Argand diagram, this point represents the complex number $x + iy$, where r and θ are respectively the modulus and amplitude (anomalus or azimuth) of the complex number; conversely, the Cartesian coordinates of (r, θ) are $(x, y) = (r \sin \theta, r \cos \theta)$.

POLAR COVALENT BOND: A covalent bond in which there is unequal sharing of electrons between atoms in a molecule. Examples of molecules having polar covalent bonds are water and amines. The biological consequence of polar covalent bonds is that it can lead to the formation of a weak bond called a hydrogen bond.

POLAR DECOMPOSITION: In mathematics, the representation of an invertible matrix with complex entries as the composition of a positive semidefinite matrix and a unitary matrix or the similar representation of a normal bounded linear operator on a complex Hilbert space.

POLAR DECOMPOSITION THEOREM: In mathematics, the theorem that any invertible matrix over the complex numbers may be written as the product of a unitary matrix and a positive definite Hermitian matrix.

POLAR DISTANCE: The angular distance of an object from a celestial pole. It is equal to 90° minus the object's declination and is sometimes used as an alternative coordinate to declination since it avoids negative values.

POLAR EASTERLIES: Dry, cold, prevailing winds that blow from east to west between 60° and 90° latitude in both hemispheres.

POLAR EFFECT: For a reactant molecule RY , the polar effect of the group R comprises all the processes whereby a substituent may modify the electrostatic forces operating at the reaction centre Y , relative to the standard. These forces may be governed by charge separations arising from differences in the electronegativity of atoms (leading to the presence of dipoles), the presence of unipoles or electron delocalisation. It is synonymous with electronic effect or electrical effect of a substituent as distinguished from other substituent effects, for example, steric effects. Sometimes, however, the term 'polar effect' is taken to refer to the influence, other than steric, that non-conjugated substituents exert on reaction rates, i.e. effects connected with electron delocalisation between a substituent and the molecular framework to which it is attached are excluded. Polar effect is then not synonymous with electronic effect

POLAR EQUATION: The equation defining an algebraic curve expressed in polar coordinates. An example is the equation of conics section.

POLAR EQUATION OF A CONIC SECTION: The polar equation of any conic section where the pole is at a focus and the polar axis is perpendicular to the directrix. It can be written as $r = \frac{2ep}{1 + e \cos \theta}$,

where e is the eccentricity and $2p$ is the distance from the focus to the directrix.

POLAR FLAGELLATION: The presence of flagella at one or both ends of a bacterial cell.

POLAR GROUP: A hydrophilic or water loving group.

POLAR MOLECULE: A molecule that has a permanent dipole moment or a molecule in which there is some separation of charge in the chemical bonds so that one part of the molecule has a positive charge and the other a negative charge. The dipole moments of polar molecules result in stronger intermolecular attraction, increased viscosities, higher melting and boiling points and greater solubility in polar solvents than in nonpolar molecules. For example, a molecule of water (H_2O) is polar because of the unequal sharing of its electrons between oxygen and hydrogen in which the former has larger electronegativity than the latter. Another example is the hydrogen fluoride (HF) molecule, because the covalent bond electrons are displaced towards the more electronegative fluorine atom. Some others are ammonia (NH_3), ozone (O_3) and sugars such as sucrose, which have many polar oxygen-hydrogen ($-OH$) groups and are highly polar.

POLAR MUTATION: A mutation in one gene that reduces the expression of downstream genes or operons. It can also affect the expression of the gene in which it occurs, if it occurs in a transcribed region. Polar mutations are found only in organisms containing polycistronic mRNA.

POLAR NUCLEI: Either of two haploid nuclei in the centre of the flowering plant embryo sac that eventually fuse with a male gamete nucleus to form the triploid endosperm nucleus, which later divides to form the endosperm.

POLAR ORBIT: An orbit with an inclination of 90° . A spacecraft following such an orbit has access to virtually every point on Earth's (or some other planet's) surface, since the planet effectively rotates beneath it. This capability is especially useful for mapping or surveillance missions. An orbit at another inclination covers a smaller portion of the Earth, omitting areas around the poles. A polar orbit covers the entire globe every 14 days.

POLAR REVERSAL: Alternation in polarity of the Earth's magnetic poles.

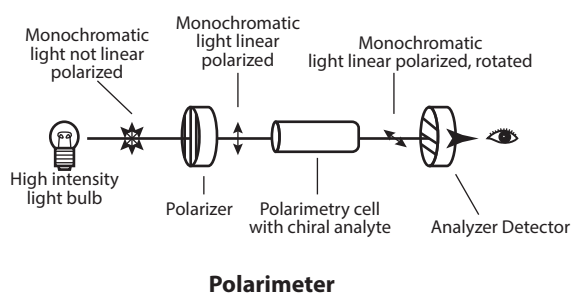
POLAR SET: In mathematics, the set of vectors, denoted S^0 , such that $\langle x, s \rangle \leq 1$ for all s in a given set S in real Hilbert space; it is closed, convex and contains zero. When S is a vector subspace, S^0 coincides with the orthogonal complement and when S is a cone S^0 coincides with the polar cone or normal cone of vectors such that $\langle x, s \rangle \leq 0$ for all s in S .

POLAR SOLVENT: A compound, such as water or liquid ammonia that is composed of polar molecules. Polar solvents can dissolve ionic compounds or covalent compounds that ionise. **Nonpolar solvents**, such as benzene, will only dissolve hydrocarbons, such as oils, grease and waxes.

POLAR STAR (POLARIS): The northern pole star, composed of a binary star and a Cepheid variable. It is visible from the Northern Hemisphere toward which Earth's axis points. Because it always indicates due north from an observer anywhere on the Earth, the Polar Star is important for navigation.

POLAR VECTOR: A vector in two dimensions specified as a magnitude or length and a direction or angle. It reverses its sign when the coordinate system is changed to a new system by a reflection in the origin, for example, $x'_i = -x_i$.

POLARIMETER: A scientific instrument for determining the rotation of polarisation of light as the light passes through a solution containing an optically active substance. Some chemical substances are optically active and light will rotate either to the left (counter-clockwise) or right (clockwise) when passed through these substances. The amount by which the light is rotated is known as the angle of rotation. The polarimeter is made up of two Nicol Prisms (the polariser and analyser). The polariser is fixed and the analyser can be rotated. The prisms may be compared to as slits S_1 and S_2 . The light waves may be considered to correspond to waves in the string. The polariser S_1 allows only those light waves which move in a single plane. This causes the light to become plane polarised. When the analyser is also placed in a similar position it allows the light waves coming from the polariser to pass through it. When it is rotated through the right angle, no waves can pass through the right angle and the field appears to be dark. If now a glass tube containing an optically active solution is placed between the polariser and analyser the light now rotates through the plane of polarisation through a certain angle, the analyser will have to be rotated in some angle. Polarimeters are used in the sugar industry for determining the quality of both juice from sugar cane and the refined sucrose. It can also be used to measure the ratio of enantiomers in solutions.



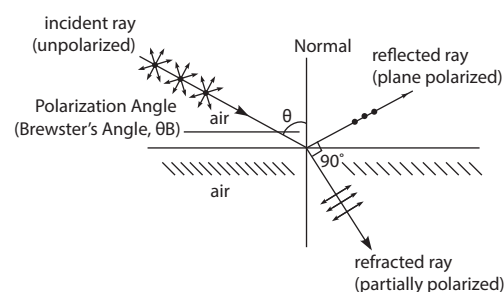
POLARIMETRY: The measurement of optical activity or, in general, the state of polarisation of a beam of electromagnetic radiation. Polarimetry is performed with a polarimeter or polariscope and is used in chemical analysis (notably for measuring sugar concentrations), to study molecular configurations and to study the strain in materials showing double refraction.

POLARISABILITY (POLARIZABILITY): The relative tendency with which a charge distribution, like a particle's electron cloud can be deformed by an external electric field, which may be caused by the

presence of a nearby ion or dipole. Its symbol is α . It is a measure of the response of a molecule to an external electric field. When a molecule is placed in an external electric field, the displacement of electric charge induces a dipole in the molecule. If the electric field strength is denoted by E and the electrical dipole moment induced by this electric field is p , the polarisability α is defined by $p = \alpha E$. Polarisability is an important fundamental property of subatomic particles; and determines the dynamical response of a bound system to external perturbations and provide insight into the internal strong interaction structure.

POLARISATION (POLARIZATION): 1. A special property of light in which the waves of light or other electromagnetic radiation are restricted to certain directions of vibration, usually specified in terms of the electric field vector. 2. Electronic polarisation of an atom is the displacement of the centre of the negatively charged electron cloud relative to the positive nucleus, which is induced by an electric field. 3. Ionic polarisation is polarisation as a result of the displacement of anions and cations in opposite directions. 4. Polarisation orientation is polarisation resulting from the alignment (by rotation) of permanent electric dipole moments with an applied electric field.

POLARISATION ANGLE (BREWSTER'S ANGLE; POLARIZATION ANGLE): An angle of incidence at which light with a particular polarisation is perfectly transmitted through a transparent dielectric surface such as glass, with no reflection. In other words, it is an angle of incidence that produces a 90° angle between the reflected and refracted ray. When unpolarised light is incident at this angle, the light that is reflected from the surface is perfectly polarised. At other angles the reflected light is partially polarised.



Polarization Angle (Brewster's Angle)

POLARISATION ERROR: An error which arises from the fact that the absorbance of a sample (especially a solid) could depend on the polarisation of the incident radiation and from the polarisation dependent response of the photodetector.

POLARISATION IDENTITY: 1. In mathematics, the identity for a complex inner product space that reconstructs the inner product from the norm. 2. The identity for a real inner product space.

POLARISATION OF LIGHT (PLANE-POLARISED LIGHT):

The process of confining the vibrations of the electric vector of light waves to one direction. Some of the processes by which light can be polarised are: polarisation by transmission, polarisation by reflection, polarisation by refraction and polarisation by scattering.

POLARISER (POLARIZER): A device such as Nicol prisms or Polaroid used to convert a beam of electromagnetic waves (light) of undefined or mixed polarisation into a beam with well-defined polarisation. If a polariser is placed in front of a source of unpolarised light, the transmitted light is plane-polarised. In a specific direction which the human eye cannot detect that light is polarised, it is necessary to use an analyser to detect the direction of polarisation. The common types of polarisers are linear polarisers and circular

polarisers. Polarisers are used in many optical techniques and instruments and polarising filters find applications in photography and liquid crystal display technology.

POLARISING POWER: The ability of a cation to polarise an anion.

POLARISCOPE (POLARIMETER): An instrument for ascertaining, measuring or exhibiting the properties of polarised light or for studying the interactions of polarised light with optically transparent media. The simplest type of instrument consists of a light source, collimator, polariser and analyser. The specimen is placed between polariser and analyser, so that any rotation of the plane of polarisation of the light can be assessed by turning the analyser.

POLARITY: 1. In physics, the condition in a system of having opposite characteristics at different points, especially with respect to electric charges or magnetic properties. For example, a magnet has a polarity; in one end is the 'north' and the other is the 'south'; and electric charge has a polarity of either positive or negative. 2. In mathematics, positivity or negativity. For example, $y = r^2$ has two roots of equal magnitude and opposite polarity.

POLAROGRAPHY: An electro analytical technique in which information about the analyte is derived from the measurement of current as a function of applied potential under conditions that encourage polarisation of an indicator or working electrode. The apparatus consists of a potentiometer for adjusting the potential, a galvanometer for measuring current and a cell which contains two electrodes, a reference electrode whose potential is constant and an indicator electrode which is commonly the dropping mercury electrode. It is used to analyse oxidisable and reducible species in solution.

POLAROID: 1. A type of filter that allows transverse waves vibrating in one plane (direction) only to pass through. 2. A trademark for a camera that produces pictures and develops within seconds of being taken or the film used in such a camera. 3. A trademark for a specially treated transparent plastic sheet which is used to polarise light and also used to reduce glare in sunglasses.

POLARON: A coupled electron-ion system that arises, when an electron is introduced into the conduction band of crystalline insulator or semiconductor inducing lattice polarisation around itself. The polaron comprises the electron plus its surrounding lattice deformation. Polarons can also be formed from holes in the valence band.

POLDER: An area of flat reclaimed land that used to be covered by a river, lake or sea or flood plains separated from the sea or river by a dike or marshes separated from the surrounding water by a dike and consequently drained.

POLE: 1. Either of the geographic North and South points on whose axis the Earth rotates. 2. A region of electric or magnetic field where the force is strongest. 3. The geometric centre of a lens or mirror surface. 4. In mathematics, it defines any of the following: (i) an isolated singularity of an analytic function that is neither removable nor essential. For this, there exists a function g that is analytic and non-zero around the point a , such that for the given function f , $f(z) = g(z)(z - a)^{-m}$ for some non-negative integer m that is the order of the pole. The pole is simple when $m = 1$. (ii) the origin of a system of polar coordinates. (iii) in the stereographic projection of a sphere, it is the point that generates the image of a variable point as the intersection with a plane of the line joining the pole to the variable point.

POLE AND POLAR: In mathematics, the pole and polar of a conic point, called **the pole of the line** and the line, called **the polar of the point** that is the locus of the point of intersection of the tangents to a given conic at the two points at which a secant through the pole cuts the conic. These are the harmonic conjugates of the pole with respect to the secant.

POLED STOCK: Animals which are naturally hornless or whose horns have been removed.

POLIOMYELITIS (POLIO; INFANTILE PARALYSIS): An acute infectious disease caused by a virus in which the spinal cord motor cells are destroyed resulting in paralysis. It is characterised by a cold with fever and sore muscles leading to inability to use limbs, to breathe or to swallow and speak.

POLISH NOTATION (PREFIX NOTATION): A system of symbolic notation of mathematical operations used in arithmetic, logics and computer programming language which omits parentheses and writes the operators before their arguments or operands. This eliminates the need for rules of operator precedence. For example, P or Q ($P \vee Q$) is written Apq and if P then Q ($P \rightarrow Q$) is written Cpq ; and $(5 - 6) \times 7$ can be written as $\times -567$. Lisp, a computer programming language uses Polish notation.

POLISH SPACE: In mathematics, the homeomorphic image of a complete separable metric space that has a countable dense subset. Examples are the real line, any separable Banach space, the Cantor space and Baire space.

POLL: 1. The back of the head. 2. To dehorn an animal. 3. A survey of people's opinions or intentions.

POLLEN (POLLEN GRAINS): Male sex cells in flowering plants or the grains of seed plants that contain the male gamete and which usually appear as a fine dust formed in the stamens in seed plants. It is transported by various means to the pistil, where fertilisation occurs. The outer layer of a pollen grain resists disintegration which may be caused by treatment with intense heat, strong acids or strong bases. Pollen grains become airborne forming part of the atmosphere. Some people are allergic to certain airborne pollen particles, which may cause hay fever or trigger an asthmatic attack.

POLLEN BEETLE: A pest of brassicae which makes buds wither. The beetle feeds on flower parts and buds.

POLLEN SAC: The microsporangium or structure in seed plants in which pollen is produced. There are usually two pairs of pollen-producing sacs in an immature, developing stamen, each divided by a partition to make four compartments. In gymnosperms, variable numbers of pollen sacs are borne on the microsporophyll.

POLLEN TUBE: Outgrowth from a pollen grain through the style that transports sperm cells from the pollen grain, either from the stigma of flowering plants or angiosperms to the ovules at the base of the pistil or directly through ovule tissue in some gymnosperms.

POLLEX: The innermost digit on the forelimb of a tetrapod vertebrate, having only two phalanges and appposable to the four fingers of the hand.

POLLINATION: The transfer of matured pollen grains from the anther of a flower to the stigma of the same kind of plant. There are two types of pollination: self pollination and cross pollination. Self pollination involves the transfer of pollen from the anther of a flower to the stigma of the same flower or to the stigma of another flower on the same plant. Cross pollination involves the transfer of pollen from the anther of a flower of one plant to the stigma of another flower of another plant, but of the same kind. Pollinators are responsible for pollination. They include wind, animals such as birds and butterflies, water or by explosive mechanisms.

POLLING: 1. Also known as **polled operation**, it is a technique for transferring data from a terminal to the central computer of a multi-user system as a synchronise activity. It also refers to when a device is repeatedly checked for readiness and if it is not, the computer returns to a different task. Polling is most often used in terms of

input/output (I/O). 2. A process generally used for refining copper and tin in which the crude metal has its own oxide as impurity and the molten metal in a container is stirred with logs of green wood. In the process, the green wood becomes decomposed by the heat from the molten metal and liberates hydrocarbon gases which reduce the oxide impurity into the metal.

POLLINIUM: A cohering group of pollen grains, which are the product of only one anther that is transported as a whole or as a single unit during pollination. It is typical of orchids and milkweeds (Asclepiadoideae).

POLLINOSIS (HAY FEVER): Inflammation of the nose and the eye caused by allergy to the pollen of certain plants, fungal spores or dust in the atmosphere, which if not properly treated may develop into develop asthma.

POLLUTANT: Any substance produced and released into the environment by human activities that can damage living organisms.

POLLUTER PAYS PRINCIPLE (POLLUTER-PAYS PRINCIPLE; PPP): The environmental policy principle that whoever causes pollution is responsible for the cost of repairing any damage.

POLLUTION: The action of degrading an environment by discharging harmful substances into the air, soil or water or noise to an unacceptably high level. Although there are some natural pollutants such as volcanoes, pollution generally occurs because of human activity. Examples are smoke, chemical and radioactive elements in the air, natural constituents of the atmosphere such as carbon dioxide and other greenhouse gases that are present in excessive amounts, the activities of DDT or carbon monoxide released from motor vehicles. Some other **pollution** problems include acid rain; photochemical smog; increasing levels of human waste; damage to the ozone layer by nitrogen oxides; chlorofluorocarbons (CFCs), halons and pollution of inland waters by agricultural fertilisers and sewage effluent, causing eutrophication. The factors determining the severity of a pollutant are its chemical nature, its concentration and its persistence. Some pollutants are biodegradable and therefore, will not persist in the environment in the long term.

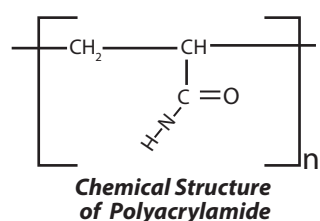
POLONIUM: A very rare radioactive metallic element with the chemical symbol Po, melting point (527 K), boiling point 962°C (1235 K), atomic number 84 and relative atomic mass 210. It is obtained from uranium ores. It is used in the removal of static electricity and the production of neutron sources.

POLY-: 1. A prefix meaning more than one, for example, polyatomic. 2. Any of a series of water-soluble polymers with the general formula $\text{HO}-(\text{CH}_2-\text{CH}_2-\text{O})_n-\text{H}$. Examples are polyacrylamide $(-\text{CH}_2\text{CHCONH}_2-)$ and polyethylene $(-\text{CH}_2\text{CH}_2-)$.

POLYACID: In polymer terminology, it is polyelectrolyte composed of macromolecules containing acid groups on a substantial fraction of the constitutional units.

POLYADIC: In mathematics, the polyadic of a relation, operation, etc., is one that has several argument places.

POLYACRYLAMIDE: A polymer formed from 2-propenamide with the chemical formula $(\text{CH}_2\text{CHCONH}_2)_n$. Polyacrylamide gels are made using a cross-linking agent to form three-dimensional matrices. Polyacrylamide is used to coagulate solids in a liquid, which is applied in wastewater treatment and processes like paper making. In its cross-linked form it is used as polyacrylamide gel electrophoresis and in manufacturing soft contact lenses. In the



straight-chain form, it is used as a thickener and suspending agent. More recently, it has been used as subdermal filler for aesthetic facial surgery. In its straight-chain form, it is also used as a thickener and suspending agent.

POLYADDITION: A polymerisation in which the growth of polymer chains proceeds by addition reactions between molecules of all degrees of polymerisation.

POLYADELPHOUS: Having stamens joined by their filament into three or more groups or clusters. On the other hand, monadelphous means having stamens that are united by their filaments to form a tube around the gynoecium. The stamens of flowers of leguminous plants are usually monadelphous.

POLYAMIDES: Polymers which contain monomers of amides joined by peptide bonds. They are condensation products of primary amine and carboxylic acid or a synthetic polymer that has a recurring amide group such as nylon. In nature they occur as proteins, such as wool and silk. They can also be made artificially through step-growth polymerisation or solid-phase synthesis, for example, nylons, aramids and sodium poly (aspartate). Polyamides are very durable and used in textiles, automobiles, carpet, etc.

POLYAMINE: An organic compound or a hydrocarbon containing more than one amino group. They include ethylene diamine $(\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{NH}_2)$, 1,3-diaminopropane $(\text{H}_2\text{N}-(\text{CH}_2)_3-\text{NH}_2)$ and hexamethylenediamine $(\text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2)$. Polyamine also includes some substances that play important roles in both eukaryotic and prokaryotic cells such as putrescine $(\text{H}_2\text{N}-(\text{CH}_2)_4-\text{NH}_2)$, cadaverine $(\text{H}_2\text{N}-(\text{CH}_2)_5-\text{NH}_2)$, spermidine $(\text{H}_2\text{N}-(\text{CH}_2)_4-\text{NH}-\text{CH}_2-\text{CH}_2-\text{NH}_2)$ and spermine $(\text{H}_2\text{N}-(\text{CH}_2)_4-\text{NH}-\text{CH}_2-\text{CH}_2-\text{NH}-\text{CH}_2-\text{CH}_2-\text{NH}_2)$.

POLYANDROUS: Having many stamens, usually more than ten.

POLYANTHOUS: Having many flowers, especially when clustered together in an involucre.

POLYATOMIC ION (MOLECULAR ION): An ion that contains two or more atoms covalently bonded or, of a metal complex that can be considered as acting as a single unit in the context of acid and base chemistry or in the formation of salts. Examples are hydroxide ion $(\text{OH})^-$ and ammonium ion $(\text{NH}_4)^+$.

POLYATOMIC MOLECULE: A molecule that consists of three or more atoms.

POLYBASIC: Of, relating to or referring to an acid that has two or more hydrogen atoms that can be replaced by basic atoms or radicals. An example is phosphorus(V) acid (H_3PO_4) .

POLYBASIC ACID: A Bronsted-Lowry acid with more than one replaceable hydrogen atom. Examples include the dibasic sulfuric acid (H_2SO_4) and tribasic phosphoric(V) orthophosphoric acid (H_3PO_4) . If all the hydrogens are replaced by metal atoms they form normal salts, but form acid salts otherwise.

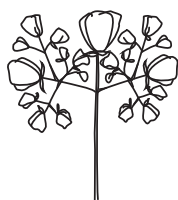
POLYCARBONATES: A group of transparent, thermoplastic polymers which have resistance to heat and other chemicals. They are generally prepared by the reaction of an aromatic difunctional phenol with either phosgene or an aromatic or aliphatic carbonate. They are used in safety spectrum, lenses and babies' bottles. They have very good electrical properties and high impact strength.

POLYCARPOUS: Having many carpels.

POLYCEPHALOUS: 1. With many heads. 2. A plant which is branched or growing in clumps with many heads, as in Asteraceae.

POLYCHAETA: A class of annelid worms in which each body segment has a pair of flattened fleshy lobes called parapodia bearing numerous bristles called chaetae, which are made of chitin.

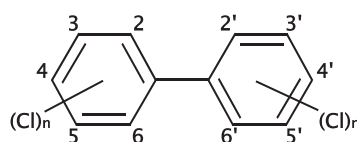
POLYCHASIUM: A cymose inflorescence in which each relative main axis produces more than two lateral axes.



Polychasium

POLYCHLORINATED BIPHENYL (PCB):

Any of a number of derivatives of biphenyl ($C_6H_5)_2$, with a distinctive pleasant smell in which some of the hydrogen atoms on the benzene rings have been replaced by chlorine atoms. Polychlorinated biphenyls are used in the manufacture of certain polymers as electrical insulators. They are highly toxic and are suspected to be carcinogenic. Their increasing use has caused concern because they have been shown to accumulate in the food chain.



Polychlorinated Biphenyl

POLYCHLOROETHENE (POLYVINYL CHLORIDE; PVC): A tasteless, odourless, synthetic thermoplastic which is a polymer of vinyl chloride with the chemical formula H_2CCHCl . It is insoluble in most organic solvents. It is made from chloroethene by heating in an inert solvent, using benzoyl peroxide as an initiator or by the free radical mechanism initiated by heating chloroethene in water with potassium persulfate or hydrogen peroxide. PVC has a wide range of uses, including vinyl flooring, food containers, fibres, guttering, bottles, windows, upholstery, cables, etc.

POLYCHOTOMOUS VARIABLES: Variables that can take more than two possible values, but since this would include all but binary variables, the term is conventionally used for categorical variables with more than two categories.

POLYCHROMATIC RADIATION: Electromagnetic radiation that consists of a mixture of different wavelengths.

POLYCHROME: Having many colours.

POLYCISTRONIC: Of or referring to a type of messenger RNA that can encode several different polypeptide chains. Bacterial messenger RNA is generally polycistronic.

POLYCONDENSATION: A polymerisation in which the growth of polymer chains proceeds by condensation reactions between molecules of all degrees of polymerisation.

POLYCRYSTALLINE GRAPHITE: A graphite material with coherent crystallographic domains of limited size regardless of the perfection and preferred orientation (texture) of their crystalline structure.

POLYCYCLIC: 1. Having two or more atomic rings in a molecule as in steroids. Polycyclic compounds may contain single rings as in phenylbenzene ($C_6H_5C_6H_5$) or fused rings as in naphthalene ($C_{10}H_8$). 2. In botany, having many whorls.

POLYCYCLIC AROMATIC HYDROCARBONS, PAHs (POLY-AROMATIC HYDROCARBONS): A group of hydrocarbons made up from multiple interconnected 6-carbon (benzene) rings. They are molecules produced when carbon-rich fuels (in this case, charcoal) burn incompletely. They can be found in soot, smoke, cigarette smoke, burning hydrocarbons, grilled meats and smoked fish. Examples are naphthalene (the simplest) and anthracene. PAHs are among the most stable of organic molecules. More complex PAHs may contain elements such as nitrogen and sulfur other than carbon and hydrogen.

POLYDENTATE: Ligands with more than one donor atom attached to the central atom in a coordination complex.

POLYDISPERSE MEDIUM: A colloidal system in which many particle sizes occur.

POLYELECTROLYTE (POLYMER ELECTROLYTE; POLYMERIC ELECTROLYTE): A polymer composed of macromolecules in which a substantial portion of the constitutional units contains ionic or ionisable groups or both. Polyelectrolytes can be either synthetic or natural. Nucleic acids, proteins, teichoic acids, somopolypeptides and some polysaccharides are examples of natural polyelectrolytes.

POLYELECTROLYTE COMPLEX: A neutral polymer complex composed of macromolecules carrying charges of opposite sign causing the macromolecules to be bound together by electrostatic interactions.

POLYELECTROLYTE GEL: A polymer gel in which the polymer network contains ionic or ionisable groups in a significant fraction of its constitutional units.

POLYELECTROLYTE NETWORK: A polymer network containing ionic or ionisable groups in a substantial fraction of its constitutional units. A polyelectrolyte network in contact with a solution of a salt is able to exchange counter-ions (cations or anions) with ionic species in the solution and act as an ion exchanger. Therefore, a polyelectrolyte network is frequently described as an ion-exchange polymer.

POLYEMBRYONY: The formation of more than one embryo in a plant seed and in humans from a single fertilised egg. This produces identical twins in humans. Often one embryo develops from the fertilised egg, while the others form asexually from other tissues in the ovule. It is best known among parasitic insects of the orders Hymenoptera, families Encyrtidae, Dryinidae, Platygasteridae and Braconidae.

POLYENE: Poly-unsaturated hydrocarbon containing two or more double carbon-carbon bonds in its molecule. These double carbon-carbon bonds interact in a process known as **conjugation**, which results in an overall lower energy state of the molecule.

POLYESTER: A synthetic resin or polymer formed by the condensation of polyhydric alcohols with dibasic acids. Polyesters are noted for the ability to resist wrinkling and to spring back into shape when creased. They have good dimensional stability, wash and dry quickly and are easily taken care of. An example is terylene. Polyesters are used in casement curtains, throw rugs and as a cushioning or insulating material.

POLYEXTREMOPHILE: An organism that is tolerant to multiple extreme environments, such as high temperature and high acidity.

POLYGAMO-DIOECIOUS: Having unisexual flowers with the staminate and pistillate flowers borne on different trees, but also having some perfect flowers on each tree.

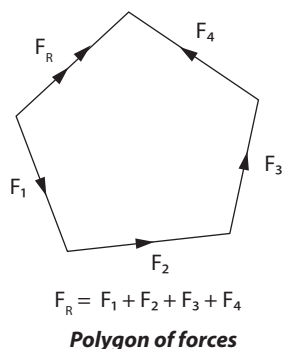
POLYGAMO-MONOECIOUS: Plants with unisexual flowers that possess both stamens and pistils with some bisexual flowers.

POLYGAMOUS: Having unisexual and bisexual flowers on the same plant.

POLYGENE (MULTIPLE FACTOR; MULTIPLE GENE INHERITANCE; QUANTITATIVE GENE): Any of a group of genes that together influence a phenotypic trait such as height, skin colour and weight in man. Polygenic inheritance occurs when one characteristic is controlled by two or more genes at different loci on different chromosomes. The genes also contribute to type II diabetes.

POLYGON: A plane or two dimensional figure formed by joining three or more straight line segments. Where the two lines meet is called a **vertex**. The sum of the interior angles with n sides is $(n - 2)180^\circ$. If the sides of a polygon are equal in length and the angles are equal, the polygon is **regular**; if not it is **irregular**. Examples of polygons indicating the number of sides are triangle (3-sided), quadrilateral (4-sided), hexagon (6-sided), octagon (8-sided) and decagon (10-sided).

POLYGON OF FORCES: A number of forces acting simultaneously on a body and represented in both magnitude and direction by the sides of a polygon, taken in that order so that the vector or side necessary to complete the figure represents the resultant of those forces. When a body is in equilibrium under the action of a number of forces, the polygon of forces will close and the sum of forces on the body will be zero.



POLYGONAL CONVEX SET: A region formed when a system of linear inequalities is graphed so that the intersection set is a convex polygon and every line segment between two non adjacent vertices of the polygon is strictly interior to the polygon except at its endpoints and every internal angle is less than or equal to 180 degrees.

POLYGONAL NUMBER: The number of equally spaced dots needed to draw a polygon. Polygonal numbers, which are a type of figurate number, include square numbers, triangular numbers, pentagonal numbers and hexagonal numbers. Polygonal numbers form an arithmetic sequence of the second order. The general term of these sequences is as follows: $z_n = \frac{n}{2}[2a + (n-1)d]$, where d is one of the numbers 1, 2, 3, etc. With $d = 1$, the triangular numbers include: 1, 3, 6, 10, 15, etc. With $d = 2$ the square numbers include: 1, 4, 9, 16, 25, etc. With $d = 3$ the **pentagonal numbers** include: 1, 5, 12, 22, 35, etc.

POLYGRANULAR CARBON: A carbon material composed of grains, which can be clearly distinguished by means of optical microscopy. Industrial carbon materials such as electrodes are mostly polygranular, but special grades are agranular materials, such as glass-like carbon, carbon fibres or pyrolytic carbon. Such materials are covered by the term agranular carbon.

POLYGRANULAR GRAPHITE: A graphite material composed of grains which can be clearly distinguished by means of optical microscopy.

POLYGYNOUS: Having many pistils or styles.

POLYGYNY: 1. In animals, the mating of a male with more than one female during any single breeding season. 2. In humans, the practice of having more than one wife known generally as **bigamy** or **polygamy**. In flowers, the possession by a plant of many pistils or styles.

POLYHEDRAL ANGLE: A geometric figure formed by the intersection of three or more planes at a common vertex, such as the faces of a polyhedron.

POLYHEDRAL SYMBOL: The polyhedral symbol indicates the geometrical arrangements of the coordinating atoms about the central atom. It consists of one or more capital italic letters derived from common geometric terms (tetrahedron, square plane, octahedron, etc.) which denote the idealised geometry of the ligands around the coordination centre and an Arabic numeral that is the coordination number of the central atom. The polyhedral symbol is used as an affix, enclosed in parentheses and separated from the name by a hyphen. Examples are *T-4*, *SP-4*, *TBPY-5*, *SPY-5*, *OC-6* and *CU-8*.

POLYHEDRANES: Polycyclic hydrocarbons of the $(CH)_n$ series having skeletons corresponding to the regular and semi-regular geometrical solids such as cubane.

POLYHEDRON: In Euclidean geometry, a three-dimensional figure formed of many faces meeting at line segments called **edges**, which meet at points called **vertices**. In a **regular polyhedron** all the faces are congruent regular polygons. There are only five regular

polyhedrons, called the platonic solids. The names of polyhedrons such as the cube (6 square faces), tetrahedron (four triangular faces) and octahedron (8 faces), etc, are indicative of the number of faces.

POLYHYDRIC ALCOHOL (POLYALCOHOL; POLYOL): An alcohol that has several hydroxyl groups per molecule or a hydrogenated form of carbohydrate with the general formula $H(CHCHO)_{n+1}H$, whose carbonyl group (aldehyde or ketone) reducing sugar has been reduced to a primary or secondary hydroxyl group. They are used in foodstuffs commercially as sweeteners. Examples of polyhydric alcohol are glycol, glycerol, xylitol, arabitol, mannitol, etc.

POLYHYDRIC PHENOL: A phenolic compound such as diphenol ($C_6H_4(OH)_2$) containing two or more hydroxyl groups.

POLYAMOND: A shape made from identical equilateral triangles that have been joined at their edges.

POLYKETIDES (ACETOGENINS; KETIDES): Natural compounds containing alternating carbonyl and methylene groups (β -polyketones), biogenetically derived from repeated condensation of acetyl coenzyme A (via malonyl coenzyme A) and usually the compounds derived from them by further condensations.

POLYLOGARITHM (JONQUIÈRE'S FUNCTION): In mathematics, the special function related to logarithm that is defined for integral $n \geq 2$ and z in the unit disk and written as

$$Li_n(z) = \sum_{k=1}^{\infty} \frac{z^k}{k^n}, \text{ where } Li_2 \text{ is called the } \textbf{dilogarithm} \text{ and } Li_3 \text{ is}$$

called the **trilogarithm**.

POLYMER: A substance containing chain-like molecules made by joining monomer molecules end to end or a compound made up of a large long chain or branching composed of many repeated single units (monomers). Polymers may be organic, inorganic or organometallic and synthetic or natural in origin. Natural inorganic polymers include diamonds, graphite, sand, asbestos, agates, mica, quartz, etc. Natural organic polymers include polysaccharides or polycarbohydrates such as starch, cellulose, nucleic acids and proteins. Synthetic inorganic polymers include boron nitride, concrete, many high-temperature superconductors and a number of glasses. Siloxanes or polysiloxanes represent synthetic organometallic polymers. Polymers are used in industries as adhesives, building materials, paper, cloths, fibres, coatings, plastics, ceramics, concretes, liquid crystals, photoresists, etc. They also form major components in soils and plant and animal life. They are also important in nutrition, engineering, biology, medicine, computers, space exploration, health and the environment.

POLYMER ALLOY: Polymeric material, exhibiting macroscopically uniform physical properties throughout its whole volume that comprises a compatible polymer blend, a miscible polymer blend or a multiphase copolymer.

POLYMER BLEND: Macroscopically homogeneous mixture of two or more different species of polymer.

POLYMER CATALYST: Polymer that exhibits catalytic activity. Certain synthetic polymer catalysts can behave like enzymes. Poly(4-vinylpyridine) in its basic form and sulfonated polystyrene in its acid form, are examples of polymers that can act as catalysts in some base- and acid-catalysed reactions, respectively.

POLYMER COMPATIBILIZER: Polymeric additive that, when added to a blend of immiscible polymers, modifies their interfaces and stabilises the blend.

POLYMER COMPLEXATION: A process that results in the formation of a polymer-polymer complex or a complex composed of a polymer and a low-molar-mass substance.

POLYMER CRYSTAL: A crystalline domain usually limited by well-defined boundaries.

POLYMER CRYSTALLITE: A small crystalline domain. A crystallite may have irregular boundaries and parts of its constituent macromolecules may extend beyond its boundaries.

POLYMER CYCLISATION/ CYCLIZATION: A chemical reaction that leads to the formation of ring structures in or from

polymer chains. Examples of cyclisation along polymer chains are: (i) cyclisation of polyacrylonitrile, (ii) acetalization of poly(vinyl alcohol) with an aldehyde, (iii) cyclisation of polymers of conjugated dienes, such as polyisoprene or polybutadiene, leading to macrocycles. Examples of cyclisation of polymer molecules are: (i) cyclisation of poly (dimethylsiloxane), (ii) back-biting reaction during ionic polymerisations of heterocyclic monomers.

POLYMER DEGRADATION: Chemical changes in a polymeric material that usually result in undesirable changes in the in-use properties of the material. In most cases, for example, in vinyl polymers, polyamides, degradation is accompanied by a decrease in molar mass. In some cases, for example, in polymers with aromatic rings in the main chain, degradation means changes in chemical structure. It can also be accompanied by crosslinking. Usually, degradation results in the loss of or deterioration in, useful properties of the material. However, in the case of biodegradation (degradation by biological activity), polymers may change into environmentally acceptable substances with desirable properties.

POLYMER DRUG: Polymer that contains either chemically-bound drug molecules or pharmacologically active moieties. A polymer drug is usually used to provide drug-delivery targeted to an organ and controlled release of an active drug at the target organ.

POLYMER FUNCTIONALIZATION: Introduction of desired chemical groups into polymer molecules to create specific chemical, physical, biological, pharmacological or other properties.

POLYMER MEMBRANE: A thin layer of polymeric material that acts as a barrier permitting mass transport of selected species.

POLYMER PHASE-TRANSFER CATALYST: Polymer that acts as a phase-transfer catalyst and thereby causes a significant enhancement of the rate of a reaction between two reactants located in neighbouring phases owing to its catalysis of the extraction of one of the reactants across the interface to the other phase where the reaction takes place.

POLYMER REACTANT: A reactant (reagent) that is or is attached to a high-molar-mass linear polymer or a polymer network. The attachment may be by chemical bonds, by weaker interactions such as hydrogen bonds or simply by inclusion.

POLYMER REACTION: A chemical reaction in which at least one of the reactants is a high-molar-mass substance.

POLYMER SOLVENT: A polymer that acts like a solvent for compounds of low molar mass. An example of a polymer solvent is poly(oxyethylene); it can dissolve various inorganic salts by complexation.

POLYMER SORBENT: A polymer that adsorbs a certain substance or certain substances from a liquid or a gas. A polymer sorbent may be a polymer adsorbent or a polymer absorbent. The former acts by surface sorption and the latter by bulk sorption.

POLYMER SUPPORT: A polymer to or in which a reagent or catalyst is chemically bound, immobilised, dispersed or associated. A polymer support is usually a network polymer. A polymer support is usually prepared in bead form by suspension polymerisation. The location of active sites introduced into a polymer support depends on the type of polymer support. In a swollen-gel-bead polymer support, the active sites are distributed uniformly throughout the beads, whereas in a macroporous-bead polymer support they are predominantly the internal surfaces of the macropores.

POLYMER SURFACTANT: A polymer that lowers the surface tension of the medium in which it is dissolved or the interfacial tension with another phase or both.

POLYMER-METAL COMPLEX: A complex comprising a metal and one or more polymeric ligands.

POLYMER-POOR PHASE (DILUTE PHASE): The phase of a two-phase equilibrium system, consisting of a polymer and low molecular weight material, in which the polymer concentration is lower.

POLYMER-RICH PHASE (CONCENTRATED PHASE): The phase of a two-phase equilibrium system, consisting of a

polymer and low molecular weight material, in which the polymer concentration is higher.

POLYMER-SOLVENT INTERACTION: The sum of the effects of all intermolecular interactions between polymer and solvent molecules in solution that are reflected in the Gibbs and Helmholtz energies of mixing.

POLYMER-SUPPORTED CATALYST: Catalyst system comprising a polymer support in which catalytically active species are immobilised through chemical bonds or weaker interactions such as hydrogen bonds or donor-acceptor interactions.

POLYMER-SUPPORTED REACTION: A chemical reaction in which at least one reactant or a catalyst is bound through chemical bonds or weaker interactions, such as hydrogen bonds or donor-acceptor interactions, to a polymer. The easy separation of low-molar-mass reactants or products from the polymer-supported species is a great advantage of polymer-supported reactions. Typical examples of polymer-supported reactions are: (i) reactions performed by use of polymer supported catalysts, (ii) solid-phase peptide synthesis, in which intermediate peptide molecules are chemically bonded to beads of a suitable polymer support.

POLYMERASE: Any enzyme whose central function is associated with polymers of nucleic acids such as RNA and DNA. In the processes of replication and transcription, RNA polymerases (types I to III) catalyse the synthesis of RNA using as a template either an existing DNA strand (DNA-dependent RNA polymerase) or an RNA strand. Type I is responsible for the synthesis of ribosomal RNA, type II for messenger RNA synthesis and type III for transfer RNA synthesis.

POLYMERASE CHAIN REACTION (PCR): A technique used in gene amplification, which replicates a fragment of DNA producing a large amount of that sequence. It uses a method called **thermal cycling**, which are cycles of repeated heating and cooling of the reaction for DNA melting and enzymatic replication of the DNA. Short DNA fragments known as **primers** containing sequences complementary to the target region along with a DNA polymerase are key components to enable selective and repeated amplification. As PCR progresses, the DNA generated is itself used as a template for replication, setting in motion a chain reaction in which the DNA template is exponentially amplified. PCR is used in research, clinical diagnostics and forensic science.

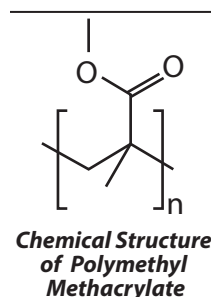
POLYMERISATION (POLYMERIZATION): The chemical reaction in which a compound is made into a polymer by the addition or condensation of monomers to form complex molecules (polymers) of high molecular mass. The two major types of polymerisation processes are condensation polymerisation and addition polymerisation. In **condensation polymerisation**, the chain growth is accompanied by elimination of small molecules such as H₂O or CH₃OH. Addition polymerisation, has the polymer formed without the loss of other materials.

POLYMEROUS: Having many parts, as in a floral whorl with many members.

POLYMETHANAL: A solid polymer of methanol, formed by evaporation of an aqueous solution of methanal.

POLYMETHYL METHACRYLATE (PMMA): A transparent, colourless thermoplastic with the chemical formula (C₅O₂H₈)_n, melting point 160 °C (433 K) and boiling point 200 °C (573 K). It does not contain the potentially harmful bisphenol-A subunits. It has a high transmission of light and high internal reflection, great strength and is economical, making it much preferred to polycarbonate.

POLYMER: 1. Also known as **polymorphonuclear leucocyte**, it is any white blood cell that has a lobed nucleus granular cytoplasm. Polymorphs constitute about 70% of the white blood cells and function as phagocytes involved in



the immune response system. Examples are neutrophils, basophils and eosinophils. 2. An organism that has different forms at different stages of its development. 3. A form of a substance such as polymers, minerals and metals that can crystallise in different forms.

POLYMORPHIC: Having many forms.

POLYMORPHIC TRANSITION: A reversible transition of a solid crystalline phase at a certain temperature and pressure (the inversion point) to another phase of the same chemical composition with a different crystal structure.

POLYMORPHISM: 1. A form of genetic variation, in which distinct different forms exist within a plant or an animal population. Examples are the human blood groups, colour variations in some butterflies and the caste system of social insects in which there are the workers, drones and queens. 2. The ability of a substance to adopt different internal structures and external forms, in response to different conditions of temperature and/or pressure. For example, diamond and graphite are both forms of the element carbon, but they have very different properties and appearance.

POLYMORPHONUCLEAR LEUCOCYTE (POLYMORPH): Any white blood cell that has a lobed nucleus granular cytoplasm. Polymorphs constitute about 70% of the white blood cells and function as phagocytes involved in the immune response system. Examples are neutrophils, basophils and eosinophils.

POLYNOMIAL: 1. A taxonomic name of a plant or animal that has more than two terms, for example, one giving a genus, species and subspecies. 2. An algebraic expression that has one or more variables denoted by letters or a term describing a mathematical expression that has more than two terms, each of which contains constants and variables raised to a positive integral power. In general, a polynomial is represented by $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, where $a_0, a_1, \dots, a_n$ are constants. An example is $4x^3 + 3x^2 + 2x + 1$.

POLYNOMIAL DOMAIN: An Euclidean domain of polynomials over a field, where the valuation function $v(p(x))$ is the degree of the polynomial $p(x)$.

POLYNOMIAL EQUATION: A polynomial that is set equal to zero or an equation of the form $g(x) = 0$, where $g(x)$ is a polynomial. An example is $3x^3 + 9x^2 - 7x + 4 = 0$.

POLYNOMIAL FUNCTION: A function of the form $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, where $a_0, a_1, \dots, a_n$ are constants is a polynomial function of degree n . A polynomial function of degree one is a linear function, that of degree two is a quadratic function and degree three is a cubic function, etc.

POLYNOMIAL REGRESSION: A linear model in which powers and possibly cross-products of the explanatory variables are included. An example is $y = \beta_0 + \beta_1 x + \beta_2 x^2$.

POLYNOMIAL RING: 1. In mathematics, the ring, denoted $R[X]$, of formal polynomials in X over a ring R , where X is a commuting indeterminate, that is, the ring of expressions of the form $\sum_{j=0}^n a_j X^j$, where a_j is an element of R and n is a natural number. 2. The ring, denoted $R[X]$, of rational polynomials over X , that is, of ratios of elements of $R[X]$.

POLYNOMIAL TIME ALGORITHM: In mathematics, an algorithm that solves an instance of a given computational problem in a time that is a polynomial in the size of the input, such as the max-flow min-cut algorithm. In the solution of a linear-programming problem, the number of computational steps grows as a power of the number of variables, rather than exponentially; this allows the solution of inaccessible problems. Mathematical operations such as addition, subtraction, multiplication and division, as well as computing square roots, powers and logarithms, can be performed in polynomial time.

POLYNUCLEOTIDE: A polymer made up of many nucleotides covalently bonded together. An example is DNA structure which is made up of many repeating monomer units of nucleotides consisting of a pentose sugar, a phosphate and a nitrogenous base.

POLYOL (POLYHYDRIC ALCOHOL; POLYALCOHOL): A hydrogenated form of carbohydrate with the general formula $H(HCHO)_{n+1}H$, whose carbonyl group (aldehyde or ketone) reducing sugar has been reduced to a primary or secondary hydroxyl group. Examples of polyhydric alcohol are glycol, glycerol, xylitol, arabitol and mannitol. They are used in foodstuffs commercially as sweeteners.

POLYOMINO: A planar geometric figure consisting of congruent squares joined edge-to-edge.

POLYP: 1. The sedentary stage in the life cycle of the coelenterate such as a coral or jellyfish or the free swimming medusa. 2. A benign tumour occurring in areas lined with mucous membrane such as the nose and gastrointestinal tract especially the colon and the uterus.

POLYPEPTIDE: A chain of chemically bonded amino acids or a peptides containing between 10 and 100 amino acids. Examples are small proteins. Polypeptides that contain 100-300 amino acids are proteins.

POLYPETALOUS: Of or referring to a flower such as a rose or carnation in which the corolla consists of separate petals.

POLYPHENISM (ENVIRONMENTAL POLYMORPHISM): The occurrence within a species of distinct morphological types or morphs in response to environmental stimuli, which may overlap in space and/or time. Factors, such as nutrition, temperature, day length, etc, interact with a single genotype to switch to discrete phenotypes. The resultant morphs are better adapted to their environmental conditions to fulfil a certain role in a social group.

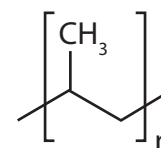
POLYPHYLETIC: Any group of organisms, whose members have originated from several different ancestors. An example is the warm-blooded animals, which contain mammals and birds, but the most recent common ancestor of mammals and birds was cold-blooded.

POLYPHYDONT: A type of dentition in which the teeth are continuously shed and replaced during the lifetime of the animal. Examples are found in sharks and frogs.

POLYPLOID: A nucleus that contains multiple sets (diploid) of chromosomes. Polyploidy is far more common in plants than in animals. Polyploidy can be induced artificially with the drug colchicine and used in plant breeding. Examples of polyploidy in animals occur in parthenogenetic species or in species which reproduce asexually by fission (the division of a single-celled organism into two equal parts, each part growing into a complete organism).

POLYPOD: Having many legs or similar appendages such as a caterpillar with unjointed legs on the abdomen and a large number of jointed legs on the thorax or this stage of larval development.

POLYPROPENE, PP (POLYPROPYLENE): A soft flexible thermoplastic made by polymerisation of propene or a thermoplastic substance that is a synthetic polymer of propylene. It can be moulded to make a wide variety of objects including pipes, industrial fibres, packaging, textiles, stationery, plastic parts and reusable containers, laboratory equipment, loudspeakers, automotive components, etc.



Chemical Structure of Polypropene, PP

POLYPROTIC ACID: An acid such as phosphoric acid (H_3PO_4) that can form two or more hydronium ions per molecule.

POLYRIBISOME (POLYSOME): A cluster of ribosomes connected by a strand of messenger RNA molecule during the translation process of protein synthesis. In eukaryotes, polyribosomes are attached to the surface of the rough endoplasmic reticulum and the outer membrane of the nucleus. In bacteria, they are found free in the cytoplasm.

POLYSACCHARIDES (GLYCANS): Polymeric carbohydrate structures such as starch, cellulose and glycogen made up of many sugar molecules joined in a long chain. The general formula is $C_x(H_2O)_y$ where x is usually a large number between 200 and 2500. They are generally not sweet and are insoluble in water.

POLYSEPALOUS: A flower such as rose and camellia that has distinctly separate sepals.

POLYSOME: A group of two or more ribosomes along a single strand of messenger RNA (mRNA) that translates the genetic information coded in the mRNA simultaneously, during protein synthesis.

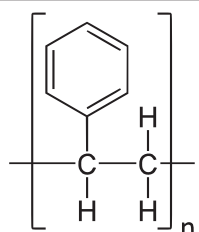
POLYSPERMY: The entry of several sperms into the egg during fertilisation, although only one sperm nucleus actually fuses with the egg nucleus. Polyspermy cell contains three or more copies of each chromosome, one from the egg and one each from multiple sperm resulting in an inviable zygote.

POLYSTACHYOUS: Having many spikes.

POLYSTEMONOUS: Having many stamens.

POLYSTICHOUS: Arranged into several rows.

POLYSTYRENE: An aromatic polymer made from the aromatic monomer styrene, a liquid hydrocarbon that is commercially manufactured from petroleum. Its molecular formula is $(C_8H_8)_n$. It is solid at room temperature, but flows if heated above its glass transition temperature. It is a clear glass-like thermoplastic usually manufactured as a white rigid form and used for heat insulation and packaging. It is used in kitchen utensils or in an expanded form in insulation and ceiling tiles.



Chemical Structure of Polystyrene

POLYSULFANES (POLYSULFANES):

Compounds having an unbranched chain of sulfur atoms (S_2 or higher) terminating in H: HS_nH . Some chemists exclude disulfane, HS_2H , from the class polysulfanes.

POLYSYNAPTIC REFLEX: A reflex action that involves an electrical impulse being transferred from afferent (sensory) neurone to an efferent (motor) neurone via one or more interneurons in the spinal cord. For example, stimulation of pain receptors in the skin initiates a withdrawal reflex, which occurs relatively quickly by activating spinal motor neurons without the delay of routing signals through the brain, although the brain will receive sensory input while the reflex action occurs and results in the removal of the organism or part from the stimulus.

POLYTENY: The condition of a chromosome, nucleus or cell in which the DNA has repeatedly replicated, without subsequent division of the cell or its nucleus. This results in a cable-like giant chromosome, which is hundreds of times larger than normal chromosomes, often showing conspicuous transverse banding. Polyteny has been observed in the cells of certain insects of the order Diptera, notably in the salivary glands of *Drosophila*, *Chironomus* and in some plant cells. Giant chromosomes are characterised by specificity of location of the cross-bands, which permits compilation of chromosome maps and study of the functional activities of each band or group of bands.

POLYTETRAFLUOROETHYLENE [PTFE; POLY(DIFLUOROMETHYLENE)]: A fluoropolymer, which is a high-molecular-weight compound consisting wholly of carbon and fluorine with the chemical formula $(C_2F_4)_n$, density 2200 kg/m³ and melting point 327°C (600 K). It is made from the monomer tetrafluoroethene (CF_2CF_2). It is a thermosetting plastic with a high melting point that is used to produce non-stick surfaces on pans and as a lubricant to coat bearings.

POLYTHENE (POLYETHYLENE; POLYETHENE): Any of the various polymers of ethylene with the chemical formula $(-CH_2-CH_2-)_n$, which is produced at high pressures and temperatures in the presence of any of several catalysts, depending on the desired properties for the finished product. Polyethylene is resistant to water, acids, alkalis and most solvents. Some of the production processes are radical polymerisation, anionic addition polymerisation, ion coordination polymerisation or cationic addition polymerisation, resulting in different types of polyethylene. The finished product is classified into several different categories based on its density and branching such as Ultra high molecular weight polyethylene (UHMWPE), Ultra low molecular weight polyethylene (ULMWPE or PE-WAX), High molecular

weight polyethylene (HMWPE), Medium density polyethylene (MDPE), Linear low density polyethylene (LLDPE), Low density polyethylene (LDPE), Very low density polyethylene (VLDPE), etc. They are widely used all over the world as packaging film, plastic bags, unbreakable bottles, pipes, pails, drinking glasses and insulation for wire and cable, etc.

POLYTHETIC: Of or referring to a type of classification in which membership of a taxon is based on its constituent organisms sharing a large number of characteristics.

POLYTHIONATE: A salt of polythionic acid, containing an anion or group of the form $S_nO_6^{2-}$ or $-S_nO_6^-$.

POLYTHIONIC ACID: Oxoacids having a chain of sulfur atoms with the general formula $H_2S_nO_6$, where $n = 0 - 4$.

POLYTOPAL REARRANGEMENT: Stereo isomerisation interconverting different or equivalent spatial arrangements of ligands about a central atom or of a cage of atoms, where the ligand or cage defines the vertices of a polyhedron. For example, pyramidal inversion of amines, Berry pseudorotation of PF_5 , rearrangements of polyhedral boranes.

POLYTOPE: A higher-dimensional analogue of a polygon or polyhedron. The number of possible regular polytopes depends on the number of dimensions. In two dimensions there are infinitely many possible regular polygons; in three dimensions there are five possible regular polyhedra; in four dimensions there are six possible regular polytopes; and in each number of dimensions higher than four, there are just three possible regular polytopes, analogous to the three-dimensional tetrahedron, cube and octahedron. A four-dimensional polytope is also sometimes called a **polyhedroid** or a **polychoron**. Just as a polygon has vertices and edges and a polyhedron has vertices, edges and faces, a four-dimensional polytope has vertices, edges, faces and cells, where a cell is a three-dimensional figure.

POLYTOPIC: Of or referring to a taxon that has risen and exists in more than one place, which may be an indication that a more derived species has evolved in an intervening area or that a parallel evolution has occurred to make the two or more separated populations appear alike.

POLYTYPIC: Of or referring to a species that exists in a variety of forms or subspecies that inhabit different geological regions. Having several variant forms, especially subspecies or varieties.

POLYUNSATURATED FAT (POLYUNSATURATED FATTY ACID): Fatty acids such as oil, fat and waxes containing a high proportion of triglyceride molecules whose fatty-acid chains contain several double bonds, which assume a *cis* or *trans* conformation depending on the geometry of the double bond. The molecule of polyunsaturated fatty acid has more than one pair of hydrogen atoms missing from its structure. Because of this the molecule's surface reduces the strength of the compound's intermolecular forces, which makes the melting point of the compound to be much lower (liquid at room temperature) than saturated fatty acids (solid at room temperature) such as animal fat. Examples of polyunsaturated fatty acids are vegetable oils such as safflower oil, sunflower oil, soya bean oil and corn oil. They are much favoured than saturated fatty acids by nutritionists. Polyunsaturated fatty acids are known to lower cholesterol levels, while saturated fatty acids raise cholesterol levels.

POLYURETHANE: Polymers made from the monomer urethane that may be either thermosetting (rigid when heated) or thermoplastic (softens when heated) and made into both flexible and rigid foams. The rigid foam is used for the thermal insulation of refrigerators, trucks and buildings. It is also used in the furniture industry to mould furniture parts. The flexible foam is used in furniture and automobile cushions, in mattresses and for carpet backings.

POLYVALENT: 1. Species that are not univalent or monovalent (having a valency of more than one), for example, the Ca^{2+} cation. 2. Vaccine that is effective against more than one strain of a microorganism, antigen, toxin or antibody.

POLYVINYL: Plastics and resins produced by the polymerisation of vinyls, such as polyvinyl chloride.

POLYVINYL ACETATE (PVA; PVAc):

A thermoplastic with the formula $(C_4H_6O_2)_n$ belonging to the polyvinyl esters family produced by the polymerisation of vinyl acetate (ethanoate), $CH_2=CHOOCCH_3$. It is used in adhesives and for coating paper and fabrics.

POLYVINYL ALCOHOL: A water-soluble odourless, nontoxic synthetic polymer with the chemical formula $[CH_2CH(OH)]_n$, density 1.19-1.31g/cm³, melting point 230 °C (493 K) and decomposes at 200 °C (473 K). It is resistant to oil, grease and solvent. It is made from polyvinyl acetate (PVA) by hydrolysis with sodium hydroxide. It has excellent film forming, emulsifying and adhesive properties and used for making adhesives that are miscible with water, as a sizing for textiles, for making synthetic fibres and numerous other uses.

POLYVINYL CHLORIDE, PVC (POLYCHLOROETHENE):

Polymer of vinyl chloride with the chemical formula $(H_2CCHCl)_n$. It is tasteless, odourless and insoluble in most organic solvents and a member of the family of vinyl resins. It is made by polymerising vinyl chloride and used for drain pipes, floor tiles, audio discs, upholstery, shoes and handbags.

POLYMER (ANOMALOUS WATER):

Liquid water generally formed by condensation of water vapour in tiny glass or fused-quartz capillaries and with properties very different from those well established for ordinary water; for example, lower vapour pressure, lower freezing temperature, higher density and viscosity, higher thermal stability.

POLYYNE: An unsaturated, hydrocarbon found in interstellar molecular clouds, where hydrogen is scarce that contains two or more triple carbon-carbon bonds in its molecule. An example is diacetylene also known as butadiyne, with the molecular formula C_4H_2 , the first in the series of polyynes.

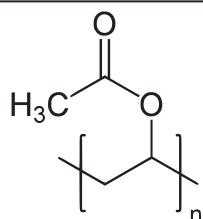
POLYZOA (BRYOZOA): A phylum of freshwater or marine animals that reproduces by budding. Bryozoans often form colonies on the sea bottom or attached to seaweed.

POME: A type of fleshy fruit produced by flowering plants in the subfamily Maloideae of the family Rosaceae such as apples and pear. The flesh of the fruit develops from the receptacle of the flower which completely encloses the fused carpels. **Pomaceous** means pertaining to a pome.

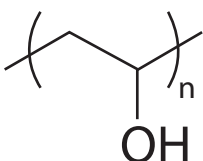
POMERON: In physics, it is a hypothetical object having energy and momentum, but no colour or electrical charge, produced when an electron strikes a proton. It was first suggested in 1958 by Russian physicist, Isaac Pomeranchuk.

POMOLOGIST: A fruit grower or scientist that studies and cultivates pome fruit, particularly from the genera Malus, Prunus and Pyrus belonging to the Rosaceae. **Pomology** is the branch of botany that studies fruits and fruit growing.

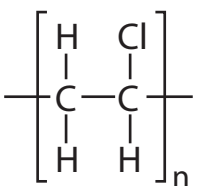
POND: 1. A small body of shallow, standing water, either natural or artificial with sunlight reaching to the bottom. It is usually smaller than a lake, and because it is shallow enough, it permits the growth of marsh, aquatic plants and other organisms. Its life forms include microscopic bacteria, insects, amphibians (toads, frogs, etc.) fish, small animals (beaver, mole, etc.), and birds (duck, herons, etc.). Some ponds are built for special purposes, for example, for breeding fish, irrigation, watering livestock, etc. 2. A type of pseudocarp or false fruit typical of certain plants belonging to the rosaceae family.



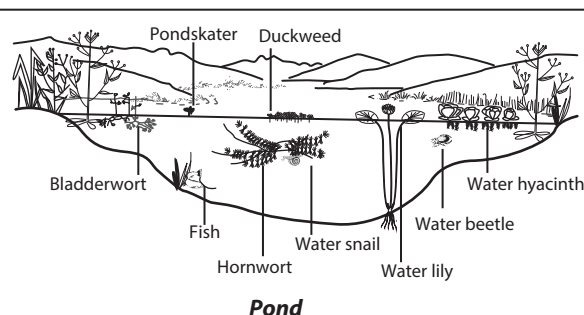
Chemical Structure of Polyvinyl Acetate



Chemical Structure of Polyvinyl Alcohol



Chemical Structure of Polyvinyl Chloride



Pond

POND SKATER (WATER STRIDER): Any of various insects of the family Gerridae, which is a slender long-legged predator insect that walks on water and feeds on dead insects.

PONS (PONS VAROLI): A thick tract of nerve fibres in the brain stem that links the medulla oblongata to the midbrain. It conducts signals from the cerebrum down to the cerebellum and medulla and carries the sensory signals up into the thalamus. It was named after Costanzo Varolio, a 16th-century Italian anatomist and surgeon.

PONTRYAGIN'S MAXIMUM PRINCIPLE (PONTRYAGIN'S MINIMUM PRINCIPLE):

In mathematics, one of many extensions of the principle that, at the optimal solution of a control problem, a Hamiltonian system is solvable and the corresponding Hamiltonian is maximised.

POOTER: A device, which works by suction and commonly used by entomologists for field and laboratory work to collect insects, crustaceans or other small, fragile organisms, usually for scientific purposes.

POP (POST OFFICE PROTOCOL; POP MAIL): 1. In telecommunications, it is a protocol used to receive e-mail on many e-mail clients. There are two versions of POP. The first, called **POP2**, requires Simple Mail Transfer Protocol (SMTP) to send messages. The newer version, **POP3**, can be used with or without SMTP. **Internet Message Access Protocol (IMAP)** is a protocol that allows users to retrieve e-mail, but still uses SMTP for sending e-mail messages. IMAP4 is the latest version of IMAP. 2. **PoP** is the acronym for **point of presence**, which is a point of access provided by an Internet service provider (ISP) that enables users to establish a connection with the Internet through communication devices such as a telephone. It consists of servers, modems, routers, ATM switches and other communication equipment that are permanently connected to the network. An Internet access provider may have several PoPs distributed at different locations in their area of operation to enable remote users to connect to the Internet.

POP-UP: An advertisement Window that opens on top of all other Windows and typically appears when a user visits a page or closes a window. These ads are produced with JavaScript code that is inserted into the HTML of a Web page. These ads have become increasingly annoying as they have become more popular with many web pages.

POPULATION: 1. A group of human, animal or plant species living in a certain area and able to interbreed, who by their increasing numbers and corresponding increasing needs can seriously affect the global ecosystem. Some factors that influence population change are availability of food, predators, diseases and climate. 2. Also known as universe, in statistics, it is the entire collection of individuals that have some properties in common from which a sample is obtained for analysis.

POPULATION BOTTLENECK: An event having little to do with the usual forces of natural selection that drastically reduces the size of a population and many genetic lineages within that population, thus decreasing genetic diversity. The remaining population has a very low level of genetic diversity, which means that the population as a whole has few genetic characteristics. The bottleneck may be caused by various events, such as an environmental disaster, the hunting of

a species to the point of extinction or habitat destruction that results in the deaths of organisms.

POPULATION CYCLE: The regular fluctuations in the size of a population, over a predictable period of time. Some factors which influence population change are availability of food, predators, diseases and climate.

POPULATION DENSITY: The number of individuals of a population per unit area or volume of living space, usually expressed as population divided by land area in square kilometres.

POPULATION DYNAMICS: The study of the short- and long-term changes in the size and age composition of populations and the biological and environmental processes influencing those changes, including the way populations are affected by birth and death rates and by immigration and emigration.

POPULATION GENETICS: The study of the distribution of inherited variation among a group of organisms of the same species as a result of natural selection, genetic drift, mutation and gene flow. It also includes understanding adaptation and speciation.

POPULATION GROWTH MODEL: A model used in predicting the growth of human populations. In the short term geometric or exponential growth model is given by

$P_t = p_0(1 + r)^t$ or $p_t = p_0 e^{rt}$, where p_0 is population at time 0, p_t is population at time t , r is the growth rate and t is time.

POPULATION GROWTH RATE (PGR): The fractional rate increase in the population that occurs when the birth rate is higher than the death rate or when immigration exceeds emigration or when a combination of these factors is present over a unit time period, often expressed as a growth rate or as a ratio of the number of individuals in the population at the beginning of that period. PGR is written as:

$$PGR = \frac{\text{pop. at end of period} - \text{pop. at start of the period}}{\text{pop. at start of the period}}. \text{ It can also be}$$

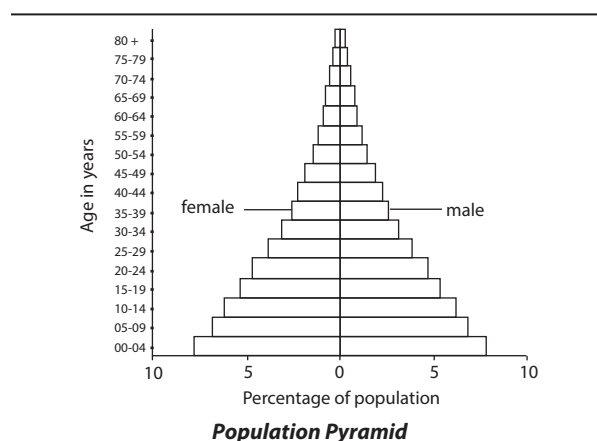
expressed as growth rate = crude birth rate – crude death rate + net immigration rate, or $\frac{\Delta P}{P} = \left(\frac{B}{P}\right) - \left(\frac{D}{P}\right) + \left(\frac{I}{P}\right) - \left(\frac{E}{P}\right)$, where P is

the total population, B is the number of births, D is the number of deaths, I is the number of immigrants and E is the number of emigrants. The most common way to express population growth is as a ratio. Growth ratio is expressed as: Growth ratio = Growth rate $\times 100$. A positive growth ratio (or rate) indicates that the population is increasing, while a negative growth ratio indicates the population is decreasing. A growth ratio of zero indicates the number of people at the two given times have neither increased nor decreased (net difference between births, deaths and migration is zero). It can also be obtained by plotting population size against time.

POPULATION INVERSION: A population inversion occurs when a system such as a group of atoms or molecules exists in a state with more members in an excited state than in lower energy states. The concept is of fundamental importance in laser science because the production of a population inversion is a necessary step in the workings of a standard laser.

POPULATION PYRAMID (AGE STRUCTURE DIAGRAM):

A diagram representing various age groups in a human population with the population plotted on the horizontal axis and the age on the Y-axis and often indicating the ages of males on left side and females on the right side. It can also be used to represent any two population statistical data. A population pyramid whose base is broad with narrow apex has high fertility. Population pyramids are often viewed as the most effective way to graphically depict the age and sex distribution of a population.



POPULATION TYPE: A method of classifying stars of type S as either population I or population II bodies devised in 1944 by Wilhelm Baade (1893–1960). Type S stars number about 700. **Population I** stars, also known as the **young disk population** are the young metal-rich highly luminous stars found in the spiral arms of galaxies. **Population II** stars are older metal-deficient stars that occur in the centres of galaxies.

POPULATION VIABILITY ANALYSIS (PVA): A method of identifying the threats faced by a species and predicting whether or not a species will persist in a particular environment and for a given time into the future. PVA is often aimed at the conservation and management of rare and threatened species, with the goal of applying the principles of population ecology to improve their chances of survival.

PORANDROUS: Having anthers that open by pores.

PORCELAIN: A high-grade ceramic ware made from quartz, white kaolin, marble and feldspar by first moulding the plastic paste and then firing. Some properties of porcelain are high strength, a white colour, very low absorption, good translucency and a hard glaze.

PORE: 1. A tiny opening or a small hole through which liquids or gases can pass. 2. **Sweat pore** is an anatomical structure of the skin of humans and other mammals used for secretion of sweat. 3. **Canal pore** is an anatomical structure that is part of the lateral line sense system of some aquatic organisms. 4. An opening across both inner and outer bacterial membranes, a part of many Gram-negative bacterial secretion systems. 5. One of the openings communicating with the skin surface at the terminus of lactiferous ducts in milk-producing mammals. 6. **Nuclear pore** is a large protein complex that penetrates the nuclear envelope in eukaryotic cells. 7. **Ion channel pore** is the ion-selective opening in the membrane of a eukaryotic cell formed by members of the ion channel family of proteins. 8. A water-selective opening (water channel) in the membrane of an eukaryotic cell formed by assemblies of the protein aquaporin.

PORE SIZE DISTRIBUTION: The distribution of pore volume with respect to pore size; alternatively, it may be defined by the related distribution of pore area with respect to pore size.

PORE SPACE: The total space not occupied by soil particles in a bulk volume of soil that can be filled by water and/or air.

PORE-SIZE DISTRIBUTION: The volume of the various sizes of pores in a soil sample. They are expressed as percentages of the bulk volume (soil plus pore space).

POROUS SOIL: A soil that is made from coarse particles that leave large gaps known as pores. The pores allow the movement of air and water through the soil. Porous soil allows water to move through it easily.

PORICIDAL: A type of dehiscence in which the pollen is released through pores at the tip of the anther, as in a poppy capsule.

PORIFERA: The phylum of marine and freshwater invertebrates comprising the sponges which live permanently attached to rocks or other surfaces.

PORIN: Any of a class of barrel proteins that form channels across cell membranes and act as a pore through which molecules can diffuse. In the outer membrane of Gram negative bacteria, most porins consist of three identical subunits that associate and extend fully through the membrane to form channels about 1 nm in diameter. These permit the entry and exit of hydrophilic low-molecular-weight substances. Some contain a specific binding site and allow the passage only of certain substances. The pores can be opened and closed, which has significance for antibiotic resistance in pathogenic bacteria. They are present in the outer membrane of Gram-negative bacteria and some Gram-positive bacteria of the group Mycolata, the mitochondria and the chloroplast.

PORISM: A type of mathematical proposition discussed by Euclid, but no longer identifiable; it is thought to have affirmed the possibility of finding conditions under which a problem is indeterminate or has innumerable solutions.

POROSE: Having pores.

POROSITY: The ratio of the void spaces in a material, expressed as a percentage or fraction. The term is used in several fields including geology, ceramics, metallurgy, materials, manufacturing, earth sciences and construction.

POROUS: A term describing a solid that is full of tiny holes and passages through which fluids can pass. An example is sandstone.

PORPHYRIA: A metabolic disease due to disordered haemoglobin synthesis. Porphyria runs in families and may cause episodic abdominal pain, skin changes, neuritis and mental changes. Certain drugs can precipitate acute attacks.

PORPHYRIN: An important class of water-soluble, nitrogen-containing organic pigments derived from four pyrrole rings linked by additional carbon atoms to form a large flat ring. Many form complexes with metal ions, in substances such as chlorophyll, cytochrome and haemoglobin. In medicine, porphyrins are used in conjunction with light, often a laser beam, to induce reactions in the body against cancer and other diseases.

PORPHYRINOGENS: Hexahydroporphyrins in which the nitrogen atoms and four *meso* positions are saturated.

PORPHYRY: An igneous rock containing large crystals in a finer matrix. Porphyritic texture forms when a rock has had two different stages of cooling resulting in two distinctly different crystal sizes.

PORRECT: 1. Stretched forward. 2. Resembling the beak of a parrot, as shown in the diagram.

PORT: 1. In computer hardware, it refers to a socket that enables a computer processor to communicate with an external device. 2. A term used to describe the process of taking a program that has been written for specific operating systems and moving it to another operating system. An example is taking a program written for Microsoft Windows and moving it to Linux. 3. An Internet port number that indicates what kind of protocol a server on the Internet is using. A commonly used and accessed port is port 80, which is the http port.

PORTABLE: A term used to describe an object that can be easily moved, such as a portable computer.

PORTABLE DOCUMENT FORMAT (PDF): A way of storing computer files so that they can be easily read when they are moved from one computer to another. It is a multi-platform file format developed by Adobe Systems. A PDF file captures document text, fonts, images and even formatting of documents from a variety of applications. It contains colour-accurate information and prints the same way they look on a screen. If e-mailed, a PDF document will look the same way on PC operating system. Adobe Reader, a free application program distributed by Adobe Systems is needed to view

a PDF file. Adobe also makes an Acrobat Plug-in for Web browsers that enables PDF files to be viewed inside a browser window.

PORTABLE NETWORK GRAPHICS (PNG): An open source raster graphics file format in which graphical images are stored as a bit map (arrangement of dots) and transmitted.

PORTAL (INTERNET PORTAL): A website or service that offers a broad range of services such as e-mail, games, quotes, search, news, stocks, etc. The idea of a portal is to offer a broad range of commonly accessed services at one location, causing the user to visit the site more frequently and commonly stay for a longer period of time. Portals typically include search engines and large directories of websites. Examples of portals are Yahoo, Excite, Lycos, Netscape, AltaVista, MSN and AOL.com.

PORTAL VEIN (PORTAL CIRCULATION; PORTAL SYSTEM): Any vein that collects blood from one network of capillaries and transports it directly to a second capillary network in another region of the body without returning it to the heart. Both capillary beds and the blood vessels that connect them are considered part of the portal venous system.

PORTULACA: A genus of mostly annual herbs in the family Portulacaceae consisting of about 125 species found in tropical, subtropical and some temperate regions. The roots are tuberous, fleshy, fibrous or small taproots. The stems are erect to prostrate, branched, fleshy or suffrutescent. Leaves alternate or subopposite, congested and involucre-like immediately proximal to inflorescence; blade terete, subterete or flattened. Inflorescences are terminal in clusters or axillary on short branches. Flowers are sessile or subsessile and usually open only in sunshine; sepals are broadly clasping at base. Capsules are membranaceous, chartaceous and dehiscence circumscissile. It has many reniform to cochleate brown to black or grey seeds. *Portulaca quadrifolia* is a succulent plant with alternate/opposite leaf arrangement with axillary scales. The leaves are sessile with entire leaf margin. The inflorescence is surrounded by an involucre and are usually terminal. Flowers are bisexual and radially symmetrical. The leaves are used as vegetables. *Portulaca oleracea* with the common names purslane, verdolaga, pigweed, little hogweed, pursley or moss rose is an annual succulent, which may reach 40 centimetres in height. It is used as a leaf vegetable; provides ground cover to create a humid microclimate for nearby plants; and its leaves are used to treat insect or snake bites, boils, sores, pain from bee stings, bacillary dysentery, diarrhoea, haemorrhoids, postpartum bleeding and intestinal bleeding.

POS TERMINAL (POINT OF SALE TERMINAL): A computer terminal at a point of sale that processes data such as to record and track customers orders, process credit and debit cards, print out a fully itemised receipt for the customer, connect to other systems in a network and manage inventory.

POSET: Acronym for **partially ordered set**, a set endowed with a partial order. A partially ordered set formalises and generalises the intuitive concept of an ordering, sequencing or arrangement of the elements of a set. An example of a partially ordered set is the set of subsets of a given set, where $A \subset B$ for sets A and B to mean A is a proper subset of B .

POSITION ANGLE (PARALLACTIC ANGLE): The direction of an imaginary arrow in the sky, measured in degrees from north through east. In other words, it is the angle between a body's hour circle and its vertical circle. Applied to a binary star system, it is the angle between the primary and secondary components of a binary system. Applied to a galaxy, it measures the object's tilt, that is, the angle between the galaxy's long axis and a line from its centre headed north.

POSITION CIRCLE: A circle that can be measured both from a chart and from the surface of the Earth for the purpose of position

fixing. The circles can be measured by sextant. Two overlapping position circles can be used to give a position fix.

POSITION ISOMERISM: In position isomerism a functional group changes position on the chain, the basic carbon skeleton remains unchanged.

POSITION LINE: Line of position at which the observer is located at a particular time.

POSITION VECTOR: A vector that takes its roots from the origin. A position vector of a point $P(a,b)$ is $\overrightarrow{OP} = \begin{pmatrix} a \\ b \end{pmatrix}$. The components

of the position vector are the coordinates of the point.

POSITIONAL CLONING:

The identification of a gene of unknown function responsible for a genetic disease, in the absence of any information about its product. The phenotype is correlated with a chromosomal site, the DNA of that region is cloned and compared with that in normal individuals. With reference to an existing genetic map, the markers indicate the chromosomal region that contains the disease gene. An example is the identification of cystic fibrosis, Huntington's disease and Friedreich's ataxia.

POSITIONAL INFORMATION: Signals, to which genes regulating development respond, indicating a cell's position relative to other cells in an embryonic structure of the organism. An example is the digit formation in the limb bud of vertebrates.

POSITIVE: 1. A quantity greater than zero, it has a plus sign (+). 2. Indicating the presence of a particular organism or component in the results of a test or examination. 3. Having an electrical charge of an opposite polarity to that of an electron and the same polarity as that of a proton. 4. Having a higher electrical potential than the ground or the defined neutral point. 5. Having an overall positive electrical charge, sometimes caused by the loss of one or more electrons. 6. Making a parallel beam of light converge. 7. Of, relating to or referring to photographic images that have colours or values of dark and light corresponding to the subject. 8. Indicating growth response or movement towards a stimulus such as light. 9. Of, relating to or referring to a mechanical action or device having little or no play.

POSITIVE BINOMIAL DISTRIBUTION: A probability distribution that arises from the binomial distribution by omission of the zero value.

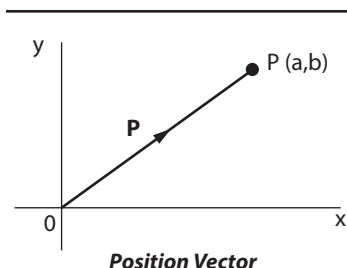
This is given by $P(X = x) = \frac{n!}{x!(n-x)!} \frac{p^x q^{n-x}}{(1-q^n)}$, where $x = 1, 2, \dots, n$.

POSITIVE CHARGE: A charge with more protons than electrons or a charge that exists in a body that has fewer electrons than protons.

POSITIVE CONTROL: A system that is turned on by the presence of a regulatory protein.

POSITIVE DEFINITE: In mathematics, the positive definite of a matrix or a self-adjoint operator on Hilbert space has $\langle Ax, X \rangle > 0$ for all $X \neq 0$.

POSITIVE FEEDBACK: The use of part of the output to enhance the input. In negative feedback, the output is used to reduce the input. A positive feedback is a closed system as it occurs in a feedback loop. It means the response of the system relies on the feedback signal to function. Alarm or panic can spread by positive feedback among a herd of animals to cause a stampede. Positive feedback often leads to exponential divergences or exponential growth of oscillations. Under strong positive feedback and a lack of stabilising forces, systems will typically accelerate towards a non-linear region, which may stabilise the system or, conversely, destroy it. Norbert Wiener was the first person to observe this in his 1948 work on cybernetics.



POSITIVE HYPERGEOMETRIC DISTRIBUTION: A probability distribution that arises from the hypergeometric distribution by omission of the zero. This is given by:

$$P(X = x) = \frac{\binom{r}{x} \binom{N-r}{n-x}}{\binom{N}{n} - \binom{N-r}{n}}, \text{ where } x = 1, \dots, \min(n, r).$$

POSITIVE ION: An atom, radical, molecule or molecular moiety which has lost one or more electrons thereby attaining an electrically positive charge.

POSITIVE POISSON DISTRIBUTION: A probability distribution that is derived from Poisson distribution by the omission of zero value

$$\text{given by, } P(x-n) = \frac{e^{-\lambda^n}}{x!(1-e^{-\lambda})!}, \text{ where } x = 1, 2, 3, \dots$$

POSITIVE POWER OF TEN: A number that can be written in the form 10^a , where a is a positive integer. In other words, it is ten multiplied by itself a certain number of times. They are numbers such as 10, 100, 1,000 that can be written using only 10s as factors.

POSITIVE ORTHANT: In mathematics, it is the orthant in which all coordinates are positive.

POSITIVE RATIONAL NUMBERS: Rational numbers that are greater than 0. Examples are 1/4, 2/9, -7/-11, -3/-13, and 5/12.

POSITIVE REACTION: In botany, it is a tropism in which the plant moves, or the plant member grows, from a region where the stimulus is weaker to one where it is stronger.

POSITIVE SEMIDEFINITE: In mathematics, it is the positive semidefinite of a matrix or self-adjoint operator on Hilbert space has $\langle Ax, X \rangle \geq 0$ for all $X \neq 0$.

POSITIVE SET: In mathematics, a set, P , of elements of an ordered field that is closed under addition and multiplication and has the property that for every non-zero member, x , of the field, either x or $-x$ is in P .

POSITIVELY HOMOGENEOUS: In mathematics, positively homogeneous function is homogeneous, but only for positive scalars $f(\lambda x) = \lambda f(x)$ for every scalar $\lambda > 0$. If $f(\lambda x) = \lambda^p f(x)$ for some $p > 0$, then f is positively homogeneous of degree p .

POSITRON: The antimatter counterpart of the electron, essentially an electron with a positive charge equal in magnitude to the electron's negative charge and the same mass. Positrons have the same spin and statistics as that of electrons and like electrons, appear as decay products of many heavier particles. Electron-positron pairs are produced by high-energy photons in matter.

POSITRON EMISSION TOMOGRAPHY (PET): A medical imaging technique that can produce three dimensional pictures of certain metabolic or biochemical changes within the body, in the brain and other organs by tracking the movement and concentration of a radioactive tracer injected into the bloodstream. It is used in medicine to diagnose brain tumours and strokes and to locate the origins of epileptic activity. In psychiatry, it is used to examine brain function in mental illnesses and in neuropsychology to study the functions of the brain. The technique is based on the detection of photons produced by the decay of short-lived radioactive isotopes, such as fluorine-18 in labelled fluoro deoxyglucose injected into the body. The isotopes emit positrons, which almost collide with electrons to produce a pair of photons that travel in opposite direction.

POSITRON MICROSCOPE: An instrument used to study solids, similar in principle to a scanning electron microscope, but using positrons rather than electrons.

POSITRONIUM: A bound state consisting of an electron and a positron. The electron and positron spins can be aligned in the same direction called **singlet states** or in opposite directions called **triplet states**. There are two types of positronium: (i) **orthopositronium** (the lowest triplet state), in which the spins of the two constituents are parallel and decays into three photons in about 1.5×10^{-7} s; and (ii) the lowest-energy singlet state called **parapositronium**, in which the spins are anti-parallel and decays into two photons in a shorter

time of about 10^{-10} s. Positronium has a hydrogen-like spectrum, but with different values of the frequencies since a positron is much lighter than a proton.

POSSIBLE: 1. In mathematics, able to exist or be true; not entailing any contradiction. 2. In logic, a possible statement or formula is one that is capable of being true under at least some interpretation, in some possible world or in some circumstances. That P is possibly true is usually written 'MP'.

POSSIBLE WORLD: In logic, it is a semantic device in modal logic that formalises the notion of what the world might have been like; that is, a complete world-description. A statement is necessary (necessarily true) if and only if it is true in every possible world and possible (possibly true) if and only if it is true in at least one.

POST OFFICE PROTOCOL (POP; POP MAIL): 1. In telecommunications, it is a protocol used to receive e-mail on many e-mail clients. There are two versions of POP. The first, called **POP2**, requires Simple Mail Transfer Protocol (SMTP) to send messages. The newer version, **POP3**, can be used with or without SMTP. **Internet Message Access Protocol (IMAP)** is a protocol that allows users to retrieve e-mail, but still uses SMTP for sending e-mail messages. IMAP4 is the latest version of IMAP. 2. **POP** is the acronym for **point of presence**, which is a point of access provided by an Internet service provider (ISP) that enables users to establish a connection with the Internet through communication devices such as a telephone. It consists of servers, modems, routers, ATM switches and other communication equipment that are permanently connected to the network. An Internet access provider may have several PoPs distributed at different locations in their area of operation to enable remote users to connect to the Internet.

POST SYNAPTIC MEMBRANE: The cell membrane at the opposite side of the synapse from the synaptic terminal of the stimulating neuron that contains receptor proteins and degradative enzymes for the neurotransmitter. It conducts impulses away from a synapse.

POST-FILTER EFFECT: This effect arises when the exciting beam does not fill the cell completely and luminescence is absorbed by the analyte and interfering impurities in the non-illuminated region facing the detector.

POST-GLACIAL REBOUND: The vertical upwards movement of land which had been beneath ice sheets, following the disappearance and shrinking of the ice sheets. Such lands return to their original shapes slowly. For example, the rebound lands beneath the former ice sheets, around Hudson Bay and central Scandinavia, are still rising by over a centimetre a year, since 19,000 years ago, when the ice sheets of North America and northern Europe began to melt, and the land which had been beneath ice sheets began to move vertically upwards.

POST-MULTIPLICATION: In mathematics, post-multiplication by a matrix or ring element is the multiplication on the right rather than on the left (pre-multiplication); this is important in any non-commutative setting.

POST-NEWTONIAN APPROXIMATION: An approximation used in the general theory relativity for finding solution of the Einstein equations for the metric tensor. The post-Newtonian approximations are expansions in a small parameter, which is the ratio of the velocity of matter, forming the gravitational field to the speed of light or the speed of gravity.

POSTCAVAL VEIN: The inferior (posterior or ascending) vena cava carrying blood to the heart from posterior parts of the body. Vena cava is either of two major veins that carry de-oxygenated blood from the posterior half of the body into the right atrium of the heart. It enters the right atrium at the lower right, back side of the heart. It brings blood from below the chest or drains blood from the trunk and hind limbs.

POSTEMERGENCE HERBICIDE: A herbicide applied on the leaves after the weed has emerged.

POSTERIOR: 1. Parts of the body at or near the hind end of an animal or designating the part of an animal that is to the rear. 2. The

side of a flower or auxiliary bud that faces towards the flower stalk or main stem respectively.

POSTERIOR PROBABILITY: The posterior probability of an event is the probability of the event computed following the collection of new data. It is the probability of event A occurring given that event B has occurred. Bayesian inference utilises prior probabilities and data to estimate the posterior probabilities, which are the results.

POSTERIOR VENA CAVA (INFERIOR VENA CAVA): The large vein through which deoxygenated blood passes and empties into the right atrium of the heart. It is posterior to the abdominal cavity and runs alongside of the vertebral column on its right side. It enters the right atrium at the lower right, back side of the heart.

POSTFIX NOTATION (REVERSE POLISH NOTATION): A notation that does not use parenthesis and writes operators such as logical constants after their arguments. For example, P or Q is written pqA and $(1 + 2) * 3$ is written as $1\ 2 +\ 3\ *\$ or $3\ 1\ 2 +\ *\$.

POSTMASTER: In computer science, is a term that is especially used for people who manage the e-mail system in a local area network.

POSTPRECIPITATION: The subsequent precipitation of a chemically different species upon the surface of an initial precipitate usually, but not necessarily, including a common ion.

POSTPROCESSOR: A hardware device or software program that utilises a set of instructions already in the software program or hardware device. For example, a machine may have a postprocessor that it utilises in developing a product or a part used in a product.

POSTREPLICATIVE REPAIR (RECOMBINATIONAL REPAIR): A form of DNA repair that fills large gaps in a new DNA strand resulting from failure of replication. Example genes in humans are BRCA2 and BRCA1, BLM and NBS1. DNA damage prevents the normal enzymatic synthesis of DNA by the replication fork. At damaged sites in the genome, both prokaryotic and eukaryotic cells utilise a number of postreplication repair (PRR) mechanisms to complete DNA replication. Chemically modified bases can be by-passed by either error-prone or error-free translesion polymerases or through genetic exchange with the sister chromatid. The replication of DNA with a broken sugar-phosphate backbone is most likely facilitated by the homologous recombination proteins that confer resistance to ionising radiation. The activity of PRR enzymes is regulated by the SOS response in bacteria and may be controlled by the post replication checkpoint response in eukaryotes.

POSTSCRIPT: A page description language (PDL) developed by Adobe Systems in 1984 that describes a page's text and graphical content. It can be used to define the appearance of graphics and text for both screen and print. PostScript has become a standard method of printing in desktop publishing.

POSTSYNAPTIC NEURON: A neuron (nerve cell) found on the cell body or dendrite to which an electrical impulse is transmitted across a synaptic cleft by the release of a chemical neurotransmitter (i.e., acetylcholine, norepinephrine, GABA (gamma aminobutyric acid) and glutamate from the axon terminal of a presynaptic neuron. In other words the postsynaptic neuron receives the neurotransmitter from the presynaptic neuron.

POSTULATE (AXIOM): In logic or mathematics, it is a statement that is taken to be true without proof. It can be used in the proof of other propositions by using logical deduction.

POSTZYGOTIC BARRIER: A type of reproductive isolating mechanisms (RIMs) that prevents gene flow in the rare event that fertilisation occurs between two closely related species. Types of postzygotic barrier include hybrid sterility and hybrid breakdown. It often prevents the hybrid zygote from developing into a viable, fertile adult. It is one of the two general categories of reproductive isolating mechanisms that prevent the formation of hybrids between members of different populations. The other category is the prezygotic mechanism that prevents fertilisation; postzygotic ones take effect afterward fertilisation.

POTASH: The general name for any potassium containing mineral, especially potassium carbonate (K_2CO_3) or potassium hydroxide (KOH).

Potassium carbonate is a white crystalline material formerly obtained from wood ashes. It is used to make special types of glass, potassium silicate, pigments, printing inks and soft soaps and as a laboratory reagent and general-purpose food additive. Potassium hydroxide is frequently called **caustic potash**; in the fertiliser industry, potassium oxide is called **potash**.

POTASSIUM: A soft, wax-like, silvery-white, low-melting, highly reactive metallic element with the symbol K, atomic number 19, atomic weight 39.102, melting point 63.65 °C (336.8 K), boiling point 774 °C (1047 K) and specific gravity 0.862. It is one of the alkaline metals, below sodium and above rubidium with a low density. It is used as a coolant in nuclear reactors and in fertilisers.

POTASSIUM BROMIDE: A white crystalline salt, soluble in water with the chemical formula KBr, density 2.74 g/cm³, melting point 734 °C (1007 K) and boiling point 1435 °C (1708 K). It is used as an antiepileptic medication for dogs and cats and in photography. It may be prepared by the action of bromine on hot potassium hydroxide solution or by the action of iron(III) bromide or hydrogen bromide on potassium carbonate solution.

POTASSIUM CARBONATE (POTASH; PEARL ASH): A crystalline, deliquescent substance that forms a strongly alkaline water solution with the chemical formula K₂CO₃, density 2.29 g/cm³, melting point 891 °C (1164 K) and decomposes at boiling point. It is prepared commercially by electrolysis of potassium chloride to form potassium hydroxide, which is then carbonated. It is available commercially as a white, granular powder and used in the manufacture of glass, soap, explosives, ceramics, fertilisers, etc.

POTASSIUM CHLORATE: A colourless crystalline compound, containing potassium, chlorine and oxygen with the chemical formula KClO₃, relative density 2.32, melting point 356°C (639 K) and decomposes above 400°C (673 K) giving off oxygen. It is soluble in water and moderately soluble in ethanol. It is prepared mainly by the fractional crystallisation of a solution of potassium chloride and sodium chlorate but it may also be prepared by electrolysis of hot concentrated solutions of potassium chloride. It is a powerful oxidising agent finding applications in weedkillers and disinfectants. Because of its ability to produce oxygen, it is used in explosives, pyrotechnics and matches.

POTASSIUM CHLORIDE: A colourless or white crystalline salt with the chemical formula KCl, density 1.984 g/cm³, melting point 770°C (1043 K) and boiling point 1420°C (1693 K). It occurs naturally as a mineral in sylvite (KCl) and a carnallite. It is produced industrially by fractional crystallisation of these deposits or of solutions from lake brines. It is used as a fertiliser and as dietary salt substitute.

POTASSIUM CYANIDE: A very poisonous white crystalline salt with the chemical formula KCN, molar mass 65.12 g/mol, density 1.52 g/cm³, melting point 634.5 °C (907.6 K), boiling point 1,625 °C (1,898 K). It can be produced by treating hydrogen cyanide with aqueous solution of potassium hydroxide, followed by evaporation of the solution in a vacuum: $\text{HCN} + \text{KOH} \rightarrow \text{KCN} + \text{H}_2\text{O}$ or by treating formamide with potassium hydroxide $\text{HCONH}_2 + \text{KOH} \rightarrow \text{KCN} + 2\text{H}_2\text{O}$. It is used in the extraction of gold and silver from ore, in photography and as an insecticide.

POTASSIUM DICHROMATE (POTASSIUM BICHROMATE; BICHROMATE OF POTASH; DICHROMIC ACID, DIPOTASSIUM SALT; CHROMIC ACID, DIPOTASSIUM SALT): An orange ionic crystalline solid monoclinic or triclinic with the chemical formula K₂Cr₂O₇, density 2.676 g/cm³ (solid), melting point 398 °C (671 K) and decomposes at 500 °C (773 K). It is soluble in water and insoluble in alcohol. It is a strong oxidising agent in the presence of dilute sulfuric acid. It is prepared by acidification of crude potassium chromate solution. The addition of a base to solutions of potassium dichromate reverses this process. The compound is

used industrially as an oxidising agent in the chemical industry and in dyestuffs manufacture, in electroplating, pyrotechnics, glass manufacture, glues, tanning, photography and lithography and in ceramic products.

POTASSIUM FERRICYANIDE (POTASSIUM HEXACYANOFERRATE(III); RED PRUSSIAN; PRUSSIAN RED): A red crystal with the chemical formula K₃Fe(CN)₆, density 1.89 g/cm³ (solid), melting point 300 °C (573 K) and decomposes at boiling point. It is soluble in water and its solution showing green-yellow fluorescence. It is produced by passing chlorine through a solution of potassium ferrocyanide. It is used as a chemical reagent and in the manufacture of pigments.

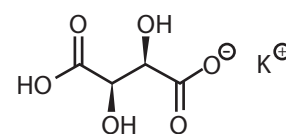
POTASSIUM FERROCYANIDE [POTASSIUM HEXACYANOFERRATE(II)]: A yellow crystal with the chemical formula K₄Fe(CN)₆·3H₂O, density 1.85 g/cm³ (trihydrate), melting point 69 -71 °C (342-344 K) and decomposes at boiling point of 400 °C (673 K). It forms lemon-yellow monoclinic crystals at room temperature. It is produced industrially from hydrogen cyanide, ferrous chloride, calcium hydroxide, the combination of which gives Ca₂[Fe(CN)₆]·11H₂O. This solution is then treated with potassium salts to precipitate the mixed calcium-potassium salt CaK₂[Fe(CN)₆], which in turn is treated with potassium carbonate to give the tetrapotassium salt. It is a strong reducing agent and used in medicines and to manufacture of explosives.

POTASSIUM FIXATION: The process of converting exchangeable or water-soluble potassium to moderately soluble potassium; that is, to a form not easily exchanged from the adsorption complex with a cation of a neutral salt solution. This occurs when potassium ions form a surface complex with oxygen atoms in the interlayers of certain silicate clay minerals. Potassium fixation plays a significant role in the soil-plant system influencing the effectiveness of fertilisation.

POTASSIUM HYDRIDE: A white or greyish white crystalline solid with the symbol KH, which is a hydride of potassium. It is prepared by passing hydrogen over heated potassium and marketed as a light grey powder dispersed in oil. The solid decomposes on heating and in contact with moisture and is an excellent reducing agent. Potassium hydride is a fire hazard because it produces hydrogen on reaction with water as follows: $\text{KH} + \text{H}_2\text{O} \rightarrow \text{KOH} + \text{H}_2$.

POTASSIUM HYDROGEN CARBONATE (POTASSIUM BICARBONATE; POTASSIUM ACID CARBONATE): A white colourless, odourless crystalline solid, with the chemical formula KHCO₃. It is soluble in water and insoluble in ethanol. It decomposes between 100 °C - 120 °C (373 K - 393 K) into potassium carbonate, water and carbon dioxide. It occurs naturally as calcinitite and is prepared by passing carbon dioxide into saturated potassium carbonate solution. It is used in baking, soft-drinks manufacture and in CO₂ fire extinguishers, used in crops, especially for neutralising acidic soil, added to some detergents, used as a laboratory reagent and also as a gastric antacid.

POTASSIUM HYDROGEN TARTRATE (CREAM OF TARTAR; POTASSIUM ACID TARTRATE; MONOPOTASSIUM TARTRATE): A white, crystalline powder with the chemical formula KC₄H₅O₆, which is a by-product of wine making. It is used in many food preparations such as stabilising egg whites, increasing their heat tolerance and volume, preventing sugar syrups from crystallising and reducing discolouration of boiled vegetables.



Chemical Structure of Potassium Hydrogen Tartrate

POTASSIUM HYDROXIDE (CAUSTIC POTASH; POTASH LYE POTASSIA; POTASSIUM HYDRATE): A white deliquescent solid with the chemical formula K(OH), density 2.044 g/cm³, melting point 406 °C (679 K) and boiling point 1327 °C (1600 K). It is often sold as pellets, flakes

or sticks. It is soluble in water giving off much heat and forming a strongly alkaline, caustic solution. It is also soluble in ethanol and very slightly soluble in ether. It is prepared industrially by the electrolysis of concentrated potassium chloride solution but it can also be made by heating potassium carbonate or sulfate with slaked lime, Ca(OH)_2 . It closely resembles sodium hydroxide in its chemical properties and has similar uses but is more soluble and is, therefore prepared as an absorber for carbon dioxide and sulfur dioxide. It is used in the manufacture of soft soap, in bleaching, other potassium salts and in Ni-Fe and alkaline storage cells. Potassium hydroxide is extremely corrosive to body tissues and especially damaging to the eyes.

POTASSIUM IODATE [IODIC ACID; POTASSIUM SALT; POTASSIUM TRIOXIODATE(V)]: A white monoclinic crystalline solid with the chemical formula KIO_3 , density 3.89 g/cm^3 and decomposes at melting point of 560°C (833 K). It is soluble in water and insoluble in ethanol. Two processes of production include by the reaction of iodine with hot concentrated potassium hydroxide:

$\text{HIO}_3 + \text{KOH} \rightarrow \text{KIO}_3 + \text{H}_2\text{O}$ or by adding iodine to a hot, concentrated solution of potassium hydroxide:

$3\text{I}_2 + 6\text{KOH} \rightarrow \text{KIO}_3 + 5\text{KI} + 3\text{H}_2\text{O}$. It is an oxidising agent and used as an analytical reagent. Some potassium iodate is used as a food additive.

POTASSIUM IODIDE: A white, cubic crystalline solid with the chemical formula KI , density 3.123 g/cm^3 , melting point 681°C (954 K) and boiling point 1330°C (1603 K). It has a strong bitter taste, soluble in water, ethanol and acetone. It may be prepared by the reaction of iodine with hot potassium hydroxide solution followed by separation from the iodate by fractional crystallisation. In solution iodine dissolves to form the triiodide ion I_3^- , which is brown. Potassium iodide is widely used as an analytical reagent, in photography and also as an additive to table salt to prevent goitre and other disorders due to iodine deficiency.

POTASSIUM MONOXIDE: A grey crystalline solid ionic compound of potassium and oxygen with the chemical formula K_2O , which decomposes at 350°C (623 K). It may be prepared by the oxidation of potassium metal with potassium nitrate: $2\text{KNO}_3 + 10\text{K} \rightarrow 6\text{K}_2\text{O} + \text{N}_2$. It can also be prepared by the reaction of oxygen and potassium, producing potassium peroxide which is treated with potassium: $\text{K}_2\text{O}_2 + 2\text{K} \rightarrow 2\text{K}_2\text{O}$. It reacts with ethanol to form potassium ethoxide (KOC_2H_5) and with liquid ammonia to form potassium hydroxide and potassamide (KNH_2).

POTASSIUM NITRATE (SALTPETRE; NITRATE OF POTASH VESTA POWDER): A colourless rhombohedral or trigonal solid with the chemical formula KNO_3 , density 2.109 g/cm^3 (16°C), melting point 334°C (833 K) and decomposes at 400°C (673 K). It is soluble in water and insoluble in alcohol. It occurs naturally as the mineral nitre and is a natural solid source of nitrogen. It may be prepared by the reaction of sodium nitrate with potassium chloride followed by fractional crystallisation. It is a powerful oxidising agent (releases oxygen on heating) and is used in gunpowder and fertilisers, rocket propellants and fireworks.

POTASSIUM NITRITE: A white or slightly yellow deliquescent solid with the chemical formula KNO_2 , soluble in water and insoluble in ethanol and may explode at 873 K (600°C). Potassium nitrite is prepared by the reduction of potassium nitrate. It reacts with cold dilute mineral acids to give nitrous acid and is also able to behave as a reducing agent if oxidised to the nitrate or as an oxidising agent if reduced to nitrogen. It is used in organic synthesis and in detecting the presence of the amino groups in organic compounds. It is also used in the manufacturing of heat transfer salts and also as a food additive known as E249 in the European Union.

POTASSIUM PERMANGANATE (POTASSIUM MANGANATE(VII)): An inorganic chemical compound with the

chemical formula KMnO_4 and density 2.703 g/cm^3 . It forms purple crystals with a metallic sheen. It is soluble in water giving intense purple solution, acetone and methanol. It is decomposed by ethanol from 100°C to 240°C (373 K to 514 K). The compound is prepared by fusing manganese(IV) oxide with potassium hydroxide to form the manganate and electrolysis of the manganate solution using iron electrodes at about 60°C (333 K). It can also be produced with sodium manganate by oxidation with chlorine and sulfuric acid, then treated with potassium chloride to crystallise the required product. It is used as an oxidising agent and disinfectant and in deodorisers and dyes.

POTASSIUM SULFATE [SULFATE OF POTASH; ARCANITE; POTASSIUM TETRAOXOSULFATE(VI)]: A non-flammable white crystalline salt with the chemical formula K_2SO_4 , density 2.66 g/cm^3 , melting point 1069°C (1342 K) and boiling point 1689°C (1962 K). It is soluble in water and insoluble in ethanol. It occurs naturally as schonite (Strassfurt deposits) and in lake brine, from which it is separated by fractional crystallisation. It can be produced by the Hargreaves process, using sulfur dioxide, oxygen and water and potassium chloride as the starting materials. In the Leblanc process, it is synthesised by reaction of potassium chloride with sulfuric acid: $2\text{KCl} + \text{H}_2\text{SO}_4 \rightarrow 2\text{HCl} + \text{K}_2\text{SO}_4$. Potassium sulfate is used in cements, in glass manufacture, as a food additive and as a fertiliser (source of K^+) for chloride-sensitive plants, such as tobacco and citrus.

POTASSIUM SULFIDE (DIPOTASSIUM MONO SULFIDE; DIPOTASSIUM SULFIDE; POTASSIUM MONO SULFIDE): A deliquescent solid with the chemical formula K_2S , density 1.8 g/cm^3 , melting point 840°C (1113 K) and decomposes at boiling point. It is colourless when pure and yellowish brown when impure. It is soluble in water and in ethanol but insoluble in diethyl ether and moderately flammable. It is made industrially by reducing potassium sulfate with carbon at high temperatures in the absence of air. In the laboratory, it can be synthesised by combining a solution of potassium in anhydrous ammonia with elemental sulfur. Solutions are strongly alkaline due to hydrolysis. It is used as an analytical reagent, in medicine and depilatories.

POTASSIUM SULFITE [POTASSIUM TRIOXOSULFATE(IV)]: A white crystalline solid with the chemical formula K_2SO_3 , soluble in water and very sparingly soluble in ethanol and decomposes on heating. It is a reducing agent and is used as such in photography and in the food and brewing industries, where it prevents oxidation and also as a food additive known as E225 in the European Union. Potassium sulfide is generally regarded as a hazardous chemical with a fire risk.

POTASSIUM SUPEROXIDE (POTASSIUM DIOXIDE): A yellow paramagnetic solid with the chemical formula KO_2 , density 2.14 g/cm^3 , solid and decomposing at melting point of 560°C (833 K). It is produced by burning molten potassium in an excess of oxygen. It is very soluble by reaction in water, soluble in ethanol and slightly soluble in diethyl ether. When treated with cold water or dilute mineral acids, hydrogen peroxide is obtained. The compound is a powerful oxidising agent and on strong heating releases oxygen with the formation of the monoxide, K_2O . Potassium superoxide is used as an oxidising agent in industrial chemistry as a carbon dioxide scrubber; water dehumidifier; and also as oxygen generator in rebreathers, spacecraft, submarines and spacesuit.

POTASSIUM THIOCYANATE: A colourless hygroscopic solid with the chemical symbol KSCN , which decomposes at 500°C (773 K). It is an important salt of the thiocyanate anion, one of the pseudohalides. It is used in solution to test for iron(III) compounds, which gives a blood-red colour.

POTASSIUM-ARGON DATING (K-AR DATING): A method of isotopically dating the ages of rocks based on the radioactive decay of potassium-40 (^{40}K) to the stable isotope argon-40 (^{40}Ar). Ages are based on the known half-life of ^{40}K and the ratio of ^{40}K

to ^{40}Ar . The method is used to measure a wide variety of ages of about 100,000 to 30 million years old.

POTATO: A common name for herbaceous perennials in the genus *Solanum* or its tuber in the nightshade family with about a thousand different varieties. The stem can grow to about 60 cm high, sprouting spirally arranged compound leaves. They bear white, pink, red, blue or purple flowers with yellow stamens. The tubers have spirally arranged buds (eyes) that may remain dormant after the tuber is fully grown for up to 10 weeks or more. The tubers are a major source of starch as well as amino acids, protein, vitamin C and B vitamins and used widely as food. An example is *Solanum tuberosum*, commonly known as Irish or white potato and "earth apple" that grows to about 60 cm high, having flowers with different shades of colours depending on the variety. Flowers are usually cross-pollinated by insects such as the bumblebees. It produces green fruits after flowering, with the fruits containing about 30 seeds. The sweet potato (*Ipomoea batatas*) does not belong to the nightshade family. It is a dicotyledonous plant that belongs to the family Convolvulaceae, with large, starchy, sweet-tasting, edible, tuberous roots.

POTENTIAL: A scalar quantity associated with a field or to the vector potential. Examples are the gravitational potential and the electric potential, from which the motion of gravitating or electrically charged bodies may be obtained. Specific forces which have associated potentials are the Coulomb potential, the van der Waals potential, the Lennard-Jones potential, the Yukawa potential, thermodynamic potential, Galvani potential and Volta potential.

POTENTIAL ACIDITY: In a soil, the amount of exchangeable hydrogen ion that can be made free or active in the soil solution by cation exchange. It is usually expressed in milliequivalents per unit mass of soil.

POTENTIAL BARRIER (POTENTIAL HILL): A region in a field of force containing a maximum of potential that opposes a particle on one side of it from passing to the other side. According to classical theory, a particle must possess energy in excess of the height of the potential barrier to pass it.

POTENTIAL CATION EXCHANGE CAPACITY: The total number of cations that a soil can adsorb at a specified pH.

POTENTIAL DIFFERENCE: 1. **Electrical potential difference** with the Symbol V , is a measure of the work done by a unit of charged particle as it flows between two points in a static electric field. Its unit is the volt (joule per coulomb). 2. **Gravitational potential difference** between two points on Earth is related to the energy needed to move a unit mass from one point to the other against the Earth's gravitational field. Its unit is joules per kilogram.

POTENTIAL DIVIDER (VOLTAGE DIVIDER): A simple linear circuit that produces an output voltage that is a fraction of its input voltage and diverts a voltage among the components of the divider. A simple example is a potentiometer with two resistors connected in series in an electric circuit in order to obtain a fraction of the potential difference or voltage across a battery or electrical source. It is commonly used to create a reference voltage and may also be used as a signal attenuator at low frequencies.

POTENTIAL ELECTRIC: The relative electrical state of an object.

POTENTIAL ENERGY: The energy of an object resulting from its position such as in electric, gravitational or magnetic field or because of its configuration rather than motion relative to other objects that it interacts with. For example, a raised weight, coiled spring or charged battery has potential energy. Water dropping down a waterfall, a mass held above the ground and a compressed spring all have potential energies equal to the work done against the force of gravity which changes to kinetic energy when they fall back to the ground or in the case of the spring, when the tension is released. Potential Energy, PE

is expressed as: mgh , where m is mass, g is acceleration due to gravity and h is height. This term was coined by the 19th century Scottish engineer and physicist William Rankine. Some types of potential energy are gravitational potential energy, chemical potential energy, and electrical potential energy. **Gravitational potential energy (GPE)** is the energy an object has due to its position in a gravitational field relative to some reference point. It is the energy stored in an object as the result of its vertical position or height. The energy is stored as the result of the gravitational attraction of the Earth for the object. Its unit is joule and the symbol is J. GPE is related to height and the higher up an object is placed, the more GPE it has. Its equation is $EP = mgh$, where EP is the potential energy (measured in joules, J), m is mass of the object (measured in kg), g is gravitational acceleration ($9.8 \text{ m}\cdot\text{s}^{-2}$), and h is the perpendicular height from the reference point (measured in metres, m). **Chemical potential energy** is the energy stored in the bonds of atoms and molecules of a substance and released during a chemical reaction. For example, the chemical potential energy stored in coal is released as heat when it is burnt. **Electrical potential energy** also known as **electrostatic potential energy** is the energy that charged particles such as electrons and protons have because of their own electric charge and their relative position to other charged particles. For example, when a circuit is connected to a power source it must store electric potential energy and supply it to heaters, lights, motors, etc.

POTENTIAL FLOW: A motion of a body in which the velocity is given by the gradient of a scalar field that is characterised by an irrotational velocity field due to the curl of the gradient of a scalar always being equal to zero. Examples of practical applications are the outer flow field for aerofoils, water waves, electroosmotic flow and groundwater flow.

POTENTIAL FUNCTION: If a vector field $f(x, y, z)$ is the gradient of a scalar function $U(x, y, z)$ then U is said to be the potential function for the field f . For example, if U represents potential energy, then the force acting on a body is the negative of the gradient of U . A potential function of f can be found only if the curl of f is zero.

POTENTIAL TEMPERATURE: The temperature that a dry air parcel would have if lowered or raised adiabatically to a level of 1000 mbar pressure (or other arbitrary standard pressure).

POTENTIAL-DETERMINING IONS (P.D. IONS): Those species which by virtue of their electron distribution between the solid and liquid phase (or by their equilibrium with electrons in the solid) determine the difference in Galvani potential between these phases. This definition requires that adsorbed p.d. ions are part of the adsorbent and belong to the category of surface ions.

POTENTIAL-ENERGY CURVE: A graph of the potential energy of electrons in a diatomic molecule, in which the potential energy of an electronic state is plotted vertically and the internuclear distance is plotted horizontally, with the minimum of the curve being the average internuclear distance. The Morse potential gives a good analytic description of a potential-energy curve. The dissociation energy and certain quantities of interest in vibrational spectroscopy can be found from the potential-energy curve.

POTENTIAL-ENERGY SURFACE: A multidimensional surface used within the adiabatic or Born–Oppenheimer approximation in quantum mechanics and statistical mechanics to model chemical reactions and interactions in simple chemical and physical systems. A potential-energy curve is the simplest type of potential-energy surface in which the total potential energy of an atom arrangement can be represented as a curve or multidimensional surface, with atomic positions as variables. To model an actual chemical reaction, a potential energy surface must be created to take into account every possible orientation of the reactant and product molecules and the electronic energy of each of these orientations.

POTENTIATION: The interaction of two substances, such as drugs, muscles or events to produce an effect that is greater than the sum of the effects of each one alone.

POTENTIOMETER: A three-terminal resistor with a sliding contact that forms an adjustable voltage divider, typically used as a brightness or volume control on audio equipment. If operated by a special mechanism they can be used as position transducers. As illustrated, when a cell or source of electromotive force is connected across a uniform meter wire (1m long), AB, there is a potential gradient along the wire from one end to the other. A potential difference (p.d.) between one end and any point on the wire can be tapped off. (Figure a). The p.d., V , is

$$\frac{AC}{AB} \times \text{p.d. between A and B. The}$$

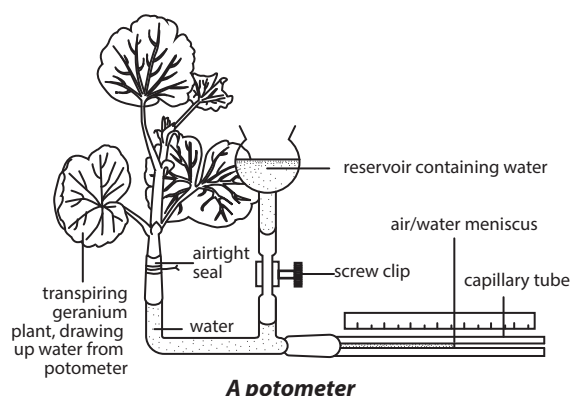
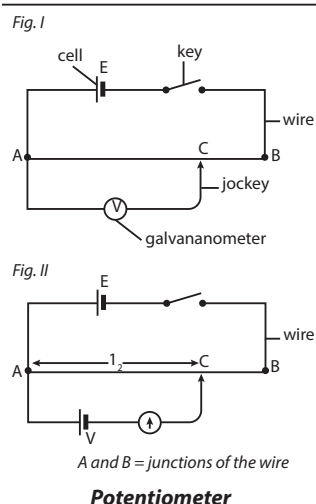
p.d., V , increases as AC increases.

If a second p.d. exactly equal to V is placed to act against the p.d. across AC, then no current flows through a galvanometer (V) in the circuit since the p.d.s are the same in value and acting against each other (fig. b).

POTENTIOMETRIC DETECTION METHOD: A detection method in which the potential of a cell is related to the concentration (activity) of a reactant which is a component of the cell fluid.

POTENTIOMETRIC TITRATION (CONSTANT-CURRENT TITRATION): A solution titration in which changes in the voltage or potential difference across an electrode immersed in the solution is measured to determine the end-point of the titration reaction.

POTOMETER: A device used to measure the rate of water loss of a plant shoot, due to transpiration. There are two main types of potometers: the bubble potometer and the mass potometer.



POTTED PLANTS (INDOOR PLANTS): Plants that are planted and grown in containers rather than in the soil for decorative purposes, as well as for indoor air purification. Examples of such plants include aloe vera, cacti, orchid cacti, opuntia, prickly pear, christmas cactus and jade plant.

POUCH: 1. A bag-like pocket of skin in which a female marsupial such as a kangaroo carries her young. 2. A bag-like pocket of skin in the cheeks of some rodents such as hamsters, in which they store and carry food. 3. An area of bag-like loose skin such as under the eyes of a sick person.

POULT: A young fowl, of turkey, pheasant, or a similar bird.

POULTRY: Any domestic bird such as chicken, turkey, goose and duck being raised for eggs and meat.

POUND: 1. The unit of mass in the avoirdupois system or the foot, pound and second (f.p.s.) system. It is equal to 453.59237 grams. 2. A unit of apothecary weight, which is equal to 373.242 grams used only for the measurement of precious metals. 3. A British unit of force equal to the weight of a standard one-pound mass where the local acceleration of gravity is 9.817 metres per second per second. 4. Also called the pound sterling, it is the basic monetary unit of the United Kingdom based on the decimal system. One hundred pennies make one pound. Its symbol is £.

POUND-REBKA EXPERIMENT: Experiments conducted in 1960 by Robert Pound and his graduate student Glen A. Rebka Jr., American physicists, of the decrease in frequency of a photon as it climbs in a gravitational field to test Albert Einstein's theory of general relativity. It is an experiment which measures the redshift of light moving in a gravitational field, or, equivalently, a test of the general relativity prediction that clocks should run at different rates at different places in a gravitational field.

POUR PLATE: In growth culture, it is a method of performing a plate count of microorganisms.

POUNDAL: The unit of force in foot pound second (f.p.s.) system, which is equal to the force required to impart an acceleration of one foot per second to a mass of one pound.

POURBAIX DIAGRAM: A diagram showing how oxidation-reduction behaviour for compounds of a given element depends on pH. Predominant ion boundaries are represented by lines and can be read much like a standard phase diagram with a different set of axes. The diagrams are named after Marcel Pourbaix (1904-1998), Russian-born Belgian chemist who invented them.

POWDER METALLURGY: A process in which powdered metals or alloys are fabricated into a variety of shapes at high temperatures rather than casting from molten metal or forging at softening temperatures.

POWER: 1. A representation by exponent or index, denoted by a superior small numeral or superscript. It indicates the number of times a number multiplies itself. 2. The power of a point with respect to a circle is the quantity $d^2 - R^2$, where R is the radius of the given circle and d is the distance from the given point to the centre of the circle.

3. The sum of integrals $P(\mathbf{R}_t) = \int_V \rho \mathbf{b} \cdot \mathbf{v} dv + \int_{\partial V} \mathbf{t} \cdot \mathbf{v} dv$, respectively over the volume and the surface area of the configuration at time t of a sub-body, $\mathbf{R}$, of which ρ is the density, $\mathbf{b}$ is the body force density, $\mathbf{v}$ is the velocity and $\mathbf{t}$ is the contact force density. 4. In statistics, the probability of rejecting the null hypothesis in a statistical test when it is in fact false; the power of a test of a given null clearly depends on the particular alternative hypothesis against which it is being tested. 5. The rate of doing work or consuming energy. It is measured in watts (joules per second). Its symbol is P . 6. The product of applied potential difference or voltage (V) and current (I) in a direct-current circuit system, expressed as IV . 7. The product of the effective values of the voltage (V) and current (I) with the cosine of the phase angle ($\cos\theta$) between current and voltage in an alternating current (a.c.) circuit. It is expressed as $IV\cos\theta$ or I^2R , where R is the resistance of the circuit. For the steady-state a.c. circuit under sinusoidal voltage and current, $\cos\theta$ is the power factor (p. f.). 8. A measure of the amount by which a lens will deviate light rays.

POWER FACTOR (P.F.): The ratio of the real power flowing to the load to the apparent power in an a.c. circuit. It has no dimensions and the value is between 0 and 1 and frequently expressed as a percentage, for example, 0.5 pf = 50% p.f. For the steady-state a.c. circuit under sinusoidal voltage and current, $\text{p.f.} = \cos\theta$, where θ is the phase angle between the voltage and current. Real power is the capacity of the circuit for performing work in a particular time. Apparent power is the product of the current and voltage of the circuit. Due to energy stored in the load and returned to the source or due to a non-linear load that distorts the wave shape of the current drawn from the source, the apparent power will be greater than the real power. Power factor is related to the real power in kilowatt (kw)

and apparent power in kilovolt ampere (kva) as p.f. = kw/kva. In an electric power system, a load with a low power factor draws more current than a load with a high power factor for the same amount of useful power transferred. Linear loads with low power factor such as induction motors can be corrected with a passive network of capacitors or inductors.

POWER MANAGEMENT UNIT (PMU): A microcontroller or integrated circuit, that controls the power functions of Macintosh computers. PMU contains several parts, including memory, software, firmware and its own CPU.

POWER MEAN (GENERALISED MEAN; HÖLDER MEAN): In mathematics, a family of functions for aggregating sets of numbers that include as special cases the Pythagorean means (arithmetic, geometric and harmonic means).

POWER REACTOR: A nuclear reactor designed to produce electrical power for the propulsion of ships and power plants. The energy released in nuclear fission process generates heat, which is used to boil water to produce pressurised steam which drives a steam turbine that generates electricity

POWER RULE: The specialisation of the chain rule to the case of an arbitrary power of a given function.

If $f(x) = g(x)^r$, then $\frac{df(x)}{dx} = [rg(x)^{r-1}] \frac{dg(x)}{dx}$.

POWER SERIES: A series whose terms contain ascending positive integral powers of a variable, such as $a_0 + a_1x + a_2x^2 + \dots + a_nx^n$, where a_0, a_1 and a_2 are constants. For example, any polynomial can be easily expressed as a power series around any centre c , even though it is one with most coefficients equal to zero.

POWER SET: In mathematics, a set of which the elements are all the subsets of a given set. It is written $P(S)$ or 2^S and has strictly more members than the given set.

POWER SUPPLY UNIT (PSU): A component that converts AC to DC to power computers and and plugs into standard wall sockets. In desktop computers it is located inside the systems unit.

POWER TOWER: A way of representing very large numbers in terms of stacks of exponents.

POWERFUL NUMBER (SQUAREFULL NUMBER): A positive whole number n such that for every prime number p dividing n , p^2 also divides n . Every powerful number can be written as a^2b^3 , where a and b are positive integers. The first 20 powerful numbers are 1, 4, 8, 9, 16, 25, 27, 32, 36, 49, 64, 72, 81, 100, 108, 121, 125, 128, 144 and 169. There are pairs of consecutive powerful numbers, such as (8,9), (288,289) and (675,676). However, there are no three consecutive powerful numbers.

POWERPOINT: Microsoft software presentation program included in the standard Office suite along with Microsoft Word and Excel. The software allows users to create anything from basic slide shows to complex presentations. It is often used to create business presentations, but can also be used for educational or informal purposes. The presentations are comprised of slides, which may contain text, images and other media, such as audio clips and movies. Sound effects and animated transitions can also be included to add extra appeal to the presentation.

POXVIRUS: An oval-shaped DNA-containing virus responsible for a wide range of pox diseases in humans and other animals that cause pus-filled blisters or pustules that form on the skin or one of a group of DNA containing viruses, often enclosed in an outer membrane that typically produce skin lesions in vertebrates. They include viruses of smallpox, cowpox, etc.

POYNTING VECTOR: A vector that gives a measure of the flow of energy in an electromagnetic field. Its Symbol is S . It is equal to the cross product of the electric-field strength and the magnetic-field strength: $S = EH$, where E is the electrical field strength and H is the magnetic field strength.

PPb: Abbreviation for **parts per billion**. It is a unit used for concentration of very small values. It denotes one part per 1,000,000,000 parts. For example, one part per billion is one part of solute per one billion parts solvent. It is also equivalent to microgram per kilogram or microgram per litre.

PRACTICAL EXTRACTION AND REPORT LANGUAGE

(PERL): A script programming language written by Larry Wall that has good text manipulation facilities and can also handle binary files.

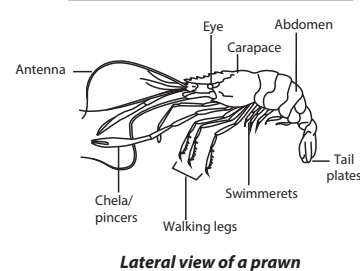
PRACTICAL NUMBER (PANARITHMIC NUMBER): A positive integer n such that all smaller positive integers can be represented as sums of distinct divisors of n . For example, 12 is a practical number because all the numbers from 1 to 11 can be expressed as sums of its divisors 1, 2, 3, 4 and 6. As well as these divisors themselves, we have $5 = 3 + 2$, $7 = 6 + 1$, $8 = 6 + 2$, $9 = 6 + 3$, $10 = 6 + 3 + 1$ and $11 = 6 + 3 + 2$. The first few practical numbers are 1, 2, 4, 6, 8, 12, 16, 18, 20, 24, ... All perfect numbers are practical.

PRAEMORSE: Abruptly truncated, as if bitten off.

PRAIRIE: A major regional biome found in North American in which grasses are the dominant vegetation. Tropical grasslands are known as **savannah** located near the equator, between the Tropic of Cancer and the Tropic of Capricorn. Grasslands found in the temperate regions of Eurasia are known as **steppes**.

PRASEODYMIUM: A soft, ductile, silver-white, malleable metallic element belonging to the rare-earth group or of the lanthanide series with the symbol Pr, atomic number 59, relative density 6.77, relative atomic mass 140.907, liquid density at melting point 6.50 g/cm^3 , melting point 935°C (1208 K) and boiling point 3520°C (3793 K). It is used to colour glass and ceramics yellow, as a core material for carbon arcs and in metallic alloys.

PRAWN: Any of the decapod crustaceans of the suborder Dendrobranchiata consisting of 540 extant species, in seven families. They differ from shrimp by the branching form of the gills and by the fact that they do not brood their eggs, but release them directly into the water. Prawns are of varying size and may reach a length of over 330 millimetres and weigh up to 450 grams. They are widely fished and farmed for their sweet, firm, meaty flesh.



Lateral view of a prawn

PRE-CAVAL VEIN: The anterior or superior vena cava, which is a short vein with a large diameter that carries blood from the head and forelimbs to the right atrium of the heart.

PRE-EMERGENCE HERBICIDE: Herbicide that is applied to the soil before the weeds have emerged. This can be used to prevent certain weeds, such as crabgrass, annual bluegrass, purslane, prostrate pigweed, knotweed and others from germinating.

PRE-MULTIPLICATION: In mathematics, the pre-multiplication by a matrix or ring element is multiplication on the left rather than on the right, called post-multiplication, this is important in any non-commutative setting.

PRE-SYNAPTIC TERMINAL: The region of the neuron conducting signals towards the synapse.

PREADAPTATION: The possession of a character that allows an organism to exploit a pre-existing structure inherited from an ancestor for a potentially unrelated function. Examples of preadaptation include dinosaurs having used feathers for insulation and display before using them to fly and sweat glands in mammals being transformed into mammary glands. Another example is the hypothesis proposed that before early humans became bipedal, they began engaging in

squat feeding or turning over rocks and leaves to find insects, worms, snails and other food.

PREAMPLIFIER: An amplifier in a radio, record player, etc, which prepares an electronic signal for further amplification or processing. It improves the signal-to-noise ratio of the whole system.

PREBIOTIC: Not yet alive; a chemical system or environment that is a precursor to life. **Prebiotic evolution** is the development of increasingly complex molecules and aggregates of molecules which, if continued, might lead to living systems.

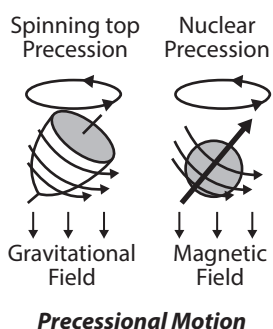
PRECAMBRIAN: The time from the formation of the Earth, believed to be about 46 million years ago to the beginning of the Cambrian period, some 570 million years ago.

PRECAUTION: A safety measure taken in advance to prevent something dangerous, unpleasant, or inconvenient from happening. For example, in a science laboratory, a few of the precautions are: (i) do not touch any equipment, chemicals, or other materials in the laboratory area until instructed to do so; (ii) do not eat food, drink beverages, or chew gum; (iii) wear safety goggles anytime chemicals, heat, or glassware are used and laboratory coat or smock should be worn during laboratory experiments; (iv) treat every chemical as if it were hazardous; (v) never return chemicals to reagent bottles; (vi) Obtain permission before operating any high voltage equipment; etc.

PRECEDENCE: In mathematics, the order in which a sequence of operations are evaluated. By convention, formulae are evaluated in the following sequence exponentiation, multiplication and division and addition and subtraction. These can be modified by the use of parentheses so that expressions in parentheses are evaluated first. In the case of operations of the same level of precedence, evaluation proceeds from left to right. For example, in $[(2 + 3) \times 5]$, the addition has precedence over the multiplication. The brackets also have precedence over one another. The greater the precedence of an operator, the narrower its scope.

PRECESSION OF THE EQUINOXES: The slow westward motion of the equinoxes (points where the Sun crosses the equator) about the ecliptic as a result of Earth's precessional motion. The equinoxes move round the ecliptic with a period of 25,800 years caused by the gravity of the Sun and the Moon acting on the Earth's equatorial bulge.

PRECESSIONAL MOTION: A form of motion that occurs when a torque is applied to a rotating body in such a way that it tends to change the direction of its axis of rotation. Precessional effects occur in atoms in magnetic fields. Precessional motion also refers to the apparent movement of stars with regard to the Earth's axis caused by the gravity of the Sun and the Moon acting on the Earth's equatorial bulge. The axial north pole revolves around the ecliptic north pole. The Earth goes through one complete precession cycle in a period of approximately 25,800 years. The shift is 1 degree in 180 years.



PRECIPITATE (PPT): 1. The solid substance formed by mixing two solutions in a double decomposition reaction and which settles to the bottom of the container in which the reaction takes place. 2. To form an insoluble compound either by reacting two salts or by changing the temperature to affect the solubility of the compound.

PRECIPITATION: 1. The process of a solid separating from a suspension, sol or solution during a chemical reaction. For example, in a sol the particles are precipitated by coagulation and a solution of salt may be concentrated by evaporation until the salt crystallises. The separation of particles of a solid or liquid suspended in a gas is also a precipitation. 2. The formation of rain, snow or hail from moisture in the air or rain, snow or hail, all of which are formed by condensation of moisture in the atmosphere that fall to the ground.

PRECIPITATION FRACTIONATION: A process in which a polymeric material, consisting of macromolecules differing in some characteristic affecting their solubility, is separated from solution into fractions by successively decreasing the solution power of the solvent, resulting in the repeated formation of a two-phase system in which the less soluble components concentrate in the polymer-rich phase.

PRECIPITATION INTERCEPTION: The stopping, interrupting, and temporary holding of precipitation in any form by tree crowns, and other vegetative bodies, before it reaches the soil. Part of the precipitation that does not reach the soil is returned to the atmosphere through evaporation and sublimation.

PRECIPITIN: Any antibody that combines with a specific soluble antigen to form a precipitate.

PRECISION: 1. In mathematics, the accuracy with which a given calculation is performed. 2. Exactness or accuracy. For example, precision instruments measure exactly to the specification required. 3. In computer science, precision of a numerical quantity is a measure of the detail in which the quantity is expressed measured in bits, but sometimes in decimal digits. On most computers **single precision** is somewhere between 9 and 16 decimal digits; **double precision** refers to calculation of twice that length and **multiple, high or extended precision** refer to calculations involving greater accuracy.

PRECOCIAL SPECIES: Any species of bird or mammal in which hatching or birth occurs at a relatively mature stage of development. The young of such species are capable of locomotion and leave the nest soon after hatching. Examples are herons, hawks, woodpeckers, owls, domestic chicken, cattle, sheep and horses.

PRECOCIOUS: 1. Unusually advanced or maturing in development. 2. In botany, it is flowering before developing leaves.

PRECOMPACT: In mathematics, the precompact of a set A in a locally convex space is such that, for any neighbourhood U of zero, there is a finite set F for which A is a subset of F + U.

PRECONCENTRATION: An operation or process as a result of which the ratio of the concentration or the amount of microcomponents (trace constituents) and macrocomponents (matrix) increases.

PRECONDITION: A condition that must be true of the parameters of a method and conditions a given problem into a form that is more suitable for solution prior to beginning a computation. For example, one may precondition a matrix by transforming it in order to increase the number of near unit eigenvalues.

PRECONSOLIDATION PRESSURE: The greatest effective vertical overburden stress to which a soil has ever been subjected. This quantity is important in geotechnical engineering, particularly for finding the expected settlement of foundations and embankments.

PRECURSOR: 1. In chemistry, it is a compound which participates as starting material or an intermediate in producing another compound through a chain of chemical reactions. For example, isoprene (2-methyl-1, 3-butadiene) is a precursor for the synthesis of terpenoid compounds. 2. In physics, precursors are characteristic wave patterns caused by dispersion of an impulse's frequency components as it propagates through a medium. 3. In climate change, precursors are atmospheric compounds which themselves are not greenhouse gases or aerosols, but which have an effect on greenhouse gas or aerosol concentrations by taking part in physical or chemical processes regulating their production or destruction rates.

PREDATION: A way of life, in which one type of wildlife known as the predator hunts and eats another type called the prey or the relationship between two groups of animals in which one species hunts, kills and eats the other.

PREDATORY RELEASE: The removal of a predator species from a prey species such as by great reduction in the predator's population size or by the migration of the prey species to an area without major predators. The removal of the predator releases the prey from one of the factors limiting its population size.

PREDECESSOR: In mathematics, the predecessor of an ordinal is a number having the given number as successor. For example, the predecessor of 1 is 0; as a consequence, ω has no predecessor while being the predecessor of the number $\omega + 1$.

PREDICATE: In logic, a property, characteristic or attribute that may be true or false depending on the values of its variables. A predicate with more than one subject is a **relation**. The categorical statement "all men are mortal" relates two predicates, "... is a man" and "... is mortal". Also, the term of a categorial proposition that is affirmed or denied of its subject; for example, all men are mortal" "men" is the subject and "is mortal" is the predicate.

PREDICATE CALCULUS (FUNCTIONAL CALCULUS): The system of symbolic logic concerned not only to represent the logical relations between sentences or propositions as wholes, but also to consider their internal structure in terms of subject and predicate. The primitive terms are individual names, predicates and variables that may be bound by quantifiers. If quantification is restricted to individuals, it is lower (or first order) predicate calculus (LPC) and is consistent, complete, but not decidable.

PREDICT: To guess or use specific knowledge or understanding to estimate a future event, for example, to predict that there would be drought the following year.

PREDICTED VARIABLE (DEPENDENT VARIABLE; RESPONSE VARIABLE): In statistics, it is the variable whose values are measured or observed in an experiment for different values of the independent variable.

PREDICATIVE: In logic, predicative is a term or expression given in terms that do not require quantification over entities of the same type as that which is thereby defined.

PREDICTION LINE: A line on a graph of data that is used to predict values that are not in the data set. In statistics, prediction lines can be fit to data using a technique called **regression analysis**.

PREDISSOCIATION: Dissociation occurring by tunnelling from a 'bound' to an 'unbound' vibronic state. In an absorption spectrum of a molecular entity, the appearance of a diffuse band region within a series of sharp bands is called predissociation, since irradiation with frequencies within the diffuse region leads to effective dissociation. The energy of the band is smaller than that of the dissociation continuum of the bound state.

PREDUAL SPACE: In mathematics, a normed space of which some given space is the dual Banach space.

PRE-ECLAMPSIA: Agents or substances containing properties used to treat hypertension during pregnancy. Severe pre-eclampsia can cause major problems with the liver and cause blood clotting that can cause serious health complications and even be fatal to both mother and baby. Certain herbs are used to treat pre-eclampsia which include lemon, ginger, carrot, garlic and beet.

PREEN GLAND (UROPYGIAL GLAND): The oil-secreting gland found on the back, at the base of the tail in most birds; used for preening (cleaning and arranging feathers using the beak) which is part of feather care activities.

PREFERENCE (PREFERENCE ORDER): In mathematics, a partial order or other similar relation especially in an economic or related setting.

PREFERENTIAL SORPTION (SELECTIVE SORPTION): An equilibrium phenomenon, operative in polymer solutions in multicomponent solvents and in polymer networks swollen by multicomponent solvents that produces differences in solvent composition in the polymer-containing region and in the pure solvent which is in thermodynamic equilibrium with that region.

PREFIX: In scientific notation, a method of expressing value of units that uses some power of 10. For example, the prefix for kilogram (kg) is kilo. One kg is equal to 1 000 g or 1×10^3 g. Grouping the 10^3 and the g together, 10^3 can be replaced with the prefix k (kilo). Some important examples of prefixes are: yotta (10^{24}) with the symbol Y,

yocto with the symbol y for 10^{-24} , zetta with the symbol Z for 10^{21} , zepto with the symbol z for 10^{-21} , exa with the symbol E for 10^{18} , atto with the symbol a for 10^{-18} , peta with the symbol P for 10^{15} , femto with the symbol f for 10^{-15} , tera with the symbol T for 10^{12} , pico with the symbol p for 10^{-12} , giga with the symbol G for 10^9 , nano with the symbol n for 10^{-9} , mega with the symbol M for 10^6 , micro with the symbol μ for 10^{-6} , hecto with the symbol h for 10^2 , centi with the symbol c for 10^{-2} , deca with the symbol da for 10^1 and deci with the symbol d for 10^{-1} .

PREFIX NOTATION (POLISH NOTATION): A system of symbolic notation of mathematical operations used in arithmetic, logics and computer programming language which omits parentheses and writes the operators before their arguments. This eliminates the need for rules of operator precedence. For example, P or Q ($P \vee Q$) is written Apq and if P then Q ($P \rightarrow Q$) is written Cpq ; and $(5 - 6) \times 7$ can be written as $\times - 5 6 7$. Lisp, a computer programming language uses Polish notation.

PREGNANCY: The period, during which the embryo grows within the uterus, averaging 266 days (38 weeks) from conception to childbirth in humans. Interruption of menstruation and change in structure and shape of the mammary glands (breasts) are early signs. Morning sickness, which may be mild or incapacitating, is a common symptom. Later an increase in abdominal size is seen and other abdominal organs are pushed up by the enlarging uterus (womb). Ligaments and joints become more flexible in preparation for delivery.

PREHENSILE: Adapted for grasping or holding. Tendrils in plants exhibit this behaviour. Some arboreal creatures such as monkeys have prehensile tails which are used to hold and manipulate objects. It also aids them in finding and eating food on trees. Such a tail aids animals to anchor the body in order to dangle from a branch.

PREHISTORIC: Of, relating to or belonging to the era before recorded history. It is about 3.5 million years ago to the time when humans began to keep written records about 3500 BC.

PREIMAGE: 1. The original figure in a transformation. 2. Also called **inverse image**, it is a set of all elements that map into a given subset of the codomain of a function.

PREIMPLANTATION GENETIC DIAGNOSIS (PGD): A diagnostic procedure, used in genetic screening in which a single cell is removed from an embryo two or three days after it has been conceived through in vitro fertilisation. A single cell is removed from an eight-stage embryo and subjected to genetic testing for a specific disease allele using either a gene probe and fluorescence in situ hybridisation or the polymerase chain reaction. The procedure can also be used to determine the gender of the embryo. PGD has been successful in screening for such diseases as cystic fibrosis and Tay-Sachs as well as for genes that predispose a person to breast or colon cancer.

PREKESE (*Tetrapleura tetraptera*): A flowering plant species in the pea family, which is native to West Africa (Ghana) with the local name, Prekese. It is deciduous and towers up to about 25 centimetres in height with sessile, mostly opposite and sometimes alternate leaves. It has slender bole with fairly small crown. The bark is smooth, thin and grey-brown. It produces fruits which are slightly curved and are very persistent, hanging at the ends of branches on stout stalks. The fruits are shiny, glabrous and dark purple-brown. The bark of the tree is used for medicinal purposes and also as dietary supplement, rich in vitamins. They are also used to spice dishes.

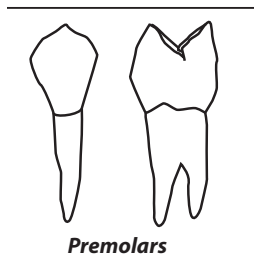
PREMISE (PREMISS): A term used in logic to represent an initial proposition or a statement that is known or assumed to be true and on which a logical argument is based.

PREMIUM COKE: An extremely well graphitizing carbon with a high degree of optical anisotropy (isochromatic areas of optical texture above about $100 \mu\text{m}$). It is characterised by a combination of the following properties which differ significantly from those of regular

coke: high real density, low reversible thermal expansion and low ash content combined, in most cases, with low sulfur content. Premium coke is mainly produced from tars or residues from petrochemistry by the delayed coking process. Also, refined coal tar pitches are used as precursors for premium coke production.

PREMIX BURNER: A burner in which fuel and oxidant are thoroughly mixed inside the burner housing before they leave the burner ports and enter the primary-combustion or inner zone of the flame. This type of burner usually produces an approximately laminar flame and is commonly combined with a separate unit for nebulising the sample.

PREMOLAR (BICUSPIDS): Transitional teeth located between the canine and molar teeth. In humans, there are two premolars per quadrant, a total of eight in the mouth, namely: maxillary first premolar, maxillary second premolar, mandibular first premolar and the mandibular second premolar. They usually have one or two roots. They are adapted for grinding in herbivores. In carnivores they may be carnassials.



PREMOTOR CORTEX: A region of cerebral cortex in the vertebrate brain situated immediately anterior to the primary motor cortex, which controls motor skills of a repetitious or patterned nature. It contains neurones that coordinate the movements of several muscles by sending impulses to the primary motor cortex and other motor centres in the brain such as the basal ganglia. This activity is important to the sensory guidance of movement and control of proximal and trunk muscles of the body. It is thought to play a crucial role, with other areas of the cortex, in performing complex movements and it stores learned motor patterns, such as that employed in writing one's signature.

PREMUTATION: In disorders caused by trinucleotide repeat expansions, an abnormally large allele that is not associated with clinical symptoms but that can expand into a full mutation when transmitted to offspring.

PRENEX NORMAL FORM: In logic, a formula of predicate calculus of the form $(Qx_1)(Qx_2) \dots (Qx_n)B$, where each (Qx_i) is an existential or universal quantifier, the variables are distinct and B is an open sentence. Each formula may be shown equivalent to one in prenex form by use of the prenex operations.

PRENEX OPERATION: In logic, any of the operations by which any well-formed formula of predicate calculus can be transformed into equivalent formulae in prenex normal form; for example, $\neg(\exists x)Fx \equiv (\forall x) \neg Fx$ and $(\forall x)Fx \equiv (\exists x)Fx$.

PREONS: Hypothetical entities that are postulated as subcomponents of quarks and leptons.

PREPOLYMER: Polymer or oligomer whose molecules are capable of entering, through reactive groups, into further polymerisation and thereby contributing more than one structural unit to at least one type of chain of the final polymer.

PRESBYOPIA: A condition which develops in human eyes, associated with aging, which causes progressive reduction in the eye's ability to focus on near objects, leading to difficulty in reading at the normal distance. It normally starts at middle age and is due to age-related loss of elasticity of the lens.

PRESENTATION: In mathematics, a set, X, of generators together with a set, R, of relations, such that the group generated by X subject to the relations of R is isomorphic to a given group. A presentation of a group is usually denoted (X, R) .

PRESERVATION: The process of keeping food from decay by micro-organisms, enzymes and oxidation by pickling, freezing, drying, pasteurisation, smoking, salting, etc.

PRESERVATIVE: Chemical additives such as sodium and calcium propionate, sorbic acid, butylated hydroxytoluene, and tetracyclines,

which are added to commercial food in order to prevent or slow down food spoilage due to the growth of bacteria, yeast, mould and other microorganisms and therefore extending its shelf-life.

PRESSURE: The ratio of the force acting on a body to the area of the surface, expressed as Force/Area. The SI unit of pressure is pascal (newton per square metre). **Pressure law** is a law that relates volume and amount when pressure and temperature are held constant. It states that the pressure of a fixed mass of gas at constant volume is directly proportional to its absolute temperature. This may be expressed as $P \propto T$, if V is fixed:

$$\frac{P}{T} = \text{constant or}$$

$$P = \text{constant} \times T.$$

It can also be expressed as

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}, \text{ where } P_1 \text{ and } T_1 \text{ are}$$

the initial pressure and absolute temperature of a gas and P_2 and T_2 are its final pressure and temperature.

PRESSURE COOKER: A closed pot in which food is cooked by steam produced by boiling water under pressure, making the food cook at higher temperatures than normal (100 °C, 373 K). It is normally a cooking pot with a locking, airtight lid and a valve system to regulate internal pressure.

PRESSURE FLOW HYPOTHESIS (MASS FLOW HYPOTHESIS):

A hypothesis explaining the movement of food through the phloem. According to this hypothesis, the solution containing nutrient sugars moves from sources such as the mesophyll layer of leaves, creating a diffusion gradient (osmotic gradient) and moves through the phloem until it reaches a sink such as a growing root, where they are needed. The movement in phloem is bidirectional; in xylem cells, it is unidirectional or moves upward.

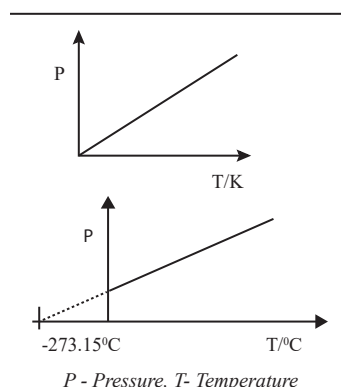
PRESSURE GAUGE: Any instrument used to measure the pressure of a fluid whether liquid or gas that states the force the fluid applies to a unit area at rest. The reading on the gauge is called the gauge pressure and is the difference between two pressures. Examples of pressure gauges are liquid-column gauge such as the mercury barometer and the manometer, the expanding element gauge such as the bourdon gauge and the aneroid barometer and the electrical transducer.

PRESSURE MEMBRANE: A membrane, permeable to water and only very slightly permeable to gas when wet, through which water can escape from a soil sample in response to a pressure gradient.

PRESSURE POTENTIAL: An important component of the total water potential within plant cells due to the hydrostatic pressure, which helps plants retain the structural rigidity of the cell wall. The pressure potential in a turgid plant cell is usually positive as the entry of water causes the protoplast to push against the cell wall and in xylem cells pressure potential is almost zero, because of transpiration. Water at atmospheric pressure has a pressure potential of zero.

PRESSURE-SENSITIVE DETECTOR: A detector which employs the pressure change which results from the absorption of radiation.

PRESSURISED WATER REACTOR (PWR): One of the two types of light water reactors (LWR) in which the heat generated in the nuclear core is removed by the primary coolant, water, circulated under high pressure to prevent it from boiling to the reactor core. The heated water transfers thermal energy to a steam generator, which



P - Pressure, T- Temperature

Graphical illustration of Pressure law

can be used to generate electricity. The other type of light water reactor (LWR) is boiling water reactor (BWR).

PRESUMPTIVE: In embryology, pertaining to embryonic tissue in the course of normal development, that is not yet determined but which will eventually develop into a particular structure by virtue of its position in the embryo.

PRESUMPTIVE TEST: An analysis of a sample of a compound or class of compounds in which colour changes occur when certain reagents are added to them. The reagents added usually react with some functional group present in the compound which is accompanied by definite colour changes. A second test may be necessary to confirm presumptive test results. Examples are the Duquenois-Levine test for marijuana, Cobalt thiocyanate test for cocaine and Kastle-Meyer test for blood.

PRESUPPOSE: In logic, to require a condition to be satisfied as a precondition for a statement to be either true or false or for a speech act to be felicitous; for example, the question "have you stopped hitting your wife?" presupposes that the person to whom the question is directed both has a wife and used to beat her.

PRESYMPTOMATIC SCREENING: A test which detects genetic disorders such as Huntington disease that manifests later in life.

PRESYNAPTIC MEMBRANE: The cell membrane or muscle fibre of a neurone that releases neurotransmitter into the synaptic cleft between the nerve cells.

PRESYNAPTIC NEURON: A neuron (nerve cell) from the axon terminal of which an electrical impulse is transmitted across a synaptic cleft to the cell body or one or more dendrites of a postsynaptic neuron by the release of a chemical neurotransmitter.

PRETREATMENT: Following the preparation of a catalyst or following its insertion into a catalytic reactor, it is often subjected to various treatments before the start of a catalytic run. The term pretreatment may, in general, be applied to this set of treatments. In some cases the word activation is used. It implies that the material is converted into a catalyst or into a very much more effective one by the pretreatment.

PRETTY GOOD PRIVACY (PGP): A very secure public key data encryption and decryption software used over the Internet. It was developed in 1991 by Philip Zimmerman.

PREVAILING WIND: The direction from which wind most commonly blows in a locality observed during a given period. The periods most often used are the observational day, month, season and year.

PREVALENCE: A measure of the number of people in a population who have a particular disease at a given point in time. It can be measured in two ways: point prevalence (pt. prev.) and period prevalence (period prev.):

$$\text{Pt. prev.} = \frac{\text{no. of cases at a particular moment}}{\text{no in population at that moment}} \text{ and}$$

$$\text{Period prev.} = \frac{\text{no. of cases during a specified time period}}{\text{no in population at midpoint of period}}.$$

PREVERNAL: 1. Appearing in late winter or early spring as some crocus flowers. 2. Plants that unfold their leaves or flowers before the other plants in the same area.

PREVOST'S THEORY OF EXCHANGES: A theory proposed by Pierre Prévost (1751–1839) in 1791 in which a body is constantly exchanging and radiating an amount of energy at equal rates when it is in equilibrium with its surroundings and the temperature remains constant. However, if the body is not at the same temperature as its surroundings there is a net flow of energy between the surroundings and the body because of unequal emission and absorption.

PREY: An animal caught, killed and eaten by a predator.

PREZYGOTIC REPRODUCTIVE ISOLATING MECHANISMS (PREZYGOTIC RIMs):

One of the two categories of reproductive isolating mechanisms that prevent the formation of hybrids between members of different populations through ecological, temporal, ethological (behavioural), mechanical and gametic isolation. The other category is postzygotic barrier. Prezygotic mechanisms take effect before fertilisation and postzygotic ones take effect afterward fertilisation.

PRIBNOW BOX (MINUS 10 SEQUENCE; PRIBNOW-SCHALLER BOX):

The sequence of six nucleotides, TATAAT (thymine-adenine-thymine-...) occurring in the promoter region of prokaryote genes. It is also called the -10 sequence, because it is centred roughly 10 base pairs upstream from the site of initiation of transcription. The promoter region of DNA, which is the only place RNA polymerase can initiate binding to the DNA has 6 bases occurring in the sequence 5' TATA AT 3'. The promoter region is also rich in adenine and thymine to permit easy base pairs separation during prokaryotic transcription to allow access to the template strand, which is the strand of DNA that directs the synthesis of mRNA. The promoters are called Pribnow box in prokaryotes and TATA box (Hogness box) in eukaryotes. The Pribnow box or Pribnow-Schaller box is named after David Pribnow and Heinz Schaller.

PRICKLE: 1. A pointed process of a plant arising from the epidermal tissue and containing no woody or conducting tissue. Because it does not contain any strengthening vascular tissue it can easily be broken off or peeled off with the epidermis. It occurs in rose, *Mimosa pudica*, etc. Prickle is not the same as spines or thorns, which are small and sharp modified plant organs such as shoots and leaves respectively that have vascular bundles. 2. A sharp process or projection, as from the skin of an animal; a spine.

PRIDE OF BARBADOS (BARBADOS FLOWERFENCE; PEACOCK FLOWER; MEXICAN BIRD OF PARADISE; DWARF POINCIANA; *Caesalpinia pulcherrima*):

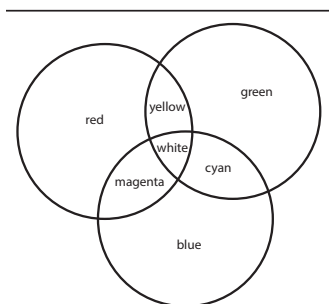
A shrub that grows up to about 4.5 metres if untrimmed. It is a member of the pea family, Fabaceae and can be found in most tropical countries. It grows commonly by roads and in hedges. It has bipinnately compound leaves with 6-10 pairs of leaflets. It blooms all year round with the flowers borne in racemes, with each flower having five petals which are orange, yellow or red in colour. It has ten stamens and a long carpel which projects from the centre of the flower. It is the national flower of the Caribbean island of Barbados.

PRIESTLY, JOSEPH (1733-1804): British theologian, political theorist and physical scientist, who in 1767, experimented with carbon dioxide from a nearby brewery and invented soda water. He was elected to the French Academy of Sciences in 1772 and to the St. Petersburg Academy in 1780. He emigrated to the USA in 1794. He is considered the discoverer of nitrogen, carbon monoxide, ammonia and several other gases. In 1774, he became the first person to isolate oxygen in its gaseous state, although Carl Wilhelm Scheele and Antoine Lavoisier also had a claim to the discovery.

PRIMAL DUAL METHODS: In mathematics, algorithms for solving constrained optimisation problems that explicitly use both primal information (information about the original problem) and dual information (information concerning the Lagrangian and associated multipliers).

PRIMARY CELL: An electrical cell in which the electrochemical reaction is not reversible and cannot be recharged. Examples are the Daniell cell, Leclanché cell, the Weston cell or any disposable battery. The reaction in primary cells cannot be reversed by running a current into the cell because batteries use up the materials in one or both of their electrodes.

PRIMARY COLOUR: Any of the three coloured lights consisting of red, blue and green that gives white light when mixed together in equal amounts or any colour that can be made by mixing together different primary colours. For additive combination of colours such as in overlapping projected lights or in CRT displays, the primary colours normally used are red, green and blue. For subtractive combination of colours such as in mixing of pigments or dyes, for example, in printing, the primaries normally used are cyan, magenta and yellow, though the set of red, yellow and blue is popular among artists. In the illustration, different secondary colours can be obtained by mixing primary colours. When red and green are mixed, yellow colour is obtained. When red and blue are mixed together, magenta colour is formed and when green and blue are mixed, cyan colour is obtained. To obtain white colour, the three primary colours (red, green and blue) are mixed together.



Mixing the three primary colours

PRIMARY COMPOUNDS: Chemicals such as carbohydrates, lipids, nucleotides and peptides made by plants essential for the plant's own metabolism.

PRIMARY CONSUMER: An organism such as butterfly described as herbivore, which is found near the very bottom of the food chain which consumes plant material.

PRIMARY CRYSTALLISATION: The first stage of crystallisation, considered to be ended when most of the spherulite surfaces impinge on each other. In isothermal crystallisation, primary crystallisation is often described by the Avrami equation.

PRIMARY ELECTRONS (PE): Electrons generated by a thermal or field emission source, used to bombard the specimen for generation of the analytical signals.

PRIMARY ENDOSPERM NUCLEUS: In flowering plants (angiosperms), the result of the fusion of a sperm nucleus and the (usually) two polar nuclei.

PRIMARY ENERGY: Energy in natural resources such as coal, crude oil, sunlight, and uranium that has not undergone any conversion or transformation by man and can be used directly, as it appears in the natural state. Some other such resources include the Sun, the wind, tides, and rivers. In contrast, **secondary energy** must undergo some conversion by man before it can be used. For example, petrol is derived from the refinement of crude oil and electric energy is obtained from the conversion of mechanical energy in the form of hydroelectric and wind plants.

PRIMARY FEATHER: The long, stiff main flight feather on the outer half of a bird's wing or tail. The wing feathers are called **remiges** and the tail ones are called **rectrices**.

PRIMARY GERM LAYERS: The three layers consisting of the ectoderm, mesoderm and endoderm of the late gastrula, into which most animal embryos differentiate and from which the organs and tissues of the body develop.

PRIMARY GROWTH: The elongation of a stem or stalk of plants that results from the activity of the apical (tip) meristems, producing subsequent expansion of cells.

PRIMARY IMMUNE RESPONSE: The initial immune response to an antigen by a mammal. It has specific antibodies appearing in the blood after a multiple day latent period. It is generally slower

and weaker than the secondary immune response. Depending on the nature of the antigen and the site of entry this response can take up to 14 days to resolve and leads to the generation of memory cells with a high specificity for the inducing antigen.

PRIMARY IMMUNODEFICIENCY: A group of rare inherited disorders of the immune system, including adenosine deaminase deficiency, severe combined immunodeficiency (SCID) and ataxia-telangiectasia.

PRIMARY ISOTOPE EFFECT: A kinetic isotope effect attributable to isotopic substitution of an atom to which a bond is made or broken in the rate-controlling step or in a pre-equilibrium step of a specified reaction. The corresponding isotope effect on the equilibrium constant of a reaction in which one or more bonds to isotopic atoms are broken, is called a 'primary equilibrium isotope effect'.

PRIMARY MACRONUTRIENTS: Elements such as nitrogen, phosphorus and potassium required by plants in relatively large quantities.

PRIMARY MEMORY (PRIMARY STORAGE; MAIN MEMORY; STORAGE): A medium that holds memory for short periods of time while a computer is working. Its performance is fast even though it has a much lower access time. This refers mainly to the random access memory (RAM), but may also include several types of memory, such as the processor cache and system read-only memory (ROM).

PRIMARY MINERAL: A mineral that formed during the original solidification (crystallisation) of a rock by igneous, hydrothermal, or pneumatolytic processes, and retains its original composition and form and has not been altered chemically. In contrast, **secondary minerals** form at a later time through processes such as weathering and hydrothermal alteration.

PRIMARY MIXTURE: A mixture obtained directly from two or more components intended for the preparation of more dilute calibration mixtures (called secondary or tertiary mixtures).

PRIMARY PHLOEM: The type of phloem derived from the primary meristem during the development of a vascular plant. The primary phloem is comprised of protophloem and metaphloem. Both monocots and dicots have primary phloems. Phloem is the main food-conducting tissue of plants, through which nutrients and photosynthate are conveyed between different sites.

PRIMARY PHOTOCHEMICAL PROCESS (PRIMARY PHOTOREACTION): Any elementary chemical process undergone by an electronically excited molecular entity and yielding a primary photoproduct.

PRIMARY POLLUTANT: A pollutant emitted directly into the air from identifiable sources, for example, sulfur dioxide and hydrocarbons. Secondary pollutants, such as ozone, are generated within the atmosphere through chemical changes which occur in primary pollutants.

PRIMARY PRODUCERS: Green plants which convert energy from sunlight into chemical energy, for example, herbivores are secondary producers as they utilise energy from green plants and supply energy to carnivores.

PRIMARY PRODUCT (PHOTO PRODUCT): The first observable chemical entity which is produced in the primary photochemical process and which is chemically different from the reactant.

PRIMARY PRODUCTIVITY: The total amount of organic matter synthesised by organisms in an ecosystem that produce biomass from inorganic compounds, mainly through the process of photosynthesis. The organisms responsible for primary production are known as **primary producers** or **autotrophs**. Examples are green plants and cyanobacteria, which form the base of the food chain.

PRIMARY SAMPLE: The collection of one or more increments or units initially taken from a population. The portions may be either

combined (composited or bulked sample) or kept separate (gross sample). If combined and mixed to homogeneity, it is a blended bulk sample. The term 'bulk sample' is commonly used in the sampling literature as the sample formed by combining increments. The term is ambiguous since it could also mean a sample from a bulk lot and it does not indicate whether the increments or units are kept separate or combined.

PRIMARY SEXUAL CHARACTERISTICS: Body structures which are directly responsible for reproduction. Examples are **testis** in males and **ovaries** in females.

PRIMARY STANDARD: 1. A substance of high purity, high stability, high equivalent weight, etc, that undergoes one invariable reaction with another. Examples are sodium chloride for standardisation of silver nitrate solutions; and sulfanilic acid for standardisation of sodium nitrite solutions. 2. A level that is accurate enough to be accepted as the norm and not calibrated by or subordinate to other standards. It is used to calibrate other standards.

PRIMARY STRUCTURE: The sequence of amino acid in a protein, linked together by a peptide bond (protein polymer), that exists between adjacent amino acids in the polypeptide chain specified by the nucleotide sequence of the segment of DNA containing the gene that codes for that protein. The primary structure of a protein determines how the protein folds into a unique three-dimensional structure, which in turn determines the biological function of the protein.

PRIMARY SUCCESSION: A type of ecological succession that occurs in an area, where there were originally no organisms. It is one of two types of biological and ecological succession, the other type is secondary succession. An example is the initial development of plant or animal communities in an area where no soil initially exists (such as caused by a lava flow following volcanic eruption or severe landslide that covered the land). The primary succession is important in pioneering the area to create conditions favourable for the growth of other forms of plants and animals.

PRIMARY WAVE (P-WAVE; COMPRESSIONAL WAVE):

A compressional wave in which the displacement of the medium's particles is in line with or parallel to the direction of travel of the wave motion. It shakes the ground back and forth in the same direction and the opposite direction as the direction the wave is moving. It is the first wave to be recorded by a seismograph. It is the fastest of all seismic waves, travelling at a velocity of 5-7 kilometres per second in the crust and 8-9 kilometres per second in the upper mantle.

PRIMARY WINDING: The winding on the input side of a transformer or induction coil that receives signal energy or alternating-current power from a source.

PRIMASE: An RNA polymerase enzyme that creates the short RNA segments (primer sequences) needed for the discontinuous replication of DNA strand. Primase is the only known DNA polymerase that can initiate the synthesis of a DNA strand without an initial RNA or DNA primer. It consists of 10 to 12 nucleotides, attached to the lagging strand of the unwound DNA molecule. Each primer sequence provides the free 3'-OH group required for the initiation of an Okazaki fragment by DNA polymerase. These primers, which are subsequently removed from the Okazaki fragments are united into a continuous new DNA strand. In bacteria, primase binds to the DNA helicase forming a complex called the **primosome**.

PRIMATES: An order of the class Mammalia comprising monkeys, apes and humans. They are characterised by traits such as large brain, specialised behaviour, poor sense of smell, complex five-digit hands and feet possessing flat nails instead of claws and sensitive binocular vision with depth perception due to forward-facing eyes. They also have prolonged development, which includes long term pregnancy and a long period of growth and development to the adult form.

PRIME: 1. In mathematics, of an integer having no factors except itself and unity. 2. Of a non-zero, non-unit element p of an integral domain such that whenever p divides ab , then p divides a or b . Any

such element is irreducible. 3. of a ring, such that the product of a pair of ideals is zero only if one of the ideals is itself zero. Any simple: ring is prime. 4. A small superscript symbol used to distinguish entities otherwise designated by the same letter such as a change of variables from (x, y, z) to (x', y', z') or $f'(x)$, the derivative of the function (x) ; x' is read 'x-prime'.

PRIME FACTOR: Integers that are multiplied together to give another integer or integers that divide another integer exactly, without leaving a remainder. For example, 2 and 3 are factors of 6. Every factor is a **prime number**.

PRIME IDEAL: In mathematics, an ideal I such that if ab is an element of I then either a is in I or b is in I . A non-zero non-unit element is prime precisely when the ideal it generates is.

PRIME MERIDIAN: An imaginary line of longitude which is defined to be 0° that runs from the North Pole to the South Pole. Together, a prime meridian and its anti-meridian (the 180th meridian in a 360° -system) form a great circle. This great circle divides a spheroid, such as Earth, into two hemispheres, Eastern Hemisphere and the Western Hemisphere. The Earth's prime meridian runs through Greenwich, England and is also called Greenwich Meridian. The distance covered by Earth's prime meridian from the North Pole to the South Pole is 20,000 km.

PRIME NUMBER: A whole number that can only be divided exactly by itself and the number 1 and has no other factor. Examples are 2, 3, 5, 7, 11, 13, 17, 19, 23.

PRIME NUMBER THEOREM: In mathematics, the theorem that $\pi(x)$, the number of primes less than or equal to a given number x , is asymptotic or is approximately equal to $x/\ln x$, where $\ln x$ is the natural logarithm of x .

PRIME SUBFIELD: In mathematics, the subfield that is contained in all subfields of a field F or the intersection of all subfields of F . It is isomorphic either to the field of rational numbers or to the field of integers modulo some prime p .

PRIME SYMBOL: Symbols used to n pairs of terms, where one is a development of another. Examples are (") for double prime symbol and (") for triple prime symbol.

PRIMER: A structure that serves as a growing point for polymerisation.

PRIMINE: The outer integument of an ovule.

PRIMITIVE: 1. In biology, it refers to a structure or an organism that is not specialised or at an early stage of evolution or development. 2. Of a term, statement etc., given in the initial specification of the theory. For example, the axioms are the primitive statements of any theory. 3. In mathematics, a function of which the derivative is a given function. A continuous function $F(x)$ is called a primitive for a function $f(x)$ on a segment X , if for each $F'(x) = f(x)$. For example, the function $F(x) = x^3$ is a primitive for the function $f(x) = 3x^2$ on the interval $(-\infty, +\infty)$, because $F'(x) = (x^3)' = 3x^2 = f(x)$ for all $x \in (-\infty, +\infty)$.

PRIMITIVE CHANGE: One of the conceptually simpler molecular changes into which an elementary reaction can be notionally dissected. Such changes include bond rupture, bond formation, internal rotation, change of bond length or bond angle, bond migration, redistribution of charge, etc. The concept of primitive changes is helpful in the detailed verbal description of elementary reactions, but a primitive change does not represent a process that is by itself necessarily observable as a component of an elementary reaction.

PRIMITIVE POLYNOMIAL: A polynomial over an integral domain, such that its greatest common divisor of its coefficients is 1. An example is $4x^2 + 5x - 1$ because the greatest common divisor of 1, 4 and 5 is 1. All primitive polynomials are irreducible.

PRIMITIVE PYTHAGOREAN TRIANGLE: A right triangle whose sides are relatively prime integers. If a , b and c are positive integers and $a^2 + b^2 = c^2$ then the solution (a, b, c) is a Pythagorean triple. If $\gcd(a, b, c) = 1$ then (a, b, c) is a primitive (or reduced) Pythagorean triple and the equation $a^2 + b^2 = c^2$ is a primitive Pythagorean triangle.

PRIMITIVE ROOT: A primitive root for a prime number p is one whose powers generate all the non-zero integers modulo p . For example, 3 is a primitive root modulo 7 since $3 = 3^1$, $2 = 3^2 \bmod 7$, $6 = 3^3 \bmod 7$, $4 = 3^4 \bmod 7$, $5 = 3^5 \bmod 7$, $1 = 3^6 \bmod 7$.

PRIMITIVE ROOT OF UNITY: The complex number z that equals 1, when raised to some integer power n : ($z^n = 1$) but z_k is not equal to 1 for any positive integer k less than n .

PRIMITIVE STREAK: A structure that forms during the early stages of avian, reptilian and mammalian embryonic. It forms the basis of bilateral symmetry, determines the site of gastrulation and initiates germ layer formation. The cells in the primitive streak proliferate rapidly to form mesoderm and notochord cells, which migrate to the interior of the embryo.

PRIMITIVE TERM: Any of the expressions explicitly stated to be a term of a theory, such as names, predicates and variables in the predicate calculus. Although the primitive terms refer to the undefined elements of a theory, they are implicitly defined by its axioms.

PRIMOCANE: A new cane on a bramble fruit that will flower and fruit the following year, as seen in raspberry and blackberry.

PRIMORDIAL BLACK HOLE: A hypothetical black hole that formed in the very early Universe soon after the Big Bang from a dense concentration of matter.

PRIMORDIUM: An organ or tissue that represents the initial stages in the development of a plant organ or embryonic development, found when the dividing cells in the fertilised ovum first differentiate. In plants, it is critical to the proper positioning and development of plant organs. Root and shoot primordia are present in a young plant embryo while leaf primordia are seen as small bulges just below the shoot apex.

PRIMORIAL (PRIME FACTORIAL): The product of all prime numbers that are less than or equal to a given prime p ; it is denoted $p\#$. For example,

$3\# = 2 \times 3 = 6$, $5\# = 2 \times 3 \times 5 = 30$ and

$13\# = 2 \times 3 \times 5 \times 7 \times 11 \times 13 = 30030$.

PRIMOSOME: A complex of protein enzymes creating RNA primers on single stranded DNA to initiate DNA replication in prokaryotes such as *E. coli*. It is composed of seven proteins: DnaG primase, DnaB helicase, DnaC helicase assistant, DnaT, PriA, Pri B and PriC.

PRINCIPAL: 1. In mathematics, also known as **principal amount**, it is the original amount of money invested or borrowed before interest is calculated. 2. In architecture, it is either of a pair of inclined timbers forming the sides of a triangular truss for a pitched roof.

PRINCIPAL AXIS: 1. In mathematics, any of the axes of a conic or quadric surface with respect to which the figure has an equation that is the sum of multiples of square. 2. In physics, also known as **optical axis**, it is the line passing through the optical centre or the centre of curvature of a lens. 3. In physics, any of the three mutually perpendicular axes about which the moment of inertia of a body is maximum.

PRINCIPAL CURVATURES: In mathematics, the principal curvatures at a point of a surface are the maximum and minimum normal curvatures. The principal curvatures measure the maximum and minimum bending of a regular surface in different directions at each point.

PRINCIPAL DIAGONAL (LEADING DIAGONAL; MAIN DIAGONAL; PRIMARY DIAGONAL): The elements of a matrix running from the top left corner and proceeding down to right edge of the bottom. The main diagonal of a square matrix is the diagonal which runs from the top left corner to the bottom right corner. The main diagonal of a rectangular matrix is the diagonal which runs from the top left corner and steps down and right, until the right edge or

the bottom edge is reached. For example, the matrix in the diagram

has 1s down its main diagonal:
$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

PRINCIPAL DIRECTIONS: The principal directions at a point of a surface are the directions in which the radius of normal curvature is maximised or minimised.

PRINCIPAL FOCUS: The point at which an incident ray is close to and parallel to the principal axis of a lens converge or appear to diverge, after refraction.

PRINCIPAL GROUP: The characteristic group chosen for citation at the end of a name by a suffix or a class name.

PRINCIPAL IDEAL: In mathematics, a proper ideal in a ring that is generated by a single element. A **principal ideal domain** or **principal domain** has an integral domain in which every ideal is principal. There are nine imaginary quadratic fields $Q(\sqrt{d})$ whose subring of algebraic integers yield principal ideal domains, those with $d = 1, 2, 3, 7, 11, 19, 43, 67$ or 163 .

PRINCIPAL IDEAL RING: A ring in which every ideal is a principal ideal. Examples include the ring of integers and the ring of polynomials over a field.

PRINCIPAL INVARIANTS: In mathematics, the principal invariants for a matrix are the coefficients in the characteristic polynomial of the given matrix.

PRINCIPAL ION: A molecular or fragment ion which is made up of the most abundant isotopes of each of its atomic constituents.

PRINCIPAL MINOR: In mathematics, the determinant of definiteness of a principal submatrix of an $n \times n$ square matrix by deleting some rows and columns with the same numbers.

PRINCIPAL MOMENT OF INERTIA: In physics, a moment of inertia of a rigid body with respect to principal axes. The principal moments of inertia are given by the entries in the diagonalised moment of inertia matrix and for a solid are denoted A, B and C in order of decreasing magnitude.

PRINCIPAL NORMAL: The principal normal to a curve at a point in Euclidean space is the unit derivative of the unit tangent vector (with respect to arc length) at the given point. It is denoted N.

PRINCIPAL PART: 1. In mathematics, the restriction of a multi-valued function to arguments for which it takes its principal value. 2. Any of the sides and interior angles of a triangle.

PRINCIPAL PLANE: 1. The plane that is perpendicular to the optical axis of a lens and that passes through the optical centre such that objects in one plane form images in the other with a lateral magnification of unity. 2. In a quadric surface, a plane that passes through the midpoints of all the chords perpendicular to it.

PRINCIPAL POINT: Any of two points on the principal axis of a thick lens from which simple related distances can be measured as from the optical centre of a thin lens.

PRINCIPAL QUANTUM NUMBER: One of the four quantum numbers characterising the electron configuration of an atom. The principal quantum number determines the size and (in hydrogen atoms) the energy of an orbital. The principal quantum number, n , is used to label electron shells; n may take on integer values from 1 to infinity.

PRINCIPAL SERIES (CHIEF SERIES): In mathematics, a normal series $(G_0, \dots, G_n)$ such that for all k between 0 and $n - 1$ inclusive, G_{k+1} is maximal in the set of normal subgroups of G properly contained in G_k .

PRINCIPAL SOLUTION MATRIX: A fundamental matrix, Y , for which $Y(t_0)$ is the identity matrix for a given argument t_0 .

PRINCIPAL STRETCHES: The eigenvalues of the positive definite symmetric tensor, $\mathbf{U}$, such that, for $\mathbf{F}$ the deformation gradient of the motion of a given body, there is an orthogonal tensor $\mathbf{R}$ such that $\mathbf{F} = \mathbf{R}\mathbf{U}$.

PRINCIPAL STRETCHING: The latent roots of the Eulerian strain rate.

PRINCIPAL SUBMATRIX: A submatrix of a given square matrix, $n \times n$, obtained by deleting some rows and columns with the same numbers.

PRINCIPAL VALUE: The unique element of a set of values a variable can take, for example, finding the inverse of trigonometric functions or the square root and the values returned by scientific calculators. For example, if $x^2 = 4$, x could be 2 or -2, but a calculator

will give $\sqrt{4} = 2$. Also, if $\cos 60^\circ = \sin^{-1}(0.5) = 30^\circ$, the calculator will give $\cos^{-1}(0.5) = 60^\circ$.

PRINCIPLE: 1. A basic general truth that underlines something such as a subject or a general or scientific law shown in the way a thing works or used as the basis for constituting a machine, etc. 2. An ingredient of a substance that gives the substance a special quality or which produces a certain quality or effect, for example, bitterness, odour, etc.

PRINCIPLE OF ALLOCATION: A basic concept in ecology that each organism has an energy budget or organisms take in a limited amount of total energy and then use that energy to survive and to reproduce. If an organism allocates energy to one function, such as growth or reproduction, it reduces the amount of energy available to other functions, such as defence.

PRINCIPLE OF ANGULAR MOMENTUM: In physics, the principle that for any mechanical system, the torque is equal to the rate of change of the angular momentum.

PRINCIPLE OF CONSERVATION OF LINEAR MOMENTUM: It states that if no external forces act on system of colliding objects, the total momentum of the object in a given direction before collision equals total momentum after collision in the same direction.

PRINCIPLE OF DUALITY: The duality of lines and points in projective geometry.

PRINCIPLE OF EQUIVALENCE (EQUIVALENCE PRINCIPLE): A law in physics which states that there is no difference between gravitational and inertial forces.

PRINCIPLE OF EXCLUDED MIDDLE (EXCLUDED MIDDLE LAW; LAW OF EXCLUDED MIDDLE): In logic, one of the three classic laws of thought, which states that for any proposition, either that proposition is true or its negation is true: there is neither a third truth value nor do any statements lack truth value.

PRINCIPLE OF INDIFFERENCE: The principle that in the absence of any reason to the contrary, each possible outcome of an experiment is to be treated as equiprobable.

PRINCIPLE OF INDEPENDENT ASSORTMENT: The law that during gamete formation in diploid organisms, alleles in one gene pair segregate into gametes independently of the alleles of other gene pairs as long as the gene pairs are on separate chromosomes. As a result, if enough gametes are produced, the collective group of gametes will contain all combinations of alleles possible for that organism.

PRINCIPLE OF INDUCTION (PRINCIPLE OF MATHEMATICAL INDUCTION): The principle that if a given integer, n , has a certain property and if it can be proven that if any integer, m , no smaller than n has the property, so has $m + 1$, then all integers greater than n have it.

PRINCIPLE OF LEAST ACTION (PRINCIPLE OF STATIONARY ACTION): In physics, it is an alternative expression of Newton's laws of motion that states that a particle moving between two points in a conservative dynamical system under the influence of a force will follow the path along which its total action is least. Action is a quantity

related to the average difference between the kinetic energy and the potential energy of the particle along its path. The principle is only true where no energy is lost from the system; for example, an object moving in free fall in a gravitational field. It is closely related to Fermat's principle of least time, which governs the path taken by a ray of light.

PRINCIPLE OF LEAST NUCLEAR MOTION: The hypothesis that for given reactants, the reactions involving the smallest change in nuclear positions will have the lowest energy of activation. It is also often simply referred to as principle of least motion.

PRINCIPLE OF MICROSCOPIC REVERSIBILITY: In a reversible reaction, the mechanism in one direction is exactly the reverse of the mechanism in the other direction. This does not apply to reactions that begin with a photochemical excitation.

PRINCIPLE OF MOMENT: It states that for a system to be in equilibrium, the sum of anticlockwise moments about any point equals the sum of clockwise moments about the same point: Σ anticlockwise moments about a point = Σ clockwise moments about the same point.

PRINCIPLE OF PARSIMONY: The principle that the most acceptable explanation of an occurrence, phenomenon or event is the simplest.

PRINCIPLE OF SEGREGATION (MENDEL'S LAWS): The first of the two genetic principles established by Gregor Mendel, which states that the genetic characteristics of a species are represented in the somatic cells by a pair of units called genes that separate during meiosis so that each gamete receives one member of a pair.

PRINCIPLE OF SUPERPOSITION (SUPERPOSITION THEOREM): It states that the resultant displacement at any point in a region through which two or more forces act on a particle at the same time, the resultant force is the vector sum of the two. This principle can apply to many physical systems such as voltages and currents in electricity, optics, acoustics, etc.

PRINT SCREEN KEY (PRT SCR): A key found on most computer keyboards, usually located in the same section as the break key and scroll lock key. In MS-DOS, it is used to transfer the screen contents to the standard printer port. In Windows and other graphic programs, it transfers the current on-screen image to the clipboard or to a graphics file.

PRINCIPAL SERIES: 1. A series that occurs in the line spectra of many atoms and ions with between one to three electrons in the outer shell, in which the total orbital angular momentum quantum number changes from 1 to 0. 2. Also known as **chief series**, a series ($G_0, \dots, G_n$) such that for all k between 0 and $n - 1$ inclusive, G_{k+1} is maximal in the set of normal sub-groups of G properly contained in G_k .

PRINT SPOOLER: A software program that enables the formatted output to be stored on a disk and responsible for managing all print jobs currently being sent to the computer printer or print server. The program allows multiple printing requests to be queued and to manage the print jobs currently waiting to be printed.

PRINTED CIRCUIT BOARD (PCB): An electrical circuit created by electronic components using conductive pathways, tracks or signal traces etched from copper sheets laminated onto an insulating board. The electronics industry's PCB design, assembly and quality control needs are set by standards that are published by the IPC organisation.

PRINTER: A computer output device that reproduces data such as words, numbers and diagrams onto paper. Examples of a printer are line printer, daisy wheel printer, dot-matrix printer, inkjet printer, laser printer and impact printer.

PRINTING: The reproduction of an image such as text or illustrative material on a surface such as paper or plastics. The five classes of printing are: 1. **relief printing**, which includes letterpress and flexography; 2. **planographic printing**, which includes offset lithography, screenless lithography, collotype and waterless printing;

3. **intaglio**, which includes gravure, steel-die and copper-plate engraving; 4. **stencil** and **screen printing**; and 5. **electronic printing**, which includes electrostatic, magnetographic, ion or electron deposition and ink-jet printing.

PRION: An exceptionally small disease-causing microorganism, many times smaller than virus, which causes certain diseases of the nervous system in humans and livestock. It is an abnormal form of a normally harmless protein found in mammals and birds. The prion can enter the brain through infection or it can arise from a mutation in the gene that encodes the protein and interacts with normal protein, causing it to accumulate in the brain and progressively damage and destroy brain cells. Prion is responsible for causing diseases such as Creutzfeldt-Jakob disease in humans, mad cow disease or Bovine Spongiform Encephalopathy (BSE) in animals and scrapie.

PRIOR DISTRIBUTION: A product state distribution calculated on the basis of some physical model for the reaction. The distribution usually referred to is that of the products at the instant of their formation.

PRIOR PROBABILITY (A PRIOR; PRIOR PROBABILITY DISTRIBUTION): In statistics, it is the probability of the event computed before the collection of new data. In other words, it is the probability assigned to a parameter or to an event in advance of any empirical evidence, often subjectively or on the assumption of the principle of indifference. It is based upon deductive reasoning rather than research or calculation and is the original probabilities of an outcome, which will be updated with new information and used in Bayesian inference to estimate posterior probabilities, which are the results.

PRISM: 1. A polyhedron of which two faces are congruent polygons in parallel planes and the other faces are parallelograms, or a three dimensional (3-D) geometric figure with two congruent and parallel faces. A **triangular prism** is a prism composed of two triangular bases and three rectangular sides, a pentahedron. A right prism is one whose bases are perpendicular to all of the edges that connect the two bases. Various forms of prism are shown in the diagrams. **Rectangular prism** is a 3-dimensional object which has six faces that are rectangles. A **hexagonal prism** is a prism composed of two hexagonal bases and six rectangular sides, an octahedron. Rectangular prism is just like a cuboid. A right rectangular prism is a prism in which the angles between the base and sides are right angles. 2. A solid transparent material such as glass, plastic and silica, commonly used to reflect a ray of light or split a beam into spectral colours.

PRISM BINOCULARS: A simple binocular with two telescopes mounted side by side. It employs a Porro prism erecting system both to erect the image and to reduce the length of the instrument. The use of prisms gives high magnification and an upright image without much increase of size. Also, the object lenses are further apart than the users' eyes, helping to improve binocular vision.

PRISMANE: A polycyclic saturated hydrocarbon with the chemical formula C_6H_6 , in which the six carbon atoms are arranged at the corners of a triangular prism.

PRISMATIC: 1. Relating to or resembling a prism. 2. Of, relating to or describing light that shows the colours of the spectrum, as refracted by a prism.

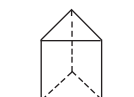


Fig 1:
triangular prism

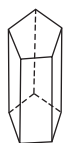


Fig 2:
pentagonal prism

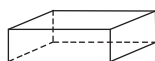


Fig 3:
rectangular prism

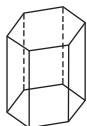


Fig 4:
hexagonal prism
Prisms

PRISMATOID: A polyhedron with all vertices in one or the other of two parallel planes. All lateral faces are therefore either triangular or quadrilateral. Examples of prismatoids are cube, cuboid, pyramidal frustum, prism and pyramid. Its volume can be calculated from the formula $V = \frac{1}{6}h(B_1 + B_2 + 4M)$, where h is the altitude, B_1 and B_2 are the areas of the bases and M is the area of the midsection.

PRISMOID: A prismatoid, whose two planes have the same number of vertices and the lateral faces are either parallelograms or trapezoids.

PRIVATE BRANCH EXTENTION, PBX (PRIVATE BRANCH EXCHANGE): A local telephone switching system consisting mainly of several branches of telephone systems connected to one external line, with each phone having a different number, used to serve a limited area such as a business. Users of the PBX, usually connect to another phone in the network by dialling a three- or four-digit number called an extension.

PROBABILITY: In statistics, it is the likelihood that an event will occur, expressed as the ratio of the number of favourable outcomes in the set of outcomes divided by the total number of possible outcomes. It is measured on a scale from 0 to 1, with 0 indicating impossibility and 1 indicating certainty. In other words, if an event is absolutely going to happen, its probability of occurring is 1 or 100%. If an event absolutely will not happen, its probability of occurring is 0 or 0%. More precisely, it is the proportion of favourable outcomes to the total number of possible outcomes.

PROBABILITY DENSITY FUNCTION (PDF; DENSITY FUNCTION): In statistics, it is a function representing the relative distribution of frequency of a continuous random variable from which parameters such as its mean and variance can be derived. It has the following properties: (i) Since the continuous random variable is defined over a continuous range of values called the **domain** of the variable, the graph of the density function will also be continuous over that range. (ii) The area bounded by the curve of the density function and the x-axis is equal to 1, when computed over the domain of the variable. (iii) The probability that a random variable assumes a value between a and b is equal to the area under the density function bounded by a and b .

PROBABILITY FUNCTION: 1. Also known as **probability mass function (PMF)**, it is the function whose values are the probabilities of the possible values for a discrete random variable. 2. Also known as **probability distribution function**, it is a function that constitutes a probability measure, which describes all the possible values and likelihoods that a random variable can take within a given range. Some of its classifications are the chi-square and normal and binomial distributions.

PROBABILITY INTEGRAL TRANSFORM: The transformation, $F_x(X)$, of a random variable, X , with an invertible cumulative distribution function F_x , to a uniform distribution on $(0,1)$.

PROBABILITY MEASURE: A mapping, P , of a sample space, X , into the interval $[0, 1]$ subject to the conditions that P maps the whole space to 1 and is additive for the union, possibly infinite, of disjoint sets that lie in the sigma algebra of events; P is then a probability density function.

PROBABILITY SCALE: An ordered line numbered from 0 to 1. The probability of an impossible event is 0 and the probability of a certain event is 1.

PROBABILITY SPACE: In statistics, a totally finite measure space together with an associated probability measure that assigns unit measure to the entire space; the probability space is then (X, Σ, P) , where X is the sample space, Σ the sigma-algebra of events and P is the probability measure.

PROBABILITY THEORY: The branch of mathematics that deals with the possible outcomes of events and their relative likelihoods. While mathematicians agree on how to calculate the probability of

certain events and how to use those calculations in certain ways, there is disagreement as to what the numbers actually mean. Probability divides into two main concepts: (i) **aleatory probability**, which represents the likelihood of future events whose occurrence is governed by some random physical phenomenon like tossing dice or spinning a wheel; and (ii) **epistemic probability**, which represents our uncertainty of belief about past events that either did or did not occur or uncertainty about the causes of future events.

PROBABILITY TREE DIAGRAM: A drawing used to analyse the number of possible outcomes of an event that consists of two or more choices. For example, the branches of a probability tree diagram may represent the four equally likely outcomes of "heads" and "tails" when one coin is flipped two times.

PROBABLE ERROR: In statistics, the deviation from the true value within which an observation is likely to lie with probability 50%. In normal distribution this is equal to 0.674σ , where σ is the standard deviation.

PROBE: 1. A specific DNA or RNA sequence which has been labelled by radioactivity, fluorescence labels or chemiluminescence labels and which is used to detect complementary sequences by hybridisation techniques such as blotting or colony hybridisation. 2. A tool for examining a place which cannot be reached otherwise, especially a thin implement with a blunt end used by a doctor for examining a wound. 3. An unmanned spacecraft which obtains information about space and transmits it back to the Earth or a rocket sent out to the Moon or other planets.

PROBLEM (MATHEMATICAL PROBLEM): In mathematics, a question or statement that requires proof or a method of solution, for example by mathematical operation or geometric construction.

PROBOSCIDEA: The order of placental mammals that comprises elephants. They are herbivores with a muscular trunk (proboscis) used for drinking, bathing and collecting food. Presently, this order consists of only two forms: the African elephant (*Loxodonta africana*) and the Asian elephant (*Elephas maximus*), with features such as large body size, enlarged incisors that form paired tusks and a proboscis called trunk and being herbivorous.

PROBOSCIS: 1. The elongated appendage from the head of certain invertebrates such as the two-winged flies (Diptera), used for sucking. 2. The muscular and very flexible elongation of the nose of an elephant which has a finger-like extremity and is capable of picking and moving objects, taking in water, collecting food, etc.

PROCAMBIUM: A plant vascular tissue formed by the apical meristems of shoots and roots. It consists of cells elongated and parallel to the long axis of the plant.

PROCEDURE: 1. A regular order or way of doing things. 2. A small part of a computer program that performs a specific task such as clearing the screen or sorting file.

PROCESS: 1. A phenomenon by which a change takes place in a system. In physiological systems, a process may be chemical, physical or both. 2. A marked prominence or projection of a bone. 3. Any of the several photochemical or photoengraving methods. 4. In biology, it is an outgrowth of a tissue or appendage. 5. The manipulation of data in a computer. Any action taken by a computer on data, whether it is being displayed or updating. 6. A software that is running in the computer. It includes anything from a small background task, such as a spell-checker or system events handler to a full-blown application like Internet Explorer or Microsoft Word.

PROCESS CONTROL: An automatic computerised control of a manufacturing process or other processes which need to perform without a manual control and uses input variables such as analogue signals from sensors and meters that are transmitted to specialised computers which cause the temperature, pressure and flow, etc., to be continually adjusted. Process control finds application in fields such

as oil refining, chemical processing and electrical generation, where the creation of a product is based on a continuous series of processes being applied to raw materials.

PROCESS IDENTIFIER, PID (PROCESS IDENTIFICATION NUMBER): A unique number that is automatically assigned to each process when it is created to identify each of the running processes in an operating system such as Linux, Unix, Mac OS X and Microsoft Windows.

PROCESSING CYCLE: The sequence of steps performed repeatedly by a computer in the execution of a programme. It includes input, processing, storage and output.

PROCESSING DEVICES OF A COMPUTER: Components responsible for processing information within a computer or any device in a computer that handles the intermediate stage of processing the incoming data. These include Central Processing Unit, Graphics Processing Unit, Motherboard, Network card, Sound Card, Video Card, etc. When a computer receives data from an input device such as a keyboard, mouse or microphone, the data goes through an intermediate processing before it can be sent to an output device such as a monitor, printer, speaker, etc.

PROCESSOR: Another name for the central processing unit or microprocessor of a computer, which can perform operations on data and also a language processor, such as an assembler, compiler or interpreter.

PROCHIRALITY: 1. The geometric property of an achiral object (or spatial arrangement of points or atoms) which is capable of becoming chiral in a single desymmetrization step. An achiral molecular entity or a part of it considered on its own, is thus called **prochiral** if it can be made chiral by the replacement of an existing atom (or achiral group) by a different one. An achiral object which is capable of becoming chiral in two desymmetrization steps is sometimes described as **propochiral**. For example, the propochiral $\text{CH}_3\text{CO}_2\text{H}$ becomes prochiral as $\text{CH}_2\text{DCO}_2\text{H}$ and chiral as CHDTCO_2H . 2. An achiral molecule or entity which contains a trigonal system and which can be made chiral by the addition to the trigonal system of a new atom or achiral group. For example, addition of hydrogen to one of the enantiotopic faces of the prochiral ketone $\text{CH}_3\text{CH}_2\text{COCH}_3$ gives one of the enantiomers of the chiral alcohol $\text{CH}_3\text{CH}_2\text{CHOHCH}_3$; the addition of CN^- to one of the diastereotopic faces of the chiral aldehyde shown below converts it into one of the diastereoisomers of the cyanohydrin. The two faces of the trigonal system may be described as *Re* and *Si*. 3. The term prochiral also applies to a tetrahedral atom of an achiral or chiral molecule which is bonded to two stereoheterotopic groups. For example, the prochiral molecule $\text{CH}_3\text{CH}_2\text{OH}$ can be converted into the chiral molecule CH_3CHDOH by the isotopic replacement of one of the two enantiotopic hydrogen atoms of the methylene group. The carbon atom of the methylene group is called prochiral. 4. The term prochirality is also applied to the enantiotopic faces of a trigonal system.

PROCONSUL: A genus of extinct apes known from fossil remains found in 1948 in East Africa and dated as being 23 to 14 million years old. It is assigned to at least three species and now considered a subgroup of Dryopithecus. These apes were quadrupeds and appear to have been tailless.

PROCTOCTIST: A single-celled organism not distinguished as animals or plants, though having some characters common to both. Proctists include algae, slime, moulds and protozoa.

PROCTOCTISTA (PROTISTA): A kingdom into which a variety of unicellular and some simple multinuclear and multicellular eukaryotic organisms can be classified or an organism belonging, in an older classification system. It includes the algae, bacteria, fungi and protozoa.

PROCUMBENT: 1. Trailing or growing on the ground without rooting at the nodes. 2. Lying face down.

PRODRUG: A drug administered in an inactive form, which requires further metabolism in the liver before becoming biologically active. The reason for using prodrug is generally for absorption, distribution, metabolism and excretion (ADME) optimisation. An example is the immunosuppressant azathioprine, which is metabolised first to cortisone and then to the pharmacologically active compound hydrocortisone.

PRODUCER: 1. An organism such as green plants that can make its own food by photosynthesis or an organism considered as a source of energy for those above in a food chain. 2. A furnace for making producer gas.

PRODUCER GAS: A form of water gas consisting of carbon monoxide, hydrogen and nitrogen which is made in a furnace or generator by burning low grade fuel such as bituminous coal or lignite in a closed.

PRODUCT: 1. A quantity or value obtained by multiplying two or more numbers. For example, the product of 2 and 12 is 24. 2. A substance or chemical compound produced by a chemical reaction. 3. Anything or substance produced by a natural or manufacturing process.

PRODUCT MEASURE: The measure μ placed on the finite Cartesian product of measure spaces (M_i, μ_i) by defining $\mu\left(\prod_{i=1}^n S_i\right) = \prod_{i=1}^n \mu_i(S_i)$

for all products of sets S measurable in the coordinate spaces. The measure then extends uniquely to the sigma-algebra generated by sets of the form $\prod_i S_i$.

PRODUCT OF INERTIA: In physics, for a set of particles, it is the sum of the terms $\sum_i m_i p_i q_i$ where m_i is the mass of the i^{th} particle and p_i and q_i are its directed distances from the two planes with the product is associated. For a continuous body, the product of inertia is $\int p q \rho \, dV$, where ρ is the density at the point and V is the volume of the body.

PRODUCT OF POWERS PROPERTY: Multiplying powers with the same base and different exponents just by adding the exponents and the product equals the same base number raised to the sum of the exponents, for example, $5^2 \times 5^4 = 5^{2+4} = 5^6$.

PRODUCT ORDER: In mathematics, the ordering placed on the Cartesian product of ordered sets by defining $\{x_i\} \leq \{y_i\}$ if $x_i \leq y_i$ for each index order.

PRODUCT RULE: A rule for differentiating the product of two or more functions of the form $\frac{d(fg)}{dx} = f \frac{dg}{dx} + g \frac{df}{dx}$.

PRODUCT STATE DISTRIBUTION: The distribution of available energy among the vibrational, rotational and translation degrees of freedom in the product molecules. The distribution usually referred to is that of the products at the instant of their formation.

PRODUCT TOPOLOGY: The topology on the Cartesian product $X \cdot Y$ of two topological spaces whose open sets are the unions of subsets $A \cdot B$, where A and B are open subsets of X and Y , respectively.

PRODUCT-DETERMINING STEP: The step of a stepwise reaction in which the product distribution is determined. The product determining step may be identical to or occur later than, the rate-controlling step on the reaction coordinate.

PRODUCT-KEY (ACTIVATION CODE; CD-KEY; PRODUCT ID; SERIAL KEY): A unique set of numbers and letters provided with software that helps prevent software piracy and prevents the installation of a software that requires it until a valid key entry is entered. It is usually found on the back of the CD-ROM package, inside a paper manual or on a separate sheet of paper that comes with the package.

PRODUCTION DISEASES: Non metabolic disorders of animals which are caused by high levels of production.

PRODUCTION FUNCTION: A mathematical equation or graph that shows the level of physical output obtainable from a given level of resource input.

PRODUCTION RATION: The amount of feed mixture which is given to a growing, working or producing animal which is always more than the basic maintenance ration. This is mostly met by feeding it with concentrate mixtures.

PRODUCTIVE SOIL: A soil in which the chemical, physical, and biological conditions can support the economic production of the crops suited to a particular area.

PRODUCTIVITY: 1. The rate at which an organism, population or community assimilates energy or makes energy potentially available to an animal that feeds on it or the amount of material in the food chain produced by the primary producers such as plants that is available for consumption by animals in an ecosystem. 2. An economical figure denoting the mass of a product formed per unit reactor volume and unit time. Productivity is often referred to per unit of enzyme or biomass. 3. **Soil productivity** is the capacity of a soil, in its normal environment, to produce a specified plant or sequence of plants under a specified system of management. The limitations are needed because no soil can produce all crops with equal success and a single system of management cannot produce the same effect on all soils.

PRODUCTIVITY SOFTWARE (PERSONAL PRODUCTIVITY SOFTWARE): In computer science, it is a work-oriented software that is used to create or modify a document, image, audio or video clip, especially word processing, spreadsheets or presentation programs.

PROFILE: 1. The side view or the edge or outline of something seen against a background or a vertical view of soil. 2. An agreed-upon subset and interpretation of a specification.

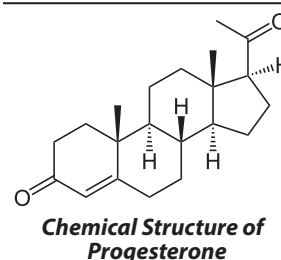
PROFUNDAL: Occurring in or designating the deep water zone of an inland lake located below the range of effective light penetration, oxygen concentration and temperature than in the surface layer. It is part of the aphotic zone.

PROGENESIS: The maturation of gametes in an organism that is still otherwise at the larval or juvenile stage of development. Organisms that are progenetic never achieve the adult form experienced by their evolutionary ancestors.

PROGENY: The offspring or a descendant of a person, animal or plant.

PROGENY TEST: The evaluation of parental genotype or the breeding value of an animal or plant variety by examining the performance of its progeny under controlled conditions.

PROGESTERONE: A steroid hormone belonging to a class of hormones called progestogens, made in the ovaries in human females which regulates the menstrual cycle and pregnancy in mammals by preparing the inner lining of the uterus for implantation of a fertilised egg cell.



PROGESTOGEN

(PROGESTAGENS; GESTAGENS): One of the groups of naturally occurring or synthetic hormones that maintains the normal course of pregnancy, for example, progesterone. It is one of the five major classes of steroid hormones. The others are oestrogens androgens, mineralocorticoids and glucocorticoids. Taken in high doses it inhibits secretion of luteinising hormone, which prevents ovulation and affects the vagina's mucus constituency and therefore, affecting conception. It is used as major constituents in oral contraceptives.

PROGLOTTID: A segment of a tapeworm's body, containing both male and female reproductive organs.

PROGRADE ROTATION: Counterclockwise rotation of a planet or moon as viewed from the North Pole of the sky or a planet; rotation in the same direction as the Sun's rotation.

PROGRAM: A set of instructions, usually called statements, that constitute an algorithm that makes a device such as a computer carry out a particular task. Processing may include the use of an assembler, a compiler, an interpreter or another translator to prepare the program for execution.

PROGRAM COUNTER (SEQUENCE-CONTROL REGISTER): The control unit of the CPU that is used to keep track of the address of the current or next instruction.

PROGRAM DOCUMENTATION: Documentation, which includes batch documents (all automated documents), interactive documents, text block correspondence and forms (forms for websites) that provide a complete technical description of a program built up as the software is written and is intended to support any later maintenance or development of the programs. Any type of document can be placed in one of these categories.

PROGRAM FILES: In computing, they are electronic files found in Microsoft Windows that contain application subfolders used by a computer program.

PROGRAM FLOW CHART: A type of flow chart used to describe the flow of data through a particular computer programme.

PROGRAM LOOP: A part of a computer program that is repeated several times. It saves writing a code several times for same task.

PROGRAM TRADING: Computerised trading using a computer program to generate orders automatically whenever there is a favourable change in prices.

PROGRAMMABLE INTERFACE CONTROLLER (PIC): A microcontroller that can be programmed to carry out specific tasks. The PIC contains a low-power microprocessor, together with a small amount of Flash RAM used to store the program and a smaller amount of working RAM in which the program data is held while running. PICs also contain all the necessary circuitry to connect to the outside world, plus timers and a section of EEPROM computer memory in which the processor's instruction set or language, is stored.

PROGRAMMABLE READ-ONLY MEMORY (PROM): An integrated circuit with a memory that can be programmed permanently and cannot be changed. They are often seen in video game consoles, mobile phones, radio-frequency identification (RFID) tags, implantable medical devices, high-definition multimedia interfaces (HDMI) and in many other consumer and automotive electronics products.

PROGRAMMED-FLOW CHROMATOGRAPHY (FLOW PROGRAMMING): A procedure in which the rate of flow of the mobile phase is changed systematically during a part or whole of the separation.

PROGRAMMED-PRESSURE CHROMATOGRAPHY (PRESSURE PROGRAMMING): A procedure in which the inlet pressure of the mobile phase is changed systematically during a part or whole of the separation.

PROGRAMMED-TEMPERATURE CHROMATOGRAPHY (TEMPERATURE PROGRAMMING): A procedure in which the temperature of the column is changed systematically during a part or the whole of the separation.

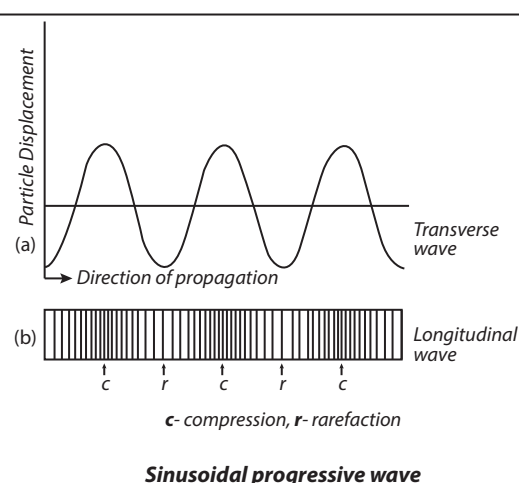
PROGRAMMER (COMPUTER PROGRAMMER): A person who writes computer software needed for any computer system or application.

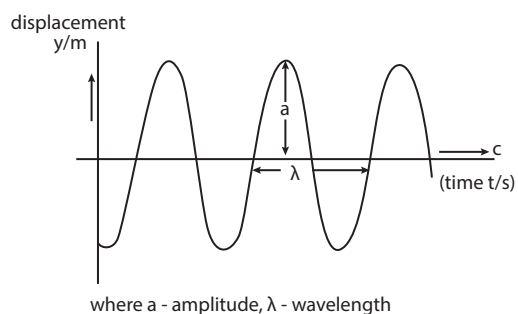
PROGRAMMING (PROGRAMMING; CODING): 1. Designing, writing, testing, debugging of computer programs or writing instructions in a programming language for the control of the computer. 2. Selecting and scheduling of television or radio programs.

PROGRAMMING LANGUAGE: A special notation in which instructions for controlling a computer is written. It consists of a unique vocabulary and set of rules that let the programmer express data processing in a symbolic manner without regard to machine-specific details. In high-level languages, which use words similar to those in English and are therefore easier to learn, one statement may represent several machine instructions and must be converted into machine code by a compiler. Low-level languages are more similar to machine code and each statement represents one machine instruction.

PROGRESSION: A series of numbers or quantities in which there is always the same relationship between each quantity and the one succeeding it. For example, an **arithmetic progression** has numbers that increase or decrease by a common difference, for example, 2, 4, 6 and 8 are increasing by the addition of 2. A **geometric progression** has numbers each bearing a fixed ratio to its predecessor, for example, 3, 6, 12, 24 are increasing by a multiple of 2. **Harmonic progression** has numbers whose reciprocals are in arithmetical progression, for example, 1, $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$,) or $\frac{1}{2}$, $\frac{1}{5}$, $\frac{1}{8}$, $\frac{1}{11}$.

PROGRESSIVE WAVE: 1. A wave or wavelike disturbance that travels from one place to another relative to the Earth's surface. 2. A wave that carries energy from one point to another in a medium without the motion of the medium. In a progressive wave, the wave profile moves along with the speed of the wave. The vibration of the particles are of the same amplitude and frequency, however the phase of the vibration changes for different points along the wave. Examples of a progressive are transverse and longitudinal waves. Progressive wave equation is given as $y = a\sin(\omega t \pm \phi)$, $\omega = 2\pi f$ and $\phi = \frac{2\pi x}{\lambda}$, where y is the displacement of the particle, x the displacement of the wave, a the amplitude of the wave, ω the angular velocity of particle executing the simple harmonic motion, t the time at any instant, ϕ the phase difference between the particle at the origin and the particle in question and λ the wavelength.





Plane progressive harmonic wave

PROGYMNOSPERMS: An extinct group of plants of woody, spore-bearing plants that flourished in the middle Devonian (360- 350 million years ago) and from which gymnosperms (conifers and cycads) descended. They were trees of up to 12 metres tall, with vegetative organs consisting of relatively loose clusters of axes.

PROJECT: Planned set of interrelated activities to be completed over a fixed period of time by small groups or by a whole class. Projects often involve collecting and analysing data and are usually cross-curricular in nature.

PROJECT GUTENBERG: The first electronic library, founded in 1971 by Michael Hart, a student at the University of Illinois to digitise (etexts), archive and distribute cultural works. The project started, when in 1971, the founder created the first electronic book (e-book) by typing the text of the U.S. Declaration of Independence into a mainframe computer at the University of Illinois. By August 2015, over 50,000 e-books had been input to Project Gutenberg, with an average of over fifty new e-books being added each week.

PROJECTED: To extend forward.

PROJECTILE: 1. Projectile motion is the curved path an object follows when thrown or propelled near the surface of the Earth under the action of gravity. Objects experiencing projectile motion have a constant velocity in the horizontal direction, and a constantly changing velocity in the vertical direction. The trajectory resulting from this combination always has the shape of a parabola. 2. An object such as an artillery shell or a rocket that can be fired or launched. 3. A term describing a part of an animal's body that can thrust forward, for example, the jaws in some types of fish.

PROJECTION: 1. The representation of a figure, solid or line on a flat surface. 2. A transformation in which a shape or set of points is mapped onto a plane. 3. A potential future evolution of a quantity or set of quantities, often computed with the aid of a model. Projections are distinguished from "predictions" in order to emphasise that projections involve assumptions concerning, for example, future socio-economic and technological developments that may or may not be realised, and are therefore, subject to substantial uncertainty.

PROJECTION FORMULA: A formal two-dimensional representation of a three-dimensional molecular structure obtained by projection of bonds (symbolised as lines) onto a plane with or without the designation of the positions of relevant atoms by their chemical symbols. A projection formula which indicates the spatial arrangement of bonds is called a stereochemical formula or stereo formula. Examples of stereo formulae are Fischer projection, Newman projection, sawhorse projection, wedge projection and zig-zag projection.

PROJECTION METHODS: In mathematics, methods of solving constrained optimisation problems by combining descent methods for unconstrained problems with an appropriate projection onto the feasible set to maintain feasibility at each iteration. For example, when

attempting to minimise a function over the orthant, if line search produces a point x with some negative coordinates, one might replace these coordinates by zeros. This replaces x by its positive part, which is equivalent to orthogonal projection onto the orthant.

PROJECTIVE: 1. Of, relating to or derived by geometric projection. 2. Of a geometric figure, having the property that it is invariant when it undergoes projection.

PROJECTIVE COORDINATES (HOMOGENEOUS COORDINATES):

1. The representation of a point with Cartesian coordinates (x, y) as (x', y', t) , where $x' = tx$ and $y' = ty$. For a given plane, this is equivalent to selecting a point not in the plane and representing any point in it by the line joining it to the reference point. This allows any point of the plane to be represented in terms of a given set of basis points as a linear combination whose coefficients are the homogeneous coordinates of the point with reference to those basis points. 2. The representation of points in a two-dimensional algebraic geometry by equivalence classes of proportional triples from the ground field and similarly for higher dimensions. The coordinates are determined by the choice of a triangle of reference and a unit point.

PROJECTIVE GEOMETRY: The branch of geometry that deals with properties of geometric figures that remain unchanged under projection. The basic elements of projective geometry are points, lines and planes. These elements retain their character under projection; for example, the projection of a line is another line and the point of intersection of two lines is projected into another point that is the intersection of the projections of the two original lines.

PROJECTIVE PLANE: In mathematics, it is an abstract geometry of two dimensions which have exactly one line through any two distinct points, have at least one point on any two distinct lines (there are no parallel lines) and have four points such that no line is incident with more than two of them. In the ordinary Euclidean plane, two lines typically intersect in one point, but there are some parallel lines that do not intersect.

PROJECTIVE RELATIVITY THEORY: A unified field theory in which projective geometry is used.

PROJECTOR: An apparatus such a video or movie projector that projects a picture on to a screen.

PROKARYOTAE: One of the five kingdoms into which organisms are classified, which includes unicellular microorganisms or infectious agents of cells (the viruses), lacking nuclei and having asexual and chromosomal reproduction and unidirectional recombination. All other organisms are eukaryotes. **Prokaryote** is a group of organisms comprising only bacteria and cyanobacteria that lack a distinct nucleus and other organelles due to the absence of internal membranes and do not develop or differentiate into multicellular forms. Bacteria are among the best-known prokaryotic organisms. A second group of prokaryotes, called archaea is genetically distinct from bacteria, often thriving in extreme environmental conditions. The lack of internal membranes in prokaryotes distinguishes them from eukaryotes.

PROKARYOTIC CELL: A cell lacking a membrane-bounded nucleus or membrane-bounded organelles. Prokaryotic cells are thus more primitive than eukaryotic cells, which evolved from them. A prokaryotic organism, such as a bacterium, consists of a single prokaryotic cell.

PROKARYOTIC FISSION (BINARY FISSION): A form of asexual reproduction used by all prokaryotes, some protozoa and some organelles within eukaryotic organisms in which a single-celled organism divides into two daughter cells, with each having the potential to grow to the size of the original cell.

PROLACTIN (LACTOGENIC HORMONE; LUTEOTROPHIC HORMONE; LUTEOTROPHIN): A peptide hormone discovered by Dr. Henry Friesen, produced by the anterior pituitary gland. In mammals the act of an infant suckling the nipple stimulates the mammary glands to produce milk and the corpus luteum of the ovary to secrete the hormone progesterone.

PROLATE SPHEROID: An ellipsoid produced by rotating an ellipse through 360° about its major axis (i.e. its longer axis).

PROLIFEROUS: 1. In botany, bearing buds or offshoots from leaves or flowers. 2. In zoology, reproducing by budding as in corals.

PROLIFIC: In biology, it refers to an animal or plant which produces a large number of offspring or fruit.

PROM (PROGRAMMABLE READ-ONLY MEMORY): An integrated circuit with a memory that can be programmed permanently and cannot be changed. They are often seen in video game consoles, mobile phones, radio-frequency identification (RFID) tags, implantable medical devices, high-definition multimedia interfaces (HDMI) and in many other consumer and automotive electronics products.

PROMETAPHASE: The phase of mitosis following prophase and preceding metaphase, in eukaryotic somatic cells in which the chromosomes become attached to the spindle fibres and align themselves at right angles to the spindle fibres across the centre of the cell.

PROMETHIUM: A radioactive, fairly short-lived metallic element of the lanthanide series with the symbol Pm, atomic number 61 and relative atomic mass 145, melting point 1042 °C (1315 K) and boiling point 3000 °C (3270 K). It is produced in nuclear reactors by the fission of uranium, thorium or plutonium. It is used in phosphorescent paints. When mixed with a phosphor, the light emitted can be used to power a photocell.

PROMINENCE: A loop of large, bright, gas extending outward from the Sun's surface. Prominences are anchored to the Sun's surface in the photosphere, and extend outwards into the Sun's corona.

PROMINENT: Conspicuous; easily noticed from its surrounding, as raised veins on the surface of a leaf.

PROMOTER: 1. In chemistry, it is a relatively small quantity of one or more substances, which when added to a catalyst improves the activity, the selectivity or the useful lifetime of the catalyst. In general, a promoter may either augment a desired reaction or suppress an undesired one. 2. In genetics, it is the region of a DNA molecule that signals the start of transcription located near the genes they regulate, on the same strand and upstream towards the 5' region of the sense strand. Promoters are called **Pribnow box** in prokaryotes and **TATA box** or **Hogness box** in eukaryotes. In eukaryote genes, the promoter typically has a core promoter very near to the transcription start site, often containing the base sequence TATA, giving it the name TATA box. Proteins called **transcription factors** bind to the promoter and ensure that the enzyme responsible for transcription RNA polymerase is correctly positioned with respect to the coding region of the gene.

PROMPT: 1. A prompt is text or symbols used to represent the system's readiness to perform the next command. A prompt may also be a text representation of where the user is currently working. 2. A message or window that appears, alerting the user or asking for a confirmation or asking a question. 3. In an FTP (File Transfer Protocol) session, prompt is a command that enables and disables the FTP prompt.

PROMPT COINCIDENCE: The occurrence of two or more events separated by a time interval which is less than a specified small value.

PROMPT NEUTRONS: The neutrons emitted during a nuclear fission process within less than a microsecond of fission in contrast to a delayed neutron decay which can occur within the same context, emitted by one of the fission products anytime from a few milliseconds to a few minutes later.

PRONATION: 1. A rotational movement of the lower forearm so that the hand faces backwards or downwards with the radius and ulna crossed. 2. The natural side-to-side movement of the foot as in walking or running. This corresponds to a anticlockwise twist for the right forearm and a clockwise twist for the left.

PRONIC NUMBER (OBLONG NUMBER; HETEROMEIC NUMBER): A number of the form $n(n + 1)$, which is the product of two consecutive integers:

$2(1 \times 2)$, $6(2 \times 3)$, $12(3 \times 4)$, $20(4 \times 5)$, etc. The pronic numbers are twice the triangular numbers and represent the lengths that produce the musical intervals: octave (1:2), fifth (2:3), fourth (3:4), major third (4:5), etc.

PRONOTUM: The upper surface of the front segment of an insect's thorax. In some species of insects, the pronotum is expanded to form a shield.

PRONUCLEUS: The nucleus of a sperm or an (egg) prior to fertilisation. Sperm and egg cells carry half the number of chromosomes of other nonreproductive cells. When the pronucleus of a sperm fuses with the pronucleus of an egg, their chromosomes combine and become part of a single nucleus in the resulting embryo, containing a full set of chromosomes.

PROOF: In logic and mathematics, a sequence of steps or statements or formulae generated according to accepted rules. Each formula is either an axiom or is derived from a previously established theorem and the last formula is the statement that has to be proved to establish the truth.

PROOF SPIRIT: A numerical scale used to indicate the alcohol (ethanol) content of an alcoholic drink and defined as a solution of alcohol of such strength that it will ignite when mixed with gunpowder. Proof spirit contains 57.07% alcohol by volume (ABV) or 49.24% by weight in the UK. Pure (absolute) alcohol is 175.25° proof in the United Kingdom or 200° proof in the USA. Alcohol content is now usually measured as % alcohol by volume.

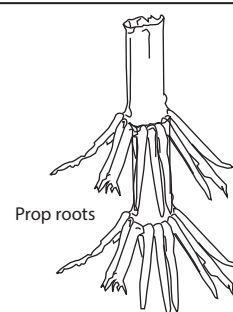
PROOF THEORY: The branch of mathematical logic that studies the syntactic properties of formal theories especially the syntactic characterisation of deductive validity. Some of its fields of studies include structural proof theory ordinal analysis, provability logic, reverse mathematics, proof mining, automated theorem proving and proof complexity.

PROMINENCE (SOLAR PROMINENCE): Solar storm in the form of huge, bright feature extending outward from the Sun's surface. Prominences are anchored to the Sun's surface in the photosphere, and extend outwards into the Sun's hot outer atmosphere, called the **corona**.

PROMOTER REGION: A region of DNA that initiates transcription factors. These factors bind to the promoter sequences, recruiting RNA polymerase, the enzyme that synthesises the RNA from the coding region of the gene.

PROP ROOT (BRACE ROOT):

Any of the modified roots that arise from the stems of certain plants and provide a prop or extra support. Prop roots originate from stems but ultimately penetrate the soil and are usually tall and slender. The prop roots develop at successively higher levels as the stem elongates, for example, as in the maize plant, serving both as anchors and as food gatherers.



Prop roots (brace roots)

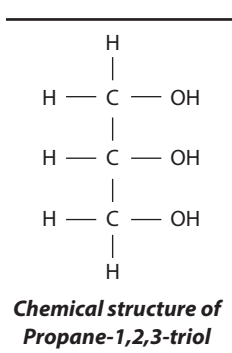
PROPAGATION: 1. Asexual reproduction in plants, using techniques such as taking cuttings, grafting, budding or layering in which parts are used to produce whole plants. In most cases the offsprings are clones, that is, genetically identical to the parent. 2. The movement of a wave away from its source such as the propagation of sound through air and of light in a vacuum.

PROPAGATOR: A closed but transparent container in which seed can be sown or cuttings grown in a humid, warm atmosphere.

PROPAGULE: Any cellular structure produced by an organism that is capable of dispersing and developing into a new individual. A propagule may be a wood cutting, leaf section or other plant parts such as a seed or spore.

PROPANAL (PROPIONALDEHYDE; PROPIONIC ALDEHYDE; PROPALDEHYDE): A colourless liquid aldehyde with the chemical formula C_2H_5CHO , density 0.81 g cm^{-3} , melting point -81°C (192 K) and boiling point $46\text{--}50^\circ\text{C}$ (319–323 K). It is a saturated 3-carbon aldehyde and is a structural isomer of acetone. It is produced industrially through hydroformylation, by combining synthesis gas (carbon monoxide and hydrogen) with ethylene using a metal catalyst: $CO + H_2 + C_2H_4 \rightarrow CH_3CH_2CHO$. In the laboratory, it is prepared by refluxing a mixture of propanol, sulfuric acid and potassium dichromate. It is used in the manufacture of plastics and rubber chemicals.

PROPANE-1,2,3-TRIOL (GLYCERINE; GLYCEROL): A colourless, sweet and sticky liquid that is soluble in water, which is a basic component of nearly all complex lipids. Its chemical formula is $C_3H_8O_3$, density 1.261 g/cm^3 , melting point 17.8°C (291.0 K) and boiling point 290°C (563 K). It is obtained from fats and oils as a by-product of soap manufacture and used as a solvent, antifreeze, a plasticiser and a sweetener and in the manufacture of dynamite, cosmetics, liquid soaps, inks and lubricants.



PROPANE: A gaseous hydrocarbon of the alkane series with the chemical formula C_3H_8 , density $2.0098 \text{ mg mL}^{-1}$ (at 0°C , 101.3 kPa), melting point -188°C (85.5 K) and boiling point -42 to -42°C (230.9–231.3 K). It is derived from other petroleum products during oil or natural gas processing and used as fuel for engines, oxy-gas torches, barbecues, portable stoves and residential central heating. It is made by fractional distillation of crude oil.

PROPANEDIOIC ACID (MALONIC ACID): A white crystalline dicarboxylic acid with the chemical formula $CH_2(COOH)_2$, density 1.619 g/cm^3 and melting point 136°C (409 K). It decomposes above its melting point to ethanoic acid. The ionised form of malonic acid, as well as its esters and salts are known as **malonates**. Propanedioic acid is used in the synthesis of other dicarboxylic acids.

PROPANOIC ACID (PROPIONIC ACID; ETHANE CARBOXYLIC ACID; PROPIONIC ACID): A colourless, naturally-occurring carboxylic acid, with a pungent odour. Its chemical formula is $C_3H_6O_2$, density 0.99 g/cm^3 , melting point -21°C (252 K) and boiling point 141°C (414 K). It is miscible with water, but can be removed from water by adding salt. In industry, propanoic acid is mainly produced by the hydrocarboxylation of ethylene using nickel carbonyl as the catalyst. It is also produced by the aerobic oxidation of propionaldehyde. It is used in the manufacture of preservatives, perfumes and artificial flavours.

PROPANOL (PROPYL ALCOHOL): A member of the homologous series of alcohols with the chemical formula C_3H_8O . It is a colourless volatile alcohol and used as a solvent or antiseptic.

PROPANONE (PROPAN-2-ONE; DIMETHYL KETONE; B-KETOPROPANE; ACETONE; 2-PROPANONE; DIMETHYL FORMALDEHYDE): A common name for acetone with the chemical formula CH_3COCH_3 , density 0.791 g cm^{-3} , melting point -95 to -93°C (178–180 K) and boiling point $56\text{--}57^\circ\text{C}$ (329–330 K). It is a colourless flammable liquid and the first member of the homologous series of aliphatic ketones. It is produced and disposed of in the human body through normal metabolic processes. It is normally present in blood and urine. People with diabetes produce it in larger amounts. It is produced industrially directly or indirectly from propylene. It is miscible with water and serves as an important solvent serving as the active ingredient in nail polish remover and as paint thinner.

PROPELLANT: 1. A combustible substance classified as liquid or as solid that is burned to give upward thrust to a rocket. 2. An explosive

charge that projects a bullet from a gun. 3. A compressed inert gas used to dispense the contents of an aerosol container when pressure is applied and released.

PROPELLER: A screw-like device with a central hub and radiating blades placed so that each forms part of a helical surface. It is used to propel some ships and aeroplanes. By its rotation in water or air, the propeller produces thrust on the blades, which gives forward motion to the vehicle.

PROPENAL (ACROLEIN; PROP-2-ENAL; ACRALDEHYDE; ACRYLIC ALDEHYDE; ALLYL ALDEHYDE; ETHYLENE ALDEHYDE): A colourless pungent liquid unsaturated aldehyde with the chemical formula C_3H_4O , density 0.839 g/ml , melting point -88°C (185 K) and boiling point 53°C (326 K). It is made from propene and is used in producing polyester, polyurethane, propylene glycol, acrylic acid, acrylonitrile and glycerol.

PROPENE (PROPYLENE; METHYLETHYLENE): A flammable gaseous hydrocarbon with the chemical formula C_3H_6 , melting point -185.2°C (88 K) and boiling point -47.6°C (226 K). It has one double bond and is the second simplest member of the alkene class of hydrocarbons. At room temperature, propene is a gas. It is obtained from petroleum and used as an organic synthesis.

PROPENOATE (ACRYLATE): A salt or ester of propenoic acid.

PROPENOIC ACID (ACRYLIC ACID; PROP-2-ENOIC ACID; ACROLEIC ACID; ETHYLENECARBOXYLIC ACID; PROPENE ACID; PROPENOIC ACID; VINYLFORMIC ACID): An unsaturated liquid carboxylic with the chemical formula $C_3H_4O_2$, density 1.051 g/ml , melting point 14°C (287 K) and boiling point 141°C (414 K). It is the simplest unsaturated carboxylic acid. It is produced from propene which is a by-product of ethylene and gasoline production:

$CH_2=CHCH_3 + 1.5 O_2 \rightarrow CH_2=CHCO_2H + H_2O$. It is miscible with water, alcohols, ethers and chloroform. It readily polymerises and it is used in the manufacture of various plastics, coatings, adhesives, elastomers, as well as floor polishes and paints.

PROPENONITRILE (2-PROPENENITRILE; CYANOETHENE; ACRYLONITRILE; VINYL CYANIDE): A colourless liquid with the chemical formula C_3H_3N , density 0.81 g/cm^3 , melting point -84°C (189 K) and boiling point 77°C (350 K). It appears yellow due to impurities. It is a monomer, made from propene and used for the manufacture of useful plastics.

PROPENYL GROUP (ALLYL GROUP): The organic group $H_2C=CHCH_2-$ derived from propylene.

PROPER FACTOR: In mathematics, it is a factor of a given number other than the number itself, including 1. For example, the proper factors of 6 are 1, 2 and 3; whilst its factors are 1, 2, 3 and 6. A perfect number is one for which the sum of the proper factors equals the number itself.

PROPER FRACTION: 1. Also known as **common fraction** or **simple fraction**, it is an expression in which the value of the numerator is less than the value of the denominator, for example, $3/5$. 2. A chemical component separated by fractionation.

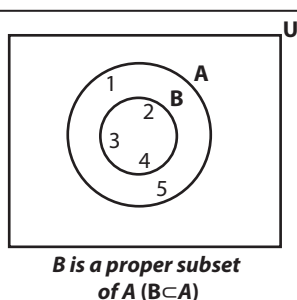
PROPER LENGTH (REST LENGTH): The length of an object measured by an observer in a frame of reference where the object is at rest. This is the greatest possible length that could be recorded for an object.

PROPER MAPPING: In mathematics (topology), it is a continuous mapping such that the inverse image of a compact set is compact.

PROPER MOTION: The apparent angular motion of a star on the celestial sphere resulting in the change of a star's apparent position relative to that of other stars. It is expressed in angular units.

PROPER POINT: In mathematics, a real point in an augmented Euclidean geometry that corresponds to a point of the Euclidean plane and at which non-parallel lines intersect.

PROPER SUBSET: Set B is a proper subset of set A , if there exists an element in A that does not belong to B . In other words, if A is a subset of B , then A is a proper subset of B if A is not equal to B . Mathematically, it is expressed as $A \subset B$. For example, if $A = \{1, 2, 3, 4, 5\}$ and $B = \{2, 3, 4\}$, it means that A is a proper subset of B .



PROPER TIME INTERVAL: The time between events as measured in a frame where the events take place at the same point in space. This is the shortest possible time that any observer could correctly record for the event and is usually the time measured in the moving frame of reference.

PROPERTIES: The characteristics that a substance possesses by virtue of its composition or the characteristic ways in which a substance behaves (reacts) and which makes it different from others. These could be the physical or chemical properties. For example, two properties of water are that it is wet and boils at 100°C .

PROPHAGE: The DNA of a temperate bacteriophage inserted as part of the linear structure of the DNA chromosome of a bacterium. The process of insertion of the viral DNA is known as **lysogeny**.

PROPHASE: A stage of mitosis, during which chromosomes contract and divide along their length into chromatids.

PROPHYLL (PROPHYLLUM): A plant structure that resembles a leaf or consists of a modified leaf.

PROPLASTID: An immature, colourless plastid. Proplastids occur in the cells of meristematic tissues. They consist of a double membrane enclosing granula stroma, multiply by division and give rise in mature cells to leucoplasts or chromoplasts.

PROPOLIS: A resinous substance collected by worker bees from certain trees which is used as glue for building and mending the hive.

PROPORTION: Two pairs of quantities a, b and c, d are in proportion if their ratios $a:b$ and $c:d$ are equal, i.e., $a:b = c:d$. This means that if a increases, b increases in a linear fashion. It is one of the four terms in a relationship of proportion between two ratios, where the first term divided by the second equals the third divided by the fourth. The quantities a, b, c are then called proportionals, a and d are called the means and b and c are the extremes. If two quantities x and y are in **direct proportion**, then x and y are related by $y = kx$, where k is the constant of direct proportion. If x and y are in **inverse proportion**, then x and y are related by the relation $xy = k$, where k is the constant of inverse proportion.

PROPORTIONAL COUNTER: A type of radiation detector for ionising radiation in which the size of the output pulse is proportional to the number of ions formed in the initial ionising event. It is used to measure α , β and x-rays. It has low sensitivity for γ -radiation.

PROPORTIONAL COUNTER TUBE: A counter tube operated under such conditions that the magnitude of each pulse is proportional to the amount of energy deposited in it.

PROPORTIONAL GAS-SCINTILLATION COUNTER: A proportional counter coupled to an ultraviolet sensitive photomultiplier tube. Initial electrons produced by the interaction of the high-energy photon with the counter fill-gas are accelerated by a high electric field where they acquire sufficient energy to excite the noble gas atoms. The resulting UV radiation is observed by a photomultiplier tube.

PROPORTIONAL LIMIT: The greatest stress a material can sustain without departure from linear proportionality of stress and strain.

PROPOSITION: In logic, it is a statement that affirms or denies something and is capable of being true or false. Also, it is the meaning expressed in such a statement, as opposed to the way it is expressed.

PROPOSITIONAL FUNCTION (OPEN SENTENCE; SENTENTIAL FUNCTION): In logic, a sentence containing one or more variables that can be either true or false depending on the value of the variable(s). Examples are $3x = 5$ and $3x + y = 4$.

PROPRIOCEPTOR: Any receptor that is sensitive to movement, pressure or stretching within the body. Sensory information from certain proprioceptors, such as those in muscles and tendons can be used by the motor system as feedback to guide postural adjustments and control of well-practiced or semiautomatic movements such as those involved in walking.

PROPROTEIN (PROTEIN PRECURSOR; PRO-PEPTIDE): A protein that is made in an inactive form, so that it requires processing to become functional by post translational modification. Protein precursors are often used by an organism when the subsequent protein is potentially harmful, but needs to be available on short notice and/or in large quantities.

PROPYL GROUP: The organic group $\text{CH}_3\text{CH}_2\text{CH}_2\cdot$ derived from propane ($\text{HCH}_2\text{CH}_2\text{CH}_3$) by removing one of the end hydrogens.

PROSENCEPHALON: One of the three sections into which the brain of a vertebrate embryo is divided. It is the forebrain section, which controls body temperature, reproductive functions, eating, sleeping and any display of emotions.

PROSIMIAN (PRIMITIVE): Any animal belonging to the suborder Strepsirrhini of primates that are not monkeys or apes. Examples are lemurs and the lorises. They are considered to have characteristics that are more primitive than those of monkeys and apes.

PROPOSITIONAL CALCULUS (SENTENTIAL CALCULUS; PROPOSITIONAL LOGIC; SENTENTIAL LOGIC): In logic, the formal theory of which the intended interpretation concerns the logical relations between sentences treated only as a whole and without regard to their internal structure. Some examples are "and" (conjunction), "or" (disjunction), "not" (negation) and "if" (but only when used to denote material conditional).

PROSOMA: The anterior or cephalic section of the body of arachnids and other arthropods of the phylum Chelicerata, which consists of the fused head and thorax (cephalothorax) and bears the chelicerae and other appendages.

PROSPECTIVE STUDY (PROSPECTIVE COHORT STUDY): A type of observational studies used in medicine, social science, actuarial science and ecology to determine the relationship between a condition and a characteristic shared by some members of a group.

PROSTACYCLIN (PROSTAGLANDIN; PGI_2): A member of the family of lipid molecules known as eicosanoids normally produced from prostaglandin H_2 by endothelial cells lining the walls of blood vessels. It inhibits blood clotting by preventing the aggregation of platelets and causes widening of blood vessels (vasodilation). Hence, it is an antagonist of thromboxane A_2 .

PROSTAGLANDIN: Any of a group of about a dozen organic compounds synthesised from essential fatty acids causing a range of physiological effects in animals. Prostaglandins are highly potent substances that are not stored but are produced as needed by cell membranes in virtually every body tissue and different ones have been found to raise or lower blood pressure and regulate smooth muscle activity and glandular secretion. Prostaglandins also control the substances involved in the transmission of nerve impulses, participate in the body's defences against infection and regulate the rate of metabolism in various tissues. Several prostaglandins are known to induce fever, possibly by participating in the temperature-regulating mechanisms in the hypothalamus. It also plays a part in causing

inflammation. One type stimulates contraction of the uterus and is used clinically to induce labour.

PROSTANOIDS: The family of natural prostaglandins and prostaglandin-like compounds.

PROSTATE (PROSTATE GLAND): A gland found in in male mammals with the size and shape of a chestnut, located directly under the bladder, in front of the rectum and opening into the urethra that is responsible for secreting an alkaline fluid into the semen during ejaculation that acts to improve the movement and viability of sperm. The fluid activates sperm motility and in neutralising the acidity of the vagina, thus enhancing fertilisation. The prostate gland is also involved in control of urine through the urethra.

PROSTATE CANCER: A slow growing cancer of the prostate gland that affects only elderly men (often of over age 65 and above) in which cells in the prostate gland, become abnormal and grow, forming tumours. Because the prostate gland is involved in control of urine through the urethra prostate cancer results in improper control of urine, hence difficulty in urination that may be accompanied with pain; and also affects normal sex drive. Its causes include age related as it commonly affects men of over 50 years; genetics; overuse of anti-inflammatory medicines, obesity, sexually transmitted diseases (STDs) such as gonorrhoea, etc.

PROSTHESIS: An artificial device inserted onto or into the body to replace or supplement the function of defective or diseased organs. It includes artificial limbs, hearing aids, false teeth and eyes.

Prosthetics is the branch of medicine or surgery that deals with the production and application of artificial body parts.

PROSTHETIC GROUP: A tightly bound non-peptide inorganic or organic component of certain protein molecules. Prosthetic groups may be lipids, carbohydrates, metal ions, phosphate groups, etc. It is usually required for biological activity, especially when the prosthetic group is complexed with an enzyme.

PROTACTINIUM: A toxic silver-grey, radioactive metallic element of the actinide series with the symbol Pa, atomic number 91 and relative atomic mass 231.036. It is the longest-lived and most abundant naturally occurring isotope. It is produced from uranium ores.

PROTAMINE: Any of a group of arginine-rich, nuclear proteins of relatively low molecular weight found in association with the chromosomal DNA of vertebrate male germ cells. They are believed to be essential for sperm head condensation and DNA stabilisation.

PROTANDRY (PROTERANDRY; PROTANDROUS): Development of the anther before the stigma, thereby releasing pollen before the stigma is matured or receptive. This is common in mints, legumes and composites.

PROTEASE: Any enzyme that takes part in the breaking down (by hydrolysis) of proteins into peptides and amino acids. Examples of proteases include pepsin and trypsin. Several proteases, acting sequentially, are normally required to break down a protein completely into its constituent amino acids.

PROTEASOME: A giant protein complex inside all eukaryotes and archaea, as well as in some bacteria that recognises and destroys proteins tagged for elimination by the small protein ubiquitin. It is hollow and has openings at both ends to allow entry of the protein to be digested. In eukaryotes, they are located in the nucleus and the cytoplasm. A human cell contains about 30,000 proteasomes.

PROTECTION OF A REACTIVE GROUP: Temporary chemical transformation of a reactive group into a group that does not react under conditions, where the non-protected group reacts.

PROTECTIVE ACTION: The phenomenon in which the addition of small amounts of hydrophilic colloid to a hydrophobic sol may make the latter more sensitive to flocculation by electrolyte. Higher

concentrations of the same hydrophilic colloid usually protect the hydrophobic sol from flocculation.

PROTECTIVE COLOURATION: The colouration of an animal that enables it to blend with its surroundings and, therefore, escape the attention of predators. Many animals have adapted their colouration as a means of defence against predators. Except where selection favours bright colouration for breeding or territorial display, higher animals are coloured in such a way that they blend in with their backgrounds. The most highly developed camouflage is found in ground-nesting birds, for example, nightjars or insects, such as walking sticks or leaf insects. Certain animals such as octopuses, cuttlefish, chameleons and some flatfishes can change the body texture and colouration to match different backgrounds.

PROTEIN: Any of a large group of macromolecules playing a central role in the structure and functioning of living cells. Together with nucleic acids, carbohydrates and lipids, proteins form the biochemical basis of life on Earth.

PROTEIN ENGINEERING: A technique used to produce proteins with altered or novel amino acid sequences. The methods used are: (i) Transcription and translation systems from synthesised lengths of DNA or RNA with novel sequences. (ii) Chemical modification of 'normal' proteins. (iii) Solid-state polypeptide synthesis to form proteins.

PROTEIN KINASE: An enzyme that transfers phosphate groups from adenosine triphosphate (ATP) to hydroxyl side chains on proteins, causing changes in function. Most phosphate on proteins of animal cells is on serine residues, less on threonine, with a very small amount on tyrosine residues. Tyrosine kinases phosphorylate proteins on tyrosine, serine/threonine kinases on serine or threonine.

PROTEIN PHOSPHATASE: Any of a group of enzymes involved in catalyzing the hydrolysis of esters of phosphoric acid and important in the absorption and metabolism of carbohydrates, nucleotides and phospholipids and in the calcification of bone. It is present in serum, semen, the kidney and the prostate. Protein phosphatases are divided into three groups: Class I enzymes are mainly specific for seryl- or threonyl-phosphate; Class II enzymes hydrolyse tyrosylphosphates and are generally designated 'phosphatase 3'; and Class III enzymes have dual specificity.

PROTEIN SYNTHESIS: The process by which some of the amino acids that an organism needs are manufactured within the cytoplasm of the cells using DNA, RNA and various enzymes. It also includes transcription and translation. Some amino acids are produced by the body. The pattern in which the amino acids are linked decides what kind of protein is produced. In turn, it is the genetic code, contained within DNA that determines the precise order in which the amino acids are linked up during protein manufacture.

PROTEINACEOUS ENDO-SPERM LAYER (ALEURONE LAYER): The single outer layer of living cells of endosperm as in maize and other grain species containing granules of protein which secretes enzymes into the endosperm to initiate digestion of stored food reserves in the endosperm during germination.

PROTEINURIA: The presence of protein in the urine, an indication of kidney disease or damage.

PROTEOGLYCAN: A subclass of protein in which the carbohydrate units are polysaccharides that contain amino sugars. The protein is glycosylated by one or more (up to about 100) glycosaminoglycans [linear polymers of up to about 200 repeating disaccharide units that consist of a hexosamine (D-glucosamine or D-galactosamine) alternating with a uronic acid (D-glucuronic or L-iduronic) or a neutral sugar (D-galactose)].

PROTEOME: The entire complement of proteins synthesised by a genome, cell, tissue or organism at a given time, for example, all of the proteins in a virus. This can be determined by analysing

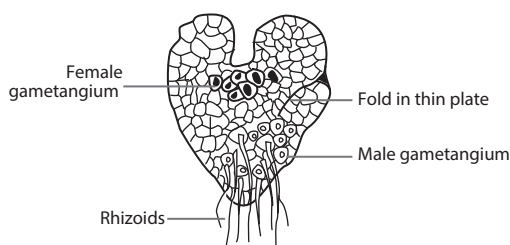
protein constituents of cell contents using techniques such as gel electrophoresis, high through-put liquid chromatography, microarrays and mass spectrometry, coupled with automated database searching to identify proteins. The proteome is larger than the genome, especially in eukaryotes, because there are more proteins than genes and unlike the genome, the proteome is constantly changing due to the influence of intracellular and extracellular factors.

PROTERANTHY (PROTERANTHOUS): Having or flowers that develop before the leaves.

PROTEROGYNY (PROTOGYNY): In botany, it is the condition whereby the pistils mature first and are therefore receptive before the anthers release pollen. This situation occurs in Arum lilies and many wind pollinated plants such as grasses.

PROTEROZOIC: A major division of geologic time spanning from 3.5 billion to 570 million years ago, which is the second division of the Precambrian era. The end of the Proterozoic coincides with the Precambrian-Cambrian boundary, the first appearance of diverse coelomate invertebrate animals.

PROTHALLUS (PROTHALIUM): A small flattened multicellular structure produced by a germinating spore and bearing sex organs that represent the independent gametophyte generation of many ferns and pteridophytes. It bears either the male or female sex organs or both.



A fern prothallus, - gametophyte

PROTHORACICOTROPIC HORMONE (PTTH): An insect hormone released by glands (corpora allata) on either side of the brain, that stimulates release of the moulting hormone, ecdysone, from the prothoracic glands. It was the first insect hormone that was discovered. PTTH isolated from the moth *Manduca sexta* is a 30-kDa protein consisting of two identical peptide chains.

PROTHORAX: The front segment of an insect's thorax, which bears the first pair of legs and no wings. The thorax is divided into three segments; the other two are mesothorax (the middle segment), and metathorax (anterior of rear segment). Each of the segments has a pair of legs and hard plates called **sclerites**.

PROTHROMBIN (FACTOR II): Carbohydrate-protein compound in plasma essential to clotting factors. It is converted by thromboplastin to thrombin, which transforms fibrinogen in plasma into fibrin. Prothrombin synthesis occurs in the liver and is dependent on adequate supplies of vitamin K, so that any conditions that impair the vitamin's absorption result in prothrombin deficiency and a tendency to prolonged bleeding.

PROTIST: Any eukaryotic organism that is mostly single-celled (although the algae tend to be multicellular) or colonial in form and lacks cellular differentiation into tissues. Protists include simple algae, lower fungi and protozoa. They display a diversity of forms which includes motility mainly by using flagella cilia or foot-like extensions called pseudopodia and cells to leaf-like or filamentous forms and sheets of cells. They can be autotrophic, saprophytic or heterotrophic or combinations. Some protists are considered the ancestors of multicellular plants, animals and fungi. Protista is a classification system that divides all living things into six kingdoms. It includes mostly single-celled organisms with cells similar to those of the Plantae, Animalia, and Fungi kingdoms.

PROTIUM (NORMAL HYDROGEN): A specific name for the atom ^1H . The cation $^1\text{H}^+$ is a proton, the species ^1H is a protide anion and ^1H is the protio group. The word proton should not be used for H^+ in its natural abundance.

PROTO-ONCOGENE: A normal cellular gene corresponding to an oncogene; a gene with a potential to cause cancer, but that requires some alteration to become an oncogene. A normal cell usually undergoes a process called apoptosis or programmed cell death but activated oncogenes can prevent these cells from dying and instead allow them to survive and proliferate.

PROTOCOL: An agreed set of standards for the transfer of data over a computer network such as local area network, Internet, Intranet, etc. Examples of protocols include PPP, TCP/IP, SLIP, HTTP and FTP.

PROTOCTISTA: A taxonomic kingdom consisting of unicellular or simple multicellular organisms that possess nuclei and cannot be classified as animals, plants or fungi.

PROTODERM (DERMATOGEN): A plant tissue, primary meristem formed by the apical meristem of shoots and roots that subsequently gives rise to the epidermis.

PROTOGENIC (SOLVENT): Capable of acting as a proton donor (strongly or weakly acidic as a Brønsted acid). The term is preferred to the synonym protic or the more ambiguous expression acidic by itself.

PROTOGYNY: The state in hermaphroditic or dioecious animals in which the female reproductive organs (carpels) of a flower reach maturity, withers and dies before those of the male (stamens) mature, ensuring that self-fertilisation does not occur.

PROTON: A positively nuclear charged particle commonly found in the nucleus of an atom. Each proton carries one unit of positive electric charge (i.e. $+1.6 \times 10^{-19} \text{ C}$). It has a charge number of +1, spin quantum number $\frac{1}{2}$ and rest mass of $1.007276470(12)\text{u}$. The proton is about 1836 times heavier than the electron, so that the proton constitutes almost the entire mass of the hydrogen atom. Most of the interior of the atom is empty space, since the sizes of the proton and the electron are very small compared to the size of the atom.

PROTON AFFINITY: The negative of the enthalpy change in the gas phase reaction (real or hypothetical) between a proton (more appropriately hydron) and the chemical species concerned, usually an electrically neutral species to give the conjugate acid of that species.

PROTON DECAY: A hypothetical form of radioactive decay in which the proton decays into lighter subatomic particles, usually a neutral pion and a positron of the type $\text{P} \rightarrow \text{e}^+ + \pi^+$, predicted to occur in Grand Unified Theories (GUTs) because baryon number is no longer conserved.

PROTON MAGNETIC MOMENT: Atomic fundamental physical constant $\mu_p = 1.41060761(47) \times 10^{-26} \text{ JT}^{-1}$.

PROTON MAGNETOGYRIC RATIO: Atomic fundamental physical constant $\gamma_p = 2.67522128(81) \text{ s}^{-1} \text{ T}^{-1}$.

PROTON NUMBER (ATOMIC NUMBER): The number of protons in the nucleus of an atom of an element and its isotopes. It is used to determine the element's position in the periodic table. Atoms with the same atomic number make up a chemical element. Its symbol is Z.

PROTON PUMP: A carrier protein or complex of proteins capable of moving protons across the membrane of a cell, mitochondrion or other subcellular compartment. In cell respiration, the pumps grab protons from the space enclosed by the two membranes, called a **matrix** within the organelle and release the protons into the inter-membrane space. Such pumps use energy, for example, adenosine triphosphate (ATP), to establish a high concentration of protons on one side of the membrane compared to the other. This proton gradient is then exploited by the cell to drive various processes, including the transport of ions and small molecules across plasma membranes. Proton pumps are also involved in the secretion of gastric juice by mucosal cells in the stomach lining.

PROTON REST MASS: Atomic fundamental physical constant $m_p = 1.6726231(10) \times 10^{-27}$ kg.

PROTON TRANSFER REACTION: A chemical reaction, the main feature of which is the intermolecular or intramolecular transfer of a proton (hydron) from one binding site to another.

PROTONATED MOLECULE: An ion formed by interaction of a molecule with a proton abstracted from an ion, as often occurs in chemical ionisation according to the reaction: $M + XH^+ \rightarrow MH^+ + X$. The symbolism $[M + H]^+$ may also be used to represent the protonated molecule. The widely used term 'protonated molecular ion' to describe the MH^+ ion is not recommended, since it suggests an associated product of a proton with a molecular ion.

PROTONEMA: Green filamentous structure developed from the spore in mosses on which the leafy plant arises as a lateral or terminal shoot.

PROTONEPHRIDIUM: An excretory system of certain invertebrates, such as the flame-cell system of flatworms, consisting of a hollow cup-shaped cell containing a bunch of cilia, whose movement draws in waste products and propels waste products into excretory tubules. It has external openings called **nephridiopores** and lacks internal openings.

PROTONIC ACID (PROTON ACID): An acid that forms positive hydrogen ions in aqueous solution.

PROTOPHILIC SOLVENT (HYDROGEN BOND ACCEPTOR/ HBA SOLVENT): A term used to describe a solvent, which is capable of acting as proton acceptor, strongly or weakly basic (as a Brønsted base).

PROTOPHLOEM: The primary phloem tissue that arises in the elongating regions of shoot or root apices.

PROTOPLASM: The granular material comprising the living contents of a cell composed of a mixture of small molecules such as ions, amino acids, monosaccharides and water and macromolecules such as nucleic acids, proteins, lipids and polysaccharides. The cytoplasm and the nucleus together make up the protoplasm.

PROTOPLAST: A spherical, osmotically-sensitive cell without its cell wall but retaining an intact cell membrane. Protoplasts are used to create hybrid cells via protoplast fusion.

PROTOSTAR: A large mass that forms by contraction out of a cloud of gas and dust in the interstellar medium and thought to develop into a star when it has collapsed sufficiently for nuclear reactions to begin. Its existence was first proposed and postulated by the Armenian physicist, Viktor Amazaspovich Ambartsumian (1908-1996). A protostar is surrounded by a dense cocoon of gas and dust that blocks visible light, but allows through large amounts of far infrared and microwave radiation. The formation into a star for a one solar-mass star lasts about 100,000 years; starting with a core of increased density in a molecular cloud and ending with the formation of a T Tauri star, which then develops into a main sequence star.

PROTOSTELE: A stele with a solid core of xylem encased by phloem or xylem interspersed with phloem and lacking pith.

PROTOSTOMES: A group of invertebrate animals in which a mouth develops from the first opening in the embryo. It is the formation of the coelom by the embryonic mesoderm having split. Examples of protostomes are molluscs, annelids and arthropods.

PROTOTHERIA: A subclass of mammals, the Monotremata comprising of only the duck billed platypus (*Ornithorhynchus anatinus*) and the spiny anteater (*Tachyglossus aculeatus*) that lay large yolky eggs.

PROTOTROPHS: Microorganisms capable of growing at or below a temperature optimum of 15 °C.

PROTOTYPE: Any of the first few original type, form or instance of something serving as a typical example, basis or standard for other things of the same category. A prototype is often used as part of the product design process to allow engineers and designers to explore design alternatives, test theories and confirm performance prior to starting production of a new product. Engineers use their experience to tailor the prototype according to the specific unknowns still present in the intended design.

PROTOXYLEM: The primary xylem tissue that is formed in the expanding region of roots and shoots produced by the differentiation of the procambium. The protoxylem of the primary stem is next to the pith, but away from the centre and next to the pericycle.

PROTOZOA (PROTOZOO): Any of a group of small single celled animals or a phylum consisting of unicellular animals such as amoeba and paramecium found in most soils, fresh water and oceans.

PROTRACTOR: An instrument for constructing and measuring plane angles, usually in the form of a semi-circle marked in degrees from 0° to 180°. Navigators use a type called the **course protractor**, to measure the angular distance between north and the course plotted on a navigation chart.

PROTUBERANCE: A rounded bulge, swelling, lump or projection.

PROTURA: An order of primitive tiny wingless insects belonging to the subclass Apterygota living in soil and leaf litter with about 175 known species. They are eyeless, lack antennae and the front pair of legs are held forwards as sensors. The mouthparts which are concealed within folds (enthognathousm) are adapted for feeding on decaying organic matter.

PROUD: In agriculture science, it is the excessive growth or development in crops or livestock.

PROUST'S LAW (LAW OF CONSTANT COMPOSITION; LAW OF CONSTANT PROPORTION; LAW OF DEFINITE PROPORTIONS): Established by Joseph Proust (1754-1826), French chemist, it states that the proportions of the amounts of the elements in a pure compound are always the same and are independent of the method by which the compound was produced or that all pure samples of the same chemical compound always contain the same elements combined in the same proportion by mass, no matter how they are made

PROUT'S HYPOTHESIS: The hypothesis put forward by the British chemist William Prout (1785 - 1850). In 1815, he observed that all atomic weights are whole multiples of the atomic weight of hydrogen and that the atoms of other elements were actually groupings of various numbers of hydrogen atoms. Prout's hypothesis remained influential in chemistry throughout the 1820s. However, more careful measurements of the atomic weights such as those compiled by Jöns Jakob Berzelius in 1828 or Edward Turner in 1832, disproved the hypothesis. With the present understanding of atomic structure, it has been found that Prout's hypothesis was not totally incorrect.

PROVE: In mathematics, to demonstrate the validity of a hypothesis or proposition.

PROVEN SIRE: A sire or the male parent of an animal, especially a domesticated mammal whose genetic performance have been established by the production performances of his offspring in a properly conducted progeny test.

PROVENDER: A general term used to describe dry feeds for livestock.

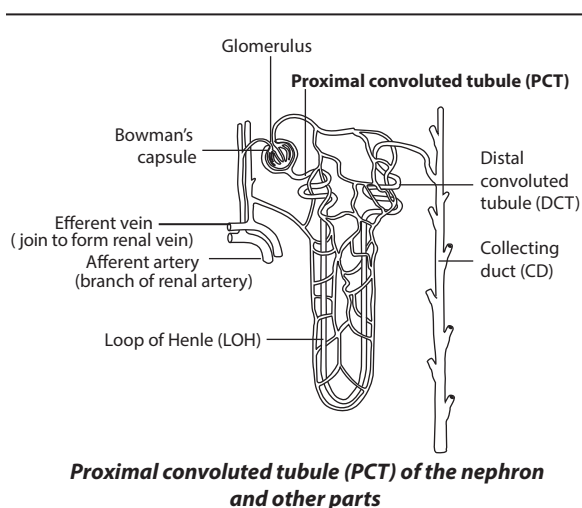
PROVENTRICULUS: 1. The anterior part of the stomach of a bird, where digestive enzymes are secreted before it progresses to the gizzard. 2. A type of gizzard occurring in many birds, invertebrates

and insects. It is situated in the anterior region of the stomach and consists of a set of bones (ossicles) and muscles that may store or commence digestion of food before it progresses to the gizzard.

PROVIRUS: A virus genome, such as retrovirus that has integrated itself into the DNA of a host cell, where it can undergo successive replications before being transcribed. In the case of retroviruses the single RNA strand of the virus is converted into double stranded DNA by the enzyme reverse transcriptase, then integrated into the host cell DNA and subsequently transcribed to form new RNA viruses. This intergration also introduces retroviral oncogenes into the host, with the attendant risk of host-cell transformation to a cancer cell. Some proviruses such as that of HIV can remain dormant for long periods before being transcribed.

PROXIMAL: 1. In biology, denoting the part of an organ that is nearest to the organ's point of attachment or the midline of the body. For example, the knuckles are at the proximal end of the fingers. 2. In mathematics, also known as **proximal**, it is nearest, for example, a proximal point in a set is a nearest point.

PROXIMAL CONVOLUTED TUBULE: The winding section of the renal tubule or nephron, which leads from the Bowman's capsule to the loop of Henle where most reabsorption of water, sodium, amino acids and sugar takes place.



PROXIMATE ANALYSIS: 1. Also known as **Weende analysis**, it is a method of chemical analysis used to determine the amounts of ash, crude fibre, crude protein, ether extract, moisture content and nitrogen-free extract in animal feed. 2. In chemical engineering, it is a method of analysis of solid fuels to predict how the fuel will behave in a furnace. The analysis consists of separating and identifying categories of compounds in a mixture, which includes moisture and ash content, the extracts of the mixture made with alcohol, petroleum ether, water, hydrochloric acid and resins, starches, reducing sugars, proteins, fats, esters, free acids, etc.

PROXIMATE CAUSES: In social behaviour, it is how the behaviour arises in animals from its ultimate cause (that is, the evolutionary history and functional utility of the behaviour). Proximate causes include hereditary, developmental, structural, cognitive, psychological and physiological aspects of behaviour. In other words, proximate causes are the mechanisms directly underlying the behaviour. For example, an animal separated from the herd may exhibit behaviours associated with fear reactions (such as elevated heart rate, shaking and hypersensitivity to sounds), which cause it to behave in ways that increase its chances of reuniting with the group. The underlying hormonal response, which is triggered by separation from the herd, is a proximate cause of these fear-based behaviours.

PROXY (PROXY CLIMATE INDICATOR): A record of the physical characteristics of the past that is interpreted, using physical and biophysical principles, to represent some combination of climate-related variations back in time. Climate-related data derived in this way are referred to as **proxy data**. Examples of proxies are tree ring records, corals, sub-fossil pollen, boreholes, lake and ocean sediments, and various data derived from ice cores.

PROXY SERVER: A computer server or software program that is part of the gateway server or another computer that separates a local network from outside networks. A proxy server will cache all pages accessed through the network allowing users who may wish to visit that same page to load it faster and reduce the networks bandwidth. When a page is accessed that is not in the proxy server's cache, the proxy server will access the page using its own IP address, cache the page and forward it to the user accessing that page.

PRUINOSE (PRUINATE): 1. Having a waxy, powdery, fine and whitish coating on the surface. 2. Conspicuously glaucous, like a prune.

PRUNIFORM: Shaped like a plum.

PRUNING: The process of trimming the shape of a tree, bush, etc, by cutting away some of the branches, etc, especially to encourage new growth or the removal of dead, broken, unwanted, diseased or insect-infected wood. A type of secateurs called **pruner** is used for pruning the trees.

PS/2 PORT: A 6-pin Mini-DIN socket used to connect a mouse or keyboard to a PC so that the serial port can be used by another device, such as a modem.

PSAMMOPHYTES: Plants that grow best in or tolerate sand, particularly fine to medium sand. In such an environment, the plants are often covered with sand, or their root system is exposed. It is also difficult for the seeds to germinate. Psammophytic trees and shrubs form strong adventitious roots on those parts of their trunks buried in the sand; adventitious buds and shoots arise on the exposed roots.

PSALTERIUM (OMASUM; MANYPLIES): The third and smallest compartment of the stomach of a cow or other ruminants, located between the abomasum and the reticulum. The inner surface has folds that break up food particles.

PSAMMOPHYTE: A plant that thrives only in sand or sandy soil.

PSEUDANTHIUM: A cluster of many small flowers which are grouped together to form a single flower-like structure. The individual flowers of a pseudanthium are commonly called florets.

PSEUDO: A prefix meaning false.

PSEUDO ACIDS: Potentially acidic compounds that require some structural reorganisation, such as keto-enol tautomerization, having a non-negligible activation energy, in order to show normal acidic properties, for example, nitroalkanes.

PSEUDO BASES: Hydroxy compounds that give salts with acids by formation of water accompanied by a change of constitution.

PSEUDO ORDER: An order of a chemical reaction that happens to differ from the true order as a result of the conditions used. Pseudo orders occur when one reactant is present in large excess. For example, a reaction of substance A undergoing hydrolysis may appear to be proportional only to [A] because the amount of water present is so large.

PSEUDO-ASYMMETRIC CARBON ATOM: The traditional name for a tetrahedrally coordinated carbon atom bonded to four different entities, two and only two of which have the same constitution but opposite chirality sense. The *r/s* descriptors of pseudo-asymmetric carbon atoms are invariant on reflection in a mirror (i.e. *r* remains *r* and *s* remains *s*), but are reversed by the exchange of any two entities (i.e. *r* becomes *s* and *s* becomes *r*). An example is C-3 of ribaric (C-3 is *r*) or xylaric acid (C-3 is *s*) or hyoscyamine (C-3 is *r*). The hyphen in pseudo-asymmetric may be omitted.

PSEUDO-CO-OLIGOMER: An irregular oligomer, the molecules of which are derived from only one species of monomer but which display a variety of structural features more appropriate for description in co-oligomer terms.

PSEUDO-COPOLYMER: An irregular polymer, the molecules of which are derived from only one species of monomer but which display a variety of structural features more appropriate for description in copolymer terms.

PSEUDO-INVERSE (GENERALISED INVERSE): In mathematics, any of several generalisations of the inverse of a matrix or a bounded linear operator with closed range on Hilbert space, often denoted A^+ for a given matrix or operator A . It is a linear operator that coincides with the inverse for an invertible operator. The most commonly encountered pseudoinverse is the Moore-Penrose matrix inverse, which is a special case of a general type of pseudoinverse known as a matrix 1-inverse.

PSEUDOPODS (PSEUDOPodium): Temporary cytoplasmic projections of the cell membrane in certain unicellular protists such as amoeba that function especially as an organ of locomotion or in taking up food or other particulate matter.

PSEUDOPRIME (CARMICHAEL NUMBER): An odd composite number ' n ' which satisfies Fermat's little theorem $b^{n-1} \equiv 1 \pmod{n}$. The first ten Carmichael numbers are 561, 1105, 1729, 2465, 2821, 6601, 8911, 10585, 15841, 29341. Only one composite pseudo-prime, 561 is less than 1000.

PSEUDOAROMATIC (ANTIAROMATIC): A cyclic compound that has a ring of atoms containing alternating double and single bonds, yet does not have the characteristic properties of aromatic compounds and is much less stable than similar nonaromatic compounds.

PSEUDOBINARY IONIC COMPOUNDS: Compounds that contain more than two elements but are named like binary compounds.

PSEUDOBULB: A swollen bulb-like part of most orchids that holds the leaves and is attached to the rhizome.

PSEUDOCARP (FALSE FRUIT; SPURIOUS FRUIT; ACCESSORY FRUIT): A fruit-like structure that incorporates tissues that are not derived from the ovary but from some adjacent tissue such as the receptacle and calyx. Examples of pseudocarp are apples, figs and pears.

PSEUDOCODE (P-CODE): A computer programming language that resembles plain English that cannot be compiled or executed, but explains a resolution to a problem.

PSEUDOCOELOM: A closed fluid-containing cavity, lying between the endoderm and the mesoderm that acts as a hydrostatic skeleton to maintain body shape, circulate nutrients and hold the major body organs. They are found in organisms such as nematodes, rotifers, acanthocephalans, kinorhynchans and nematomorphs.

PSEUDOCOPULATION: 1. Attempted copulation by male insects with a flower that serve a reproductive function for one or both participants but do not involve actual sexual union between the individuals. It results in the pollination of a flower and is common in the orchid family, where the flowers of many species resemble a particular species of female bee. 2. Also known as **amplexus**, it is a type of mating in anuran amphibians (frogs and toads) which does not involve the insertion of an intromittent organ (penis) by the male into a female and the eggs are fertilised outside the female's body. The male grasps a female with his front legs as part of the mating process and the eggs are fertilised as they are released from the female's body.

PSEUDOFASCICULATE: Clustered closely, but not really joined into a bundle.

PSEUDOGAMY: A form of parthenogenesis in which the male gamete is required to activate the development of the egg but fertilisation does not occur. Generally, the pollen or sperm is from a member of the same species, although in some cases, for example,

in the fish *Poeciliopsis*, egg development is triggered by the sperm of a closely related species. Pseudogamous plants require pollination and the endosperm undergoes fertilisation even though the diploid embryo develops without it.

PSEUDOGENE: A sequence of nucleotide in DNA that resembles a functional gene but that has lost its protein-coding ability or is otherwise no longer expressed in the cell. Pseudogenes are thought to arise by duplication of an existing gene through unequal crossing-over during meiosis, with accompanying loss of the promoter or other flanking regions required for transcription.

PSEUDOHALOGENS: A group of compounds resembling the halogen elements, in their chemistry. Examples include cyanogens $(CN)_2$, thiocyanogen $(SCN)_2$ and azidodithiocarbonate $(N_3CS_2)_2$, which are symmetrical pseudohalogens and the unsymmetrical pseudohalogens, which include cyanogen halides $(ICN, ClCN, BrCN)$, etc. These anions behave as halogens and the presence of the internal double bonds or triple bonds do not appear to affect their chemical behaviour.

PSEUDOHEART: Any one of a series of contractile blood vessels in annelid worms that pump blood from the dorsal vessel to the ventral vessel.

PSEUDOMACHINE (P-MACHINE): A computer or computer processor created through software and not hardware.

PSEUDOMONOMEROUS: A pistil which appears to be composed of a single carpel, because it has a single functional carpel, although it is actually composed of two or more carpels.

PSEUDOMORPH: A mineral or mineral compound that has replaced another in situ and has retained the appearance and dimensions of the original mineral.

PSEUDOPARENCHYMA: A plant tissue that superficially resembles parenchyma but is made up of interwoven hyphae or filaments that superficially resembles plant tissue. Examples are fungi such as mushrooms and toadstools and the thalli of certain red and brown algae.

PSEUDOPERICYCLIC: A concerted transformation is pseudopericyclic if the primary changes in bonding occur within a cyclic array of atoms at one (or more) of which nonbonding and bonding atomic orbitals interchange roles.

PSEUDOPodium (PSEUDOPods): A finger-shaped part pushed out by a unicellular organism. Two pseudopodia are used to trap food. Pseudopodia extend and contract by the reversible assembly of actin subunits into microfilaments. Filaments near the cell's end interact with myosin which causes contraction. The pseudopodium extends itself until the actin reassembles itself into a network. This is how amoebas move, as well as some cells found in animals, such as white blood cells.

PSEUDOPREGNANCY: A condition resembling pregnancy occurring in some mammals, in which many of the phenomena of pregnancy are present but no foetus develops in the uterus, usually following infertile copulation.

PSEUDORANDOM (PSEUDO-RANDOM): In mathematics, the pseudorandom of a sequence of numbers or any digital data that is generated by the random number generator of a computer by a definite mathematical procedure such as linear congruence method and satisfying one or more statistical tests for randomness. Such numbers are not truly random since their pattern recurs cyclically and they are often required, for statistical purposes, to be reproducible.

PSEUDOSCALER: A scalar quantity that changes sign when the coordinate system is changed to a new system by a reflection in the origin.

PSEUDOSCAPE: A false scape, where the leaves apparently seem to originate from the base, but they are actually not so. It involves the elongation of the leaf-bearing stem in the region between the roots and existing leaves.

PSEUDOSCIENCE: A process of statements, beliefs, or practices that are claimed to be both scientific and factual in one or more ways but which deviate from the scientific method. For example, astrology is generally viewed as a pseudoscience.

PSEUDOSPHERE (PSEUDO-SPHERE): In mathematics, the surface of revolution of a tractrix about its asymptote. This produces a surface, such that each point is constant and negative (a pseudo-spherical surface). Lines on this surface do not conform to the parallel postulate, so that the possibility of constructing the surface provides a relative consistency proof for non-Euclidean geometry.

PSEUDOVERTICILLATE: Appearing to be whorled, but not actually so.

PSI FUNCTION (DIGAMMA FUNCTION): The logarithmic derivative of the gamma function, by the capital Greek letter Γ and

$$\text{defined as } \Psi(z) = \frac{\Gamma'(z)}{\Gamma(z)}.$$

PSI PARTICLES (J-PARTICLES): An elementary particle that has no charge, but has unusually long lifetime.

PSILOCYBIN (PSILOCYBINE; 4-PO-DMT): A hallucinogen, found in certain species of mushroom. The active metabolite is psilocin or 4-HO-DMT (4-hydroxyl-dimethyltryptamine), which is responsible for all of the psychoactive effect of the drug. It is produced by hundreds of species of fungi, including those of the genus *Psilocybe* such as *Psilocybe cubensis*, *Psilocybe semilanceata* and *Psilocybe cyanescens*. They are used both recreationally and traditionally, for spiritual purposes and as entheogens. They can be detected by the Marquis test or the Froede test.

PSYCHIATRY: The branch of medicine concerned with the study of mental illness. It has two major branches: (i) **psychotherapy**, which is the application of psychological techniques to the treatment of mental illnesses, where a physiological origin is unknown or does not exist; (ii) **medical therapy**, where attack is made on either the organic source of the disease or, at least, on its physical or behavioural symptoms.

PSYCHOLOGY: The systematic study of human and animal mental processes and behaviour. The issues studied by psychologists cover a wide range, which includes learning, cognition, intelligence, motivation, emotion, perception, personality and the extent to which individual differences are shaped by genetics or environment. The methods used in psychological research include observation, interviews, psychological testing, laboratory experimentation and statistical analysis.

PSYCHONEUROSIS (NEUROSIS): Any psychological emotional disorders, such as anxiety, depression and phobias with no loss of reality.

PSYCHOPHILY: Pollinated by butterflies. Flowers pollinated by butterflies are usually large and showy, pink or lavender in colour, frequently have a landing area and are scented.

PSYCHOSIS: Any mental illness, whether of neurological or purely psychological origin, which renders the individual incapable of distinguishing reality from unreality or fantasy. If the loss of mental capacity is progressive, the illness is termed a **deteriorative psychosis**.

PSYCHOSOMATIC: A physical symptom or disease thought to arise from emotional or mental factors, such as stress and anxiety. Examples of such diseases include psoriasis, eczema, stomach ulcers, high blood pressure and heart disease.

PSYCHOPATHY: A personality disorder characterised by chronic antisocial behaviour, lack of feelings of guilt, failure to learn from experience, etc.

PSYCHOPHARMACOLOGY: The branch of psychology that is concerned with the effects of drugs on mood and behaviour and their use to treat mental illness.

PSYCHOTHERAPY: The application of the theories and discoveries of psychology to the treatment of mental illness. Psychotherapy does not usually involve physical techniques, such as the use of drugs or surgery.

PSYCHOTROPIC DRUG: Any drug that affects mood, mind and emotions. The term covers a range of drugs including stimulants, antidepressants and tranquillisers. Examples include lithium for treating bipolar disorder and cocaine.

PSYCHROMETER: An instrument with wet- and dry-bulb thermometers, for measuring the water vapour content or relative humidity in the atmosphere. It consists of two identical mercury or electrical thermometers, with one of them having a wet bulb and the other a dry bulb. The wet bulb thermometer has a wet cotton or linen wick wrapping around its bulb. Evaporating water from the wick absorbs heat from the thermometer bulb, causing the temperature reading to be lower than that of the dry bulb. The difference between the dry bulb and wet bulb temperatures gives the Wet Bulb Depression and the relative humidity can now be read from the relative humidity table.

PSYCHROTROPH: An organism that is able to grow at temperatures of between 0 °C degrees and 30 °C. These organisms are major contributors in the spoilage of refrigerated foods.

PSYCHROPHILIC (CRYOPHILES): Describing an organism that lives and grows optimally at relatively low temperatures, usually below -15 °C (258 K) and cannot grow above 10 °C (283 K). Psychrophiles which consist mainly of bacteria, algae and fungi, are present in alpine and arctic soils, high-latitude and deep ocean waters, polar ice, glaciers and snowfields. For example, the ice in Polar Regions often contains dense masses of algae. Such organisms cannot survive warmth or heat.

PTERIDOPHYTA: A class of flowerless plants in the plant kingdom that includes ferns, horsetails, club mosses, quillworts and other like plants. Pteridophytes possess true stems, leaves and roots and have a vascular system.

PTERIDOSPERMALES (SEED FERNS): An extinct order of the gymnospermae that flourished in the Carboniferous and Permian periods. They possessed characteristics of both the ferns and the seed plants.

PTEROCARPOUS: Having winged fruits.

PTEROCAULOUS: Having winged stems.

PTEROPSIDA: A subdivision of the tracheophyta containing the ferns and seed plants, gymnospermae and angiospermae characterised by parenchymatous leaf gaps in the stele and leaves, which are thought to have originated in the distant past as branched stem systems.

PTEROSAURIA: An extinct order of flying reptiles, the pterodactyls that lived in the Mesozoic Era. Pterodactyls had beaked jaws and an elongated fourth finger that supported a membranous wing. They had long jointed tails, large, lightly built skull with large bird-like brains, no feathers and could probably only fly by soaring.

PTEROSPERMOUS: Having winged seeds.

PTEROSTIGMA: A small coloured spot near the leading edge on the wings of some species of insect. Its purpose is to aid gliding.

PTERYGOPOUS: Having winged peduncles.

PTERYGOTA: A subclass of insects composed of two sections: Paleoptera, which includes the mayflies, dragonflies and damselflies; and Neoptera. They typically possess wings, although some are secondarily wingless.

PTOLEMAIC ASTRONOMY: The system of astronomy originally proposed by Apollonius of Perga in the third century BC that the Earth was the centre of the universe and other objects orbit around it, which was popular in ancient Greece. In addition to this motion the orbiting bodies also described epicycles, small circles about points on

the deferent. The system gave moderately good predictions, but was completely replaced by the heliocentric astronomy of Copernicus in the 16th century.

PTOMAIN: Any of a group of toxic chemical substances (alkaloids) produced by decarboxylation of amino acids during putrefaction of proteins.

PTYALIN (SALIVA AMYLASE): An enzyme which converts starch to maltose. It is found in saliva (ptyalin) and pancreatic juice (amylase).

PUBERTY: A stage in human development when the individual becomes sexually mature characterised by the maturing of the genital organs, development of secondary sex characteristics and, in girls, onset of menstruation. It may occur at about age 12 in girls and 14 in boys.

PUBERULENCE (PUBERULOUS): Having fine and minutely short hairs, as seen in *Solidago puberula*.

PUBES: The lowest part of the front of the human trunk, especially the region surrounding the external genitalia region. In mammals and many reptiles, the pubes are united at a slightly movable joint.

PUBESCENT: 1. Attaining puberty. 2. Covered with short and soft hairs.

PUBIC: Located near or relating to the pubis or the pubes in the lower part of the abdomen, for example, pubic hair or pubic bone.

PUBIS (PUBIC BONE): One of a pair of bones in humans joined at the front of the pelvis that forms the lower section of the hipbone. In infants, this is a separate bone; however, it later fuses with the ischium and the ilium to form the hipbone.

PUBLIC DOMAIN: 1. The free use of work or object without any restrictions. This may be because of the lack of a copyright or because the work is no longer owned. 2. **Public domain software** refers to software that can be freely distributed and used, either because the company has released it to be public domain or because the company is no longer in business. Public domain software may or may not carry a copyright.

PUBLIC KEY: In public-key cryptography, a string of bits that is associated with a particular person and that may be used to decrypt messages from that person or to encrypt messages to him/her.

PUBLIC KEY INFRASTRUCTURE (PKI): Core framework required to provide public key cryptography and digital signature services for electronic commerce transactions via the Internet. A PKI provides confidentiality to keep information private, integrity to prove it has not been altered, authentication to prove the identity of a person or a computer application and non-repudiation to prevent information being disowned.

PUBLIC-KEY CRYPTOGRAPHY (ASYMMETRIC CRYPTOGRAPHY): A cryptography in which a pair of keys is used to encrypt and decrypt a message so that it arrives securely. Initially, a network user receives a public and private key pair from a certificate authority. Any other user who wants to send an encrypted message can get the intended recipient's public key from a public directory. They use this key to encrypt the message and they send it to the recipient. When the recipient gets the message, they decrypt it with their private key, which no one else should have access to.

PUDDLE CLAY: Clay with sand or gravel that has had water added and mixed thoroughly so that it becomes air tight. It is used in building and maintaining canals or reservoirs.

PUFFER FISH: Any of about 120 species of fishes belonging to the family Tetraodontidae that mostly live in tropical seas, but also in fresh or brackish waters. They are characterised by a rounded body that can be inflated as a defensive strategy when the animal is threatened by predators. They have four large teeth fused into the upper and lower jaws with a vertical gap in front used for crushing the shells of

crustaceans and molluscs, their natural prey. Examples of pufferfish are balloonfish, blowfish, bubblefish, globefish, swellfish, toadfish, toadies, honey toads and sea squab.

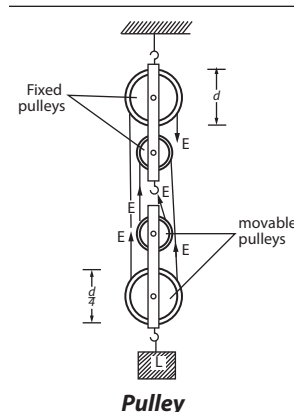
PUFFING: An irreversible expansion of some carbon artifacts during graphitisation heat treatment between 1650 and 2700 K. Puffing is caused by the release of heteroatoms, for example, sulfur atoms, from the coke in association with specific microstructural rearrangements.

PUFFING INHIBITOR: Metals or metal compounds with a high chemical affinity for the heteroatoms in the carbons. They are distributed as fine particles within the carbon materials to be graphitised. Iron and iron compounds are most frequently used as puffing inhibitors when puffing is related to sulfur.

PULL TECHNOLOGY: Transmission of information from an internal network or a cloud-based service to Internet users in which the user requests for the data. An example is downloading a Web page. The reverse in which information is transmitted automatically from the server without a user requesting for it is called **push technology**.

PULLET: A young female chicken, usually less than one year old.

PULLEY: A wheel with a groove on its rim over which a flexible rope, cord, cable, chain or belt can move to change the direction of a pulling force. An effort applied at one end of the rope moves a load attached to the other end. One or more independently rotating pulleys can be used to gain mechanical advantage, especially for lifting weights or to change the direction or size of the applied force. The pulley is one of the five simple machines.



PULLORUM DISEASE: A highly infectious disease of young poultry caused by the bacterium *Salmonella pullorum*, marked by dyspnoea and diarrhoea, a reduction in egg yield and reduction in fertility of the eggs, etc.

PULLULAN: A type of alpha glucan, which is a water-soluble polysaccharide, composed of glucose units that are polymerised in such a way as to make it viscous and impermeable to oxygen. Three glucose units in maltotriose are connected by an α -1,4 glycosidic bond, whereas consecutive maltotriose units are connected to each other by an α -1,6 glycosidic bond. It is derived from starch by the fungus *Aureobasidium pullulans*. It is used to manufacture edible films that are used in various breath freshener or oral hygiene and in adhesives, food packaging and moulded articles.

PULMONATE: 1. An order of gastropod molluscs belonging to the Pulmonata subclass and Sorbeconcha clade, characterised by breathing by means of a lunglike sac. Pulmonates include land snails, the slugs and certain aquatic snails. 2. Having lungs or lung-like organs.

PULMONARY: A term describing organs, tissues and parts of the body related to the lungs and breathing.

PULMONARY ARTERY: The artery that conveys deoxygenated blood from the right ventricle of the heart to the lungs. It begins at the base of the right ventricle. It is short and wide, about 5 cm in length and 3 cm in diameter. It then branches into two pulmonary arteries, delivering de-oxygenated blood to the left and right lungs.

PULMONARY CIRCULATON: The part of cardiovascular circulatory system of birds and mammals that transport deoxygenated blood from the right side of the heart to the lungs and returns oxygenated blood to the left side of the heart.

PULMONARY EMBOLISM: An obstruction or sudden blockage in the pulmonary artery or one of its branches, usually due to a blood clot that travels to the lung from the leg. A clot that forms in one part of the body and travels in the bloodstream to another part of the body is called an **embolus**.

PULMONARY TRUNK: One of the major arteries leaving the heart; which is about 5 cm long and 3 cm in diameter. The pulmonary trunk lies within the fibrous pericardium, enclosed, with the ascending aorta, in a sheath of the serous pericardium.

PULMONARY VALVE: A valve in the heart lying between the right ventricle and the pulmonary artery. It is a semilunar valve that prevents blood returning to the ventricle from the pulmonary vein.

PULMONARY VEIN: The large blood vessel of the circulatory system that conveys oxygenated blood from the lungs to the left atrium of the heart. It is one of the four veins (two from each lung) that carry oxygen-rich blood from the lungs to the left side of the heart. They are right inferior/superior and left inferior/superior. The left superior pulmonary vein returns blood from the left superior lobe of the lung to the left atrium. The right superior pulmonary vein returns blood from the superior and middle lobes of the right lung to the left atrium. The left inferior pulmonary vein returns blood from the inferior lobe of the left lung to the left atrium. The right inferior pulmonary vein returns blood from the inferior lobe of the right lung to the left atrium.

PULP: 1. The soft inside of a fruit or vegetable. 2. A thick soft substance made by crushing wood, which can be made into paper. 3. The soft tissue that makes up the interior of the tooth. It contains blood vessels and nerves.

PULP CAVITY: The space within the central part of a tooth containing the dermal pulp. It is made up of the pulp chamber and a root canal.

PULPWOOD: Timber from some trees such as spruce, aspen, hemlock, some species of eucalyptus, acacia or pine, used in making paper and also used as the raw material for some wood products.

PULPY KIDNEY DISEASE (ENTEROTOXEMIA; OVEREATING DISEASE): A severe disease of sheep and goats caused by type D and C strains of bacteria called *Clostridium perfringens*. These bacteria are normally found in low numbers in the gastrointestinal tract of all sheep and goats but suddenly release toxins in the digestive tract of the animal. Symptoms include dysentery, not feeding, unable to stand and lying on the sides, frothing at the mouth, etc. It occurs in older lambs and can be fatal. Enterotoxemia can progress so quickly that animals may be found dead within hours with no previous signs of disease.

PULSAR (PULSATING STAR): A neutron star, which is a highly magnetised rotating celestial body producing intense short bursts of visible radiation, radio waves and x rays, emitted in brief (0.03 second to 4^{+0} seconds) regular pulses. Individual pulses can vary widely in intensity. More than 1300 pulsars have been found since its discovery in 2001.

PULSAR WIND NEBULA: A cloud created from gas and particles blown off a neutron star by pulsar winds generated by its central pulsar. They have been found to be X-ray emitters and are possibly gamma ray sources.

PULSATANCE: Angular velocity of frequency that may be associated with a periodic motion, a process, a wave system, etc., expressed in radians per second. It is equal to 2π times frequency in hertz.

PULSATOR: The part of a milking machine which controls the cyclic pressure changes necessary for the suction action and release of the milk from the udder.

PULSE: 1. Impulse transmitted by the alternate expansion and contraction of artery walls as the heart circulates blood in the body.

Artery walls are elastic and become distended by increased blood volume during **systole** or contraction of the heart. During **diastole** or relaxation of the heart, blood volume in the arteries decreases and the walls contract, propelling the blood farther along the arterial pathway. In man, it can be felt easily to the skin surface such as at the wrist. The normal rate is 70 to 90 pulsations per minute in adults and 90 to 120 in children. Various diseases may be indicated by changes in the rate, rhythm and force of the pulse. 2. Members of the Fabaceae (Leguminosae) family which includes the pea or legume and also seeds of various plants such as beans, lentil and peas that grow in pods and used as food. The leaves are usually compound and the blossoms may have an irregular butterfly-like shape. The flowers have 10 stamens and the corolla and the calyx are formed of 5 petals and 5 sepals, respectively. Some species have thorny branches. Pulses are a good source of protein.

PULSE AMPLITUDE ANALYSER (HEIGHT ANALYSER): A sub-assembly for determining the distribution function of a set of pulses in terms of their amplitude.

PULSE AMPLITUDE SELECTOR (SINGLE CHANNEL PULSE HEIGHT ANALYSER): A circuit which gives an output pulse for each input pulse whose amplitude lies within a chosen interval.

PULSE CHASE: A method of following metabolic processes using radioisotopes. A sample organism, cell or cellular organelle is exposed to a radioactively labelled compound followed by replacing the labelled compound with an excess of the same compound, which is unlabelled. The sample is then examined at various later times to determine the fate of radioactivity incorporated during the pulse. The labelled compound is called **pulse** and the unlabelled compound is called **chase** or **cold chase**.

PULSE JET ENGINE: A type of jet engine in which combustion occurs in pulses. The two main types of pulsejet engines are valved and the valveless engines, both of which use resonant combustion that harness the expanding combustion products to form a pulsating exhaust jet, which produces thrust intermittently. Pulse jet engines can be made with few or no moving parts and are capable of running statically.

PULSE MODULATION: A technique for transmitting information by means of a series of electrical pulses, with the duration, amplitude or frequency of the pulses modified to carry the information. Varying the amplitude, polarity, presence or absence of discrete pulses, duration or occurrence in time of the pulses gives rise to the four basic forms of pulse modulation: pulse-amplitude modulation (PAM), pulse-code modulation (PCM), pulse-width modulation (PWM, also known as pulse-duration modulation (PDM) and pulse-position modulation (PPM).

PULSE OXIMETRY: A laser technique for measuring the oxygen content of the blood by shining red or infrared light through a thin part of a patient's anatomy; usually used to evaluate various medical conditions that affect the function of the heart and lungs. The instrument used is called **oximeter**, a photoelectric device.

PULSE REACTOR: In a pulse reactor, a carrier gas, which may be inert or possibly one of the reactants, flows over the catalyst and small amounts of the other reactant or reactants are injected into the carrier gas at intervals. A pulse reactor is useful for exploratory work but kinetic results apply to a transient rather than to the steady state conditions of the catalyst.

PULVERACEOUS (PULVERULENT): Having a finely dusted or powdered surface.

PULVINATE (PULVINIFORM): 1. Having the form of a mat or cushion. 2. Having a swelling or enlargement at the base of a petiole. This is usually said of a leafstalk.

PULVINUS: 1. A thickening at the base of a leaf or leaflet consisting of a core of vascular tissue within a flexible, bulky cylinder of

thin-walled parenchyma cells that facilitates growth-independent movement. The movement is caused by changes in turgor pressure leading to a contraction or expansion of the parenchyma tissue. The response is initiated when sucrose is unloaded from the phloem into the apoplast. The increased sugar concentration in the apoplast decreases the water potential and triggers the efflux of potassium ions from the surrounding cells. This is followed by an efflux of water resulting in a sudden change of turgor pressure in the cells of the pulvinus. Examples for pulvinar movements include the night closure movement of legume leaves and the touch response of the sensitive plant or touch-me-not of the *Mimosa pudica*. 2. A thickened region in the stem or leaf sheath of grasses, often containing an intercalary meristem.

PUMICE: Light porous volcanic rock produced by igneous rock that cooled so rapidly there was no time for it to crystallise. When it solidified, the vapours dissolved in it were suddenly released and the whole mass swelled up into a froth that immediately consolidated. It is used as an abrasive in cleaning, polishing and scouring compounds. It is also used in precast masonry units, poured concrete, insulation and acoustic tile and plaster.

PUMP: 1. A machine that draws a fluid into itself through an entrance port and forces the fluid out through an exhaust port. It is operated by a piston or similar mechanism and can be used to raise, compress or transfer liquids or gases. An example is the **centrifugal pump**, which moves a liquid such as water by accelerating it radially outward in an impeller to a surrounding volute casing. 2. Proteins in plasma membrane that actively move substances such as ions or molecules across a cell membrane. 3. A heap of vegetables, usually potatoes, covered with earth and mulch and sometimes stored in a shed.

PUMPED STORAGE HYDROELECTRICITY: A type of hydroelectric power generation that uses surplus electricity available during off-peak hours to pump water from a lower elevation reservoir into a higher level reservoir. During the period of high demand (on-peak period), the water from the high level reservoir falls by gravity to rotate turbines, which drive the generators and convert the turbines' mechanical energy into electricity. Pumped storage is the largest-capacity form of grid energy storage now available.

PUMPKIN: Several members of *Curcubita* especially *Cucurbita pepo*, *Cucurbita mixta*, *Cucurbita maxima* and *Cucurbita moschata* consisting of vines which bear large, edible, globular fruits (also called pumpkin) with a firm yellowish-orange rind.

PUNCTATE: Characterised by pits, translucent sunken glands or coloured dots. **Punctulate** means having minute punctate. **Punctiform** means reduced to a point.

PUNCTUAL: In mathematics, consisting of or confined to a single point.

PUNCTUATED EQUILIBRIUM: Theory that evolutionary development occurs primarily through short bursts of intense speciation, followed by lengthy periods of stasis or equilibrium, showing little evolutionary change.

PUNCTURED NEIGHBOURHOOD (DELETED NEIGHBOURHOOD): In mathematics, it is a neighbourhood of a point, without the point itself; that is the point has been deleted. It is usually written with a prime to indicate deletion, as $N'(\delta, a)$ in a metric space or $U'(a)$ more generally in a topological space.

PUNGENT: 1. A powerful, sharp or bitter taste or smell as in mustard. 2. Describing a plant or animal part that ends in a sharp point.

PUNICEOUS: Having a bright red or purple colour; crimson.

PUNNETT SQUARE: A graphic technique named after Reginald C. Punnett, who devised it. It is used in genetics for determining the likely outcome of a particular genetic cross or breeding experiment to determine the probability of an offspring having a particular genotype. It summarises every possible combination of one maternal allele with one paternal allele for each gene being studied in the cross.

PUPA: The stage of an insect between the larva and the adult of an insect with complete metamorphosis, during which the insect is in a cocoon or case, stops feeding and undergoes internal changes. **Pupate** is to change from the larval to the pupal stage.

PUPIL: A circular hole or opening located in the centre of the iris of the eye through which light enters the retina. It regulates the amount of light passing to the retina by becoming larger or smaller and appears black because most of the light entering the pupil is absorbed by the tissues inside the eye.

PUPILLARY REFLEX (WESTPHAL-PILCZ REFLEX; WESTPHAL'S PUPILLARY REFLEX): A contraction of the pupil in response to stimulation of the retina by light. An increase in light intensity causes both pupils to contract even if only one retinal is directly stimulated, known as **consensual reflex**. A decrease in light intensity causes dilation of the pupils.

PURE: 1. Without harmful substances, clean, unadulterated. 2. In chemistry, a **pure substance** is a substance that cannot be separated by physical means such as filtration, evaporation, distillation or chromatography. Chemical substances commonly encountered in pure form are oxygen, hydrogen, carbon dioxide, diamond (carbon), gold, table salt (sodium chloride) and refined sugar (sucrose). Each pure substance has its own set of unique set chemical and physical properties which help to identify it. 3. In genetics, it is any organism produced by self-fertilisation or continual inbreeding.

PURE BLOOD: An organism having certain biological traits which are passed on to all subsequent generations when bred with another true breeding organism for the same traits.

PURE COVALENT BOND: A covalent bond between atoms of the same element, for example, H-H, Cl-Cl or H:Cl, Cl:Cl.

PURE CULTURE: A nutrient medium containing the growth of a single strain of an organism free from other living species or strains.

PURE GEOMETRY (SYNTHETIC GEOMETRY): The study of geometry, usually projective geometry, by the synthetic method without the use of coordinates.

PURE LINE (TRUE BREEDING ORGANISM): A population of plants or animals having a particular feature that has been retained or remained unchanged through many generations.

PURE LINE SELECTION: A method in which new variety is developed by selection of single best plant progeny among traditional varieties or land races. Pure line selection generally involves three distinct steps: (i) numerous superior appearing plants are selected from a genetically variable population; (ii) progenies of the individual plant selections are grown and evaluated by simple observation, frequently over a period of several years; and (iii) when selection can no longer be made on the basis of observation alone, extensive trials are undertaken, involving careful measurements to determine whether the remaining selections are superior in yielding ability and other aspects of performance. Any progeny superior to an existing variety is then released as a new "pure-line" variety.

PURE MATHEMATICS: The intrinsic study of mathematical systems and structures, with no consideration given as to the utility of the results for practical purposes. Some of the fields of studies include analysis, algebra and geometry.

PURE SUBSTANCE: A material that has a homogeneous composition and has consistent properties throughout. It is made of only one type of atom or only one type of molecular or formula units. Examples are gold, diamond, tin, sulfur, water, baking soda, table salt, and sucrose (sugar). Impure substance are much more difficult to break down into their pure parts. This can be done using complex chemical methods. One such method, chromatography, if used can only produce one substance at the end of the process. If a substance is impure then several substances can be seen at the end of the process.

PURE SURD (ENTIRE SURD): A surd in which all terms are irrational numbers. In other words, it is a surd having no rational factor except unity. Examples are $\sqrt{3}$, $\sqrt{7}$, $\sqrt{10}$ and $\sqrt[3]{50}$.

PURE-BREEDING: A term describing an inherited trait controlled by a homozygous pair of alleles, which as a result of successive self-crosses reappears generation after generation.

PUREBRED (PUREBREDS): Referring to an animal which is a cultivated variety or *cultivar* of an animal species, achieved through the process of selective breeding or an organism bred from members of a recognised breed, strain or kind without admixture of other blood over many generations. **Thoroughbred** refers exclusively to a specific breed of horse.

PURELY ATOMIC: In mathematics, of a measure, such that there is a countable set in the support of the measure with null complement. The measure is thus a countable combination of point masses, each based on an atom.

PURGATIVE: A strong laxative or a substance, especially a medicine that causes the bowels to empty. It includes psyllium, saline laxatives such as Epsom salts or milk of magnesia, glycerin, lubricants such as mineral oil and some vegetable oils, etc.

PURIFICATION: The removal of impure elements from something. For example, potable water must be purified from impurities before it is safe to drink.

PURINE: Any of a class of heterocyclic compounds with the chemical formula $C_5H_4N_4$, molar mass $120.11 \text{ g mol}^{-1}$ and melting point 214°C (487 K). It is sparingly soluble in water, with a two-ring structure composed of carbon and nitrogen atoms. Important biological derivatives include adenine and guanine, which occur in nucleotides and nucleic acids (DNA and RNA). Caffeine can also be derived from purine.

PURKINJE CELL (PURKINJE NEURONS): A large neuron with many branching extensions that is found in the cortex of the cerebellum of the brain. Purkinje cells send inhibitory projections to the deep cerebellar nuclei and constitute the sole output of all motor coordination in the cerebellar cortex. It was named after the discoverer, Czech anatomist, Johannes Purkinje.

PURKINJE EFFECT: The tendency of the peak sensitivity of the human eye to shift toward the blue end of the spectrum at low illumination levels. It can affect visual estimates of variable stars when using comparison stars of different colours, especially if one of the stars is red. The effect is named after its discoverer, Czech anatomist, Johannes Purkinje.

PURKINJE FIBRES: Modified cardiac muscle fibres located in the inner ventricular walls of the heart, just beneath the endocardium. They are specialised myocardial fibres that conduct an electrical stimulus or impulse from the right and left branches of an atrioventricular bundle to all parts of the ventricles of the heart that enables the heart to contract in a coordinated fashion.

PURPURESCENT: Becoming purplish.

PUS: An off-white or yellow liquid consisting of inflammatory exudate, the debris of leukocytes (white blood cells) and bacteria resulting from local inflammation, especially abscesses.

PUSH TECHNOLOGY: An automatic transmission of information from a server to Internet users without a request being made. The reverse in which a user requests for the data is called **pull technology**.

PUSH-BUTTON: In computing, in a dialogue box or toolbar, a square, oval or oblong button which presents the user with an option. By clicking or "pushing" on the button, the user opts to initiate an action such as "Save the file", "Don't Save the file" or "Cancel". Push-buttons can also be pressed using the keyboard, for example, one can press the return key instead of clicking the 'OK' button in a dialogue box.

PUSTULE: Small, blister-like projections. **Pistular** or **pustulose** means having small blisters or pustules, at the base of a hair.

PUT TO: To bring a female animal to be impregnated by a male animal.

PUTAMEN: 1. A paired structure which together forms the dorsal striatum; one of the nuclei that make up the basal ganglia. 2. The hard and stony endocarp of some fruits such as the peach.

PUTREFACTION: Decomposition of organic matter by microorganisms, usually resulting in amines such as putrescine and cadaverine, which have a putrid odour. The rate at which purification occurs depends on factors such as weather, exposure and location.

PUTTY POWDER: Impure tin(IV) oxide (SnO_2) or tin and lead mixed in various proportions used in glass and marble works, for polishing glass, etc.

PVC: Abbreviation for **polyvinyl chloride**, also known as **polychloroethene**, is a polymer of vinyl chloride with the chemical formula $(\text{H}_2\text{CCHCl})_x$. It is tasteless, odourless and insoluble in most organic solvents and a member of the family of vinyl resins. It is made by polymerising vinyl chloride and used for drain pipes, floor tiles, audio discs, upholstery, shoes and handbags.

PYCNOMETER (PYKNOMETER): An instrument for measuring the density or specific gravity of materials.

PYGMY BEETLE: A beetle pest, *Atomaria linearis*, that affects sugar beet.

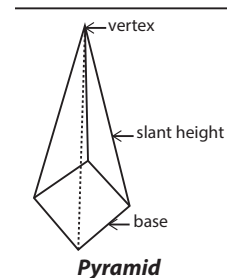
PYKNOSIS (KARYOPYKNOSIS): A nucleus of a damaged cell that has decreased in volume and become denser due to some irreversible degree of condensation of chromatin in the nucleus of a cell undergoing necrosis or apoptosis.

PYLORIC SPHINCTER (PYLORUS; SPHINCTER MUSCLE OF PYLORUS): A contractile organ which is a ring of muscles around the opening of the stomach into the duodenum which is divided into two sections: (i) the **pyloric antrum**, which connects to the body of the stomach; and (ii) the **pyloric canal**, which connects to the duodenum. It functions as a valve to control the volume of food that passes from the stomach to the small intestine and also closes to prevent large particles of food from leaving the stomach, where they are further digested chemically into smaller particles.

PYORRHEA: A flow of pus from the margins of the gums and teeth. It causes loosening of the teeth and halitosis.

PYRAMID: 1. A three dimensional solid with a polygonal base and triangular faces meeting at a common vertex.

There are many types of pyramids and they are named after the shape of their bases. Examples include the following: (i) A rectangular pyramid is solid having a rectangle for its base and four triangular faces. The pyramids in Egypt are rectangular pyramids. (ii) A triangular pyramid, which is a pyramid having a triangular base, 3 side triangular faces, 4 vertices and 6 edges. The tetrahedron is a triangular pyramid having congruent equilateral triangles for each of its faces.



(iii) The square pyramid has a square base and four triangular sides that connect at the apex. The diagram shows a square pyramid. (iv) The pentagonal pyramid has a pentagon base (five sides and five corners), 6 faces with 5 isosceles triangle side faces, 6 vertices and 10 edges. 2. A pointed or cone-shaped body part located in the brain, consisting of collections of motor nerve fibres from the spinal cord to the motor cells in the ventral horns. 3. A crystalline figure in which three or more nonparallel faces intersect all three axes of the crystal. 4. A huge stone tomb of ancient Egypt in which the Pharaohs were buried. It has a square base and triangular walls that slope to meet in a point at the top.

PYRAMID OF BIOMASS (PYRAMID OF MASS): A graphical representation quantifying the amount of biomass in each trophic level of a food chain.

PYRAMID OF ENERGY: A graphical representation of the amount of the incoming solar energy transferred into the trophic levels in a particular habitat, measured in kilojoules or watts per square metre per year (kJ/m^2 or W/m^2). A pyramid of energy indicates how much energy is lost at each trophic level through respiration, photosynthesis, etc.

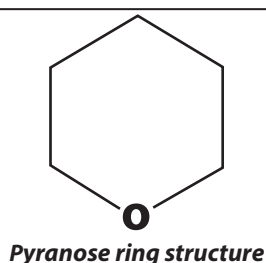
PYRAMID OF NUMBERS: A graphical representation in the form of a pyramid that relates the number of organisms in each of the trophic level in the food chain, using the feeding relationship as a basis for the numerical analysis of an ecosystem. The pyramidal shape indicates that the number of organisms or species is largest at the bottom and narrows towards the apex. The Pyramid of Numbers can be used to determine how the population of a particular species can affect another. It can also serve as a basis for the quantitative analysis of an ecosystem.

PYRAMIDAL CELL (PYRAMIDAL NEURONS): A type of neurone found in the brain in areas such as cerebral cortex, the hippocampus and in the amygdala, which are the primary excitation units of the mammalian prefrontal cortex and the corticospinal tract. Pyramidal cells have a pyramid-shaped cell body and its dendrites extend from both the body and the axon.

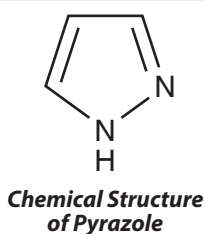
PYRAMIDAL INVERSION: A polytopal rearrangement in which the change in bond directions to a three-coordinate central atom having a pyramidal arrangement of bonds (tripodal arrangement) causes the central atom (apex of the pyramid) to appear to move to an equivalent position on the other side of the base of the pyramid. If the three ligands to the central atom are different pyramidal inversion interconverts enantiomers.

PYRAMIDAL NUMBER: The number of dots that may be arranged in a pyramid with a regular polygon as base.

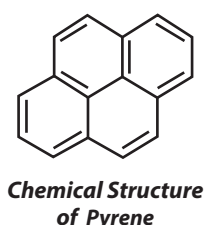
PYRANOSE: Any of a group of carbohydrates whose molecules have a six-membered heterocyclic ring consisting of five carbon atoms, ten hydrogen atoms and one oxygen atom.



PYRAZOLE: Crystalline heterocyclic aromatic organic compounds with the chemical formula $\text{C}_3\text{N}_2\text{H}_4$, melting point $66-70^\circ\text{C}$ ($339-343^\circ\text{K}$) and boiling point $168-188^\circ\text{C}$ ($441-461^\circ\text{K}$). It is composed of a 5-membered ring structure, whose ring contains three carbon atoms and two nitrogen atoms with double bonds and classified as alkaloids. They do not occur naturally and are synthetic compounds. Pyrazoles are produced synthetically through the reaction of α,β -unsaturated aldehydes with hydrazine and subsequent dehydrogenation. Derivatives of pyrazoles are used in medicine as analgesic, anti-inflammatory, antipyretic, antiarrhythmic, tranquilising, muscle relaxing, psychoanaleptic, anticonvulsant, monoamineoxidase inhibiting, antidiabetic and antibacterial agents.



PYRENE: 1. A solid polynuclear aromatic hydrocarbon extracted from coal tar. Its chemical formula is $\text{C}_{16}\text{H}_{10}$. It has four fused benzene rings and believed to be carcinogenic. Pyrene forms during incomplete combustion of organic compounds. 2. The stone or pit of certain fruits such as cherry.



PYRENOID: A spherical protein body found in the chloroplasts. It is responsible for carbon

dioxide fixation of algae and the hornworts. It is associated with the storage of starch and contains high levels of ribulose-1,5-bisphosphate carboxylase/oxygenase (RubisCO).

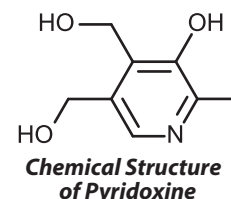
PYRETHRUM: Organic pesticide, derived from the flowers of plant species of the genus *Tanacetum*, native to South Asia. They are not toxic to plants, animals and humans and are used widely in the household as livestock sprays as well as in gardening. It is also used in insecticides.

PYREX: A borosilicate glass used in chemical and industrial apparatus and in ovenware. It is inert, a good insulator and heat-resistant (having a low coefficient of expansion and a high softening temperature).

PYREXIA (FEVER; CONTROLLED HYPERTHERMIA): A condition characterised by an elevation of temperature above the normal range of 36.5 to 37.5°C as measured in the mouth. It is a symptom of many disorders, such as infection by a virus or a bacterium and it is not itself a disease. The first signs may be chilly sensations, with associated periods of flushed or warm feelings. The temperature may rise slowly or rapidly and may fluctuate. A rise in temperature may be accompanied by shaking chills. A falling temperature may bring on heavy sweating.

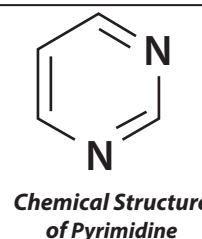
PYRIDINE: A colourless heterocyclic aromatic base, with the chemical formula $\text{C}_5\text{H}_5\text{N}$. It has an unpleasant smell and found in bone oil and coal tar. It is produced from acetaldehyde (ethanal) and ammonia. Pyridine is used as a solvent and to denature grain alcohol (ethanol). Important derivatives are pyridoxine and niacin.

PYRIDOXINE: One of the three components of vitamin B_6 with the chemical formula $\text{C}_8\text{H}_{11}\text{NO}_3$. The others are **pyridoxal** and **pyridoxamine**. It is derived from pyrimidine and found in yeast, liver, fish and cereals. Its functions in the body include stabilising sodium and potassium and promoting red blood cell production. It also decreases the formation of homocysteine in the cardiovascular health system; and helps stabilise hormonal changes in women and aid the immune system. Lack of pyridoxine may cause anaemia, nerve damage, seizures, skin problems and sores in the mouth.



PYRIFORM: Shaped like a pear.

PYRIMIDINE: Any organic nitrogenous base with the chemical formula $\text{C}_4\text{H}_4\text{N}_2$, having a ring structure of four carbon and two nitrogen atoms. It is sparingly soluble in water and gives rise to a group of biologically important derivatives such as cytosine, thymine, uracil and vitamin B_1 (thiamine).



PYRITE (IRON PYRITE; FOOL'S GOLD): Iron sulfide with the chemical formula FeS_2 . It is a pale yellow-gold or brassy metallic lustre, resembling minerals such as arsenopyrite, chalcopyrite and pyrrhotite, in physical outlook and appearance. It is an ore of iron and a source of iron and sulfur. It occurs naturally in some types of crystals. One type, auriferous pyrite resembles gold so much that it is exploited commercially as gold.

PYRO-: 1. A prefix used to designate fire, heat or high temperature, in the formation of compound words such as pyrolysis. 2. A prefix designating compounds formed by heating a compound, usually with the elimination of water, carbon dioxide or other simple molecule, for example, pyroglutamic acid from glutamic acid.

PYROCLASTIC: Composed chiefly of rock fragments or particles such as pumice, obsidian or volcanic ash ejected during a volcanic eruption.

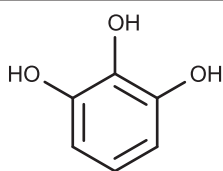
PYROCLASTIC FLOW (PYROCLASTIC DENSITY CURRENT):

A term describing fragments resulting from some volcanic eruptions, which travel away at great speeds from the site of eruption as currents of hot gas and rock known as **tephra**. The gas can reach temperatures of about 1,000°C (1,273 K). It solidifies into sizes from fine ash to large boulders, including rocks that are formed by consolidation of such materials.

PYROELECTRIC DETECTOR: A detector, based on the temperature dependence of ferroelectricity in some crystals. It produces an electrical signal proportional to the energy flux on the collector surface.

PYROELECTRICITY: The property of certain crystals such as tourmaline, lithium sulfate monohydrate, cane sugar and ferroelectric barium titanate which lack a centre of symmetry and also have polar directions to acquire opposite electrical charges on opposite faces when heated.

PYROGALLOL: A solid poisonous, white, crystalline, aromatic compound with a bitter taste with the chemical formula $C_6H_6O_3$, melting point 130 °C (400 K) and boiling point 309 °C (513 K). It dissolves in concentrated solution of sodium hydroxide. It is an active reducing agent and used in photographic developing and in the manufacture of some dyes.



Chemical Structure of Pyrogallol

PYROGEN: Any substance, typically produced by a bacterium that causes a rise in temperature in a human or animal body.

PYROLUSITE: A soft, grey-black oxide mineral composed of manganese(IV) oxide (MnO_2). It is a secondary mineral of aqueous origin crystallising in the tetragonal system. It is of widespread occurrence and the chief ore of manganese.

PYROLYSIS: The thermochemical decomposition of a substance by heating it to a high temperature under pressure and at various operating temperatures in the absence of oxygen, although a small amount of oxygen is present. The process is used in dry distillation, destructive distillation or cracking to produce charcoal, activated carbon and methanol, to convert ethylene dichloride into vinyl chloride to make PVC, to produce coke from coal, to convert biomass into syngas, to turn waste into safely disposable substances, baking, frying, grilling, etc.

PYROLYSIS-GAS CHROMATOGRAPHY: A version of reaction chromatography in which a sample is thermally decomposed to simpler fragments before entering the column.

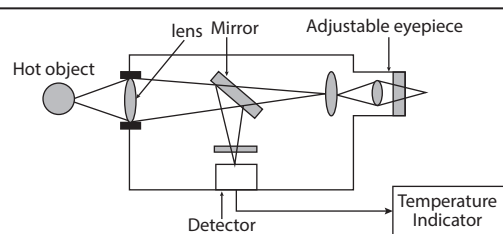
PYROLYTIC CARBON: A carbon material deposited from gaseous hydrocarbon compounds on suitable underlying substrates (carbon materials, metals, ceramics) at temperatures ranging from 1000 to 2500 K (chemical vapour deposition).

PYROLYTIC GRAPHITE: A graphite material with a high degree of preferred crystallographic orientation of the *c*-axes perpendicular to the surface of the substrate, obtained by graphitisation heat treatment of pyrolytic carbon or by chemical vapour deposition at temperatures above 2500 K.

PYROMETALLURGY: Metallurgy involved in the winning and refining of metals where heat is used, as in roasting and smelting.

PYROMETER: A device used for measuring high temperatures. Platinum resistance thermometers and pyrometers operating on the principle of the thermocouple have the disadvantage that they must be in contact with the hot body, but optical and radiation pyrometers can be used at a distance. Optical pyrometers estimate temperature from the light intensity in a narrow band of the visible spectrum by

optical comparison of a glowing filament with an image of the hot body. Radiation pyrometers focus the body's heat radiation on a responsive thermal element such as a thermocouple.



Block diagram of radiation pyrometer

PYROMETRIC CONES: A series of cones used to indicate the temperature inside a furnace or kiln. The cones are made from different mixtures of clay, limestone, feldspars, etc and each one softens at a different temperature. The drooping of the vertex is an indication that the known softening temperature has been reached and thus the furnace temperature can be estimated.

PYROMETRY: The measurement of high temperature from the amount of radiation emitted using a pyrometer.

PYROPHONIC: Pyrophonic alloys are alloys that give sparks when struck.

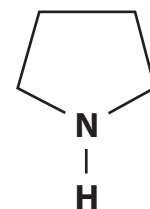
PYROPHORIC: Of or relating to a substance such as iron sulfide that will ignite spontaneously in air or a substance, which gives sparks when struck. Pyrophoric materials are often water reactive and will ignite when they contact water or humid air. They can be handled safely in atmospheres of argon or nitrogen. Examples of pyrophoric materials include alkyl aluminium and alkyl lithium compounds dissolved in hydrocarbon solvents such as hexane or heptane. Other examples include alkali metals, which are usually stored under hydrocarbons.

PYROXENE: Any of a group of dark, dense, rock-forming silicate minerals rich in calcium, iron and magnesium and commonly found in basalt. Pyroxenes are monoclinic or orthorhombic, have a chain structure, with prismatic cleavage close to 90° and are related to the amphiboles. The group includes augite, hypersthene and diopside, with composition varying as a mixture of $FeSiO_3$, $MgSiO_3$ and $CaSiO_3$. **Pyroxenite** is an igneous rock composed largely of pyroxene.

PYRRHOTITE: A bronze-brown iron sulfide mineral with the niccolite structure, occurring in basic igneous rocks in Scandinavia and Ontario. Pyrrhotite is nonstoichiometric with composition $Fe_{1-x}S$. Improved smelting techniques now make it possible to extract the iron.

PYRROLE (1-AZACYCLOPENTA-2,4-DIENE): A colourless, heterocyclic and aromatic organic compound having a five membered ring with the molecular formula C_4H_5N , molar mass 67.09 g mol⁻¹, density 0.967 g cm⁻³, melting point -23 °C (250 K) and boiling point 129 °C to 131 °C (402 K to 404 K). It is very volatile and darkens readily, when exposed to air. Its derivatives such as chlorophyll and chlorins are important biologically.

PYRROLIDINE: A saturated heterocyclic compound having one nitrogen atom in a five-membered ring, with the chemical formula C_4H_9N . It is found in certain plants and the ring structure is present in many alkaloids.



Chemical Structure of Pyrrolidine

PYRUVIC ACID: A colourless liquid organic acid, which smells like vinegar and can be formed by the distillation of tartaric

acid. It has the chemical formula CH_3COCOOH . A derivative of pyruvic acid known as phenylpyruvic acid occurs in the urine of people who suffer from phenylketonuria. This is a disorder in which pyruvate is an important intermediate in metabolism, being produced during glycolysis and converted to acetyl coenzyme A, required for the citric acid cycle (Kreb's cycle). Under anaerobic conditions pyruvate is converted to lactate or ethanol.

PYTHAGORAS OF SAMOS (C.569 BC-C.500 BC): Greek philosopher, mathematician and astronomer who founded a secretive philosophical and religious school, the Pythagorean school, in Croton, southern Italy. Pythagoras left no writings and virtually nothing is known about him as an individual. The theorem now known as Pythagoras's theorem which bears his name was known to the Babylonians 1000 years earlier but he may have been the first to prove it.

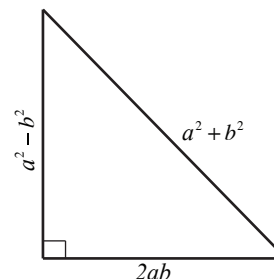
PYTHAGOREAN MEANS: They consist of the arithmetic mean (A), the geometric mean (G) and the harmonic mean (H) with the following properties: value preservation, first order homogeneity, invariance under exchange and averaging.

PYTHAGORAS' THEOREM: A proved geometric proposition stating that the square of the longest side (the hypotenuse) of a right-angled triangle is equal to the sum of the squares of the other two sides or a theorem that states that in a right-angled triangle, the area of the square on the hypotenuse (the longest side) is equal to the sum of the areas of the squares drawn on the other two sides. If the hypotenuse is c units long and the lengths of the other sides are a and b , then $c^2 = a^2 + b^2$. It is named after the Greek mathematician Pythagoras (c. 570 BC- c. 495 BC)

PYTHAGOREAN TRIANGLE: A right triangle whose sides are integers and whose side lengths (a, b, c) form a Pythagorean triple or a right triangle whose sides form a Pythagorean triple. A Pythagorean triangle with greatest common divisor or $\text{GCD}(a, b, c) = 1$ is known as a primitive right triangle. The inradius r of a Pythagorean triangle is always a whole number since $r = \frac{1}{2}(a + b - c)$. The area (A) of such a triangle is also a whole number since for primitive Pythagorean triples, one of a or b must be even and for imprimitive triples, both a and b are even, so $A = \frac{1}{2}ab$ is always a positive integer.

PYTHAGOREAN TRIPLES: A set of three whole numbers that satisfies Pythagoras's theorem, in that the squares of two of the numbers add up to the square of the third number.

Examples include (3, 4, 5), (5, 12, 13) and (7, 24, 25). These are called primitive triplets because they have no common divisors. If the members of a primitive triplet are multiplied by the same integer the result is a new (but not primitive) triplet. If a, b are integers, where $a > b$ and both a and b are greater than zero, Pythagorean triples can be generated as $2ab$, $a^2 - b^2$ and $a^2 + b^2$ being the three sides of the right-angled triangle.



PYTHON: 1. The common name for a group of non-venomous constricting snakes belonging to Pythonidae family, found in tropical and subtropical Asia, Africa and Australia in both wet and dry environments. Pythons range in size from 1 to 10 metres. Some pythons are capable of swimming and climbing trees. Pythons feed on wild animals such as lizards, monkeys, deers and antelopes. They kill their prey by constriction and asphyxiation and the prey is swallowed whole. A python's teeth are angled backward (recurved) to keep hold on prey during swallowing. The larger the meal, the longer it takes to digest. The entire animal is digested in the snake's stomach except for fur or feathers. Pythons will not usually attack humans unless startled or provoked, although females protecting their eggs can be aggressive. Python skin is used to make clothing such as vests, belts, boots, shoes or fashion accessories such as handbags. 2. A general purpose high-level programming interpreted language designed to be easily read and simple to implement. It supports many programming paradigms such as object oriented, procedural, functional and imperative. It is open source, which means it is free to use, even for commercial applications. Python can run on Mac, Windows and Unix systems and has also been ported to Java and .NET virtual machines.

PYXIS (PYXIDIUM): Dehiscing or splitting along a circumference with the upper part coming off as a lid.



Q: Symbol for the set of rational numbers.

QR ALGORITHM: In mathematics, it is a stable, reasonably efficient, numerical method of calculating the eigenvalues and eigenvectors of a matrix.

Q FEVER (QUERY FEVER): An infectious disease caused by a small bacterium *Coxiella burnetii*. It is characterised by fever, headache and often dry cough and chest pain. It is transmitted by ticks from various farm animals and is common among farm workers and veterinarians.

Q-SWITCHED LASER: A laser in which the state of the device introducing important losses in the resonant cavity and preventing lasing operation is suddenly switched to a state, where the device introduces very low losses. This increases rapidly the Quality factor of the cavity, allowing the build-up of a short and very intense laser pulse. Typical pulse durations are in the nanoseconds range. The Q-switching may be active (a rotating mirror or electro-optic device) or passive (a saturable absorber).

Q-VALUE: The total energy released in a reaction. It is the difference between the kinetic energy of the products and the kinetic energy of the original particles (reactants) in any nuclear reaction. It is related with the following equations: $Q = E(\text{reactants}) - E(\text{products})$. When the Q-value is positive, the reaction is **exothermic** and energy is released. If the Q-value is negative the reaction is **endothermic** and energy is absorbed. The Q-value of a decay process is, therefore, equal to the energy released in the decay process. Q-value also finds application in particle physics, where Q-value is the kinetic energy released in the decay of the particle at rest. For example, the decay value for neutron can be expressed as $Q = m_n - m_p - m_e$, where m_n is the mass of the neutron, m_p is the mass of the proton, m_e is the mass of the electron antineutrino and m_e is the mass of the electron.

Q.931: A signalling protocol used over voice over IP communications. It includes all the parameters required for the establishment of the communication link, setup, call proceedings, destination, ring alert, etc.

QED: An acronym for *quod erat demonstrandum*, which is Latin for "which was to be shown". It is put at the end of a proof to signify that the proof has been completed.

QEF: An acronym for *quod erat faciendum*, which is Latin for "that which was to be done". This is often written at the end of a construction to indicate that whatever was required has been achieved.

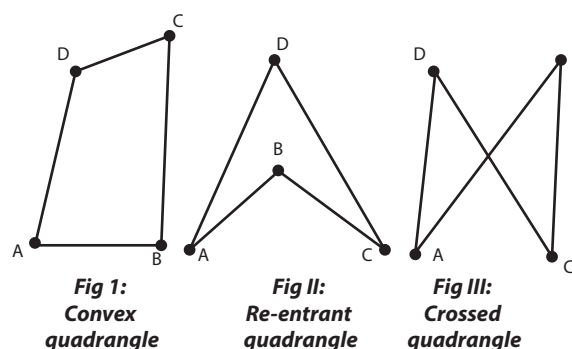
QUAD-: 1. A prefix denoting the numeral four. For example, a quadrangle is two-dimensional figure that consists of four points connected by straight lines. 2. In chemistry, denoting four kinetically important substrates or products in an enzyme mechanism. The numbers of kinetically important substrates or products in an enzyme mechanism are designated by Uni (1), Bi (2), Ter (3) or **Quad** (4).

QUAD SPEED: In computing, a CD-ROM drive that transfers data at the rate of 600 kilobytes per second, which is four times as fast as the original models. The data access time is about 125 milliseconds (ms).

QUAD-CORE CPU: A CPU having four independent processing cores in a single chip. It is similar to a dual-core CPU, but has four separate processors rather than two, which can process instructions at the same time. Examples include the Intel Core 2 Quad, Intel Nehalem and AMD Phenom X4 processors.

QUADBIT (NIBBLE): One of the 16 possible 4 bit combinations (four binary digits or half a byte). It is used in some communication signals.

QUADRANGLE: A two-dimensional figure such as a rectangle that consists of four points connected by straight lines. It is **convex** if both diagonals lie inside, **re-entrant** if one lies outside and **crossed** if both lie outside. Examples are given below:



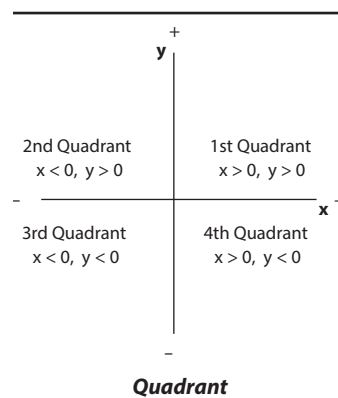
QUADRANGULAR: Having four angles.

QUADRANGULAR PRISM: A prism whose base is quadrilateral or a polygon having four sides.

QUADRANGULAR PYRAMID: A pyramid whose base is a quadrilateral (polygon having four sides).

QUADRANT: 1. In mathematics, it refers to any of the following:

(i) A 90° arc representing one fourth of the circumference of a circle. (ii) Any one of the four sections into which the perpendicular axes of a coordinate system divides an area of a two-dimensional surface. (iii) Any of the four approximately equal parts into which an area or a surface is divided by two real or imaginary perpendicular lines. 2. In astronomy, it is an instrument with a movable sighting mechanism attached to a 90° arc, formerly used by navigators and astronomers for measuring altitude and angles of celestial bodies. 3. In engineering, it is a part of a machine that converts horizontal reciprocating motion to vertical reciprocating motion.



QUADRANTAL ANGLE: An angle in standard position whose terminal side coincides with one of the coordinate axes.

QUADRAT: 1. A square frame or structure with sides 1 metre long, used in sampling for the purpose of studying the distribution of plants and/or animals in a particular place or area such as a field, shore or mountain side. 2. A small plot of land set aside for plant and/or animal population studies.

QUADRATE: 1. A bony or cartilaginous structure connecting the lower and upper jaws of vertebrates that articulates with the lower jaw at the side of the skull, found in birds, fishes, reptiles and amphibians. In mammals, this structure has evolved into the incus, a small bone of the middle ear. 2. Having four equal sides and four angles; examples are a square, a cube or a rectangle.